Advanced Algebra

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Chapter 0

Elementary Definitions and Conventions

0.1 Rings and Ideals

Definition 0.1.1. Rings
A *ring* is an abelian additive group with a multiplicative operation $(a, b) \mapsto ab$ satisfying the associativity and distributivity. Moreover, we say that a ring R is

- *commutative* if for any $a, b \in R$, $ab = ba$;
- *unital* if there exists an multiplicative identity 1_R. Furthermore, an element u in a unital ring R is called a *unit* (or an *invertible element*) if there exists $v \in R$, s.t.: $uv = 1_R$, which is denoted as $v = u^{-1}$ and called the *inverse* of u. We denote by $U(R)$ the set of all units in R and then we get the definition of *field*: a commutative unital ring F is a *field* if $F \setminus \{0_F\} = U(F)$.

Convention: In this book, every ring is treated as a commutative unital ring. Furthermore, in this book, unless otherwise specified, the letter "R" represents a (commutative unital) ring.

Definition 0.1.2. Ideals

- A subset $I \subseteq R$ of R is called an *ideal* of R, denoted as $I \lhd R$, if I is additive subgroup of R and if for any $r \in R$, any $x \in I$, we have $rx \in I$.
- An ideal $I \lhd R$ is said to be *generated by a subset* $S \subseteq R$ if for any $x \in I$, we can write x in the form
$$x = \sum_{i=1}^{n} r_i s_i$$
where $r_i \in R, s_i \in S$. Then we denote I as (S).
Convention: We write as 0 the ideal generated by the empty set $\emptyset$.
- An ideal is *principal* if it can be generated by one single element.
- An ideal $I \lhd R$ is *prime* if I is proper, i.e.: $I \neq R$, and if $xy \in I$ implies $x \in I$ or $y \in I$.

- An ideal is *maximal* if it is proper and if it is NOT contained in any other proper ideal.

Proposition 0.1.3. *Prime Ideal-Maximal Ideal Proposition*

1. $I \triangleleft R$ *is prime* $\Leftrightarrow \forall$ *ideals* $J, K \triangleleft R$ *with* $J, K \subseteq I$, *we have* $J \subseteq I$ *or* $K \subseteq I \Leftrightarrow R/I$ *is an ID.*
2. $I \triangleleft R$ *is maximal* $\Leftrightarrow R/I$ *is a field.*

Proof. Omitted. $\square$

Definition 0.1.4. Domain
R is called a *domain* if $0 = (\emptyset)$ is prime.

Definition 0.1.5. Prime Elements
An element $p \in R$ if *prime* if (p) is a prime ideal, or equivalently, if $p \notin U(R)$ and if $p \mid xy \Rightarrow p \mid x$ or $p \mid y$.

Definition 0.1.6. Ring Homomorphisms
A map $\varphi : R \to S$ from a ring R to a ring S is called a *ring homomorphism* if it preserves addition and multiplication. Furthermore, if $\varphi(1_R) = 1_S$, then we say φ is *unital.*

Convention: In this book, we will omit the adjectives "unital" and "ring", i.e.: "homomorphism" means "unital ring homomorphism".

Definition 0.1.7. Let $r, s, r_1, \cdots, r_n \in R$ be elements of R.

- r is an *idempotent* if $r^2 = r$.
- r and s are *orthogonal* if $rs = 0$.
- $\{r_1, \cdots, r_n\}$ is a *complete set of orthogonal idempotents* in R if every r_i is an idempotent in R, if $\forall i \neq j \in \{1, \cdots, n\}$, $r_i r_j = 0$, i.e.: they are orthogonal, and if $\sum_{i=1}^{n} r_i = 1_R$.

Example: $\{(1_{R_1}, 0, \cdots, 0), (0, 1_{R_2}, 0, \cdots, 0), \cdots, (0, \cdots, 0, 1_{R_n})\} \subseteq R_1 \times \cdots \times R_n$ is a *complete set of orthogonal idempotents* in $R_1 \times \cdots \times R_n$.

Proposition 0.1.8. *Complete Set of Orthogonal Idempotents-Decomposition Proposition*
If $\{e_1 \cdots, e_n\}$ *is a complete set of orthogonal idempotents in* R, *then*

$$R \cong Re_1 \times \cdots \times Re_n$$

is a direct product decomposition.

Proof. We only deal with the case $n = 2$. Consider the map $\varphi : R \to Re_1 \times Re_2$ given by $\varphi(r) := (re_1, re_2)$.

- φ *is a homomorphism:* For any $r, s \in R$, let $r = r_1e_1 + r_2e_2$ and $s = s_1e_1 + s_2e_2$.
  - $\varphi(r + s) := (r_1e_1 + s_1e_1, r_2e_2 + s_2e_2) = (r_1e_1, , r_2e_2) + (s_1e_1, s_2e_2) := \varphi(r) + \varphi(s)$.
  - $\varphi(rs) := ((rs)e_1, (rs)e_2) = ((re_1)(se_1), (re_2)(se_2)) = (re_1, se_1)(re_2, se_2) := \varphi(r)\varphi(s)$.

- φ *is injective:* $\ker\varphi := \{r \in R \mid re_1 = 0, re_2 = 0\} = \{r \in R \mid r_1 = 0, r_2 = 0\} = \{0\}$.
- φ *is surjective:* $\forall (x,y) \in Re_1 \times Re_2$, $\exists r_1, r_2 \in R$, s.t.: $x = r_1e_1, y = r_2e_2$. Then $\varphi(r_1e_1 + r_2e_2) := ((r_1e_1 + r_2e_2)e_1, (r_1e_1 + r_2e_2)e_2) = (r_1e_1, r_2e_2) = (x,y)$.

$\square$

Remark: Conversely, if $R \cong R_1 \times \cdots \times R_n$ can be decomposed as a direct product of finitely many (nontrivial) subrings, then $\{e_i := (0, \cdots, 0, \underbrace{1}_{ith\ position}, 0, \cdots, 0)\}_{i=1}^n$ is a complete set of orthogonal idenpotents.

Definition 0.1.9. Commutative Algebra

- A *commutative algebra over R* (or *commutative R-algebra*) is a ring S together with a homomorphism $\alpha : R \to S$, namely (S, α).

  e.g.:

  - Any ring S together with the homomorphism $\alpha : n \mapsto \begin{cases} \overbrace{1 + \cdots + 1}^{n-times} & , if\ n > 0 \\ 0 & , if\ n = 0 \\ \underbrace{(-1) + \cdots + (-1)}_{(-n)-times} & , if\ n < 0 \end{cases}$

    is a commutative $\mathbb{Z}$-algebra. Furthermore, (S, α) is the *unique* commutative $\mathbb{Z}$-algebra.
  - $(R[x_1, \cdots, x_n], {}_{r \mapsto r + 0x_1 + \cdots + 0x_n}^{R \to R[x_1, \cdots, x_n]})$ is a commutative R-algebra.
- We usually omit the homomorphism α and write $rs := \alpha(r)s$ for $r \in R, s \in S$.
- A *subalgebra* of S is a subring S' of S that contains $\alpha(R)$.

  e.g.: $R[x_1^2, x_2^3]$ is a R-subalgebra of $R[x_1, x_2]$.
- A *homomorphism of R-algebra* $\varphi : S \to T$ is a homomorphism of rings such that $\varphi(rs) = r\varphi(s)$ $\forall r \in R, s \in S$. More precisely, $\varphi : (S, \alpha_S) \to (T, \alpha_T)$ is a homomorphism of rings from S to T such that $\varphi(\alpha_S(r)s) = \alpha_T(r)\varphi(s)$ $\forall r \in R, s \in S$.

  e.g.: Given $r \in R$, $\varphi : {}_{f \mapsto f(r)}^{R[x] \to R}$ is a homomorphism of R-algebras.

Definition 0.1.10. Polynomial Rings
Let $R[x_1, \cdots, x_n]$ be a *polynomial ring over R*.

- The elements in R are referred to as *scalars*.
- A *monomial* is a product of variables, whose *degree* is defined as the number of these factors (counting repeats), e.g.: $\deg(x_1^2x_2^3) = \deg(x_1x_1x_2x_2x_2) = 5$. By *convention*, we regard 1 as the empty product, i.e.: 1 is the unique monomial of degree 0.
- A *term* is a scalar timing a monomial, i.e.: of the form:

$$r \cdot x_1^{a_1} \cdots x_n^{a_n} \quad where\ r \in R, a_i \in \mathbb{N} \cup \{0\}$$

- $f \in R[x_1, \cdots, x_n]$ is said to be *homogeneous* if $f = 0$ or if the monomials in the terms of f all have the same degree. We also use the word "*form*" to mean "homogeneous polynomial".

0.2 Unique Factorization

Convention: In this book, we say that R is *factorial* if it is a UFD.

Theorem 0.2.1. *PID $\Rightarrow$ FD $\Rightarrow$ UFD.*

Proof. Outline: PID $\Rightarrow$ ACCP $\Rightarrow$ FD. $\square$

Chapter 1

Introduction to Module Theory

1.1 Basic Definitions and Examples

Definition 1.1.1. Modules
An *R-module* M is an abelian additive group with an action of R on M, that is a map $\substack{R\times M \to M \\ (r,m) \mapsto rm}$, satisfying that: for any $r, s \in R$, any $m, n \in M$,

- $r(sm) = (rs)m$
- $(r + s)m = rm + sm$
- $r(m + n) = rm + rn$
- $1_R m = m$

Remark: Modules over field F and vector spaces over a field F is the same.
Convention: In this book, unless otherwise specified, the letter "M" represents an R-module.

Definition 1.1.2. Submodules
A R-submodule of M is a subgroup N of M which is closed under the action of ring elments, i.e.: for any $r \in R$, any $n \in N$, we have $rn \in N$. Moreover, we call 0 the trivial R-submodule.

Example:

- $M = R$ is an R-module, where the action is just the usual multiplication in R. In particular, every field can be considered as a (1-dimensional) vector space over itself. Furthermore, when R is considered as an R-module, then the submodules of R are precisely the ideals of R.
- Let $R = F$ be a field and let $n \in \mathbb{N}_+$. Consider

  $$F^n := \{(a_1, \cdots, a_n) \mid a_i \in F, \forall i\}$$

  which is called an *affine n-space over F*. Make F^n into an n-dimensional vector space over F by defining addition and scalar multiplication componentwise: for any $a_i, b_i \in F, \alpha \in F$, we define:

  - addition by $(a_1, \cdots, a_n) + (b_1, \cdots, b_n) := (a_1 + b_1, \cdots, a_n + b_n)$ and
  - multiplication by $\alpha(a_1, \cdots, a_n) := (\alpha a_1, \cdots, \alpha a_n)$.

- Let $n \in \mathbb{N}_+$. Define

$$R^n := \{(a_1, \cdots, a_n) \mid a_i \in R, \forall i\}$$

  with the componentwise addition and multiplication similar as above, which is called the *free module of rank n over R*. An obvious submodule of R^n is given by the ith component := the set of all n-tuples with arbitrary ring elements in the ith position and zeros in the jth position for all $j \neq i$.

- If S is a subring of R, then an R-module M is automatically a S-module, e.g.: $\mathbb{R}$ is an $\mathbb{R}$-module, a $\mathbb{Q}$-module and a $\mathbb{Z}$-module.

- Let $I \lhd R$ be an ideal that *annihilates* M, i.e.: for any $a \in I$, any $m \in M$, $am = 0$. Then M is an (R/I)-module with an action of R/I on M defined as follows: $\forall m \in M, \forall r + I \in R/I$, we define

$$(r + I)m := rm$$

  whose well-defined-ness is easy to check by the hypothesis that I annihilates M. In particular, if I is a maximal ideal of R, then M is a vector space over R/I.

- We record the next two examples as the later two subsections.

Proposition 1.1.3. *Submodule Criterion*
A subset $N \subseteq M$ of an R-module M is a submodule of M iff

1. *$N \neq \emptyset$*
2. *$\forall x, y \in N, \forall r \in R$, we have $x + ry \in N$.*

Definition 1.1.4. R-algebra
An R-algebra is a ring A with a homomorphism $\varphi : R \to A$ such that

$$\varphi(R) \subseteq Z(A)$$

where $Z(A) := \{a \in A \mid ar = ra, \forall r \in A\}$ denotes the center of A.

Remark: If A is an R-algebra, then it is easy to check that A is an R-module with the action of R on A defined by

$$ra := \varphi(r) \cdot_A a$$

Definition 1.1.5. Algebra Homomorphisms and Isomorphisms
Let A, B be R-algebras. An *R-algebra homomorphism* (or *isomorphism*) is a homomorphism (respectively isomorphism) $\varphi : A \to B$ such that $\varphi(r \cdot a) := r \cdot \varphi(a)$ for all $r \in R, a \in A$.

Example:

- Any ring with identity (may **NOT** commutative) is a $\mathbb{Z}$-algebra.

- A commutative unital ring A is an R-algebra for any subring R of A.

1.1.1 $\mathbb{Z}$-modules

Let $R = \mathbb{Z}$ and let $(A, +)$ be any abelian additive group. Define the action of $\mathbb{Z}$ on A by

$$na := \begin{cases} \overbrace{a + \cdots + a}^{n-times} & , if\ n > 0 \\ 0 & , if\ n = 0 \\ \underbrace{(-a) + \cdots + (-a)}_{(-n)-times} & , if\ n < 0 \end{cases}$$

and hence A is a $\mathbb{Z}$-module. Conversely, every $\mathbb{Z}$-module, by def, is an abelian group. Hence,

$$\mathbb{Z}-modules \quad \Leftrightarrow \quad abelian\ groups$$

Furthermore, we also have

$$\mathbb{Z}-submodules \quad \Leftrightarrow \quad abelian\ subgroups$$

If A is an abelian additive group containing an element x of order $n < \infty$, then $nx = 0$. Hence, a $\mathbb{Z}$-module may have nonzero elements x such that $nx = 0$ for some nonzero ring element n. In particular, if A has order m, then by Lagrange's Theorem, for any $x \in A$, $mx = 0$. Therefore, A is a $(\mathbb{Z}/m\mathbb{Z})$-module. Furthermore, if $m = p$ is prime, then A is a vector space over the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

1.1.2 $F[x]$-modules

Let F be a field, let $R = F[x]$ be the polynomial ring, let V be a vector space over F and let T be a linear transformation from V to V. Then we're already seen that V is an F-module.

Proposition 1.1.6. *Vector Spaces-Polynomial Rings-Modules Proposition*
V can be regarded as an $F[x]$-module.

Proof. We only find the "legal" action of $F[x]$ on V here and omit the check of axioms of modules. We need some preparations:

1. For any $n \in \mathbb{N} \cup \{0\}$, we define

$$T^n := \begin{cases} id_V & , if\ n = 0 \\ T \circ T \circ \cdots \circ T\ (n\ times) & , if\ n \in \mathbb{N}_+ \end{cases}$$

   which is again a linear transformation from V to V.

2. It's easy to check that the linear combinations of linear transformations are again linear transformations.

Now we can define the action of $F[x]$ on V: for any $a_i \in F$, any $v \in V$, we define:

$$(a_n x^n + \cdots + a_1 x + a_0)v := (a_n T^n + \cdots + a_1 T^1 + a_0 T^0)(v) \overset{2}{=} a_n T^n(v) + \cdots + a_1 T(v) + a_0 v$$

□

Remark:

- Note that F is naturally the subring of $F[x]$ (the constant polynomials) and by the first preparation in the proof of Vector Spaces-Polynomial Rings-Modules Proposition, the action of $F[x]$ on V is consistent with the action of F on V. Hence, the def of the action of $F[x]$ on V is an extension of the action of F on V.

  Example: Let $V = F^n$ be the affine n-space, defined in the second example of modules and submodules, and let T be the "shift operator"

  $$T(a_1, a_2, \cdots, a_n) := (a_2, a_3, \cdots, a_n, 0)$$

  Then given $k \in \mathbb{N} \cup \{0\}$ and $i \in \{1, \cdots, n\}$,

  $$T^k(e_i) := \begin{cases} e_{i-k} & , if\ i > k \\ 0 & , if\ i \leq k \end{cases}$$

  and hence, for example, if $m < n$,

  $$(a_m x^m + \cdots + a_1 x + a_0)e_n := a_m T^m(e_n) + \cdots + a_1 T(e_n) + a_0 e_n = (0, \cdots, 0, a_m, \cdots, a_1, a_0)$$

  From this, we can determine the action of any polynomial on any vector in F^n.

  Conversely, if V is an $F[x]$-module, then V is an F-module and the action of the ring element x on V is a linear transformation from V to V. Hence, there is a bijection between the collection of $F[x]$-modules and the collection of all pairs $(V, T : V \to V)$:

  $$\{V \text{ an } F[x]\text{-module}\} \longleftrightarrow \left\{\begin{matrix} V \text{ a vector space over } F \\ \text{and} \\ T : V \to V \text{ a linear transformation} \end{matrix}\right\}$$

  given by the element x acts on V as the linear transformation T.

- Consider the $F[x]$-submodules of V where V is any $F[x]$-module and T is the linear transformation:

$V \to V$ given by the action of x. An $F[x]$-submodule W of V must be an F-submodule of V, i.e.: W must be a vector subspace of V. For any $w \in W$, $T(w) \in W$, i.e.: W is *T-stable* (or *T-invariant*) and hence $T^n(W) \subseteq W \ \forall n \in \mathbb{N} \cup \{0\}$. Therefore, if W is *T-stable* (or *T-invariant*), then the action of $F[x]$ on W is closed in W. So we see that

$$the\ F[x] - submodules\ of\ V \Leftrightarrow the\ T - stable\ vector\ subspaces\ of\ V$$

i.e.: we have the bijection:

$$\{W \text{ an } F[x]\text{-submodule of } V\} \longleftrightarrow \left\{\begin{matrix} W \text{ a vector subspace of } V \\ \text{and} \\ W \text{ is } T\text{-stable} \end{matrix}\right\}$$

e.g.: If T is the shift operator defined on affine n-space above and $k \in \{0, 1, \cdots, n\}$, then the subspace

$$U_k := \{(a_1, \cdots, a_k, 0, \cdots, 0) \mid a_i \in F\}$$

is clearly T-stable and hence is an $F[x]$-submodule of V.

Definition 1.1.7. Annihilators
The *annihilater* of M is defined as

$$annM := \{r \in R \mid rM = 0\}$$

Example: R/I is an R-module and $annR/I = I$.

Definition 1.1.8.

- Let $I, J \lhd R$ be ideals of R. We define the *ideal quotient* by

$$(I : J) := \{r \in R \mid rJ \subseteq I\}$$

(which can be supposed to suggest division: $I = (i), J = (j)$ with i nonzerodivisor).

- Then we extend the ideal quotients to submodules M, N of R-module P and write

$$(M : N) := \{r \in R \mid rN \subseteq M\}$$

- Let $I \lhd R$ be an ideal of R and let M be a submodule of R-module P. Then we define

$$(M : I) = (M :_P I) := \{p \in P \mid Ip \subseteq M\} \subseteq P$$

which is a submodule of P.

1.2 Module Homomorphisms

Definition 1.2.1. Module Homomorphisms

- A *homomorphism of R-modules* is a homomorphism $\varphi : M \to N$ of abelian groups such that $\varphi(rm) = r\varphi(m)$.
- If φ is injective, then we say it is a *monomorphism*; if φ is surjective, then we say it is a *epimorphism*; if φ is bijective, then we say it is a *isomorphism*.
- Similarly as the ring homomorphisms, we define the kernel and image of module homomorphisms.
- Let M, N be R-modules. Define $Hom_R(M, N)$ to be the set of all R-module homomorphisms from M into N.

Proposition 1.2.2. *Kernels and images of R-module homomorphisms are submodules.*

Example:

- R-module homomorphisms need NOT be ring homomorphisms and ring homomorphisms need NOT be R-module homomorphisms:
  - $R = \mathbb{Z}$. The $\mathbb{Z}$-mordule homomorphism $x \mapsto 2x$ is NOT a ring homomorphism.
  - $R = F[x]$. The ring homomorphism $f(x) \mapsto f(x^2)$ is NOT an $F[x]$-module homomorphism.
- Let $n \in \mathbb{N}_+$ and let $M = R^n$. For any $i \in \{1, \cdots, n\}$, the projection map $\pi_i : R^n \to R$ given by

$$\pi_i(x_1, \cdots, x_n) := x_i$$

  is a surjective R-module homomorphism with $\ker \pi_i :=$ the submodule of n-tuples which have a zero in position i.
- If $R = F$ is a field, then F-module homomorphisms are called linear transformations.
- Recall that $\mathbb{Z}$-modules $\Leftrightarrow$ abelian groups. Similarly, we have:

$$\mathbb{Z}\text{-module homomorphisms} \Leftrightarrow \text{abelian group homomorphisms}$$

- Let $I \lhd R$ be an ideal of R and let M, N be R-modules annihilated by I, i.e.: $\forall i \in I, m \in M, n \in N$, we have $im = 0, in = 0$. Any R-module homomorphism from M to N is then automatically a homomorphism of (R/I)-modules. In particular, if A is an abelian additive group of order p prime, then any group homomorphism from A to itself is a $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$-module homomorphism, i.e.: is a linear transformation over the field $\mathbb{F}_p$. In particular, the group of all (group) automorphisms of A is the group of invertible linear transformations from A to itself: $GL(A)$.

Proposition 1.2.3. *Module Homomorphisms-Properties Proposition*
Let M, N, L be R-modules.

1. *A map $\varphi : M \to N$ is an R-module homomorphism iff $\forall x, y \in M, r \in R$, we have*

$$\varphi(rx + y) = r\varphi(x) + \varphi(y)$$

2. *Let $\Phi, \Psi \in Hom_R(M, N)$. Consider the map $\Phi + \Psi$ given by*

$$(\Phi + \Psi)(m) := \Phi(m) + \Psi(m), \forall m \in M$$

   Then $\Phi + \Psi \in Hom_R(M, N)$ and hence $(Hom_R(M, N), +)$ is an abelian group. Furthermore, given $r \in R$, define $r\Phi$ by

$$(r\Phi)(m) := r(\Phi(m)), \forall m \in M$$

   Then $r\Phi \in Hom_R(M, N)$ and hence this action of R on $Hom_R(M, N)$ gives that $Hom_R(M, N)$ is an R-module.

3. *Let $\varphi_1 \in Hom_R(L, M), \varphi_2 \in Hom_R(M, N)$. Then $\varphi_2 \circ \varphi_1 \in Hom_R(L, N)$.*
4. *With the addition as above and multiplication defined as map composition, $Hom_R(M, N)$ is a ring (commutative unital ring).*
5. *$Hom_R(M, M)$ is an R-algebra.*

Proof.

1. - ($\Rightarrow$) holds immediately by definition.
   - ($\Leftarrow$): Take $r = 1$ to check that φ is additive and take $y = 0$ to check that φ commutes with the action of R on M.
2. Omitted.
3. For any $r \in R$, any $x, y \in L$, we have

$$(\Psi \circ \Phi)(rx + y) = \Psi(\Phi(rx + y)) \stackrel{1}{=} \Psi(r\Phi(x) + \Phi(y)) \stackrel{1}{=} r(\Psi \circ \Phi)(x) + (\Psi \circ \Phi)(y)$$

   Hence, by 1, we're done.
4. Omitted.
5. Omitted.

$\square$

Definition 1.2.4. Endomorphisms
The ring $Hom_R(M, M)$ is called the *endomorphism ring* of M, denoted as $End_R(M) := Hom_R(M, M)$ and the elements of $End_R(M)$ are called *endomorphisms*.

Remark: There is a *natural map* from R to $End_R(M)$ given by $r \mapsto r \cdot id_R$. If the image of R is contained in $Z(End_R(M)) :=$ the center of $End_R(M)$, then $End_R(M)$ is an R-algebra. Note that $R = \mathbb{Z}, M = \mathbb{Z}/2\mathbb{Z}, r = 2 \Rightarrow rm = 0, \forall m \in M$. Hence, the natural map from R to $End_R(M)$, which can be shown that it is a ring homomorphism, may NOT be injective. But when $R = F$

is a field, this map is injective and the copy of R in $End_R(M)$ is called the (subring of) *scalar transformations.*

Proposition 1.2.5. *Quotient Module Proposition*
Let N be an R-submodule of an R-module M. The abelian (additive) quotient group M/N can be made into an R-module by defining an action of R on M/N by

$$r(x+N) := (rx) + N, \forall r \in R, \forall x + N \in M/N$$

The natural projection map $\pi : M \to M/N$ given by $x \mapsto x + N$ is an R-module homomorphism with kernel N.

Proof.

- Since M is an abelian additive group, then the quotient group M/N is defined and is an abelian additive group.
- The action of R on M/N is well-defined: $\forall x + N = y + N \in M/N$, i.e.: $y - x \in N$, since N is an R-submodule, then $ry - rx = r(y - x) \in N$ and hence $rx + N = ry + N$.

Therefore, M/N is an R-module.

- π is an R-module homomorphism, i.e.: $\pi(rm) = r\pi(m)$:

$$\pi(rm) := (rm) + N := r(m + N) := r\pi(m)$$

- The kernel of any module homomorphism is the same as its kernel when viewed as a homomorphism of the abelian group structures.

□

Definition 1.2.6. Sum of Submodules
Let A, B be R-submodules of an R-module M. The *sum* of A and B is defined as

$$A + B := \{a + b \mid a \in A, b \in B\}$$

Proposition 1.2.7. *Sum of Submodules-Property Proposition*
The sum of two submodules B is again a submodule and is the smallest submodule containing them.

Proof. The proof of the first statement is going to omitted. Let A, B be two submodules and let N be any submodule that contains both A and B. Then $\forall a \in A \subseteq N, b \in B \subseteq N$, by the additive closed-ness of submodule, $a + b \in N$ and hence $A + B \subseteq N$. □

Theorem 1.2.8. *Isomorphism Theorem for Modules*

- ***The First Isomorphism Theorem for Modules:*** *Let M, N be R-modules and let $\varphi : M \to N$ be an R-module homomorphism. Then $\ker\varphi$ is a submodule of M and we have the isomorphism:*

$$M/\ker\varphi \cong im\varphi$$

- ***The Second Isomorphism Theorem for Modules:*** *Let A, B be R-submodules of an R-modules M. Then we have the isomorphism:*

$$(A + B)/B \cong A/(A \cap B)$$

- ***The Third Isomorphism Theorem for Modules:*** *Let A, B be R-submodules of an R-module M such that $A \subseteq B$. Then we have the isomorphism:*

$$(M/A)/(B/A) \cong M/B$$

- ***The Fourth** (or **Lattice**) **Isomorphism Theorem for Modules:*** *Let N be an R-submodule of an R-module M. Then there is a bijection between the submodules of M containing N and the submodules of M/N*

1.3 Generation of Modules

Definition 1.3.1. Generation of Modules
Let $N_1, \cdots, N_n$ be R-submodules of an R-module M.

- We generalize the sum of two submodules into the n submodules case:

$$N_1 + \cdots + N_n := \{a_1 + \cdots + a_n \mid a_i \in N_i\}$$

  is the *sum* of $N_1, \cdots, N_n$.

- For any subset $A \subseteq M$ of M, we define the *submodule of M generated by A* as

$$RA := \{r_1a_1 + \cdots + r_ma_m \mid r_1, \cdots, r_m \in R, a_1, \cdots, a_m \in A, m \in \mathbb{N}_+\}$$

  (where by convention $RA := \{0\}$ if $A \neq \emptyset$). If A is the finite set $\{a_1, \cdots, a_n\}$, then we write:

$$RA = Ra_1 + \cdots + Ra_n$$

  If N is a submodule of M and if $N = RA$ for some $A \subseteq M$, then we call A a *set of generators* or *generating set for N* and we say that N is *generated by A*.A submodule N of M is *finitely generated* if N is generated by some finite subset of M.

- A submodule N of M is *cyclic* if there exists $a \in M$, s.t.: $N = Ra := R\{a\}$.

Remark:

- These definitions do **NOT** require that R is unital, but this condition ensures that A is contained in RA.
- It's easy to see using the Submodule Criterion that for any subset $A \subseteq M$ of M, RA is a submodule of M, and is the smallest submodule of M that contains A.

- In particular, for submodules $N_1, \cdots, N_n$ of M, $N_1 + \cdots + N_n$ is the submodule generated by the set $N_1 \cup \cdots \cup N_n$ and is the smallest submodule of M that contains N_i for all i. If $N_1, \cdots, N_n$ are generated by sets $A_1, \cdots, A_n$, respectively, then $N_1 + \cdots + N_n$ is generated by $A_1 \cup \cdots \cup A_n$.
- A submodule N of an R-module M may have many different generating sets (e.g.: the above remark). If N is finitely generated, then there is a smallest non-negative integer d such that N is generated by d elements (and no fewer), where every generating set consisting of d elements is called a *minimal set of generators for N* (it is NOT unique in general). If N is NOT finitely generated, then it need NOT have a minimal generating set.

Example:

- Let $R = \mathbb{Z}$ and let M be any $\mathbb{Z}$-module, that is, any abelian group. If $a \in M$, then $\mathbb{Z}a$ is precisely the cyclic subgroup of M generated by a, say $\langle a \rangle$.
- Let $M = R$. Note that R is a finitely generated, in fact cyclic, R-module since $R = R1$. Hence, we say I is a cyclic R-submodule of R-module R is precisely that I is a principal ideal of R. Therefore, a PID is a commutative unital ID with every R-submodule of R cyclic.
- Consider a vector space over F as an $F[x]$-module wia T. V is a cyclic $F[x]$-module with generator $v \Leftrightarrow V = \{p(x)v \mid p(x) \in F[x]\} \Leftrightarrow$ every elements of V can be written as an F-linear combination of elements of $\{T^n(v) \mid n \in \mathbb{N} \cup \{0\}\} \Leftrightarrow \{v, T(v), T^2(v), \cdots\}$ spans V as a vector space over F.

  For instance, if $T = id_V$ is the identity linear transformation from V to V or if $T = 0$ is the zero linear transformation, then $\{v, T(v), T^2(v), \cdots\}$ consists of one single elements v or 0, respectively. Hence, if $\dim_F(V) > 1$, then V can **NOT** be a cyclic $F[x]$-module corresponding $T = id_V$ or $T = 0$.

  For another example, let $V = F^n$ be an affine n-space and let T be the "shift operator". Recall that

  $$T^k(e_n) = \begin{cases} e_{n-k} & , if\ 1 \le k < n \\ 0 & , if\ k \ge n \end{cases}$$

  Hence, V is spanned by the elements $e_n, T(e_n), T^2(e_n), \cdots, T^{n-1}(e_n)$, that is, V is a cyclic $F[x]$-module with generator e_n.

1.4 Direct Products, (Direct Sums,) and Free Modules

Definition 1.4.1. Direct Product

Let $M_1, \cdots, M_k$ be R-modules. We define the *direct product* of $M_1, \cdots, M_k$ by

$$M_1 \times \cdots \times M_k := \{(m_1, \cdots, m_k) \mid m_i \in M_i\}$$

with addition and action of R defined componentwise.

Remark:

- The direct product of a collection of R-modules is again an R-module.
- In the finite case, the direct product is also referred to as the *(external) direct sum*, i.e.:

$$M_1 \times \cdots \times M_k = M_1 \oplus \cdots \oplus M_k$$

Proposition 1.4.2. *Module-Isomorphic-to-Direct Product of its Submodules Proposition*
Let $N_1, \cdots, N_k$ be submodules of an R-module M. The followings are equivalent:

1. *The map $\pi : N_1 \times \cdots \times N_k \to N_1 + \cdots + N_k$ given by*

$$\pi(a_1, \cdots, a_k) := a_1 + \cdots + a_k$$

is an isomorphism of R-modules.

2. *For any $j \in \{1, \cdots, k\}$, we have*

$$N_j \cap (N_1 + \cdots + \hat{N}_j + \cdots + N_k) = 0$$

3. *Every $x \in N_1 + \cdots + N_k$ can be uniquely written in the form $a_1 + \cdots + a_k$ with $a_i \in N_i$.*

Proof.

- $(1 \Rightarrow 2)$: Suppose on the contrary that 2 fails for some j and then we can pick $0 \neq a_j \in (N_1 + \cdots + \hat{N}_j + \cdots + N_k) \cap N_j$. Then $\forall i \neq j$, there exists $a_i \in N_i$, s.t.:

$$a_j = a_1 + \cdots + a_{j-1} + a_{j+1} + \cdots + a_k$$

Hence, there is a contradiction to the injectivity of π:

$$0 = a_1 + \cdots + a_{j-1} + (-a_j) + a_{j+1} + \cdots + a_k = \pi(a_1, \cdots, a_{j-1}, -a_j, a_{j+1}, \cdots, a_k) \neq \pi(0)$$

- $(2 \Rightarrow 3)$: It suffices to prove the uniqueness of the form. Suppose that $a_1+\cdots+a_k = b_1+\cdots+b_k$ for some $a_i, b_i \in N_i$. Then given any fixed j, we have

$$N_j \ni a_j - b_j = (b_1 - a_1) + \cdots + (b_{j-1} - a_{j-1}) + (b_{j+1} - a_{j+1}) + \cdots + (b_k - a_k) \in (N_1 + \cdots + \hat{N}_j + \cdots + N_k)$$

Hence, by 2, for any j, $a_j - b_j = 0$.

- $(3 \Rightarrow 1)$: Note that π is clearly a surjective R-module homomorphism. Since 3 implies the injectivity of π, then 1 holds.

□

Remark: If an R-module $M = N_1 + \cdots + N_k$ is the sum of submodules $N_1, \cdots, N_k$ of M satisfying the equivalent conditions of the Module-Isomorphic-to-Direct Product of its Submodules

Proposition, then M is said to be the *(internal) direct sum* of $N_1, \cdots, N_k$, written

$$M = N_1 \oplus \cdots \oplus N_k$$

(The reason why we identity the external direct sum and the interval direct sum is the Module-Isomorphic-to-Direct Product of its Submodules Proposition.1.)

Definition 1.4.3. Free Modules
An R-module F is said to be *free* on the subset $A \subseteq F$ of F if $\forall 0 \neq x \in F, \exists! 0 \neq r_1, \cdots, r_n \in R, \exists! a_1, \cdots, a_n \in A$, with some $n \in \mathbb{N}_+$, s.t.:

$$x = r_1 a_1 + \cdots + r_n a_n$$

In this situation, we say that A is a *basis* or *set of free generators for* F. And then we define the *rank* of $F :=$ the cardinality of A.

Remark: We need to pay attention on the difference of the uniqueness property of direct sums and the uniqueness property of free modules. Namely,

- in the direct sum of two modules, say $N_1 \oplus N_2$, each element can be written *uniquely* as $n_1 + n_2$, where the uniqueness is on the *module elements* n_1, n_2,
- while in the case of free modules, the uniqueness is on the *ring elements and the module elements*.

For example, let $R = \mathbb{Z}$ and let $N_1 = N_2 = \mathbb{Z}/2\mathbb{Z}$. Then each element of $N_1 \oplus N_2$ has a unique representation in the form $n_1 + n_2$, **but** $n_1 = 3n_1 = 5n_1 = \cdots$ and hence each element does NOT have a unique representation in the form $r_1 a_1 + r_2 a_2$. Therefore, $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ is NOT a free $\mathbb{Z}$-module on the set $\{(1,0),(0,1)\}$. Similarly, it is NOT free on any set.

Theorem 1.4.4. *Universal Property of Free Modules*
For any set A, there exists a free R-module $F(A)$ on the set A and $F(A)$ satisfies the following universal property: if M is any R-module and $\varphi : A \to M$ is any map of sets, then there exists an R-module homomorphism $\Phi : F(A) \to M$, s.t.: $\Phi(a) = \varphi(a), \forall a \in A$, i.e.: the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, hook] \arrow[dr, "\varphi"'] & P \arrow[d, "\Phi"] \\ & X_i \end{tikzcd}$$

When $A = \{a_1, \cdots, a_n\}$ is finite, $F(A) = Ra_1 \oplus \cdots \oplus Ra_n \cong R^n$.

Proof. If $A = \emptyset$, then let $F(A) := \{0\}$. Otherwise, if $A \neq \emptyset$, then let $F(A)$ be the collection of all set functions $f : A \to R$ such that $f(a) = 0$ for all but finitely many $a \in A$. Make $F(A)$ into an R-module by pointwise addition of functions and pointwise multiplication of a ring elements times a function, i.e.: for any $a \in A$, any $r \in R$, any $f, g \in F(A)$, we define:

$$(f+g)(a) := f(a) + g(a), (rf)(a) := r(f(a))$$

(The details of checking the axioms of R-module are omitted.) Identity A as a subset of $F(A)$ by $a \mapsto f_a$ where f_a is the function defined by

$$f_a(x) := \begin{cases} 1 & , if\ x = a \\ 0 & , if\ x \in A \setminus \{a\} \end{cases}$$

Then, in this way, we can refer to $F(A)$ as all finite R-linear combinations of elements of A by identifying each function f with the sum $r_1 a_1 + \cdots + r_n a_n$ where f takes on the value r_i at a_i and zero at all other elements of A. Moreover, each element of $F(A)$ has a unique expression as such a formal sum.

Now define $\Phi : F(A) \to M$ by

$$\Phi(\sum_{i=1}^{n} r_i a_i) := \sum_{i=1}^{n} r_i \varphi(a_i)$$

By the uniqueness of the expression for the elements of $F(A)$ as linear combinations of a_i, we see that Φ is a well-defined R-module homomorphism (where the details are omitted). Then by def, $\Phi|_A = \varphi$.

As for the uniqueness of Φ, since $F(A)$ is generated by A and since we know the values of an R-module homomorphism on A, then its values on every element of $F(A)$ are uniquely determined.

When $A = \{a_1, \cdots, a_n\}$ is finite, the Module-Isomorphic-to-Direct Product of its Submodule Proposition.3 shows that $F(A) = Ra_1 \oplus \cdots \oplus Ra_n$. Since the map: $R \overset{\cong}{\to} Ra_i$ given by $r \mapsto ra_i$ is an isomorphism for every i, then $F(A) \cong R^n$. $\square$

Corollary 1.4.5. *of the Universal Property of Free Modules*

1. *If F_1 and F_2 are free modules on the same set A, then there exists an unique isomorphism between F_1 and F_2 which is the identity map on A.*
2. *If F is any free R-module with basis A, then $F \cong F(A)$. In particular, F enjoys the same universal property with respect with A as $F(A)$ does in the Universal Property of Free Module.*

Remark: When $R = \mathbb{Z}$, the free module on a set A is called the *free abelian group on A*. If $|A| = n$, then $F(A)$ is called the free abelian group of *rank n* and is isomorphic to $\underbrace{\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}}_{n\ times}$.

1.5 Tensor Products

1.5.1 Tensor Products of Modules

Definition 1.5.1. Bilinear Maps of Modules

Let M, N, P be R-modules. A map $\alpha : M \times N \to P$ is called *bilinear* if for any $m \in M$, any $n \in N$, the maps

$$\alpha_n := \alpha(\cdot, n) : M \to P \quad \textit{and} \quad \alpha_m := \alpha(m, \cdot) : N \to P$$

are R-linear.

Definition 1.5.2. Tensor Product of Modules
Let M, N be R-modules. A *tensor product* of M and N over R is an R-module T together with a bilinear map $\tau : M \times N \to T$ satisfying the following *universal property*:
for any bilinear map $\alpha : M \times N \to P$ to any module P, there exists a unique linear map $\Phi : T \to P$ such that $\alpha = \Phi \circ \tau$, i.e.: the following diagram commutes:

$$\begin{tikzcd} M \times N \arrow[r, "\alpha"] \arrow[d, "\tau"'] & P \\ T \arrow[ur, dashed, "\exists!\Phi"'] & \end{tikzcd}$$

The elements of a tensor product are called *tensors*.

Proposition 1.5.3. *Uniqueness of Tensor Products of Modules*
A tensor product is unique up to unique isomorphism in the following sense:
if T_1 and T_2 together with linear maps $\tau_1 : M \times N \to T_1$ respectively $\tau_2 : M \times N \to T_2$ are two tensor products for M and N over R, then there is a unique R-module isomorphism $\varphi : T_1 \to T_2$, s.t.: $\tau_2 = \varphi \circ \tau_1$.

Proof. For $T = T_1, P = T_2$, we have the following commutative diagram:

$$\begin{tikzcd} M \times N \arrow[r, "\tau_2"] \arrow[d, "\tau_1"'] & T_2 \\ T_1 \arrow[ur, dashed, "\exists!\varphi"'] & \end{tikzcd}$$

where φ is the unique R-module homomorphism from T_1 to T_2. And for $T = T_2, P = T_1$, we have the following commutative diagram:

$$\begin{tikzcd} M \times N \arrow[r, "\tau_1"] \arrow[d, "\tau_2"'] & T_1 \\ T_2 \arrow[ur, dashed, "\exists!\psi"'] & \end{tikzcd}$$

where ψ is the unique R-module homomorphism from T_2 to T_1. Now for $T = P = T_1$, we have the following commutative diagram:

$$\begin{tikzcd}
M \times N \arrow[r, "\tau_1"] \arrow[d, "\tau_1"'] & T_1 \\
T_1 \arrow[ur, dashed, bend left, "\mathrm{id}_{T_1}"] \arrow[ur, dashed, bend right, "\psi\circ\varphi"'] &
\end{tikzcd}$$

i.e.: $\tau_1 = \psi \circ \varphi \circ \tau_1$ and $\tau_1 = id_{T_1} \circ \tau_1$. Hence, by the uniqueness part of the universal property, we can conclude that $\psi \circ \varphi = id_{T_1}$. Similarly for $\varphi \circ \psi = id_{T_2}$. Therefore, φ is an R-module isomorphism. $\square$

Proposition 1.5.4. *Existence pf Tensor Products of Modules*
Any two R-modules have a tensor product of them.

Proof. Let M, N be R-modules. Denote by F the R-module of all finite linear combinations of elements of $M \times N$, i.e.:

$$F := \{a_1(m_1, n_1) + \cdots + a_k(m_k, n_k) \mid k \in \mathbb{N}_+, a_i \in R, (m_i, n_i) \in M \times N\}$$

More precisely, F can be referred to as the set of maps from $M \times N$ to R that have non-zero values at most at finitely may elements, which has the R-module structure with pointwise addition and scalar multiplication.
Note that $\forall m \in M, n \in N, a \in R$, $a(m, n), 1(am, n)$ and $1(m, an)$ are different elements of F. Let G be the submodule of F generated by the elements of the forms:

$$(m_1 + m_2, n) - (m_1, n) - (m_2, n), (am, n) - a(m, n),$$
$$(m, n_1 + n_2) - (m, n_1) - (m, n_2), (m, an) - a(m, n)$$

$\forall a \in R, m, m_1, m_2 \in M, n, n_1, n_2 \in N$. Then set $T := F/G$. Then the map $\tau : M \times N \to T$ given by

$$\tau(m, n) := \overline{(m, n)} := (m, n) + G$$

is R-bilinear obviously. Now we claim that T together with τ is a tensor product of M and N over R. To check the universal property, let $\alpha : M \times N \to P$ be an R-bilinear map to any R-module P.

- For the existence, define a homomorphism $\varphi : T \to P$ by $\varphi(\overline{(m, n)}) := \alpha(m, n)$ and then extend this by linearity, i.e.:

  $$\varphi(a_1\overline{(m_1, n_1)} + \cdots + a_k\overline{(m_k, n_k)}) := a_1\alpha(m_1, n_1) + \cdots + a_k\alpha(m_k, n_k)$$

  By the bilinearity of α, we can easily see the well-defined-ness of φ. Then we find the existence part of the universal property of tensor products.

- The unique part is easy to check so we omit it here.

$\square$

Remark:

- We denote by $M \otimes_R N$ the tensor product of M and N over R and write $\tau(m, n)$ as $m \otimes n$, called the *tensor product* of m and n.
- Tensors in $M \otimes_R N$ that are of the form $m \otimes n$ for $m \in M, n \in N$ are called *pure* or *monomial*. Note that this def is **NOT** "meaningless" since the general element of $M \otimes_R N$ can be written as a finite linear combination

$$a_1(m_1 \otimes n_1) + \cdots + a_k(m_k \otimes n_k)$$

- By the proof of Existence of Tensor Products Proposition, we have the operations:

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n, (am) \otimes n = a(m \otimes n),$$
$$m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2, m \otimes (an) = a(m \otimes n)$$

  for all $a \in R, m, m_1, m_2 \in M, n, n_1, n_2 \in N$. In fact, we can get these operations also by the def saying that τ is bilinear.
- Using multilinear maps instead of bilinear maps, we can also define tensor products $M_1 \otimes \cdots \otimes M_k$ of more than two modules in the same way as above.

Proposition 1.5.5. *Tensor Product of Modules-Properties Proposition*
For any R-modules M, N, P, there are some isomorphisms:

1. $M \otimes_R N \cong N \otimes_R M$
2. $M \otimes_R R \cong M$
3. $(M \oplus N) \otimes_R P \cong (M \otimes_R P) \oplus (N \otimes_R P)$
4. $M \otimes_R N \otimes_R P \cong (M \otimes_R N) \otimes_R P \cong M \otimes_R (N \otimes_R P)$

Proof. 1,3,4 hold immediately by the universal property of tensor product of modules. For 2, we also use universal property first and get the following commutative diagram:

$$\begin{tikzcd} M \times R \arrow[r, "(m,r)\mapsto rm"] \arrow[d, "(m,r)\mapsto m\otimes r"'] & M \\ M \otimes R \arrow[ur, dashed, "\exists!\varphi:(m\otimes r)\mapsto m"'] & \end{tikzcd}$$

Consider the map $\psi : M \to M \otimes R$ given by $m \mapsto m \otimes 1$. Since $(\varphi \circ \psi)(m) := \varphi(m \otimes 1) := m$ and since $(\psi \circ \varphi)(m \otimes r) := (rm) \otimes 1 = r(m \otimes 1) = m \otimes r$, then $\psi \circ \varphi = id_{M \otimes R}$ and $\varphi \circ \psi = id_M$. Hence, $\varphi : M \otimes R \xrightarrow{\cong} M$ is an isomorphism. □

Example:

- Let M and N be *free* R-modules of ranks m respectively n. Then $M \cong R^m$ and $N \cong R^n$.

Hence, by the Tensor Product-Properties Proposition.2,3, one has

$$M \otimes N \cong R^m \otimes R^n \cong R^m \otimes (\underbrace{R \oplus \cdots \oplus R}_{n\ times}) \overset{3}{\cong} (R^m \otimes R)^n \overset{2}{\cong} (R^m)^n = R^{mn}$$

So after a choice of bases of elements of M and N, the elements of $M \otimes N$ can be described as mn-matrices over R.

- Let $I, J \lhd R$ be coprime. Then $\exists a \in I, b \in J$, s.t.: $a + b = 1$. Hence, in the tensor product $R/I \otimes_R R/J$, $\forall$ pure tensors $\overline{r} \otimes \overline{s}$ with $r, s \in R$, we have

$$\overline{r} \otimes \overline{s} = 1(\overline{r} \otimes \overline{s}) = (a+b)(\overline{r} \otimes \overline{s}) = \overline{ar} \otimes \overline{s} + \overline{r} \otimes \overline{bs}$$

  Since I and J are ideals and $a \in I, b \in J$, then $ar \in I, bs \in J$ and hence $\overline{ar} = 0 \in R/I, \overline{bs} = 0 \in R/J$. Therefore, $\overline{r} \otimes \overline{s} = 0$ and hence $R/I \otimes_R R/J = 0$.
  For example, $\forall$ primes $p \neq q$,

$$\mathbb{Z}_p \otimes_{\mathbb{Z}} \mathbb{Z}_q = 0$$

  This shows that (in contrast to the previous example) a tensor product space need **NOT** be "bigger" than its factors and might be 0 even if none of its factors are.

- Consider the tensor product $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2$, we have

$$2 \otimes \overline{1} = 2(1 \otimes \overline{1}) = 1 \otimes \overline{2} = 0$$

  However, if we consider $2 \otimes \overline{1}$ as an element of $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2$, the above computation is invalid since $1 \notin 2\mathbb{Z}$. So it induces a question: is it still true for $2 \otimes \overline{1} = 0 \in 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2$?
  The answer is NO! Since $2\mathbb{Z} \cong \mathbb{Z}$ as a $\mathbb{Z}$-module by isomorphism $2 \mapsto 1$, then by the Tensor Product-Properties Proposition.2,3,

$$2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2 \overset{2}{\cong} \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2 \overset{3}{\cong} \mathbb{Z}_2$$

  by the isomorphism $\varphi : 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2 \xrightarrow{\cong} \mathbb{Z}_2$ sending $(2a) \otimes \overline{b}$ to $\overline{ab}$. Now we see that $\varphi(2 \otimes \overline{1}) = \overline{1} \neq 0$ and hence $2 \otimes \overline{1} \neq 0 \in 2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}_2$.

1.5.2 Tensor Products of Module Homomorphisms

Definition 1.5.6. Tensor Products of Module Homomorphisms
Let $\varphi : M \to N$ and $\varphi' : M' \to N'$ be two R-module homomorphisms. Then the map $\substack{M \times M' \to N \otimes N' \\ (m, m') \mapsto \varphi(m) \otimes \varphi'(m')}$ is bilinear and hence the universal property of tensor product of modules gives rise to a unique homomorphism

$$\varphi \otimes \varphi' : \substack{M \otimes M' \to N \otimes N' \\ m \otimes m' \mapsto \varphi(m) \otimes \varphi'(m')}$$

$\forall m \in M, m' \in M'$. We refer to $\varphi \otimes \varphi'$ as the *tensor product* of φ and φ'.

Remark:

- We can make an interpretation on this def in the view of tensor product of modules: since $\underset{(\varphi,\varphi')\mapsto\Phi_{\varphi,\varphi'}}{Hom_R(M,N)\times Hom_R(M',N')\to Hom_R(M\otimes M',N\otimes N')}$ where $\Phi_{\varphi,\varphi'}(m\otimes m') := \varphi(m)\otimes\varphi'(m')$ is bilinear, then the universal property gives rise to the following commutative diagram:

$$\begin{tikzcd} Hom_R(M,N)\times Hom_R(M',N') \arrow[r, "(\varphi,\varphi')\mapsto\Phi_{\varphi,\varphi'}"] \arrow[d, "(\varphi,\varphi')\mapsto\varphi\otimes\varphi'"'] & Hom_R(M\otimes M',N\otimes N') \\ Hom_R(M,N)\otimes Hom_R(M',N') \arrow[ur, dashed, "\exists!\Phi:\varphi\otimes\varphi'\mapsto\Phi_{\varphi,\varphi'}"'] & \end{tikzcd}$$

  Then we can identity $\Phi_{\varphi,\varphi'}$ directly as $\varphi\otimes\varphi'$.

- One application of tensor products of modules and module homomorphisms is to extend the ring of scalars for a given module and we record it as the later subsection:

1.5.3 Extension of Scalars

Motivation: Suppose that the ring R is a subring of the ring S. Then any S-module is immediately an R-module. More generally, if $f : R \to S$ is a homomorphism, then the action of R on N can be defined as

$$rn := f(r)n, \forall r \in R, n \in N$$

In this situation, S can be considered as an *extension* of the ring R and the resulting R-module is said to be obtained from N by *restriction of scalars* from S to R.
Now suppose that R is a subring of S and we try to reverse this, namely we start with an R-module N and attempt to define an S-module structure on N that extends the action of R on N to an action of S on N (hence "extending the scalars" from R to S). In general, this is impossible, even in the simplest situation: the ring R itself is an R-module but is usually NOT an S-module. For example, $\mathbb{Z}$ is a $\mathbb{Z}$-module but it can NOT be made into a $\mathbb{Q}$-module (otherwise, then $\frac{1}{2}\circ 1 = z$ would be an element of $\mathbb{Z}$ with $z + z = 1$ which is impossible in $\mathbb{Z}$). However, although $\mathbb{Z}$ itself can NOT be made into a $\mathbb{Q}$-module, it is *contained* in a $\mathbb{Q}$-module, namely $\mathbb{Q}$ itself, i.e.: there's an injection (also called an embedding) of $\mathbb{Z}$-module $\mathbb{Z}$ into the $\mathbb{Q}$-module $\mathbb{Q}$. This raises the question of whether an arbitrary R-module N can be embedded as an S-submodule of some S-module, or more generally, the question of what R-module homomorphisms exist from N to S-modules.
Example: We know this process for vector spaces very well: suppose that we're given a real vector space V with $\dim_{\mathbb{R}} V = n < \infty$ and want to study the eigenvalues and eigenvectors of a linear transformation $\varphi : V \to V$. We then usually set up the matrix $A \in M_{m\times n}(\mathbb{R})$ corresponding to φ in some chosen basis and compute its characteristic polynomial. But often it happens that this polynomial does NOT split in $\mathbb{R}$ and we then therefore consider $A \in M_{n\times n}(\mathbb{C})$ as a complex matrix. We use complexification of real vector spaces to attain this goal, which will be discussed later.

Definition 1.5.7. Extension of Scalars

Let M be an R-module and let R' be an R-algebra (so that R' is a ring as well as an R-module). For any $a \in R$, consider the multiplication map $\mu_a : R' \to R$ given by

$$\mu_a(s) := as$$

which is obviously a homomorphism of R-modules. If we set $M_{R'} := M \otimes_R R'$, then we obtain a scalar multiplication with R' on $M_{R'}$

$$\substack{R' \times M_{R'} \to M_{R'} \\ (a, m\otimes s) \mapsto a(m\otimes s) := (1\otimes \mu_a)(m\otimes s) = m\otimes(as)}$$

which turns $M_{R'}$ into an R'-module. We say that $M_{R'}$ is obtained from M by an *extension of scalars* from R to R'.

Remark: Any R-module homomorphism $\varphi : M \to N$ gives rise to an "extend" R'-module homomorphism

$$\varphi_{R'} := \varphi \otimes id : M_{R'} \to N_{R'}$$

Example (*Complexification of a real vector space*): Let V be a real vector space. Extending scalars from $\mathbb{R}$ to $\mathbb{C}$, we call $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$ the *complexification* of V. By def, this is now a complex vector space.
Let us assume for simplicity that V is finitely generated, with basis $\{b_1, \cdots, b_n\}$. Then $V \cong \mathbb{R}^n$ by an isomorphism mapping b_i to e_i for $i = 1, \cdots, n$ and consequently, $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{R}^n \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{R}^n \otimes_{\mathbb{R}} \mathbb{C} \cong (\mathbb{R} \otimes_{\mathbb{R}} \mathbb{C})^n \cong \mathbb{C}$ as R-modules by the Tensor Product of Modules-Properties Proposition and by def, also as $\mathbb{C}$-modules. Moreover, $b_i \otimes 1$ maps to e_i under this chain of isomorphisms for all i and so as expected the vectors $b_1 \otimes 1, \cdots, b_n \otimes 1$ form a basis of $V_{\mathbb{C}}$.
Finally, consider a linear transformation $\varphi : V \to V$ described by the matrix $A \in M_{n\times n}(\mathbb{R})$ with respect to $\{b_1, \cdots, b_n\}$, i.e.: $\varphi(b_i) = \sum_{j=1}^n a_{ji} b_j$, $\forall i$. Then $\varphi_{\mathbb{C}} : V_{\mathbb{C}} \to V_{\mathbb{C}}$ obtained in the def of Extension of Scalars is an endomorphism of the complexification vector space and since

$$\varphi_{\mathbb{C}}(b_i \otimes 1) = \varphi(b_i) \otimes 1 = \sum_{j=1}^n a_{ji}(b_i \otimes 1)$$

then we see that the matrix of $\varphi_{\mathbb{C}}$ with respect to the basis $\{b_1 \otimes 1, \cdots, b_n \otimes 1\}$ is precisely A again, just now considered as a complex matrix.

Proposition 1.5.8. *Extension of Scalars Proposition*
Let M, N be R-modules and let R' be an R-algebra. The extension of scalars commutes with tensor products in the sense that there exists an isomorphism of R'-modules

$$(M \otimes_R N)_{R'} \cong_{R'} M_{R'} \otimes_{R'} N_{R'}$$

Proof.

Lemma 1.5.9. *For any R'-module X, we have*

$$M_{R'} \otimes_{R'} X \cong_{R'} M \otimes_R X$$

where the R'-structure of $M \otimes_R X$ is given by X, i.e.: $r'(m \otimes x) := m \otimes (r'x)$.

Proof. of lemma: By universal property, we have the following commutative diagrams:

$$\begin{tikzcd}
M_{R'} \times X \arrow[r, "(m\otimes r',x)\mapsto m\otimes(r'x)"] \arrow[d, "(m\otimes r',x)\mapsto(m\otimes r')\otimes x"'] & M \otimes_R X \\
M_{R'} \otimes_{R'} X \arrow[ur, dashed, "\exists!\varphi:(m\otimes r')\otimes x\mapsto m\otimes(r'x)"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
M \times X \arrow[r, "(m,x)\mapsto(m\otimes 1)\otimes x"] \arrow[d, "(m,x)\mapsto m\otimes x"'] & M_{R'} \otimes_R X \\
M \otimes_R X \arrow[ur, dashed, "\exists!\psi:m\otimes x\mapsto(m\otimes 1)\otimes x"'] &
\end{tikzcd}$$

Since $(\varphi \circ \psi)(m \otimes x) := \varphi((m \otimes 1) \otimes x) := m \otimes x$ and $(\psi \circ \varphi)((m \otimes r') \otimes x) := \psi(m \otimes (r'x)) := (m \otimes 1) \otimes (r'x) = (m \otimes r') \otimes x$, then φ is an R'-module isomorphism between $M_{R'} \otimes_{R'} X \cong M \otimes_R X$ as R'-modules. $\square$

Now we return back to the *proof of the Extension of Scalars Proposition:* Let $X = N_{R'}$ and then we get:

$$\begin{aligned} M_{R'} \otimes_{R'} N_{R'} &:= (M \otimes_R R')_{R'} \otimes_{R'} N_{R'} \overset{lemma}{\cong} M \otimes_R N_{R'} := M \otimes_R (N \otimes_R R') \\ &\cong (M \otimes_R N) \otimes_R R' := (M \otimes_R N)_{R'} \end{aligned}$$

$\square$

Proposition 1.5.10. *Extension of Scalars for Free Modules*
The module obtained from the free R-module $N \cong R^n$ by extension of scalars from R to an R-slgebra R' (and hence a ring that contains R) is the free R'-module S^n, i.e.:

$$R' \otimes_R R^n \cong (R')^n$$

as R'-modules.

Proof. It immediately hold by the Tensor Product of Modules-Properties Proposition. $\square$

1.5.4 Tensor Product of Algebras

Definition 1.5.11. Tensor Product of Algebras
Let R_1, R_2 be R-algebras. The map $R_1 \times R_2 \times R_1 \times R_2 \to R_1 \otimes R_2$ given by $(s, t, s', t') \mapsto (ss') \otimes (tt')$

is multilinear and hence by using repeatedly the universal property of tensor product, it induces an R-module homomorphism

$$\begin{gathered}(R_1 \otimes R_2) \otimes (R_1 \otimes R_2) \to R_1 \otimes R_2 \\ (s \otimes t) \otimes (s' \otimes t') \mapsto (ss') \otimes (tt')\end{gathered}$$

Again by the universal property of tensor product, this now corresponds to a bilinear map

$$\begin{gathered}(R_1 \otimes R_2) \times (R_1 \otimes R_2) \to R_1 \otimes R_2 \\ (s \otimes t, s' \otimes t') \mapsto (s \otimes t) \cdot (s' \otimes t') := (ss') \otimes (tt')\end{gathered}$$

This multiplication makes $R_1 \otimes R_2$ into a ring and hence into an R-algebra.

Proposition 1.5.12. *Multivariate Polynomial Rings-Tensor Products Proposition*

$$R[x, y] \cong R[x] \otimes_R R[y]$$

are isomorphic as R-algebra.

Proof. There are R-module homomorphisms $\varphi : R[x] \otimes_R R[y] \to R[x, y]$ and $\psi : R[x, y] \to R[x] \otimes_R R[y]$ given by

$$\varphi(f \otimes g) := fg, \ \psi(\sum_{i,j} a_{ij} x^i y^j) := \sum_{ij} a_{ij}(x^i \otimes y^j)$$

Since for all $i, j \in \mathbb{N} \cup \{0\}$, $(\psi \circ \varphi)(x^i \otimes y^j) = x^i \otimes y^j$ and $(\varphi \circ \psi)(x^i y^j) = x^i y^j$ and since $x^i \otimes y^j$ and $x^i y^j$ generate $R[x] \otimes_R R[y]$ respectively $R[x, y]$ as an R-module, then we see that φ and ψ are inverse to each other. Since

$$\varphi((f \otimes g) \cdot (f' \otimes g')) := \varphi((ff') \otimes (gg')) := ff'gg' = fgf'g' = \varphi(f \otimes g) \cdot \varphi(f' \otimes g')$$

then φ is also a ring homomorphism with the multiplication in $R[x] \otimes_R R[y]$. Therefore, $R[x, y] \cong R[x] \otimes_R R[y]$ as R-algebras. $\square$

1.5.5 Exact Sequences

1.5.5.1 Exact Sequences

Intuition: One of the fundamental results for studying the structure of an algebraic object B (e.g. a group, a ring, or a module) is the First Isomorphism Theorem, which relates to the sub-objects of B (the normal groups, the ideals, or the submodules, respectively) with the possible homomorphic images of B. More precisely, consider whether, given two modules A, C, there exists a module B containing (an isomorphic copy of) A such that $B/A \cong C$, where B is said to be an *extension* of C by A. So we introduce the following definition:

Definition 1.5.13. Exact Sequences

- The pair of homomorphisms (of algebraic objects) $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$ is said to be *exact* (at Y) if $im\alpha = \ker \beta$.

- A sequence $\cdots \to X_{k-1} \to X_k \to X_{k+1} \to \cdots$ of homomorphisms (of algebraic objects) is said to be an *exact* sequence if it is exact ar each X_k between a pair of homomorphisms (of algebraic objects).

Remark: Note that if $\psi : A \to B$ is injective and if $\varphi : B \to C$, then the sequence

$$0 \to A \overset{\psi}{\to} B \overset{\varphi}{\to} C \to 0$$

is exact and that the converse also holds.

Definition 1.5.14. Short Exact Sequence
We often call the exact sequence

$$0 \to A \overset{\psi}{\to} B \overset{\varphi}{\to} C \to 0$$

a *short exact sequence.*

Remark: Every exact sequence can be written as a short exact sequence:

$$X \overset{\alpha}{\to} Y \overset{\beta}{\to} Z \text{ is exact} \Leftrightarrow 0 \to \alpha(X) \to Y \to Y/\ker\beta \to 0 \text{ is a short exact sequence}$$

Example:

- Given modules A, C, the sequence

$$0 \to A \overset{a \mapsto (a,0)}{\longrightarrow} A \oplus C \overset{(a,c) \mapsto c}{\longrightarrow} C \to 0$$

  is a short exact sequence, which follows that there always exists at least one extension of C by A. In particular, consider two $\mathbb{Z}$-modules $A = \mathbb{Z}, C = \mathbb{Z}/n\mathbb{Z}$:

$$0 \to \mathbb{Z} \overset{z \mapsto (z,0)}{\longrightarrow} \mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z} \overset{(z_1, z_2 + n\mathbb{Z}) \mapsto z_2 + n\mathbb{Z}}{\longrightarrow} C \to 0$$

  giving one extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$.

- If $\varphi : B \to C$ is any homomorphism of algebraic objects, then we have an exact sequence:

$$0 \to \ker\varphi \hookrightarrow B \overset{\varphi}{\to} im\varphi \to 0$$

  In particular, if φ is surjective, then the sequence may be extend to a short exact sequence.

- Let $a \in R$ be an element in R. Then we can get the following *short exact sequence*:

$$0 \to R/(I : a) \overset{r + (I;a) \mapsto ar + I}{\longrightarrow} R/I \twoheadrightarrow R/(I + (a)) \to 0$$

- Let S be a set of generators for M and let $F(S)$ be the free R-module on S. Then we have a short exact sequence:

$$0 \to \ker\varphi \hookrightarrow F(S) \overset{\varphi}{\to} M \to 0$$

where φ denotes the uniqueness R-module homomorphism which is the identity on S (by the universal property of free module). More generally, when M is any group (possibly non-abelian), the above short exact sequence describes a *presentation* of M, where $\ker\varphi$ is the normal subgroup of $F(S)$ generated by the *relations* defining M.

- (*Presentation of Modules*):

  As in the presentation of groups, an R-module can be given by "generators" and "relations", e.g.: if we say that M has one generator g and relations $r_1 g = r_2 g = \cdots = r_n g = 0$ for some $r_i \in R$, then $M = R/(r_1, \cdots, r_n)$. Here is an exact sequence view:

  Note that each element m of M corresponds to a homomorphism from R to M, sending 1 to m. Hence, if $\{m_i\}_{i\in I}$ generates M, then consider the corresponding homomorphism $\varphi : \oplus_{i\in I} R \to M$, sending the ith 1 to m_i, which is easy to check that φ is surjective.

  Let $\{n_j\}_{j\in J}$ be such that it generates $\ker\varphi$, which give the "relations" on m_j's. Similarly, n_j's correspond a homomorphism $\psi : \oplus_{j\in J} R \to \oplus_{i\in I} R_i$, sending the jth 1 to n_j.

  Now we see that: M is described as the module with generators $\{m_i\}_{i\in I}$ and relations $\{n_j\}_{j\in J}$ iff the sequence
  $$\oplus_{j\in J} R \xrightarrow{\psi} \oplus_{i\in I} R \xrightarrow{\varphi} M \to 0$$
  is exact. This example leads to a definition for presentation of modules but we will first introduce the main content and then introduce this concept at the end of this "subsubsection". For convenience, let's give this example a name, say "the presentation of modules".

- In contrast to the first example, another extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ is given by the short exact sequence
  $$0 \to \mathbb{Z} \xrightarrow{z\mapsto nz} \mathbb{Z} \xrightarrow{z\mapsto z+n\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \to 0$$
  Note that the modules in the middle of the previous two exact sequences are **NOT** isomorphic even though the respective "A" and "C" terms are isomorphic. Hence, there are (at least) two "essentially different" or "inequivalent" ways of extending $\mathbb{Z}/n\mathbb{Z}$, which leads to the following definition:

Definition 1.5.15. Homomorphisms of Short Exact Sequences
Let $0 \to A \to B \to C \to 0$ and $0 \to A' \to B' \to C' \to 0$ be two short exact sequences of modules.

- A *homomorphism of short exact sequences* is a triple (α, β, γ) of module homomorphisms such that the following diagram commutes:
  $$\begin{tikzcd} 0 \arrow[r] & A \arrow[r] \arrow[d, "\alpha"] & B \arrow[r] \arrow[d, "\beta"] & C \arrow[r] \arrow[d, "\gamma"] & 0 \\ 0 \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' \arrow[r] & 0 \end{tikzcd}$$
  This homomorphism is an *isomorphism of short exact sequences* if α, β, γ are all isomorphisms, in which case the extensions B and B' are said to be *isomorphic extensions*.

- The two short exact sequences are called *equivalent* if $A = A', C = C'$ and there is an isomorphism between them as in the above that is the identity maps on A and C (i.e.: $\alpha = id_A, \gamma = id_C$). In this case the corresponding extensions B and B' are said to be *equivalent extensions*.

Remark:

- Isomorphic extensions $\Rightarrow$ isomorphism of modules.
- If B and B' are isomorphic extensions, then there is an R-module isomorphism between B and B' that restricts to an isomorphism from A to A' and induces an isomorphism on the quotients C and C'.
- If B and B' are equivalent extensions of C by A, then there exists an R-module isomorphism between B and B' that restricts to the identity map on A and induces the identity map on C.
- The notion of isomorphic extensions measures how many different extensions of C by A there are, allowing for C and A to be changed by an isomorphism. The notion of equivalent extensions measures how many different extensions of C by A there are when A and C are rigidly fixed.
- It's easy to see that homomorphisms and isomorphisms between short exact sequences with composition
$$(\alpha, \beta, \gamma) \circ (\alpha', \beta', \gamma') := (\alpha \circ \alpha', \beta \circ \beta', \gamma \circ \gamma')$$
gives a group structure.
- Since: if the triple (α, β, γ) is an isomorphism (or equivalent) then $(\alpha^{-1}, \beta^{-1}, \gamma^{-1})$ is an isomorphism (or equivalent respectively) in the reverse direction,
then: "isomorphism" (or "equivalence") is an equivalence relation on any set of short exact sequence.

Example:

- Let $m, n \in \mathbb{N}_+ \setminus \{1\}$. Assume that $n \mid m$ and then let $k := m/n$. Then the following diagram commutes:
$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathbb{Z} & \xrightarrow{z \mapsto nz} & \mathbb{Z} & \xrightarrow{z \mapsto z+n\mathbb{Z}} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \alpha: z \mapsto z+k\mathbb{Z}} & & \downarrow{\scriptstyle \beta: z \mapsto z+m\mathbb{Z}} & & \downarrow{\scriptstyle \gamma = id_{\mathbb{Z}/n\mathbb{Z}}} & & \\
0 & \longrightarrow & \mathbb{Z}/k\mathbb{Z} & \xrightarrow{z+k\mathbb{Z} \mapsto nz+m\mathbb{Z}} & \mathbb{Z}/m\mathbb{Z} & \xrightarrow{z+m\mathbb{Z} \mapsto z+n\mathbb{Z}} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0
\end{array}$$
i.e.: (α, β, γ) is a homomorphism of short exact sequences.
- The following commutative diagram gives an isomorphism of the exact sequence with oneself

but NOT an equivalence of sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{z \mapsto nz} & \mathbb{Z} & \xrightarrow{z \mapsto z + n\mathbb{Z}} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow{\alpha} & & \downarrow{\beta} & & \downarrow{\gamma} & & \\ 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{z \mapsto nz} & \mathbb{Z} & \xrightarrow{z \mapsto z + n\mathbb{Z}} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \end{array}$$

where α, β, γ all map $x \mapsto -x$, since α and γ are NOT identity maps.

Theorem 1.5.16. *The short Five Lemma*
Let (α, β, γ) be a homomorphism of short exact sequences:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow{\alpha} & & \downarrow{\beta} & & \downarrow{\gamma} & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

1. *If α and γ are injective, then so is β.*
2. *If α and γ are surjective, then so is β.*
3. *If α and γ are isomorphisms, then so is β and hence the two sequences are isomorphic.*

Proof. We only prove 1 here. Let $\psi : A \to B$ and $\varphi : B \to C$ be the homomorphisms in the upper short exact sequence. For any $b \in B$ with $\beta(b) = 0$, the image of $\beta(b)$ in C' is also 0. By the commutativity of the diagram, $\gamma(\varphi(b)) = 0$. By the injectivity of γ, $\varphi(b) = 0 \Rightarrow b \in \ker \varphi = im\psi$, i.e.: there exists $a \in A$, s.t.: $b = \psi(a)$. Then again by the commutativity of the diagram, the image of $\alpha(a)$ in B' is the same as $\beta(\psi(a)) = \beta(b) = 0$. Note that α and the map: $A' \to B'$ are injective by hypothesis. Hence, $a = 0$. Therefore, $b = \psi(a) = 0$. □

Definition 1.5.17. Free Presentation of Modules
Recall the fifth example under the definition of short exact sequence, namely, the presentation of modules, we often write $F = \oplus_{j \in J} R, G = \oplus_{i \in I} R$ to describe the relations of M respectively the generators of M. Then we call the sequence $F \to G \to M \to 0$ a *free presentation* of M. Furthermore:

- In case I and J are finite, so that each of F and G is finitely generated free module over R, the sequence is called a *finite free presentation*.
- M is *finitely generated* if $|I| < \infty$.
- M is *finitely presented* if $|J| < \infty$.

1.5.5.2 Split Sequences

Intuition: We've already seen that there is always at least one extension of a module C by A, namely the direct sum $B = A \oplus C$. In this case, the module B contains a submodule $C' = 0 \oplus C \cong C$ as well as the submodule A and this complement to A "splits" B into a direct sum.

Definition 1.5.18. Split (Short Exact) Sequences
We say a short exact sequence

$$0 \to A \overset{\alpha}{\to} B \overset{\beta}{\to} C \to 0$$

is *split* if there exists an R-module homomorphism $\mu : C \to B$, s.t.:

$$\varphi \circ \mu = id_C$$

In this case, the extension B is called a *split extension* of C by A. Furthermore, any map $\nu : C \to B$, s.t.: $\varphi \circ \nu = id_C$ is called a *section* of φ and if $\nu = \mu$ is an R-module homomorphism as above, then it is called a *splitting homomorphism* for the sequence.

Remark: We want to know "split" more explicitly, namely as in intuition above but before that, we need some preparations and then we can prove the Split Sequence Proposition, which will be introduced in the following text, more easily.

Definition 1.5.19. General def for Direct Sum and Direct Product
Let $\{M_i\}_{i\in I}$ be a family of R-modules.

- *Direct Product:*

$$\Pi_{i\in I} M_i := \{(m_i)_{i\in I} \mid m_i \in M_i\}$$

- *Direct Sum:*

$$\oplus_{i\in I} M_i := \{(m_i)_{i\in I} \mid m_i \in M_i, all\ but\ finitely\ many\ m_i's\ are\ 0\}$$

Definition 1.5.20. Direct Summand of Module
Let M, P be R-modules. We say that M is a *direct summand* of P if there are homomorphisms $\alpha : M \to P$ and $\sigma : P \to M$ such that the composition $\sigma \circ \alpha = id_M$.

Proposition 1.5.21. *Direct Summand-Direct Sum-Decomposition Proposition*
If M is a direct summand of P with $\alpha : M \to P, \sigma : P \to M$, then

$$P \cong M \oplus (\ker \sigma)$$

Proof. Consider the map $\varphi : M \oplus (\ker \varphi) \to P$ given by $(m, k) \mapsto \alpha(m) + k$.

- *φ is a module homomorphism:* For any $m, n \in M$, any $k, t \in \ker \sigma$, any $r \in R$,
  - $\varphi((m, k) + (n, t)) = \varphi(m + n, k + t) := \alpha(m + n) + k + t = (\alpha(m) + k) + (\alpha(n) + t) := \varphi(m, k) + \varphi(n, t)$
  - $\varphi(r(m, k)) = \varphi(rm, rk) := \alpha(rm) + rk = r(\alpha(m) + k) := r\varphi(m, k)$
- *φ is injective:* For any (m, k) with $\varphi(m, k) = 0$, we have

$$\alpha(m)+k = 0 \Leftrightarrow \varphi(m) = -k \Rightarrow m = (\sigma\alpha)(m) = -\sigma(k) = 0 \Rightarrow \alpha(m) = 0 \Rightarrow k = 0 \Rightarrow (m, k) = 0$$

- φ *is surjective:* For any $p \in P$, set $m := \sigma(p), k := p - \alpha(\sigma(p))$.
  - **Check** that $k \in \ker \sigma$:

$$\sigma(k) := \sigma(p) - (\sigma\alpha)(\sigma(p)) = \sigma(p) - \sigma(p) = 0$$

  - $\varphi(m, k) := \alpha(m) + k = \alpha(\sigma(p)) + p - \alpha(\sigma(p)) = p$.

□

Proposition 1.5.22. *Split Sequence Proposition*

1. $0 \to A \overset{\alpha}{\to} B \overset{\beta}{\to} C \to 0$ *is split* $\Leftrightarrow B \cong A \oplus C$.
2. $0 \to A \overset{\alpha}{\to} B \overset{\beta}{\to} C \to 0$ *is split* $\Leftrightarrow$ *there exists a homomorphism* $\sigma : B \to A$ *such that* $\sigma\alpha = id_A$.

Proof. 1. • ($\Rightarrow$): Note that C is a direct summand of B. Hence, by the Direct Summand-Direct Sum-Decomposition Proposition,

$$B \cong C \oplus (\ker \beta) \overset{\ker \beta = im\alpha}{=} C \oplus (im\alpha) \overset{\alpha\ inj.}{\cong} C \oplus A \cong A \oplus C$$

- ($\Leftarrow$): Consider $C' \subseteq B$ with $\varphi(C') \cong C$. Then define $\mu := \varphi^{-1} : C \overset{\cong}{\to} C' \subseteq B$

2. • ($\Rightarrow$): Set $\sigma' := id_B - \tau\beta : B \to B$. Since

$$\beta\sigma' := \beta - \beta\tau\beta = \beta - id_C\beta = 0$$

then $im\sigma' \subseteq \ker \beta = im\alpha$. Now we define σ by the following:

For any $b \in B$, $\sigma'(b) \in im\alpha \Rightarrow \exists a \in A$, s.t.: $\alpha(a) = \sigma'(b)$. Since α is injective, then such a is unique. Hence we can define a well-defined $\sigma(b) :=$ the unique $a \in A$ such that $\alpha(a) = \sigma'(b)$ and hence $\alpha(\sigma(b)) = \sigma'(b) \Leftrightarrow \alpha\sigma = \sigma'$.

For any $a \in A$, since

$$\alpha(\sigma\alpha(a)) = \alpha\sigma(\alpha(a)) = \sigma'(\alpha(a)) := \alpha(a) - \tau\beta\alpha(a) \overset{im\alpha = \ker \beta}{=} \alpha(a) - 0 = \alpha(a)$$

then by the injectivity of α, one has

$$\sigma\alpha(a) = a \Rightarrow \sigma\alpha = id_A$$

- ($\Leftarrow$): Similarly.

□

1.6 Projective, Injective and Flat Modules

Intuition: Consider a short exact sequence of R-modules

$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$

with M an extension of N by L. It's natural to ask whether properties for L and N imply related properties for the extension M.

1.6.1 Projective Mordules ($\mathbf{Hom_R}(D, \cdot)$)

Intuition: The first situation we shall consider is whether an R-module homomorphism from some fixed R-module D to either L or N implies that there is an R-module homomorphism from D to M.

Proposition 1.6.1. *L-ψ-Inducing-Map Proposition*
Let D, L, M be R-modules and let $\psi : L \to M$ be an R-module homomorphism. Then the map $\psi' : Hom_R(D, L) \to Hom_R(D, M)$ given by

$$\psi'(f) := \psi \circ f$$

is a homomorphism of abelian group. Furthermore, if ψ is injective, then ψ' is also injective, i.e.:

$$\text{if } 0 \to L \xrightarrow{\psi} M \text{ is exact,then } 0 \to Hom_R(D, L) \xrightarrow{\psi'} Hom_R(D, M) \text{ is also exact.}$$

Proof. It's clear that ψ' is a homomorphism. Now suppose that ψ is injective. If $g \neq f \in Hom_R(D, L)$, then $\exists d \in D$, s.t.: $f(d) \neq g(d)$. Hence, by the injectivity of ψ, we have

$$(\psi \circ f)(d) \neq (\psi \circ g)(d)$$

Therefore, $\psi'(f) := \psi \circ f \neq \psi \circ g := \psi'(g)$. $\square$

Remark:

- The L-ψ-Inducing-Map Prop. can be indicated by the commutative diagram:

$$\begin{tikzcd} & D \arrow[d, "f"'] \arrow[dr, dashed, "\psi'(f)"] & \\ (0) \arrow[r] & L \arrow[r, "\psi"] & M \end{tikzcd}$$

- As for the situation for N, the question is whether there exists an R-module homomorphism $F : D \to M$ that *extends* or *lifts* f to M, i.e.: that makes the following diagram commutes:

$$\begin{tikzcd} & D \arrow[d, "f"] \arrow[dl, dashed, "F"'] \\ M \arrow[r, "\varphi"] & N \end{tikzcd}$$

However, unlike the L-ψ-Inducing-Map Prop, this may not neccesarily hold in general, i.e.: in general it may not be possible to lift a homomorphism f from D to N to a homomorphism from D to M. For example, consider the nonsplit exact sequence

$$0 \to \mathbb{Z} \overset{z \mapsto 2z}{\longrightarrow} \mathbb{Z} \overset{z \mapsto z+2\mathbb{Z}}{\longrightarrow} \mathbb{Z}/2\mathbb{Z} \to 0$$

Let $D = \mathbb{Z}/2\mathbb{Z}$ and let $f = id_{\mathbb{Z}/2\mathbb{Z}}$ be the identity map from D to N. Then any homomorphism F of D into $M = \mathbb{Z}$ must map D to 0 (since $\mathbb{Z}$ hsa no elements of order 2). Hence, $(z \mapsto z + 2\mathbb{Z}) \circ F$ maps D to 0 in N. Therefore, $(z \mapsto z + 2\mathbb{Z}) \circ F \neq f$, which shows that

$$\text{if } M \overset{\varphi}{\to} N \to 0 \text{ is exact,then } Hom_R(D, M) \overset{\varphi'}{\to} Hom_R(D, N) \to 0 \text{ is } notnecessarily \text{ exact.}$$

where $\varphi' : f \mapsto \varphi \circ f$.

Theorem 1.6.2. *Associated $Hom_R(D, \cdot)$ Sequence Theorem*
Let D, L, M, N be R-modules. Given a short exact sequence

$$0 \to L \overset{\psi}{\to} M \overset{\varphi}{\to} N \to 0$$

the associated sequence

$$0 \to Hom_R(D, L) \overset{\psi'}{\to} Hom_R(D, M) \overset{\varphi'}{\to} Hom_R(D, N) \tag{*}$$

is exact. A homomorphism $f : D \to N$ lifts to a homomorphism $F : D \to M \Leftrightarrow f \in im\varphi' \subseteq Hom_R(D, M)$. In general, $\varphi' : Hom_R(D, M) \to Hom_R(D, N)$ need NOT be surjective. Furthermore, φ' is surjective $\Leftrightarrow$ every homomorphism from D to N lifts to a homomorphism from D to M, in which case the exact sequence $(*)$ can be extended to a short exact sequence.
Moreover, $(*)$ is exact for all R-modules $D \Leftrightarrow$ the sequence

$$0 \to L \overset{\psi}{\to} M \overset{\varphi}{\to} N \text{ is exact.}$$

Proof.

- The only thing in the first statement that has not already been proved is the exactness of $(*)$ at $Hom_R(D, M)$, i.e.: $\ker\varphi' = \mathrm{im}\psi'$.
  - $\ker\varphi' \subseteq \mathrm{im}\psi'$: $\forall F \in \ker\varphi' \subseteq Hom_R(D, M)$, we have $\varphi \circ F = 0$ as homomorphisms from D to N, i.e.: $\forall d \in D$, $\varphi(F(d)) = 0$ and hence $F(d) \in \ker\varphi$. By the exactness of $0 \to L \overset{\psi}{\to} M \overset{\varphi}{\to} N \to 0$, $F(d) = \varphi(l_d)$ for some $l_d \in L$. By the injectivity of ψ, the element l_d is unique and hence this gives a well-defined map $F' : D \to L$ given by

$F'(d) := l_d$, which can be checked that F' is a homomorphism, i.e.: $F' \in Hom_R(D, L)$. Then we have

$$F = \psi \circ F' := \psi'(F') \Rightarrow F \in \mathrm{im}\psi'$$

Therefore, $\ker \varphi' \subseteq \mathrm{im}\psi'$.

– $\mathrm{im}\psi' \subseteq \ker \varphi'$: $\forall F \in \mathrm{im}\psi'$, $\exists F' \in Hom_R(D, L)$, s.t.: $F = \psi'(F')$. Hence, $\forall d \in D$, we have

$$\varphi(F(d)) = \varphi(\psi(F'(d))) = (\varphi \circ \psi)(F'(d)) \overset{\ker\varphi = \mathrm{im}\psi}{=} 0$$

$\Rightarrow (\varphi \circ F)(d) = 0, \forall d \in D$, i.e.: $\varphi'(F) = \varphi \circ F = 0 \Rightarrow F \in \ker \varphi' \Rightarrow \mathrm{im}\psi' \subseteq \ker \varphi'$.

- For the last statement in the theorem, note that the surjectivity of φ is not required for the proof that $(*)$ is exact and hence the $(\Leftarrow)$ direction has already been proven. For the converse, note that in general, $Hom_R(R, X) \cong X$ are isomorphic as R-modules for any R-module X, with the isomorphism $f \mapsto f(1)$. Taking $D = R$ in $(*)$, the exactness of the sequence $0 \to L \overset{\psi}{\to} M \overset{\varphi}{\to} N$ follows easily.

□

Remark: More precisely, the sequence $0 \to Hom_R(D, L) \overset{\psi'}{\to} Hom_R(D, M) \overset{\varphi'}{\to} Hom_R(D, N) \to 0$ is exact $\Leftrightarrow$ there is a bijection

$$\{F : D \to M \text{ homomorphisms}\} \leftrightarrow \{(g : D \to L, f : D \to N) \text{ pairs of homomorphisms}\}$$

given by $F|_{\psi(L)} = \psi'(g)$ and $f = \varphi'(F)$.

Proposition 1.6.3. *Split-$Hom_R(D, \cdot)$ Proposition*
Let D, L, N be R-modules. Then

1. $Hom_R(D, L \oplus N) \cong Hom_R(D, L) \cong Hom_R(D, N)$*, and,*
2. $Hom_R(L \oplus N, D) \cong Hom_R(L, D) \cong Hom_R(N, D)$.

Proof. We only prove 1 and the proof of 2 is similar.

- Let $\pi_1 : L \oplus N \to L, \pi_2 : L \oplus N \to N$ be the natural projections. If $f \in Hom_R(D, L \oplus N)$, then $\pi_1 \circ f \in Hom_R(D, L)$ and $\pi_2 \circ f \in Hom_R(D, N)$. Hence, $\varphi : Hom_R(D, L \oplus N) \to Hom_R(D, N)$ with $f \mapsto (\pi_1 \circ f, \pi_2 \circ f)$ is a homomorphism.
- Conversely, let $\psi : Hom_R(D, L) \oplus Hom_R(D, N) \to Homr_R(D, L \oplus N)$ be given by $(f_1, f_2) \mapsto f$ where $f(d) := (f_1(d), f_2(d)), \forall d \in D$. Then it's easy to show that $\psi \circ \varphi = id_{Hom_R(D,L\oplus N)}$ and $\varphi \circ \psi = id_{Hom_R(D,L)\oplus Hom_R(D,N)}$ and hence φ is an isomorphism between $Hom_R(D, L \oplus N) \cong Hom_R(D, L) \oplus Hom_R(D, N)$.

□

Remark:

- These results can extend immediately by induction to any finite direct sum of R-modules.

- These results are referred to by saying that Hom *commutes with finite direct sums in either variable*. For infinite direct sums, the situation is more complicated: Part 1 still holds if $L \oplus N$ is replaced by an arbitrary direct sum and the right hand side is replaced by a direct product; part 2 still holds if the direct sums on both sides are replaced by direct products.

Corollary 1.6.4. *of the Split-$Hom_R(D, \cdot)$ Proposition*
If the sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a split short exact sequence of R-modules, then

$$0 \to Hom_R(D, L) \xrightarrow{\psi'} Hom_R(D, M) \xrightarrow{\varphi'} Hom_R(D, N) \to 0$$

is also a split short exact sequence of abelian groups for every R-module D.

Remark: Converse also holds.

Definition 1.6.5. Projective Modules
An R-module P is called *projective* if it satisfies one of the following (equivalent) conditions (we will prove the equivalence of those conditions later):

1. $\forall$ R-modules L, M, N, if $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence, then so is

$$0 \to \mathrm{Hom_R}(P, L) \xrightarrow{\psi'} \mathrm{Hom_R}(P, M) \xrightarrow{\varphi'} \mathrm{Hom_R}(P, N) \to 0$$

2. $\forall$ R-modules M, N, if $M \xrightarrow{\varphi} N \to 0$ is exact, then every R-module homomorphism from P to N lifts to an R-module homomorphism to M, i.e.: given $f \in \mathrm{Hom_R}(P, N)$, there is a lift $F \in \mathrm{Hom_R}(P, M)$, s.t.: the following diagram commutes:

$$\begin{tikzcd}
 & P \arrow[dl, dashed, "F"'] \arrow[d, "f"] & \\
M \arrow[r, "\varphi"] & N \arrow[r] & 0
\end{tikzcd}$$

3. If P is a quotient of the R-module M, then P is isomorphic to a direct summand of M, i.e.: every short exact sequence $0 \to L \to M \to P \to 0$ splits.

4. P is a direct summand of a free R-module.

Proposition 1.6.6. *Projective Modules-Conditions-Equivalent Proposition*
The conditions in the definition of projective modules are equivalent.

Proof.
- $(1 \Leftrightarrow 2)$: is a restatement of a result in the Associated $Hom_R(D, \cdot)$ Sequence Theorem.
- $(2 \Rightarrow 3)$: Let $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} P \to 0$ be exact. By 2, the identity map from P to P lifts to a

homomorphism μ making the following diagram commute:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\mu"'] \arrow[d, "id_P"] & \\ M \arrow[r, "\varphi"] & P \arrow[r] & 0 \end{tikzcd}$$

Then we find that μ is a splitting homomorphism for the sequence.

- $(3 \Rightarrow 4)$: We refer to P as a set and then we have: the sequence

$$0 \to \ker\varphi \hookrightarrow F(P) \xrightarrow{\varphi} P \to 0$$

is exact where $F(P)$ denotes the free module on the set P. Hence, if 3 holds, then this sequence is split, so $F(P) \cong \ker\varphi \oplus P$.

- $(4 \Rightarrow 2)$: Let S be a set and K be an R-module, s.t.:

$$F(S) \cong P \oplus K$$

$\forall f \in \mathrm{Hom_R}(P, N)$, consider the natural projection map $\pi : F(S) \to P$ and hence $f \circ \pi \in \mathrm{Hom_R}(F(S), N)$. $\forall s \in S$, define $n_s := (f \circ \pi)(s) \in N$. By the surjectivity of φ, $\varphi(m_s) = n_s$ for some $m_s \in M$. By the universal property of free modules, $\exists!$ R-module homomorphism $F' : F(S) \to M$, s.t.: $F'(s) = m_s$, i.e. the following diagram commutes:

$$\begin{tikzcd} & F(S) \cong P \oplus K \arrow[ddl, dashed, "\exists! F'"'] \arrow[d, "\pi"] & \\ & P \arrow[d, "f"] & \\ M \arrow[r, "\varphi"] & N \arrow[r] & 0 \end{tikzcd}$$

namely, $\varphi \circ F' = f \circ \pi : F(S) \to N$. Now define a map $F : P \to M$ by

$$F(d) := F'((d, 0))$$

Since $F = (P \hookrightarrow F(S)) \circ F'$, then F is a homomorphism. Then $\forall d \in P$, we have

$$(\varphi \circ F)(d) = (\varphi \circ F')((d, 0)) = (f \circ \pi)((d, 0)) = f(d)$$

and hence $\varphi \circ F = f$. Therefore, the diagram

$$\begin{tikzcd} & P \arrow[dl, dashed, "F"'] \arrow[d, "f"] & \\ M \arrow[r, "\varphi"] & N \arrow[r] & 0 \end{tikzcd}$$

commutes.

□

Corollary 1.6.7. *of the Projective Modules-Conditions-Equivalent Proposition*

1. *Free modules are projective.*
2. *A finitely generated module is projective $\Leftrightarrow$ it is a direct summand of a finitely generated free module.*
3. *Every module is a quotient of a projective module.*

Example:

- If $R = F$ is a field, then every F-module is projective.
- $\mathbb{Z}$ is a projective $\mathbb{Z}$-module.
- Free $\mathbb{Z}$-modules have no nonzero elements of finite order, so no nonzero finite abelian groups can be isomorphic to a submodule of free $\mathbb{Z}$-module. Hence, no nonzero finite abelian group is a projective $\mathbb{Z}$-module. In particular, for $n \geq 2$, the $\mathbb{Z}$-module $\mathbb{Z}/n\mathbb{Z}$ is *not* projective.
- Since $\mathbb{Q}/\mathbb{Z}$ is a torsion $\mathbb{Z}$-module, then it is *not* a submodule of a free $\mathbb{Z}$-module and hence it is *not* projective. Note also that the exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \xrightarrow{\pi} \mathbb{Q}/\mathbb{Z} \to 0$ does not split since $\mathbb{Q}$ contains no submodule isomorphic to $\mathbb{Q}/\mathbb{Z}$.
- The $\mathbb{Z}$-module $\mathbb{Q}$ is *not* projective.
- The direct sum of two projective modules is again projective.
- Let $R = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and let P_1, P_2 be the principal ideals generated by $(1, 0)$ respectively $(0, 1)$. Then $R = P_1 \oplus P_2$ and hence P_1 and P_2 are projective R-modules.

1.6.2 Injective Modules ($\mathbf{Hom_R}(\cdot, D)$)

Intuition: In the contrast, naturally, we consider the "dual" quesrion of maps from L or N to D.

Theorem 1.6.8. *Associated $Hom_R(\cdot, D)$ Sequence Theorem*
Let D, L, M, N be R-modules. Given a short exact sequence

$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$

the associated sequence

$$0 \to Hom_R(N, D) \overset{\varphi'}{\to} Hom_R(M, D) \overset{\psi'}{\to} Hom_R(L, D) \qquad (**)$$

where $\varphi'(f) := f \circ \varphi, \psi'(f) := f \circ \psi$. *A homomorphism* $f : L \to D$ *lifts to a homomorphism* $F : M \to D \Leftrightarrow f \in im\psi' \subseteq Hom_R(L, D)$. *In general* $\psi' : Hom_R(M, D) \to Hom_R(L, D)$ *need not be surjective. Furthermore,* ψ' *is surjective* $\Leftrightarrow$ *every homomorphism from* M *to* D*, in which case the exact sequence* $(**)$ *can be extended to a short exact sequence.*
Moreover, $(**)$ *is exact for all* R*-modules* $D \Leftrightarrow$ *the sequence*

$$L \overset{\psi}{\to} M \overset{\varphi}{\to} N \to 0 \textit{ is exact.}$$

Proof. Similarly as the proof of the Associated $\mathrm{Hom_R}(D, \cdot)$ Sequence Theorem hence we omit this proof here. □

Definition 1.6.9. Injective Modules
An R-module Q is called *injective* if it satisfies one of the following (equivalent) conditions (we will *not* prove the equivalence of these conditions here):

1. $\forall$ R-modules L, M, N, if $0 \to L \overset{\psi}{\to} M \overset{\varphi}{\to} N \to 0$ is a short exact sequence, then so is

$$0 \to \mathrm{Hom_R}(N, Q) \overset{\varphi'}{\to} \mathrm{Hom_R}(M, Q) \overset{\psi'}{\to} \mathrm{Hom_R}(L, Q) \to 0$$

2. $\forall$ R-modules L, M, if $0 \to L \overset{\psi}{\to} M$ is exact, then every R-module homomorphism from L to Q lifts to an R-module homomorphism of M to Q, i.e.: given $f \in \mathrm{Hom_R}(L, Q)$, $\exists$ a lift $F \in \mathrm{Hom_R}(M, Q)$, s.t.: the following diagram commutes:

$$\begin{tikzcd} 0 \arrow[r] & L \arrow[r, "\psi"] \arrow[d, "f"'] & M \arrow[dl, dashed, "F"] \\ & Q & \end{tikzcd}$$

3. If Q is a submodule of the R-module M, then Q is a direct summand of M, i.e.: every short exact sequence $0 \to Q \to M \to N \to 0$ splits.

Theorem 1.6.10. *Baer's Criterion*
The R*-module* Q *is injective* $\Leftrightarrow \forall$ *ideal* $I \triangleleft R$ *of* R*, any* R*-module homomorphism* $g : I \to Q$ *can be extended to an* R*-module homomorphism* $G : R \to Q$.

Proof.
- The $(\Rightarrow)$ direction holds immediately by the second condition in the definition of injective module.
- Conversely, we want to prove the second condition. Suppose that $0 \to L \overset{\psi}{\to} M$ is exact and that $f : L \to Q$ is an R-module homomorphism. Consider the collection $\mathcal{S}$ of all pairs (f', L')

where $f' : L' \to Q$ are the lifts of f and L' is a submodule of M containing L. Then we define the partial order on $\mathcal{S}$ by

$$(f', L') \leq (f'', L'') \text{ if } L' \subseteq L'' \text{ and } f''|_{L'} = f'$$

Since $\mathcal{S} \neq \emptyset$, then by the Zorn's Lemma, there is a maximal element (F, M') in $\mathcal{S}$. It suffices to complete this proof by proving the following claim:

– **Claim:** $M = M'$.

– *Proof. of claim:* Suppose on the contrary that $M' \subsetneq M$, i.e.: there is an element $m \in M \setminus M'$, and then let $I = \{r \in R \mid rm \in M'\}$, which can be easily verified that I is an ideal of R. For applying the hypothesis, consider the map $g : I \to Q$ given by $g(x) := F(xm)$, an R-module homomorphism from I to Q. Then by hypothesis, $\exists$ a lift $G : R \to Q$ of g. Note that $M' + Rm$ is a submodule of M. Define a map $F' : M' + Rm \to Q$ by

$$F'(m' + rm) := F(m') + G(r)$$

Check: if $m'_1 + r_1 m = m'_2 + r_2 m$, then $(r_1 - r_2)m = m'_2 - m'_1$ which follows that

$$\begin{aligned} F'(m'_1 + r_1 m) - F'(m'_2 + r_2 m) &:= F(m'_1 - m'_2) + G(r_1 - r_2) = F(m'_1 - m'_2) + g(r_1 - r_2) \\ &:= F(m'_1 - m'_2) + F((r_1 - r_2)m) = 0 \end{aligned}$$

and hence F' is well-defined and it is then immediately an R-module homomorphism, extending f to $M' + Rm$. This contradicts the maximality of M'. $\square$

$\square$

Proposition 1.6.11. *Injective PID-Modules Proposition*
Let R be a PID and let Q be an R-module.

1. *Q is injective $\Leftrightarrow rQ = Q\ \forall 0 \neq r \in R$. In particular, a $\mathbb{Z}$-module is injective $\Leftrightarrow$ it is divisible.*
2. *Quotient modules of injective R-modules are again injective.*

Proof. 1. $\forall$ nonzero ideal $I \lhd R$, PID, $I = (r)$ for some $0 \neq r \in R$. Then an R-module homomorphism $f : I \to Q$ is completely determined by $f(r) \in Q$. Since f can be extended to an R-module homomorphism $F : R \to Q \Leftrightarrow$ there is an element q' in Q with $F(1) = q'$ satisfying $q := f(r) = F(r) = rq'$, then the Baer's Criterion holds for $Q \Leftrightarrow rQ = Q$.

2. Since a quotient of Q with $rQ = Q\ \forall 0 \neq r \in R$ has the same property, then we're done.

$\square$

Example:

- Since $\mathbb{Z}$ is *NOT* divisible, then $\mathbb{Z}$ is *NOT* an injective $\mathbb{Z}$-module, which also follows from the fact that the exact sequence $0 \to \mathbb{Z} \xrightarrow{z \mapsto 2z} \mathbb{Z} \xrightarrow{z \mapsto z + 2\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \to 0$ does *NOT* split.

- $\mathbb{Q}$ is an injective $\mathbb{Z}$-module.
- $\mathbb{Q}/\mathbb{Z}$ is an injective $\mathbb{Z}$-module.
- A direct sum of divisible $\mathbb{Z}$-modules is again divisible and hence a direct sum of injective $\mathbb{Z}$-modules is a gain injective. e.g.: $\mathbb{Q} \oplus \mathbb{Q}/\mathbb{Z}$ is injective.

Corollary 1.6.12. *of Injective PID-Moudules Proposition*
Every $\mathbb{Z}$-module is a submodule of an injective $\mathbb{Z}$-module.

Proof. Let M be a $\mathbb{Z}$-module, let A be any set of $\mathbb{Z}$-module generators of M and let $F(A)$ be the free $\mathbb{Z}$-module on the set A. Then by the universal property of free modules, $\exists$ a *surjective* $\mathbb{Z}$-module homomorphism $\varphi : F(A) \to M$ and then we can identify $M = F(A)/\ker\varphi$. Let $\mathcal{Q}$ be a free $\mathbb{Q}$-module on A. Then $\mathcal{Q} \cong$ a direct sum of a number of copies of $\mathbb{Q}$, so it is divisible and hence by the Injective PID-Modules Proposition, an injecitve $\mathbb{Z}$-module containing $F(A)$. Since $\ker\varphi$ is a $\mathbb{Z}$-module of $\mathcal{Q}$, then $\mathcal{Q}/\ker\varphi$ is injective, again by the Injective PID-Modules Proposition. Since $M = F(A)/\ker\varphi \subseteq \mathcal{Q}/\ker\varphi$, then we're done. □

Remark: This corollary can be used to prove the following more general version valid for arbitrary R-modules:

Theorem 1.6.13. *Any R-module is contained in an injective R-module.*

1.6.3 Flat Modules ($D \otimes_R \cdot$)

Theorem 1.6.14. *Tensor-Exact Sequences Proposition*
Let D, M, N, L be R-modules. Given a short exact sequence $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$, the associated sequence of R-modules

$$D \otimes_R L \xrightarrow{id_D \otimes \psi} D \otimes_R M \xrightarrow{id_D \otimes \varphi} D \otimes_R N \to 0 \qquad (***)$$

is exact. In general, the map $id_D \otimes \psi$ need NOT be surjective.
Moreover, $(***)$ is exact for all R-modules $D \Leftrightarrow$ the sequence

$$L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0 \text{ is exact.}$$

Proof. For the first statement, it remains to prove the exactness of $(***)$ at $D \otimes_R M$. By the exactness of $L \xrightarrow{\psi} M \xrightarrow{\varphi} N$, $\varphi \circ \psi = 0$ and hence we have

$$(id_D \otimes \varphi)(\sum_i (d_i \otimes (\psi(l_i)))) := \sum_i (d_i \otimes ((\varphi \circ \psi)(l_i))) = 0$$

which follows that $\operatorname{im}(id_D \otimes \psi) \subseteq \ker(id_D \otimes \varphi)$. In particular, there is a natural projection $\pi : (D \otimes_R M)/\operatorname{im}(id_D \otimes \psi) \to (D \otimes_R M)/\ker(id_D \otimes \varphi) \cong D \otimes_R N$. It suffices to show that π is an isomorphism. To show this, we shall define an inverse of π. Consider $\pi' : D \times N \to (D \otimes_R M)/\operatorname{im}(id_D \otimes \psi)$ given by $(d, n) \mapsto d \otimes m + \operatorname{im}(id_D \otimes \psi)$ where $m \in M$ is such that $\varphi(m) = n$, which is well-defined by

$\ker\varphi = \mathrm{im}\psi$. It's easy to verify that π' is bilinear and hence it induces an R-module homomorphism $\tilde{\pi} : D \otimes_R N \to (D \otimes_R M)/\mathrm{im}(id_D \otimes \psi)$ with

$$\tilde{\pi}(d \otimes n) := d \otimes m + \mathrm{im}(id_D \otimes \psi)$$

Then we find an inverse of π. □

Remark:

- If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is split, then $(***)$ can be extended to a split short exact sequence $0 \to D \otimes_R L \xrightarrow{id_D \otimes \psi} D \otimes_R M \xrightarrow{id_D \otimes \varphi} D \otimes_R N \to 0$.
- This theorem gives rise to the following definition, where the conditions are equivalent, which can be directly obtained from this theorem:

Definition 1.6.15. Flat Modules
An R-module A is called *flat* if it satisfies the following (equivalent) conditions:

1. $\forall$ R-modules L, M, N, if
$$0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$$
is a short exact sequence, then
$$0 \to A \otimes_R L \xrightarrow{id_A \otimes \psi} A \otimes_R M \xrightarrow{id_A \otimes \varphi} A \otimes_R N \to 0$$
is also a short exact sequence.
2. $\forall$ R-modules L, M, if $0 \to L \xrightarrow{\psi} M$ is an exact sequence of R-modules, then $0 \to A \otimes_R L \xrightarrow{id_A \otimes \psi} A \otimes_R M$ is an exact sequence of abelian groups.

Proposition 1.6.16. *Important Example of Flat Modules Free modules are flat. More generally, projective modules are flat.*

Proof.

- To show that a free module F is flat, it suffices to show that $\forall$ injective map $\psi : L \to M$ of R-modules, the induced map $id_F \otimes \psi : F \otimes_R L \to F \otimes_R M$ is also injective, i.e.: condition 2 holds for F.
Suppose first that $F \cong R^n$ is a finitely generated free R-module. Then $F \otimes_R L \cong R^n \otimes_R L \cong L^n$ and similarly $F \otimes_R M \cong M^n$. Hence, ψ induces a natural injective map : $L^n \to M^n$.
Suppose now that F is an arbitrary free module and that the element $\sum(f_i \otimes l_i) \in \ker(id_F \otimes \psi) \subseteq F \otimes_R L$. This means that $\sum(f_i, \psi(l_i))$ can be written as a sum of generators in the free group on $F \times M$, namely as in the proof of existence of tensor product. Since this sum of elements is finite, then f_i's all lie in some finitely generated submodule F' of F, which reduces to the finitely generated case.
- Let P be a projective module. Then by the fourth condition in the definition of projective module, P is a direct summand of a free module F, asy $F = P \oplus P'$. As in the above, we'll prove the second condition in the definition of flat modules, so we pich $\psi : L \to M$ to be

injective. We've shown that $id_F \otimes \psi : F \otimes_R L \to F \otimes_R M$ is also injective. Since $F = P \oplus P'$, then by the Tensor Product of Modules-Properties Proposition,

$$id_{P\oplus P'} \otimes \psi : (P \otimes_R L) \oplus (P' \otimes_R L) \cong (P \oplus P') \otimes_R L \to (P \oplus P') \otimes_R M \cong (P \otimes_R M) \oplus (P' \otimes_R M)$$

is injective. Hence, $id_P \otimes \psi : P \otimes_R L \to P \otimes_R M$ is injective.

□

Example:

- $\mathbb{Z}$ is projective $\mathbb{Z}$-module $\Rightarrow$ it is a flat $\mathbb{Z}$-module.
- Note that the injection $\mathbb{Z} \hookrightarrow \mathbb{Q}$ of $\mathbb{Z}$-modules induces the zero map from $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ to $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = 0$. Hence, $\mathbb{Z}/2\mathbb{Z}$ is *NOT* a flat $\mathbb{Z}$-module.
- $\mathbb{Q}$ is a flat $\mathbb{Z}$-module: Let $\psi : L \to M$ be an injective map of $\mathbb{Z}$-modules. Then every element in $\mathbb{Q} \otimes_{\mathbb{Z}} L$ can be written in the form

  $$(\frac{1}{d}) \otimes l \text{ for some } d \in \mathbb{Z} \setminus \{0\}, l \in L$$

  If $(\frac{1}{d}) \otimes l \in \ker(id_{\mathbb{Q}} \otimes \psi)$, then $(\frac{1}{d}) \otimes \psi(l) = 0 \in \mathbb{Q} \otimes_{\mathbb{Z}} M$. Hence, $\psi(c \cdot l) = c\psi(l) = 0 \in M$ for some $c \in \mathbb{Z} \setminus \{0\}$. Then by the injectivity of ψ, $c \cdot l = 0 \in L$. Therefore, $(\frac{1}{d}) \otimes l = (\frac{1}{cd}) \otimes (cl) = 0 \in \mathbb{Q} \otimes_{\mathbb{Z}} L$, i.e.: $id_{\mathbb{Q}} \otimes \psi$ is injective.
- The $\mathbb{Z}$-module $\mathbb{Q}/\mathbb{Z}$ is injective but *NOT* flat: Consider the injective map $\psi : \mathbb{Z} \to \mathbb{Z}$ given by $z \mapsto 2z$. Since

  $$(id_{\mathbb{Q}/\mathbb{Z}} \otimes \psi)((\frac{1}{2} + \mathbb{Z}) \otimes 1) = (\frac{1}{2} + \mathbb{Z}) \otimes 2 = (1 + \mathbb{Z}) \otimes 1 = 0_{\mathbb{Q}/\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}}$$

  and since $(\frac{1}{2} + \mathbb{Z}) \otimes 1 \neq 0_{\mathbb{Q}/\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}}$, then $id_{\mathbb{Q}/\mathbb{Z}} \otimes \psi$ is *NOT* injective.
- The direct sum of flat modules are flat.

Definition 1.6.17. Torsion-Free Modules
An R-module M is *torsion-free* if $rm \neq 0$ whenever r is not a zerodivisor in R and $m \neq 0$.

Proposition 1.6.18. *Flat Modules-are-Torsion Free Proposition*
Every flat R-module F is torsion-free.

Proof. Suppose on the contrary that $\exists r_0 \in R$ not a zerodivisor an $\exists m_0 \neq 0$, s.t.: $r_0 m_0 = 0$. Consider $f : R \to R$ given by $f(r) := r_0 r$. Since r_0 is not a zerodivisor, then $\ker f = \{0\}$ and hence f is injective. The following diagram commutes:

$$\begin{tikzcd}
F \otimes_R R \arrow[r, "id_F \otimes f"] \arrow[d, "\pi"] & F \otimes_R R \arrow[d, "\pi"] \\
F \arrow[r, "m \mapsto r_0 m"] & F
\end{tikzcd}$$

By the flat-ness of F, $id_F \otimes f$ is injective and hence so is the map $m \mapsto r_0 m$, contradicting to $0 \neq m_0 \in \ker(m \mapsto r_0 m)$. $\square$

Theorem 1.6.19. *Adjoint Associativity*
Let R, S be rings, let A, C be R-modules and let B be a module over both R and S. Then $\exists$ an isomorphism between

$$Hom_S(A \otimes_R B, C) \cong Hom_R(A, Hom_S(B, C))$$

Proof. Let $\varphi \in \mathrm{Hom}_S(A \otimes_R B, C)$. For any $a \in A$, we define a homomorphism $\Phi(a) : B \to C$ by $\Phi(a)(b) := \varphi(a \otimes b)$. So the map $f : \varphi \to \Phi$ gives a homomorphism from $\mathrm{Hom}_S(A \otimes_R B, C)$ to $\mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C))$. Conversely, let $\Phi \in \mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C))$. Since the map from $A \times B$ to C given by $(a, b) \mapsto \Phi(a)(c)$ is bilinear, then by the universal property of tensor products, it induces a homomorphism $\varphi : A \otimes_R B \to C$. So the map $g : \Phi \mapsto \varphi$ gives a homomorphism from $\mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C))$ to $\mathrm{Hom}_S(A \otimes_R B, C)$, which can easily be verified to be an inverse of f. $\square$

Chapter 2

Algebraic Geometry

2.1 Chain Conditions

2.1.1 Noetherian Rings

Definition 2.1.1. Noetherian Rings
A ring R is called *Noetherian* if it satisfies the *ascending chain condition (ACC) on ideals*, i.e.: there is no infinite strictly increasing chain of ideals in R, namely, every increasing chain of ideals in R

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

must stop somewhere ($I_n = I_{n+1} = I_{n+2} = \cdots$ for some n).

Proposition 2.1.2. *Quotient-Noetherian Proposition*
If I is an ideal of the Noetherian ring R, then the quotient ring R/I is a Noetherian ring. In particular, any homomorphic image of a Noetherian ring is Noetherian.

Proof. Omitted. □

Theorem 2.1.3. *Equivalent-Conditions-of-Noetherian Rings Proposition*
The followings are equivalent:

1. *R is a Noetherian ring.*
2. *Every nonempty set of ideals of R contains a maximal element under inclusion.*
3. *Every ideal of R is finitely generated.*

Proof.
- ($1 \Rightarrow 2$): Suppose on the contrary that there is a nonempty set of ideals of R Σ which has no maximal elements. For any $I_1 \in \Sigma$, since Σ has no maximal elements, then there exists an $I_2 \in \Sigma$, s.t.: $I_1 \subsetneq I_2$. Similarly, repeating this procedure, we can construct an infinite increasing chain of ideals in R, say $I_1 \subsetneq I_2 \subsetneq \cdots$, contradicting 1.

- $(2 \Rightarrow 3)$: Given any $I \triangleleft R$, consider the set Σ of all finitely generated ideals, that is containe in I, i.e.:
$$\Sigma := \{J \subseteq I \mid J \text{ is a finitely generated ideal}\}$$
Since $(0) \in \Sigma$, then Σ is nonempty and hence by 2, $\exists M \in \Sigma$, s.t.: M is a maximal element of Σ. By the definition of Σ, let $M = (m_1, \cdots, m_k) \subseteq I$. It suffices to prove the following claim:
  - **Claim:** $M = I$.
  - *Proof. of claim:* Suppose on the contrary that $M \subsetneq I$ and hence we can pick an element $x \in I \setminus M$. Then
  $$M' := (m_1, \cdots, m_k, b) \subseteq I$$
  is immediately contained in M, contradicting the maximality of M. □
- $(3 \Rightarrow 1)$: Given any increasing chain of ideals $I_1 \subseteq I_2 \subseteq \cdots$, consider $I := \bigcup_{n=1}^{\infty} I_n$, which is an ideal of R. Hence by 3, $I = (i_1, \cdots, i_m)$ is finitely generated. By the definition of I, $\forall j \in \{1, \cdots, m\}$, $i_j \in I_{n_j}$ for some $n_j \in \mathbb{N}_+$. Take $N := \max\{n_1, \cdots, n_m\}$. Then $I_N = I_n$ $\forall n \geq N$.

□

Example:

- By the third condition in the Equivalent-Conditions-of-Noetherian Rings Prop, every PID is Noetherian. In particular, $\mathbb{Z}, F[x]$ (where F is a field), $\mathbb{Z}[i]$ are Noetherian.
- $\mathbb{Z}[x_1, x_2, \cdots]$ is *NOT* Noetherian since the ideal $(x_1, x_2, \cdots)$ can *NOT* be generated by any finite set.
- $\mathcal{C}([0,1]) :=$ continuous real valued functions on $[0,1]$ is *NOT* Noetherian.

Remark: In $\mathbb{Z}$, $(2) \supseteq (4) \supseteq (8) \supseteq \cdots$ is an infinite decreasing chain. Hence, a Noetherian ring need *NOT* satisfy the *descending chain condition (DCC) on ideals*. However, we shall see that a ring satisfying DCC on ideals necessarily satisfies ACC on the ideals, i.e.: is Noetherian, which are called *Artinian* and are studied in the later sections.

Theorem 2.1.4. *Hilbert's Basis Theorem*
If R is a Noetherian ring, then so is the polynomial ring $R[x]$.

Proof. Given an ideal $I \triangleleft R[x]$ in $R[x]$, $\forall n \in \mathbb{N} \cup \{0\}$, consider

$$L_n := \{a \in R \mid \exists f(x) \in I, \text{ s.t.: } f(x) = ax^n + \text{ lower-order terms } \} \cup \{0\}$$

which is an ideal in R. Then by the Noetherian-ness of R, $\exists N \in \mathbb{N} \cup \{0\}$, s.t.: $L_N = L_n, \forall n \geq N$. By the Equivalent-Conditions-Noetherian Prop.3, L_n's are finitely generated, i.e.: $\forall 0 \leq n \leq N$, $L_n = (a_{n,1}, \cdots, a_{n,m_n})$. Then by the definition of L_n, $\forall a_{n,j} \in L_n$, $\exists$ a polynomial with leading coefficient $a_{n,j}$ and of order n. We denote this polynomial as $f_{n,j}(x)$. Now consider an ideal of $R[x]$

$$J := (\{f_{n,j}(x) \mid 0 \leq n \leq N, 1 \leq j \leq m_n\})$$

i.e.: $J \triangleleft R[x]$ is an ideal of $R[x]$ generated by the set $\{f_{n,j}(x) \mid 0 \leq n \leq N, 1 \leq j \leq m_n\}$. Since each $f_{n,j}(x)$ is in I, then we have $J \subseteq I$. It suffices to show that $I \subseteq J$. $\forall g \in I$, we do induction on $\deg(g)$.

- **Basic Case:** $\deg(g) = 0 \Rightarrow g(x) = 0 \in J$.
- **Inductive Step:** Suppose that all polynomials in I of order $\leq d-1$ are in J. For $\deg(g) = d$, let $g(x) = bx^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0$. By the definition of L_n, $b \in L_d$.
  - If $d \leq N$, then $b \in (a_{d,1}, \cdots, a_{d,m_d})$ and hence $b = r_1 a_{d,1} + \cdots + r_{m_d} a_{d,m_d}$ for some $r_j \in R$. Consider the polynomial

    $$h(x) := r_1 f_{d,1} + \cdots + r_{m_d} f_{d,m_d}(x) \in J$$

    with leading coefficient $\sum_{j=1}^{m_d} r_j a_{d,j} = b$ and of order d. Hence, $g(x) - h(x)$ is of order $d-1$. Therefore, $g(x) - h(x) \in J \Rightarrow g(x) \in J$.
  - If $d > N$, then $L_d = L_N = (a_{N,1}, \cdots, a_{N,m_N})$ and hence $b = r_1 a_{N,1} + \cdots + r_{m_N} a_{N,m_N}$ for some $r_j \in R$. Consider the polynomial

    $$h(x) := (r_1 f_{N,1}(x) + \cdots + r_{m_N} f_{N,m_N}(x)) \cdot x^{d-N} \in J$$

    with leading coefficient $\sum_{j=1}^{m_N} r_j a_{N,j} = b$ and of order d. Then apply the same argument in case $d \leq N$ and then we're done.

□

Corollary 2.1.5. *1 of Hilbert's Basis Theorem*
If R is a Noetherian ring, then so is $R[x_1, \cdots, x_n]$.

Definition 2.1.6. Let F be a field.

- Recall that a ring R is a *F-dlgebra* if $\exists$ a homomorphism $\varphi : F \to R$, s.t.: $\varphi(F) \subseteq Z(R)$. We identify F into R.
- The ring R is a *finitely generated F-algebra* if R is generated as a ring by F together with finitely many elements of R.
- Let R, S be F-algebras. A map $\psi : R \to S$ is a *F-algebra homomorphism* if ψ is a (ring) homomorphism and $\psi|_F =$ identity map.

Corollary 2.1.7. *2 of Hilbert's Basis Theorem*
The ring R is a finitely generated F-algebra $\Leftrightarrow$ there is some surjective F-algebra homomorphism $\varphi : F[x_1, \cdots, x_n] \to R$. Any finitely generated F-algebra is therefore Noetherian.

Proof.
- - ($\Rightarrow$): Let R be generated by $r_1, \cdots, r_n$. Define $\varphi : F[x_1, \cdots, x_n] \to R$ by $x_i \mapsto r_i$ and $\varphi(a) = a, \forall a \in F$.
  - ($\Leftarrow$): $\varphi(x_i)$ generates R.
- Since: field $\Rightarrow$ PID $\Rightarrow$ Noetherian, then by the corollary 1 of Hilbert's Basis Theorem, $F[x_1, \cdots, x_n]$ is Noetherian and hence by the Quotient-Noetherian Proposition, so is R.

□

2.1.2 Artinian Rings

Definition 2.1.8. Artinian Rings
A ring R is called *Artinian* if it satisfies the DCC on ideals, i.e.: there is no infinite strictly decreasing chain of ideals in R, namely, every decreasing chain of ideals in R

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

must stop somewhere ($I_n = I_{n+1} = I_{n+2} = \cdots$ for some n).

Proposition 2.1.9. *Equivalent-Conditions-of-Artinian Rings Proposition*
The followings are equivalent:

1. *R is an Artinian ring.*
2. *Every nonempty set of ideals in R contains a minimal element under inclusion.*

Proof. Similarly as the Equivalent-Conditions-of-Noetherian Rings Prop. □

Definition 2.1.10. (Krull) Dimension
For any commutative ring R, the *(Krull) dimension* of R is the maximum possible length n of a chain $P_0 \subseteq P_1 \subseteq P_2 \subseteq \cdots \subseteq P_n$ of distinct prime ideals on R. The *dimension* of R is said to be *infinite* if R has arbitrary long chains of distinct prime ideals.

Remark:

- A ring with finite dimension must satisft both ACC and DCC on *prime ideals* (not necessarily on all ideals).
- A field has dimension 0 and a PID that is not a field has dimension 1.

Definition 2.1.11. Radicals
Let $I \lhd R$ be an ideal in a commutative ring R.

- The *radical* of I, denoted by rad$I$, is the collection of elements in R some power of which lie in I, i.e.:
$$\text{rad}I := \{r \in R \mid r^k \in I \text{ for some } k \in \mathbb{N}_+\}$$
- The radical of the zero ideal is called the *nilradical* of I.

- An ideal I is called a *radical ideal* if $I = \text{rad}I$.

Definition 2.1.12. Jacobson Radical
The *Jacobson radical* of R is the intersection of all maximal ideals of R, denoted by Jac$R$.

Proposition 2.1.13. *Jacobson Radical Proposition*
Let R be a commutative ring.

1. *If I is a proper ideal of R, so is $(I, JacR)$, the ideal generated by I and $JacR$.*
2. *The Jacobson radical contains the nilradical of R, i.e.:*

$$rad0 \subseteq JacR$$

3. *$x \in JacR \Leftrightarrow 1 - rx \in U(R), \forall r \in R$.*

Proof. 1. I proper $\Leftrightarrow I \subseteq M$ for some maximal ideal M. Then by the def of Jac$R$, Jac$R \subseteq M$. Hence, $(I, \text{Jac}R) \subseteq M$.

2. - ($\Rightarrow$): Suppose on the contrary that $\exists r \in R$, s.t.: $1 - rx \notin U(R)$. Let M be a maximal ideal containing $1 - rx$. Since $1 \notin M$, then $rx \notin M$ and hence $x \notin \text{Jac}R \subseteq M$, contradiction.
   - ($\Leftarrow$): Suppose on the contrary that $x \notin J$, i.e.: $x \notin M$ for some maximal ideal M. Then $R = (x, M)$. Hence, $1 = rx + y$ for some $y \in M$. Therefore, $1 - rx = y \in M \Rightarrow 1 - rx \notin U(R)$, contradiction.

□

Theorem 2.1.14. *Nakayama's Lemma*
Let R be a commutative ring. If M is any finitely generated R-module and $(JacR)M = M$, then $M = 0$.

Proof. Suppose on the contrary that $M \neq 0$. Let n be the smallest integer such that M is generated by n elements, say $m_1, \cdots, m_n$. Since $M = (\text{Jac}R)M$, then $m_n = r_1m_1 + \cdots + r_nm_n$ for some $r_j \in \text{Jac}R$. Hence, $(1 - r_n)m_n = r_1m_1 + \cdots + r_{n-1}m_{n-1}$. By the Jacobson Radical Proposition.3, $1 - r_n \in U(R) \Rightarrow m_n \in (m_1, \cdots, m_{n-1})$, contradicting the minimality of n. □

Theorem 2.1.15. *Artinian Rings-Properties Theorem*
Let R be an Artinian ring.

1. *There're only finitely many maximal ideals in R.*
2. *The quotient $R/JacR$ is a direct product of a finite number of fields. More precisely, if $M_1, \cdots, M_n$ are the finitely many maximal ideals in R, then*

$$R/JacR \cong R/M_1 \times \cdots \times R/M_n$$

3. *Every prime ideal of R is maximal, i.e.: R has (Krull) dimension 0. The Jacobson radical of R equals the nilradical of R and is a nilpotent ideal $(JacR)^m = 0$ for some $m \in \mathbb{N}_+$.*

4. *The ring R is isomorphic to the direct product of a finite number of Artinian local rings(, where "local ring" means the ring with unique maximal ideal).*

5. *Every Artinian ring is Noetherian.*

Proof. 1. Let $\mathcal{S}$ be the set of all ideals of R that are the intersection of a finite number of maximal ideals. By the Equivalent-Conditions-of-Artinian Rings Prop, $\mathcal{S}$ has a minimal element, say $M_1 \cap \cdots \cap M_n$, which means that $\forall M$ maximal ideal in R, we have

$$M \cap M_1 \cap \cdots \cap M_n = M_1 \cap \cdots \cap M_n$$

Hence, $M_1 \cap \cdots \cap M_n \subseteq M$. Then $M_i \subseteq M$ for some i (which can easily be proved and we omit the details here). Since M_i and M are both maximal, then $M = M_i$ and thus $M_1, \cdots, M_n$ are *all* the maximal ideals of R.

2. holds immediately by the Chinese Remainder Theorem.

3. We first prove that Jac$R$ is nilpotent. By DCC, there is some $m \in \mathbb{N}_+$, s.t.: $(\mathrm{Jac}R)^m = (\mathrm{Jac}R)^{m+i}$ for all $i \in \mathbb{N}$. Suppose on the contrary that $(\mathrm{Jac}R)^m \neq 0, \forall m \in \mathbb{N}_+$. Consider the set $\mathcal{S}$ of proper ideals I such that $I(\mathrm{Jac}R)^m \neq 0$. Hence, Jac$R \in \mathcal{S}$. By the Equivalent-Conditions-of-Artinian Proposition, $\exists I_0 \in \mathcal{S}$, s.t.: I_0 is the minimal element of $\mathcal{S}$. Then there must be an element $x \in I_0$, s.t.: $x(\mathrm{Jac}R)^m \neq 0$. Note that by the minimality of I_0, $I_0 = (x)$. But
$$((x)\mathrm{Jac}R)(\mathrm{Jac}R)^m = x(\mathrm{Jac}R)^{m+1} = x(\mathrm{Jac}R)^m \neq 0$$
which means that $(x)\mathrm{Jac}R \in \mathcal{S} \Rightarrow (x)\mathrm{Jac}R \subseteq (x)R = (x) \overset{\text{min. of } I_0}{\subseteq} (x)\mathrm{Jac}R \Rightarrow (x) = (x)\mathrm{Jac}R$. Therefore, by the Nakayama's Lemma, $(x) = 0 \Rightarrow x = 0$, contradicting $x(\mathrm{Jac}R)^m \neq 0$, which follows that Jac$R$ is nilpotent, and hence Jac$R \subseteq$ rad0. Then by the Jabocson Radical Prop.2, these two ideals are equal.
Now we prove the first statement. Given any prime ideal P in R, note that

$$\mathrm{Jac}R = \mathrm{rad}0 = \mathrm{Nil}R = \bigcap_{I \lhd R,\text{ prime}} I \subseteq P$$

Hence, the image of P in $R/\mathrm{Jac}R \overset{2}{\cong} R/M_1 \times \cdots \times R/M_n$ consists of the elements that are 0 in one of the components. In particular, such a prime ideal also maximal ideal in $R/M_1 \times \cdots \times R/M_n$, which follows that P is a maximal ideal in R

4. Let $M_1, \cdots, M_n$ be all the distinct maximal ideals of R and let $(\mathrm{Jac}R)^m = 0$ as in 3. Since

$$M_1^m \times \cdots \times M_n^m \subseteq (M_1 \times \cdots \times M_n)^m \subseteq (\mathrm{Jac}R)^m = 0$$

then $M_1^m \times \cdots \times M_n^m = 0$. Then by the Chinese Remainder Theorem,

$$R \cong R/M_1^m \times \cdots \times R/M_n^m$$

and each R/M_i^m is an Artinian ring with unique maximal ideal M_i/M_i^m.

5. It suffices by 4 to show that an Artinian local ring is Noetherian. Let S be an Artininan ring with unique maximal ideal M. In this case, we have $M = \text{Jac}R$. Hence, by 3, $M^m = (\text{Jac}R)^m = 0$ for some $m \in \mathbb{N}_+$. Therefore, $R \cong R/M^m$, which is Noetherian $\Leftrightarrow$ Artinian.

□

Corollary 2.1.16. *of Artinian Rings-Properties Theorem*
The ring R is Artinian $\Leftrightarrow$ R is Noetherian and has (Krull) dimension 0*.*

Proof.
- holds immediately by the Artinian Rings-Properties Theorem.
- Omitted.

□

Example:

- Let $n > 1$ be an integer. Since $R = \mathbb{Z}/n\mathbb{Z}$ is finite, then it is Artinian. Let $n = p_1^{a_1} \cdots p_s^{a_s}$ be the unique factorization of n into distinct prime powers. Then

$$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/p_1^{a_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_s^{a_s}\mathbb{Z}$$

  Since $\mathbb{Z}/p_i^{a_i}\mathbb{Z}$ is an Artinian local ring with unique maximal ideal $(p_i)/(p_i^{a_i})$, then this is the decomposition of $\mathbb{Z}/n\mathbb{Z}$ given by the Artinian Rings-Properties Theorem.4. The Jacobson radical of R is the ideal generated by $p_1 \cdots p_s$ and $R/\text{Jac}R \cong \mathbb{Z}/p_1\mathbb{Z} \times \cdots \times \mathbb{Z}/p_s\mathbb{Z}$.
- For any field F, a F-algebra R that is finite dimensional as a vector space over F is Artinian since ideals in R are in particular F-subspaces of R. Hence, the length of any chain of ideals in R is bounded by $\dim_F R$.

2.1.3 Noetherian and Artinian Modules

Definition 2.1.17. Noetherian Modules
An R-module M is *Noetherian* if every ascending chain of submodules $M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$ stabilizes. That is, for some k, $M_{k+j} = M_k, \forall j \in \mathbb{N}$.

Proposition 2.1.18. *Equivalent-Conditions-of-Noetherian Modules Proposition*
An R-module M is Noetherian $\Leftrightarrow$ every nonempty set Σ of submodules of M has a maximal element $\Leftrightarrow$ every submodule of M is finitely generated.

Proof. Similarly as the Equivalent-Conditions-of-Noetherian Rings Prop. □

Definition 2.1.19. Artinian Modules
An R-module M is *Artinian* if every descending chain of submodules $M_0 \supseteq M_1 \supseteq M_2 \supseteq \cdots$ inside M stabilizes.

Proposition 2.1.20. *Equivalent-Conditions-of-Artinian Modules Proposition*
An R-module M is Artinian $\Leftrightarrow$ every nonempty set Σ of submodules of M has a minimal element.

Proof. Similarly as the Equivalent-Conditions-of-Noetherian Rings Prop. $\square$

Remark: A ring R is Noetherian (or Artinian) if R is a Noetherian (respectively Artinian) as an R-module.

2.2 Affine Algebraic Sets

Definition 2.2.1. Affine Spaces
Let F be a field. The set $\mathbb{A}^n$ of n-tuples of elements of F is called *affine n-space over F*.

Remark: If the variables $x_1, x_2, \cdots, x_n$ are independent over F, then every polynomial f in $F[x_1, \cdots, x_n]$ can be viewed as F-valued functions $f : \mathbb{A}^n \to F$ on $\mathbb{A}^n$ by evaluating f at the points in $\mathbb{A}^n$:

$$f : (a_1, \cdots, a_n) \mapsto f(a_1, \cdots, a_n) \in F$$

This gives the following definition:

Definition 2.2.2. Coordinate Rings of Affine Spaces
Let $\mathbb{A}^n$ be an affine n-space over F. We define the *coordinate ring of* $\mathbb{A}^n$ by

$$F[\mathbb{A}^n] := \text{ the ring of } F\text{-valued functions on } \mathbb{A}^n$$

Remark: It's easy to show that $F[\mathbb{A}^n]$ has ring structure together with the usual addition and multiplication.
Example: $F = \mathbb{R}, n = 2$. The coordinate ring of Euclidean 2-space $\mathbb{R}^2$, denoted as $\mathbb{R}[\mathbb{A}^2]$, is the ring of polynomials in two variables, say x and y, acting as real valued functions on $\mathbb{R}^2$ (the usual "coordinate functions").

Definition 2.2.3. Affine Algebraic Sets
Let $\mathbb{A}^n$ be an affine n-space over F.

- Let S be any non-empty subset of $F[\mathbb{A}^n]$. We define by $\mathcal{Z}(S)$ the set of points where all functions in S are simultaneously zero, namely

  $$\mathcal{Z}(S) := \{(a_1, \cdots, a_n) \in \mathbb{A}^n \mid f(a_1, \cdots, a_n) = 0, \forall f \in S\}$$

  Conventionally, we write $\mathcal{Z}(\emptyset) := \mathbb{A}^n$.

- A subset V of $\mathbb{A}^n$ is called an *affine algebraic set* if V is the set of common zeros of some set S of polynomials, i.e.: if $V = \mathcal{Z}(S)$ for some $S \subseteq F[\mathbb{A}^n]$. In this case, $V = \mathcal{Z}(S)$ is called the *locus* of S in $\mathbb{A}^n$.

Remark:

- If $S = \{f\}$ or $\{f_1, \cdots, f_m\}$, then we shall simply write $\mathcal{Z}(f)$ or $\mathcal{Z}(f_1, \cdots, f_m)$ and call it the locus of f or $f_1, \cdots, f_m$, respectively.
- Note that the locus of a single polynomial of the form $f - g$ is the same as the solutions in affine n-space of the equation $f = g$. Hence, affine algebraic sets are the solutions to systems of polynomial equations.

Example:

- If $n = 1$, then the locus of a single polynomial $f \in F[x]$ is the set of roots of f in F. Precisely, the affine algebraic sets in $\mathbb{A}^1$ are $\emptyset$, any finite set and F.
- The one point subsets of $\mathbb{A}^n$ for any n are affine algebraic since

$$\mathcal{Z}(x_1 - a_1, \cdots, x_n - a_n) = \{(a_1, \cdots, a_n)\}$$

  More generally, any finite subset of $\mathbb{A}^n$ is an affine algebraic set.
- One may define lines, planes, etc. in $\mathbb{A}^n$, which are *linear algebraic sets*, the loci of sets of linear (degree 1) polynomials of $F[x_1, \cdots, x_n]$. For example, a line in $\mathbb{A}^2$ is defined by $ax + by = c$ which is the locus of the polynomial $f(x, y) = ax + by - c \in F[x, y]$. In particular, the coordinate axes, coordinate plane, etc. in $\mathbb{A}^n$ are affine algebraic sets.
- In general, the affine algebraic set $\mathcal{Z}(f)$ of a nonconstant polynomial f is called a *hypersurface* in $\mathbb{A}^n$.

Proposition 2.2.4. *Affine Algebraic Sets-Properties Proposition*
Let S, T be subsets of $F[\mathbb{A}^n]$.

1. *If $S \subseteq T$, then $\mathcal{Z}(T) \subseteq \mathcal{Z}(S)$.*
2. *$\mathcal{Z}(S) = \mathcal{Z}(I)$ where $I = (S)$ denotes the ideal in $F[\mathbb{A}^n]$ generated by S.*
3. *The intersection of two affine algebraic sets is again an affine algebraic set. In particular, $\mathcal{Z}(S) \cap \mathcal{Z}(T) = \mathcal{Z}(S \cup T)$. More generally, an arbitrary intersection of affine algebraic sets is an affine algebraic set: if $\{S_j \mid j \in J\}$ is any collection of subsets of $F[\mathbb{A}^n]$, then*

$$\bigcap_{j \in J} \mathcal{Z}(S_j) = \mathcal{Z}(\bigcup_{j \in J} S_j)$$

4. *The union of two affine algebraic sets is again an affine algebraic set. In particular, $\mathcal{Z}(I) \cup \mathcal{Z}(J) = \mathcal{Z}(IJ)$, where I, J are ideals and IJ is their product.*
5. *$\mathcal{Z}(0) = \mathbb{A}^n$ and $\mathcal{Z}(1) = \emptyset$ where 0 and 1 denote constant functions.*

Proof. 1. holds directly by the particular part in 3.

2. Since $S \subseteq (S) = I$, then by 1, we have $\mathcal{Z}(I) \subseteq \mathcal{Z}(S)$. Conversely, for any $(a_1, \cdots, a_n) \in \mathcal{Z}(S)$, by def, $f(a_1, \cdots, a_n) = 0, \forall f \in S$. Hence, for any finite sum of elements in S, it vanishes at $(a_1, \cdots, a_n)$, which follows that $f(a_1, \cdots, a_n) = 0, \forall f \in I \Rightarrow (a_1, \cdots, a_n) \in \mathcal{Z}(I)$.

3. It suffices to prove the general part:

$$\begin{aligned}(a_1, \cdots, a_n) \in \bigcap_{j \in J} \mathcal{Z}(S_j) &\Leftrightarrow f(a_1, \cdots, a_n) = 0, \forall f \in S_j, \forall j \in J \\ &\Leftrightarrow f(a_1, \cdots, a_n) = 0, \forall f \in \bigcup_{j \in J} S_j \\ &\Leftrightarrow (a_1, \cdots, a_n) \in \mathcal{Z}(\bigcup_{j \in J} S_j)\end{aligned}$$

4. By 2, it suffices to prove the particular part:

$$\begin{aligned}(a_1, \cdots, a_n) \in \mathcal{Z}(I) \cup \mathcal{Z}(J) &\Leftrightarrow (a_1, \cdots, a_n) \in \mathcal{Z}(I) \text{ or } (a_1, \cdots, a_n) \in \mathcal{Z}(J) \\ &\Leftrightarrow f(a_1, \cdots, a_n) = 0, \forall f \in I \text{ or } g(a_1, \cdots, a_n) = 0, \forall g \in J \\ &\Leftrightarrow fg(a_1, \cdots, a_n) = 0, \forall fg \in IJ \\ &\Leftrightarrow (a_1, \cdots, a_n) \in \mathcal{Z}(IJ)\end{aligned}$$

5. Obvious.

□

Proposition 2.2.5. *Geometric Property-in-Affine Spaces Proposition*
Every affine algebraic set is the intersection of a finite number of hypersurfaces in $\mathbb{A}^n$, namely, $\forall$ subset $S \subseteq \mathbb{A}^n$, $\exists f_1, \cdots, f_q \in F[\mathbb{A}^n]$, s.t.:

$$\mathcal{Z}(S) = \mathcal{Z}(f_1) \cap \cdots \cap \mathcal{Z}(f_q)$$

Proof. Since by corollary 1 of Hilbert Basis Theorem, $F[x_1, \cdots, x_n]$ is a Noetherian ring, then by the Equivalent-Conditions-of-Noetherian Rings Prop.3, every ideal I in $F[x_1, \cdots, x_n]$ is finitely generated, say $I = (f_1, \cdots, f_q)$. Then by the Affine Algebraic Sets-Properties Prop.2, we're done.

□

Remark:

- By 2, every affine algebraic set is the affine algebraic set corresponding to an *ideal* of the coordinate ring. Hence, we may consider $\mathcal{Z}(\cdot)$ as a map, namely

$$\mathcal{Z} : \{\text{ideals of } F[\mathbb{A}^n]\} \to \{\text{affine algebraic sets in } \mathbb{A}^n\}$$

- If V is an affine algebraic set in affine n-space, then there may be many ideals I such that

$V = \mathcal{Z}(I)$. For example, in affine 2-space over $\mathbb{R}$, the y-axis is the locus of the ideals (x) of $\mathbb{R}[x, y]$ and also the locus of $(x^2), (x^3)$, etc. More generally, the zeros of any polynomial are the same as the zeros of all its positive powers and hence $\mathcal{Z}(I) = \mathcal{Z}(I^k), \forall k \in \mathbb{N}_+$.
While the ideal where locus determines a particular affine algebraic set V is not unique, there is a unique largest ideal that determines V, given by the set of all polynomials tht vanishes on V.

Definition 2.2.6. Let $A \subseteq \mathbb{A}^n$ be an arbitrary subset of $\mathbb{A}^n$. Define

$$\mathcal{I}(A) := \{f \in F[x_1, \cdots, x_n] \mid f(a_1, \cdots, a_n) = 0, \forall (a_1, \cdots, a_n) \in A\}$$

which is an ideal and is the unique largest ideal of functions that are identically zero in A.

Remark: This defines a correspondence

$$\mathcal{I} : \{\text{subsets in } \mathbb{A}^n\} \to \{\text{ideals of } F[\mathbb{A}^n]\}$$

Example:

- In the Euclidean plane, $\mathcal{I}$(the x-axis) is the ideal generated by y in the coordinate ring $\mathbb{R}[x, y]$.
- Over any field F, the ideal of functions vanishing at $(a_1, \cdots, a_n) \in \mathbb{A}^n$ is a maximal ideal, since it is the kernel of the surjective ring homomorphism from $F[x_1, \cdots, x_n]$ to the field F given by evaluation at $(a_1, \cdots, a_n)$. It follows that

$$\mathcal{I}((a_1, \cdots, a_n)) := (x_1 - a_1, \cdots, x_n - a_n)$$

- Let $V = \mathcal{Z}(x^3 - y^2)$ in $\mathbb{A}^n$. If $(a, b) \in V$, then $a^3 = b^2$. If $a \neq 0$, then $b \neq 0$ and we can write $a = (b/a)^2, b = (b/a)^3$. Hence,
$$V = \{(a^3, a^2) \mid a \in F\}$$
For any polynomial $f(x, y) \in F[x, y]$, we can write $f(x, y) = f_0(x) + f_1(x)y + (x^3 - y^2)g(x, y)$. For $f(x, y) \in \mathcal{I}(V)$, i.e.: $f(a^2, a^3) = 0, \forall a \in F$, we have

$$f_0(a^2) + f_1(a^2)a^3 = 0, \forall a \in F$$

Let $f_0(x) = a_r x^r + \cdots + a_1 x + x_0, f_1(x) = b_s x^s + \cdots + b_1 x + b_0$. Then

$$f_0(x^2) + f_1(x^2)x^3 = (a_r x^{2r} + \cdots + a_1 x^2 + a_0) + (b_s x^{2s+3} + \cdots + b_1 x^5 + b_0 x^3)$$

This polynomial is 0 $\forall x = a \in F$. If F is infinite, then this polynomial has infinitely many zeros, which can happen only if the coefficients are zero. Since the coefficients of the terms of even degree are the coefficients of $f_0(x)$ and the coefficients of the terms of odd degree are the coefficients of $f_1(x)$, then $f_0(x) = f_1(x) = 0$. Hence, $f(x, y) = (x^3 - y^2)g(x, y)$, so

$$\mathcal{I}(V) = (x^3 - y^2) \subseteq F[x, y]$$

However, on the other hand, if F is finite, then there may be elements in $\mathcal{I}(V)$ not lying in the ideal $(x^3 - y^2)$. For example, if $F = \mathbb{F}_2$, then V is simply the set $\{(0, 0), (1, 1)\}$ and hence $\mathcal{I}(V)$ contains the polynomial $x(x - 1)$.

Proposition 2.2.7. *$\mathcal{I}$-Properties Proposition*
Let A, B be subsets of $\mathbb{A}^n$.

1. *If $A \subseteq B$, then $\mathcal{I}(B) \subseteq \mathcal{I}(A)$.*
2. *$\mathcal{I}(A \cup B) = \mathcal{I}(A) \cap \mathcal{I}(B)$.*
3. *$\mathcal{I}(\emptyset) = F[x_1, \cdots, x_n]$ and if F is infinite, then $\mathcal{I}(\mathbb{A}^n) = 0$.*

Proof.
1. For any $f \in \mathcal{I}(B)$, $f(B) = 0 \overset{A \subseteq B}{\Rightarrow} f(A) = 0 \Rightarrow f \in \mathcal{I}(A)$.
2. $f \in \mathcal{I}(A \cup B) \Leftrightarrow f(A \cup B) = 0 \Leftrightarrow f(A) = 0, f(B) = 0 \Leftrightarrow f \in \mathcal{I}(A) \cap \mathcal{I}(B)$.
3. Obvious.

□

Proposition 2.2.8. *Relations between $\mathcal{Z}$ and $\mathcal{I}$ Proposition*

1. *Let A be any subset of $\mathbb{A}^n$. Then $A \subseteq \mathcal{Z}(\mathcal{I}(A))$. If I is any ideal, then $I \subseteq \mathcal{I}(\mathcal{Z}(I))$.*
2. *If $V = \mathcal{Z}(I)$ is an affine algebraic set, then $V = \mathcal{Z}(\mathcal{I}(V))$. If $I = \mathcal{I}(A)$, then $\mathcal{I}(\mathcal{Z}(I)) = I$. Namely,*
$$\mathcal{Z}(\mathcal{I}(\mathcal{Z}(I))) = \mathcal{Z}(I), \mathcal{I}(\mathcal{Z}(\mathcal{I}(A))) = \mathcal{I}(A)$$

Proof.
1.
   - $(a_1, \cdots, a_n) \in A \Rightarrow f(a_1, \cdots, a_n) = 0, \forall f \in \mathcal{I}(A) \Rightarrow (a_1, \cdots, a_n) \in \mathcal{Z}(\mathcal{I}(A))$.
   - $f \in I \Rightarrow f(a_1, \cdots, a_n) = 0, \forall (a_1, \cdots, a_n) \in \mathcal{Z}(I) \Rightarrow f \in \mathcal{I}(\mathcal{Z}(I))$.
2.
   - By 1, we've already have $\mathcal{Z}(I) \subseteq \mathcal{Z}(\mathcal{I}(\mathcal{Z}(I)))$. Conversely, for any $(a_1, \cdots, a_n) \in \mathcal{Z}(\mathcal{I}(\mathcal{Z}(I)))$, $f(a_1, \cdots, a_n) = 0, \forall f \in \mathcal{I}(\mathcal{Z}(I)) \Rightarrow (a_1, \cdots, a_n) \in \mathcal{Z}(I)$.
   - Similar.

□

Remark: The second relation shows that the maps $\mathcal{Z}$ and $\mathcal{I}$ act as inverses of each other provided one restricts to the collection of affine algebraic sets $V = \mathcal{Z}(I)$ in $\mathbb{A}^n$ and to the set of ideals in $F[\mathbb{A}^n]$ of the form $\mathcal{I}(V)$.

Definition 2.2.9. Coordinate Rings of Affine Algebraic Sets
Let $V \subseteq \mathbb{A}^n$ be an affine algebraic set. The quotient ring $F[\mathbb{A}^n]/\mathcal{I}(V)$ is called the *coordinate ring of V*, denoted as $F[V]$.

Remark:

- Since for $V = \mathbb{A}^n$ and F infinite, by the $\mathcal{I}$-Properties Prop.3, we have $\mathcal{I}(V) = 0$, then this def extends the previous terminology.
- The polynomials in $F[\mathbb{A}^n]$ define F-valued functions on V simply by restricting these functions on $\mathbb{A}^n$ to the subset V. Two such polynomial functions f and g define the *same* function on $V \Leftrightarrow (f-g)_V \equiv 0 \Leftrightarrow f - g \in \mathcal{I}(V)$. Hence, the cosets $\overline{f} = f + \mathcal{I}(V) \in F[V]$ are precisely the restrictions to V of ordinary polynomial functions f from $\mathbb{A}^n$ to F.
- If x_i denotes the $i$the coordinate function on $\mathbb{A}^n$ ($x_i : (\cdots, \underbrace{a_i}_{i\text{th position}}, \cdots) \mapsto a_i$), then the restriction $\overline{x_i}$ of x_i to V is an element of $F[V]$ and $F[V]$ is finitely generated as an F-algebra by $\overline{x_1}, \cdots, \overline{x_n}$ (but this need *not* be a minimal generating set).

Example: Let $V = \mathcal{Z}(xy-1)$ be the hyperbola $y = 1/x$ in $\mathbb{R}^2$. Then

$$\mathbb{R}[V] := \mathbb{R}[x,y]/(xy-1)$$

The polynomials $f(x,y) = x$ (the x-coordinate function) and $g(x,y) = x + (xy-1)$ which are different functions on $\mathbb{R}^2$, define the same function on V. In the quotient ring $\mathbb{R}[V]$, we have $\overline{f}(a,b) = f(a, 1/a)$.

Definition 2.2.10. Morphisms of Affine Algebraic Sets
Let $V \subseteq \mathbb{A}^n, W \subseteq \mathbb{A}^m$ be two affine algebraic sets. A map $\varphi : V \to W$ is called a *morphism* (or *polynomial map* or *regular map*) *of affine algebraic sets* if there are polynomials $\varphi_1, \cdots, \varphi_m \in F[x_1, \cdots, x_n]$, s.t.: for any $(a_1, \cdots, a_n) \in V$,

$$\varphi((a_1, \cdots, a_n)) = (\varphi_1(a_1, \cdots, a_n), \cdots, \varphi_m(a_1, \cdots, a_n))$$

Furthermore, φ is an *isomorphism of affine algebraic sets* if there exists a morphism of affine algebraic sets $\psi : W \to V$, s.t.:

$$\varphi \circ \psi = \mathrm{id}_W, \psi \circ \varphi = \mathrm{id}_V$$

Remark: In general, $\varphi_1, \cdots, \varphi_m$ are *not* uniquely defined. For example, both $f(x,y) = x$ and $g(x,y) = x + (xy-1)$ in the example above define the same morphism from $V = \mathcal{Z}(xy-1)$ to $W = \mathbb{A}^1$.

Theorem 2.2.11. *Morphisms-Homomorphisms Theorem*
Let $V \subseteq \mathbb{A}^n, W \subseteq \mathbb{A}^m$ be affine algebraic sets. There is a bijective correspondence

$$\left\{\begin{matrix} \textit{morphisms from } V \textit{ to } W \\ \textit{as affine algebraic sets} \end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix} F\textit{-algebra homomorphisms} \\ \textit{from } F[W] \textit{ to } F[V] \end{matrix}\right\}$$

More precisely,

1. *Every morphism $\varphi : V \to W$ induces an associated F-algebra homomorphism $\tilde{\varphi} : F[W] \to F[V]$ defined by $\tilde{\varphi}(f) := f \circ \varphi$.*

2. *Every F-algebra homomorphism $\Phi : F[W] \to F[V]$ is induced by a unique morphism $\varphi : V \to W$, i.e.: $\Phi = \tilde{\varphi}$.*

3. *If $\varphi : V \to W$ and $\psi : W \to U$ are morphisms of affine algebraic sets, then $\widetilde{\psi \circ \varphi} = \widetilde{\varphi} \circ \widetilde{\psi} : F[U] \to F[V]$.*

4. *The morphism $\varphi : V \to W$ is an isomorphism $\Leftrightarrow$ $\widetilde{\varphi} : F[W] \to F[V]$ is an F-algebra isomorphism.*

Proof.

1. $\forall$ polynomial $F \in \mathcal{I}(W)$, it's easy too check that $F \circ \varphi = F \circ (\varphi_1, \cdots, \varphi_m) \in \mathcal{I}(V)$. This gives the well-defined-ness of $\widetilde{\varphi}$. We leave to readers to check that $\widetilde{\varphi}$ is an F-algebra homomorphism.

2. Let F_i be a representative in $F[x_1, \cdots, x_n]$ for the image under Φ of $x_i \in F[W]$. Then define $\varphi := (F_1, \cdots, F_m) : \mathbb{A}^n \to \mathbb{A}^m$. It suffices to show that $\varphi(V) \subseteq \varphi(W)$, since by def φ is already defined by polynomials. Given any $g \in \mathcal{I}(W) \subseteq F[x_1, \cdots, x_m]$, we have

$$g(x_1 + \mathcal{I}(W), \cdots, x_m + \mathcal{I}(W)) = g(x_1, \cdots, x_m) + \mathcal{I}(W) = 0 \in F[W]$$

and hence $\Phi(g(x_1 + \mathcal{I}(W), \cdots, x_m + \mathcal{I}(W))) = 0 \in F[V]$. Since Φ is an F-algebra homomorphism, then

$$g(\Phi(x_1 + \mathcal{I}(W)), \cdots, \Phi(x_m + \mathcal{I}(W))) = 0 \in F[V]$$

By def, $\Phi(x_i + \mathcal{I}(W)) = F_i \text{ mod} \mathcal{I}(V)$ and hence $g \circ (F_1, \cdots, F_m) \in \mathcal{I}(V)$, which follows that $g \circ (F_1, \cdots, F_m)(a_1, \cdots, a_m) = 0, \forall (a_1, \cdots, a_m) \in V$. Then we have $\widetilde{\varphi} = \Phi$. As for the uniqueness, if $\widetilde{\varphi} = \widetilde{\varphi'} = \Phi$, then $(\varphi' - \varphi)|_V \equiv 0$ and then we're done.

3. Omitted and 4 holds then immediately.

$\square$

Example: For any infinite field F, let $V = \mathbb{A}^1$ and let $W = \mathcal{Z}(x^3 - y^2) = \{(a^2, a^3) \mid a \in F\}$. The map $\varphi : V \to W$ defined by $\varphi(a) = (a^2, a^3)$ is a morphism from V to W. Note that φ is a bijection. The coordinate rings are $F[V] = F[x]$ and $F[W] = F[x, y]/(x^3 - y^2)$ and the associated F-algebra homomorphism of coordinate rings is determined by

$$\widetilde{\varphi} : \begin{array}{c} F[W] \to F[V] \\ x \mapsto x^2 \\ y \mapsto x^3 \end{array}$$

The image of $\widetilde{\varphi}$ is the subalgebra $F[x^2, x^3] = F + x^2 F[x]$ of $F[x]$, so in particular $\widetilde{\varphi}$ is *not* surjective. Hence, $\widetilde{\varphi}$ is *not* an isomorphism of coordinate rings, which by the Morphisms-Homomorphisms Th'm.4, follows that φ is *not* an isomorphism of affine algebraic sets, even though it is a bijective map. On the other hand, the inverse map of φ is given by $\psi(0, 0) = 0$ and $\psi(a, b) = b/a$ for $a \neq 0$, which can *not* be achieved by a polynomial.

Corollary 2.2.12. *of Morphisms-Homomorphisms Theorem*

Let $\varphi : V \to W$ be a map between affine algebraic sets. Then φ is morphism $\Leftrightarrow$ $\forall f \in F[W]$, the *composite map $f \circ \varphi$ is an element of $F[V]$ (as an F-valued function on V). When φ is a morphism, $\varphi(v) = w$ with $v \in V, w \in W \Leftrightarrow \widetilde{\varphi}^{-1}(\mathcal{I}(\{v\})) = \mathcal{I}(\{w\})$.*

Proof.

- We first prove the second statement. Note that $\varphi(v) = w \Leftrightarrow$ every polynomial f vanishing at w vanishes at $\varphi(v) \Leftrightarrow$ every polynomial f vanishing at w satisfies $\widetilde{\varphi}(f)$ vanishes at $v \Leftrightarrow$ every polynomial f vanishing at w is in $\mathcal{I}(\{v\}) \Leftrightarrow \widetilde{\varphi}(f) \in \mathcal{I}(\{v\})$ for every $f \in \mathcal{I}(\{w\}) \Leftrightarrow \widetilde{\varphi}(\mathcal{I}(\{w\})) \subseteq \mathcal{I}(\{v\}) \Leftrightarrow \mathcal{I}(\{v\}) \subseteq \widetilde{\varphi}^{-1}(\mathcal{I}(\{v\})) \overset{\mathcal{I}(\{w\}),\mathcal{I}(\{v\}) \text{ are maximal}}{\Longleftrightarrow} \mathcal{I}(\{w\}) = \widetilde{\varphi}^{-1}(\mathcal{I}(\{v\}))$.
- We now prove the first statement:
  - ($\Rightarrow$): Obvious.
  - ($\Leftarrow$): Note that the composition with any map $\varphi : V \to W$ defines an F-algebra homomorphism $\widetilde{\varphi}$ from the F-algebra of F-valued functions on W to the F-algebra of F-valued functions on V. If $f \circ \varphi \in F[V], \forall f \in F[W]$, then $\widetilde{\varphi}$ is an F-algebra homomorphism from $F[W]$ to $F[V]$ and hence there exists a unique morphism $\Phi : V \to W$ which induces $\widetilde{\varphi}$, i.e.: $\widetilde{\Phi} = \widetilde{\varphi}$. Also, since $\widetilde{\varphi}$ is an F-algebra homomorphism from $F[W]$ to $F[V]$, then we've already shown that $\varphi(v) = w \Leftrightarrow \widetilde{\Phi}^{-1}(\mathcal{I}(\{v\})) = \widetilde{\varphi}^{-1}(\mathcal{I}(\{v\})) = \mathcal{I}(\{w\}) \Leftrightarrow \Phi(v) = w$. Hence, $\varphi = \Phi$ which is a morphism.

□

2.3 Radicals

Intuition: Since the zeros of a polynomial f are the same as the zeros of the powers $f^2, f^3, \cdots$, in general there are many different ideals in the ring $F[x_1, \cdots, x_n]$ whose zero locus define the same affine algebraic set V in affine n-space. This leads to the notion of the radical of an ideal, which can be defined in any commutative ring:

Definition 2.3.1. Radicals
Let $I \lhd R$ be an ideal in a commutative ring R.

- The *radical* of I, denoted as rad I, is the collection of elements in R some power of which lie in I, i.e.:
$$\text{rad } I := \{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{N}_+\}$$
- The radical of zero ideal is called the *nilradical of* R, namely
$$\text{Nil } R := \text{rad } 0$$
- I is called a *radical ideal* if $I = \text{rad } I$.

Remark: Since $a \in \text{Nil } R \Leftrightarrow a^k = 0$ for some $k \in \mathbb{N}_+$, then
$$\text{Nil } R = \text{ the set of all nilpotent elements of } R.$$

Proposition 2.3.2. *Radicals-Nilradicals Proposition*
Let $I \lhd R$ be an ideal in a commutative ring R. Then rad I is an ideal containing I, and (rad I)/I is the nilradical of R/I. In particular,

$$R/I \text{ has no nilpotent elements} \Leftrightarrow I = \text{rad } I \text{ is a radical ideal.}$$

Proof. It's clear that $I \subseteq \text{rad } I$. By def, the nilradical of R/I consists of the elements in the quotient some power of which is 0. By the Lattice Isomorphism Theorem for rings, this collection of elements corresponds to the elements of R some power of which lie in I, i.e.: rad I. It therefore suffices to show that the nilradical N of any commutative ring R is an ideal, which is easy to check. □

Proposition 2.3.3. *Radicals-Prime Ideals Proposition*

$$\text{rad } I = \bigcap_{I \subseteq P \lhd R,\ prime} P$$

In particular, Nil $R = \bigcap_{P \lhd R,\ prime} P$.

Corollary 2.3.4. *of Radicals-Prime Ideals Proposition*
Prime (and hence also maximal) ideals are radical.

Example:

- In $R = \mathbb{Z}$, the ideal (a) is a radical ideal $\Leftrightarrow$ a is square free or zero. More generally, if $a = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$ with $a_i \in \mathbb{N}_+$ is the prime factorization of the positive integer a, then

  $$\text{rad } a = (p_1 p_2 \cdots p_r)$$

  For example, rad (180) = (30). Note that $(p_1), (p_2), \cdots, (p_r)$ are precisely the prime ideals containing (a) and that their intersection is the ideal $(p_1 p_2 \cdots p_r)$. More generally, in any UFD R,

  $$\text{rad } a = (p_1 p_2 \cdots p_r)$$

  if $a = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$ is the unique factorization of a into distinct irreducibles.
- The ideal $(x^3 - y^2)$ in $F[x, y]$ is a prime ideal, and hence it is radical.
- If $l_1, \cdots, l_m$ are linear polynomials in $F[x_1, \cdots, x_n]$, then $I = (l_1, \cdots, l_m)$ is either $F[x_1, \cdots, x_n]$ or a prime ideal and hence I is a radical ideal.

Proposition 2.3.5. *Noetherian-Radicals Proposition*
If R is a Noetherian ring, then for any ideal I, some positive power of rad I is contained in I. In particular, the nilradical, Nil R, of a Noetherian ring R is a nilpotent ideal, say $N^k = 0$ for some $k \in \mathbb{N}_+$.

Proof. By the Equivalent-Conditions-of-Noetherian Rings Prop.3, rad I is finitely generated, since by the Radicals-Nilradicals Prop. rad I is an ideal. Let rad $I = (a_1, \cdots, a_m)$. Then by the def of radicals, for each i, we have $a_i^{k_i} \in I$ for some positive integer k_i. Take $k := \max\{k_1, \cdots, k_m\}$. Then the ideal $(\text{rad } I)^{km}$ is generated by elements of the form

$$a_1^{d_1} a_2^{d_2} \cdots a_m^{d_m} \text{ where } d_1 + \cdots + d_m = km$$

Note that at least one of d_i's is greater than or equal to k, say $d_j \geq k$. Hence, $a_j^{d_j} = a_j^{d_j - k} a_j^k \overset{I \triangleleft R}{\in} I$. Therefore, the generators $a_1^{d_1} a_2^{d_2} \cdots a_m^{d_m}$ of $(\text{rad } I)^{km}$ are all elements in I and hence

$$(\text{rad } I)^{km} \subseteq I$$

□

2.4 Zariski Topology

Definition 2.4.1. Algebraically Closed Field
A field F is said to be *algebraically closed* if every polynomial $p(x) \in F[x]$ has a root $\lambda \in F$.

Example: By the Fundamental Theorem of Algebra, the set of complex numbers $\mathbb{C}$ is algebraically closed.

Theorem 2.4.2. *Hilbert's Nullstellensatz Theorem*
Let E be an algebraically closed field. Then $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$ for every ideal of $E[x_1, \cdots, x_n]$. Moreover, the maps $\mathcal{Z}$ and $\mathcal{I}$ in the correspondence

$$\{\textit{affine algebraic sets}\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\textit{radical ideals}\}$$

are bijections that are inverses of each other.

Proof. We'll prove this in the later section. □

Remark:

- The maps $\mathcal{I}$ and $\mathcal{Z}$ in the Nullstellensatz are defined over any field F and as mentioned are not bijections if F is *not* algebraically closed. However, for any field F, the map $\mathcal{Z}$ is always surjective and the map $\mathcal{I}$ is always injective.
- One particular consequence of the Nullstellensatz is that for any *proper* ideal I, we have $\mathcal{Z}(I) \neq \emptyset$, since rad $I \neq F[\mathbb{A}^n]$. Hence, there always exists at least one common zero for all the polynomials contained in a proper ideal (over an algebraically closed field).

Definition 2.4.3. Zariski Topology
The *Zariski topology* on affine n-space over an arbitrary field F is the topology in which the closed sets are the affine algebraic sets in $\mathbb{A}^n$.

Remark:

- One can check that the Zariski topology is a well-defined topology by the Affine Algebraic Sets-Properties Prop.3,4,5.
- The Zariski topology is defined by the zero sets of the ideals in the coordinate ring of $\mathbb{A}^n$.

Example: For the Zariski topology on $\mathbb{A}^1$, the only closed sets are $\emptyset, F$ and the finite sets and hence the non-empty open sets are the complements of finite sets. If F is infinite, then in the Zariski topology, any two non-empty open sets in $\mathbb{A}^1$ have non-empty intersection.

Proposition 2.4.4. *Separation Axiom-of-Zariski Topology-on-$\mathbb{A}^1$ Proposition*
In the language of point-set topology, the Zariski topology on $\mathbb{A}^1$ is always T_1, but for infinite fields, the Zariski topology is never T_2 (Hausdorff).

Proof.
- Every single point subset in $\mathbb{A}^1$ can be easily checked to be an affine algebraic set in $\mathbb{A}^1$ and hence by the def of the Zariski topology, it is closed. Therefore, the Zariski topology on $\mathbb{A}^1$ is T_1.
- Directly holds by the above example.

□

Example: $F = \mathbb{R}$. A non-empty Zariski open set just the real line $\mathbb{R}$ with some finite number of points removed and any two such sets have (infinitely many) points in common. Note that the Zariski open (or closed) sets in $\mathbb{R}$ are also open (respectively closed) sets with respect to the usual Euclidean topology. The converse is *not* true: $[0, 1]$ is closed in the Euclidean topology but is not closed in the Zariski topology. Hence, the Euclidean topology on $\mathbb{R}$ is finer than the Zariski topology.

Definition 2.4.5. Zariski Closure and Zariski Density
For any subset A of $\mathbb{A}^n$,

- the *Zariski closure* of A is the smallest affine algebraic set containing A.
- If $A \subseteq V$ for some affine algebraic set V, then A is *Zariski dense* in V if the Zariski closure of A is V.

Example: $F = \mathbb{R}$. The affine algberaic sets in $\mathbb{A}^1$ are $\emptyset, \mathbb{R}$ and finite subsets of $\mathbb{R}$. The Zariski closure of any infinite set A of real numbers is then all of $\mathbb{A}^1$ and A is Zariski dense in $\mathbb{A}^1$.

Proposition 2.4.6. *Zariski Closure Proposition*
The Zariski closure of a subset A in $\mathbb{A}^n$ is $\mathcal{Z}(\mathcal{I}(A))$.

Proof. By the Relations between $\mathcal{Z}$ and $\mathcal{I}$ Prop.1, we have

$$A \subseteq \mathcal{Z}(\mathcal{I}(A))$$

For the minimality, let V be any affine algberaic set containing A. Then by the $\mathcal{I}$-Properties Prop.1, $\mathcal{I}(V) \subseteq \mathcal{I}(A)$. Hence, by the Affine Algebraic Sets-Properties Prop.1, $\mathcal{Z}(\mathcal{I}(A)) \subseteq \mathcal{Z}(\mathcal{I}(V))$. By the Relations between $\mathcal{Z}$ and $\mathcal{I}$ Prop.2, $\mathcal{Z}(\mathcal{I}(V)) = V$ and hence we're done. □

Proposition 2.4.7. *Zariski Closure-Morphism Proposition*
Let $\varphi : V \to W$ be a morphism of affine algebraic sets and let $\widetilde{\varphi} : F[W] \to F[V]$ be the associated F-algebra homomorphism of coordinate rings.

1. $\ker \widetilde{\varphi} = \mathcal{I}(\varphi(V))$.
2. *The Zariski closure of $\varphi(V)$ is the zero set in W of* $\ker \widetilde{\varphi}$. *In particular, the homomorphism $\widetilde{\varphi}$ is injective $\Leftrightarrow$ $\varphi(V)$ is Zariski dense in W.*

Proof. 1. Since $\widetilde{\varphi}(f) = f \circ \varphi$, then we have:

$$\widetilde{\varphi}(f) = 0 \Leftrightarrow (f \circ \varphi)(P) = 0, \forall P \in V \Leftrightarrow f(Q) = 0, \forall Q = \varphi(P) \in \varphi(V) \Leftrightarrow f \in \mathcal{I}(\varphi(V))$$

2. The first statement holds immediately by 1 and the Zariski Closure Prop. For the second statement:
   - ($\Rightarrow$): $\widetilde{\varphi}$ injective $\Leftrightarrow \mathcal{I}(\varphi(V)) \overset{1}{=} \ker \widetilde{\varphi} = 0 \Rightarrow$ the Zariski closure of $\varphi(V) = \mathcal{Z}(0) = W \Leftrightarrow$ $\varphi(V)$ is Zariski dense in W.
   - ($\Leftarrow$): $\mathcal{Z}(\mathcal{I}(V)) = W \Rightarrow \mathcal{I}(\varphi(V)) = \mathcal{I}(\mathcal{Z}(\mathcal{I}(\varphi(V)))) = \mathcal{I}(W) = 0$.

□

Remark: If $\varphi : V \to W$ is a morphism of affine algebra sets, then the image $\varphi(V)$ of V need *not* be an affine algebra subset of W, i.e.: need *not* be Zariski closed in W. For example, the projection of the hyperbola $V = \mathcal{Z}(xy - 1)$ in $\mathbb{R}^2$ onto the x-axis has image $\mathbb{R}^1 \setminus \{0\}$, which is *not* an affine algebra set.

2.5 Affine Varieties

2.5.1 Decompositions of Affine Algberaic Sets

Intuition: We now consider the question of whether an affine algebraic set can be decomposed into smaller affine algebraic sets and the corresponding algebraic formulation in terms of its coordinate ring.

Definition 2.5.1. Affine Varieties
A non-empty affine algebraic set V is called *irreducible* if it cannot be written as $V = V_1 \cup V_2$, where V_1 and V_2 are proper affine algebraic sets in V. An irreducible affine algebraic sets is called an *affine variety*.

Remark: Equivalently, an affine algebraic set is irreducible if it cannot be written as the union of two proper closed subsets in the sense of Zariski topology.

Proposition 2.5.2. *Affine Algebraic Sets-Decomposition Proposition*

1. *The affine algebraic set V is irreducible $\Leftrightarrow \mathcal{I}(V)$ is a prime ideal.*
2. *Every non-empty affine algebraic set V may be written uniquely in the form*

$$V = V_1 \cup V_2 \cup \cdots \cup V_q$$

where each V_i is irreducible and $V_i \not\subseteq V_j, \forall j \neq i$ (i.e.: the decomposition is "minimal" or "irredundant").

Proof.

1.
   - ($\Leftarrow$): Suppose on the contrary that $V = V_1 \cup V_2$ is not irreducible, where V_1, V_2 are proper affine algebra sets in V. Since $V_1 \neq V$, then $\exists f_1$, s.t.: f_1 vanishes on V_1 but not on V, i.e.: $f_1 \in \mathcal{I}(V_1) \backslash \mathcal{I}(V)$. Similarly, we can also find a function $f_2 \in \mathcal{I}(V_2) \backslash \mathcal{I}(V)$. Then $f_1 f_2$ vanishes on $V_1 \cup V_2 = V$, i.e.: $f_1 f_2 \in \mathcal{I}(V)$, which follows that $\mathcal{I}(V)$ is not prime, contradiction.
   - ($\Rightarrow$): Suppose on the contrary that $\mathcal{I}(V)$ is not a prime ideal, i.e.: $\exists f_1, f_2 \in F[\mathbb{A}^n]$, s.t.: $f_1 f_2 \in \mathcal{I}(V)$ but $f_1, f_2 \notin \mathcal{I}(V)$. Consider

   $$V_1 := \mathcal{Z}(f_1) \cap V, V_2 := \mathcal{Z}(f_2) \cap V$$

   Since $\mathcal{Z}(f_1), \mathcal{Z}(f_2), V$ are closed, then so are the intersections, i.e.: V_1, V_2 are affine algebraic sets. Since neither f_1 nor f_2 belongs to $\mathcal{I}(V)$, then neither f_1 nor f_2 vanishes on V and hence V_1, V_2 are proper subsets of V. But note that by assumption $f_1 f_2 \in \mathcal{I}(V)$, so by the Affine Algebraic Sets-Properties Prop.4, we have

   $$V \subseteq \mathcal{Z}(f_1 f_2) = \mathcal{Z}(f_1) \cup \mathcal{Z}(f_2) \Rightarrow V = V_1 \cup V_2$$

   contradiction.
2.
   - *Existence:* Let $\mathcal{S}$ be the collection of non-empty affine algebraic sets that cannot be written as a finite union of irreducible affine algebraic sets. Suppose on the contrary that $\mathcal{S} \neq \emptyset$. Then the corresponding set of ideals

   $$\{\mathcal{I}(V) \mid V \in \mathcal{S}\} \subseteq F[\mathbb{A}^n]$$

   has a maximal element I_0 by the Equivalent-Conditions-of-Noetherian Rings Prop.2, since $F[\mathbb{A}^n]$ is a Noetherian ring. Then $V_0 = \mathcal{Z}(I_0)$ is a minimal element of $\mathcal{S}$. By the def of $\mathcal{S}$, V_0 cannot be decomposed into finitely many irreducible affine algebraic sets, which follows that V_0 is not irreducible (otherwise V_0 itself is the finite decomposition). Hence, $V = V_1 \cup V_2$ for some proper closed subsets V_1, V_2 of V. Since V_1, V_2 are proper subsets of V, then by the minimality of V, $V_1, V_2 \notin \mathcal{S}$. Therefore, V_1, V_2 can be written as finite unions of irreducible affine algebraic sets and then so is V, contradiction.

- *Uniqueness:* Suppose that V has two decomposition into affine varieties:

$$V = V_1 \cup V_2 \cup \cdots \cup V_r = U_1 \cup U_2 \cup \cdots \cup U_s$$

Then $V_1 \subseteq U_1 \cup U_2 \cup \cdots \cup U_s$. Since $V_1 \cap U_j$ is an affine algebraic set $\forall j = 1, \cdots, s$, then we obtain a decomposition of V_1 into affine varieties:

$$V_1 = (V_1 \cap U_1) \cup (V_1 \cap U_2) \cup \cdots \cup (V_1 \cap U_s)$$

By the irreducibility of V_1, we must have $V_1 = V_1 \cap U_i$ for some $i \in \{1, \cdots, s\}$, i.e.: $V_1 \subseteq U_i$. Similarly, we have $U_i \subseteq V_{i'}$ for some $i' \in \{1, \cdots, r\}$. Hence, $V_1 \subseteq V_{i'}$, so by the hypothesis of 2, $i' = 1$ and thus $V_1 = U_i$. Applying the similar argument for each V_k, we can get $r = s$ and that

$$\{V_1, \cdots, V_r\} = \{U_1, \cdots, U_s\}$$

and then we're done.

□

Corollary 2.5.3. *of Affine Algebraic Sets-Decomposition Proposition*
An affine algebraic set V is a variety $\Leftrightarrow$ its coordinate ring $F[V]$ is an ID.

Proof. It follows immediately by the Affine Algebraic Sets-Decomposition Prop.1. □

2.5.2 Primary Decomposition of Ideals in Noetherian Rings

Definition 2.5.4. Primary Ideals
A *proper* ideal Q in the commutative ring R is called *primary* if whenever $ab \in Q$ and $a \notin Q$, we have $b^n \in Q$ for some positive integer $n \in \mathbb{N}_+$, i.e.: $b \in \text{rad } Q$.

Proposition 2.5.5. *Primary Ideals-Properties Proposition*
Let R be a ring (commutative and unital).

1. *Prime ideals in R are primary.*
2. *The ideal $Q \lhd R$ i primary $\Leftrightarrow$ every zero divisor in R/Q is nilpotent.*
3. *If Q is primary, then rad Q is a prime ideal and is the unique smallest prime ideal containing Q.*
4. *If Q is an ideal whose radical is a maximal ideal, then Q is a primary ideal.*
5. *Suppose M is a maximal ideal and Q is an ideal with $M^n \subseteq Q \subseteq M$ for some positive integer n. Then Q is a primary ideal with rad $Q = M$.*

Proof. 1. holds immediately by the def of primary ideals.

2. holds immediately by the def of primary ideals.

3. By the corollary of the Radicals-Prime Ideals Prop, it suffices to show that rad Q is prime. Then we have

$$ab \in \text{rad } Q \Rightarrow a^m b^m = (ab)^m \in Q \overset{Q \text{ primary}}{\Rightarrow} \text{WLOG, let } (b^m)^n \in Q \Rightarrow b \in \text{rad } Q.$$

4. By 2, it suffices to show that every zero divisor in R/Q is nilpotent, so we can be reduced to the situation where $Q = (0)$ and let $M = \text{rad } Q = \text{rad } (0) = \text{Nil } R$ be a maximal ideal. By the Radicals-Prime Ideals Prop, M is the unique prime ideal in R, and hence also the unique maximal ideal in R. If d is a zero divisor of R/Q, then the ideal (d) is a proper ideal, hence contained in some maximal ideal. By the uniqueness for maximal ideals of M, since $M = \text{Nil } R$, then $d \in \text{Nil } R/Q$.
5.
   - $Q \subseteq M \Rightarrow \text{rad } Q \subseteq \text{rad } M \overset{M \text{ maximal}}{=} M$.
   - $M^n \subseteq Q \Rightarrow M \subseteq \text{rad } Q$.

   Therefore, by 4, we're done.

□

Definition 2.5.6. Associated Prime Ideals of Primary Ideals
Let Q be a primary ideal. Then the prime ideal $P = \text{rad } Q$ is called the *associated prime to* Q, and Q is said to *belong to* P (or to be P*-primary*).

Proposition 2.5.7. *A finite intersection of P-primary ideals is again a P-primary ideal.*

Example:

- The primary ideals of $\mathbb{Z}$ are precisely 0 and the ideals (p^m) for p prime and $m \geq 1$.
- Let F be any field. The ideal $(x) \triangleleft F[x, y]$ in $F[x, y]$ is primary since it is a prime ideal. For any $n \in \mathbb{N}_+$, the ideal $(x, y)^n$ is primary since (x, y) is a maximal ideal.
  Furthermore, the ideal $Q = (x^2, y)$ in $F[x, y]$ is primary since $(x, y)^2 \subseteq Q \subseteq (x, y)$ (by the Primary Ideals-Properties Prop.5). Similarly, $Q' = (4, x)$ in $\mathbb{Z}[x]$ is a $(2, x)$-primary ideal.
- Primary ideal need *not* be powers of prime ideals. For instance, Q in the previous example is not the power of any prime ideal: If $(x^2, y) = P^k$ for some prime ideal P and some $k \in \mathbb{N}_+$, then $x^2, y \in P^k \subseteq P$ and hence $x, y \in P$, since P is prime. Then the only case is $P = (x, y)$. Since $y \notin (x, y)^2$, then $k = 1$. But $x \notin (x^2, y)$, contradiction.
- If R is Noetherian, and Q is a primary ideal belonging to the prime ideal P, then by the Noetherian-Radicals Prop, we have

$$P^m \subseteq Q \subseteq P$$

  for some $m \in \mathbb{N}_+$. If P is a maximal ideal, then the Primary Ideals-Properties Prop.5 show that the converse also holds. This is not necessarily true if P is prime but not maximal. For instance, consider the ideal $I = (x^2, xy)$ in $F[x, y]$. Then $(x^2) = (x)^2 \subseteq I \subseteq (x)$ and (x)

is prime, but I is *not* primary: $xy \in I$ and $x \notin I$ but no positive power of y is in I. This example also shows that an ideal whose radical is prime is not necessarily prime. However, (x^2, xy) can be written as the intersection of primary ideals:

$$(x^2, xy) = (x) \cap (x, y)^2$$

Definition 2.5.8. Primary Decomposition

- An ideal I in R has a *primary decomposition* if it may be written as a finite intersection of primary ideals:
$$I = \bigcap_{i=1}^{m} Q_i \text{ where } Q_i\text{'s are primary ideal.}$$
- The primary decomposition above is said to be *minimal* and then we call Q_i's the *primary components* of I if the following two conditions are satisfied:
  - no primary ideal contains the intersection of the remaining primary ideals, i.e.: $Q_i \not\supseteq \bigcap_{j \neq i} Q_j$ for all i, and
  - the associated prime ideals are all distinct: $\operatorname{rad} Q_i \neq \operatorname{rad} Q_j$ for all $i \neq j$.

Definition 2.5.9. Irreducible Ideals
A *proper* ideal I in the commutative ring R is said to be *irreducible* if I cannot be written nontrivially as the intersection of two other ideals, i.e.: if $I = J \cap K$, with J, K ideals, implies that $I = J$ or $I = K$.

Remark:

- It's easy to see that a prime ideal is irreducible.
- The ideal $(x, y)^2$ in $F[x, y]$ is *not* irreducible but is primary. In Noetherian ring, however, irreducible ideals are necessarily primary:

Proposition 2.5.10. *Noetherian-Irreducible Proposition*
Let R be a Noetherian ring.

1. *Every irreducible ideal in R is primary.*
2. *Every proper ideal in R is a finite intersection of irreducible ideals.*

Proof. 1. Let Q be an irreducible ideal in R and suppose that $ab \in Q$ and that $b \notin Q$. Note that for any fixed $m \in \mathbb{N}_+$, the set of elements $x \in R$ with $a^m x \in Q$ is an ideal in R, namely $A_m \triangleleft R$. Then it's clearly that we obtain an ascending chain $A_1 \subseteq A_2 \subseteq \cdots$ and since R is Noetherian, then this chain of ideals must stabilize, i.e.: $A_n = A_{n+1} = \cdots$ for some $n \in \mathbb{N}_+$. Consider the two ideals
$$I := (a^n) + Q, J := (b) + Q$$

in R, each containing $Q \Rightarrow Q \subseteq I \cap J$. For any $y \in I \cap J \subseteq I$, $y = a^n z + q$ for some $z \in R, q \in Q$. Since $ab \in Q$, then $aJ \subseteq Q$ and in particular $ay \in Q$. Hence,

$$a^{n+1}z = ay - aq \in Q \overset{\text{def of } A_m}{\Rightarrow} z \in A_{n+1} = A_n \overset{\text{def of } A_m}{\Rightarrow} a^n z \in Q \Rightarrow y \in Q$$

which follows that $I \cap J \subseteq Q$. Thus, $Q = I \cap J$. Since Q is irreducible, then $Q = I$ or $Q = J$.

- If $Q = J$, then $b \in Q$, contradicting our hypothesis.
- If $Q = I$, then $a^n \in Q$ and hence Q is primary.

2. Similar to the proof of the Affine Algebraic Sets-Decomposition Prop.2, so we omit it here.

□

Theorem 2.5.11. *Primary Decomposition Theorem / Lasker-Noetherian Decomposition Theorem Let R be a Noetherian ring. Then every proper ideal I in R has a minimal primary decomposition. If*

$$I = \bigcap_{i=1}^{m} Q_i = \bigcap_{i=1}^{n} Q_i'$$

are two minimal primary decompositions for I, then the set of associated primes in the two decompositions are the same:

$$\{rad\ Q_1, rad\ Q_2, \cdots, rad\ Q_m\} = \{rad\ Q_1', rad\ Q_2', \cdots, rad\ Q_n'\}$$

Moreover, the primary components Q_i belonging to the minimal elements in this set of associated primes are uniquely determined by I.

Proof. By the Noetherian-Irreducible Prop, we have that every proper ideal has a primary decomposition. If any of primary ideals in this decomposition contains the intersection of the remaining primary ideals, then we can remove this ideal since this will not change the intersection. Hence, the first condition in the def of minimal primary decomposition holds. By the previous prop, we know that a finite intersection of P-primary ideals is again P-primary and hence we can replace the primary ideals in the decomposition with the intersections of all those primary ideals belonging to the same prime. Therefore, the second condition in the def of minimal primary decomposition. This proves the first statement of the Primary Decomposition Theorem.
The proofs of the uniqueness of the set of associated primes and of the uniqueness of the primary components associated to the minimal primes are omitted. □

Definition 2.5.12. If I is an ideal in the Noetherian ring R, then the associated primes ideals in any primary decomposition of I are called the *associated prime ideals of I*. If an associated prime ideal P of I does not contain any other associated prime ideal of I, then P is called an *isolated prime ideal*; the remaining associated prime ideals of I are called *embedded prime ideals.*

Corollary 2.5.13. *of the Primary Decomposition Theorem*
Let I be a proper ideal in the Noetherian ring R.

1. *A prime ideal P contains the ideal $I \Leftrightarrow P$ contains one of the associated primes of $I \Leftrightarrow P$ contains one of the isolated primes of I, i.e.: the isolated primes of I are precisely the minimal elements in the set of all prime ideals containing I. In particular, there are only finitely many minimal elements among the prime ideals containing I.*
2. *The radical of I is the intersection of the associated primes of I, hence also the intersection of the isolated primes of I.*
3. *There are prime ideals $P_1, \cdots, P_n$ (not necessarily distinct) containing I such that $P_1 P_2 \cdots P_n \subseteq I$.*

2.6 Integral Extensions

Intuition: In this section, we introduce a generalization to rings of algebraic extensions of fields — integral extension of rings.

Definition 2.6.1. Integral Extension of Rings
Let R be a subring of the ring S.

- An element $s \in S$ is *integral over* R if s is the root of a monic polynomial in $R[x]$.
- The ring S is an *integral extension of* R or just *integral over* R if every $s \in S$ is integral over R.
- The *integral closure of* R *in* S is the set of elements of S that are integral over R.
- The ring R is said to be *integrally closed in* S if R is equal to its integral closure in S. The integral closure of an ID in its field of fraction is called the *normalization of* R. An ID is called *integrally closed* or *normal* if it is integrally closed in its field of fractions.

Proposition 2.6.2. *Integral Elements-Properties Proposition*
Let R be a subring of the ring S and let $s \in S$. The followings are equivalent:

1. *s is integral over R.*
2. *$R[s]$ (:= the ring of all R-linear combinations of powers of s) is a finitely generated R-module.*
3. *$s \in T$ for some subring T, with $R \subseteq T \subseteq S$, that is a finitely generated R-module.*

Proof.

- ($1 \Rightarrow 2$): Let s be a root of $x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in R[x]$. Then we have

$$s^n = -(a_{n-1}x^{n-1} + \cdots + a_1 x + a_0)$$

and hence, s^n (and then all higher powers of s) can be expressed as R-linear combinations of $s^{n-1}, \cdots, s, 1$. Hence $R[s] = R1 + Rs + \cdots + Rs^{n-1}$ is finitely generated as an R-module.

- ($2 \Rightarrow 3$): Let $T = R[x]$ and then we're done.
- ($3 \Rightarrow 1$): Let $\{v_1, v_2, \cdots, v_n\}$ be a finite generating set for T. Since T is a ring, then

$sv_i \in T, \forall i = 1, \cdots, n$. Hence, $\exists a_{ij} \in R$, s.t.:

$$sv_i = \sum_{j=1}^{n} a_{ij} v_j \Leftrightarrow 0 = \sum_{j=1}^{n} (\delta_{ij} s - a_{ij}) v_j$$

where δ_{ij} denotes the Kronecker delta. If we denote by $B, \mathbf{v}$ the $n \times n$ matrix whose (i, j)-entrt is $\delta_{ij} s - a_{ij}$ and the $n \times 1$ column vector whose entries are $v_1, \cdots, v_n$, respectively. Then we have $B\mathbf{v} = 0$. By the Cramer's Rule, $(\det B)v_i = 0, \forall i = 1, \cdots, n$. Since $1 \in T$, then $\det B = 0$. But $B = sI - (a_{ij})_{ij}$. Hence, s is a root of monic polynomial $\det(xI - (a_{ij})_{ij}) \in R[x]$, which is the characteristic polynomial of $(a_{ij})_{ij}$.

□

Corollary 2.6.3. *1 of the Integral Elements-Properties Proposition*
Let $R \subseteq S$ be as in the Integral Elements-Properties Proposition above and let $s, t \in S$.

1. *If s and t are integral over R, then so is $s \pm t$ and st.*
2. *The integral closure of R in S is a subring of S containing R.*
3. *Integrality is transitive: let S be a subring of T; if T is integral over S and S is integral over R, then T is integral over R.*

Proof. By the Integral Elements-Properties Prop.2, both $R[s]$ and $R[t]$ are finitely generate R-modules, say

$$R[s] = Rs_1 + \cdots + Rs_n, R[t] = Rt_1 + \cdots + Rt_m$$

Then $R[s, t] = Rs_1t_1 + \cdots + Rs_it_j + \cdots + Rs_nt_m$ is a ring containing $s \pm t$ and st that is also finitely generated R-module. Hence, 1 and 2 hold.
To prove 3, $\forall b \in T$, it is integral over S, so it is the root of some monic polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 \in S[x]$. Since $a_i \in S$ is integral over R, then each ring $R[a_i]$ is a finitely generated R-module and hence the ring $R_1 := R[a_0, a_1, \cdots, a_{n-1}]$ is also a finitely generated R-module. Since the monic polynomial $p(x)$ has its coefficients in R_1, then b is integral over R_1. Therefore, $R_1[b] = R[a_0, a_1, \cdots, a_{n-1}, b]$ is a finitely generated R-module. By the Integral Elements-Properties Prop.2, this means that b is integral over R. □

Remark: Corollary 1 of the Integral Elements-Properties Proposition 2 shows that taking the elements of S that are integral over R gives a (possibly larger) subring of S; 3 shows that the process of taking the integral closure stops after one step:

Corollary 2.6.4. *2 of the Integral Elements-Properties Proposition*
Let R be a subring of the ring S. Then the integral closure of R in S is integrally closed in S.

Example:

- If R and S are fields, then S is integral over $R \Leftrightarrow S$ is algebraic over R. If $s \in S$ is a root of the polynomial $p(x) \in R[x]$, then it is a root of the monic polynomial obtained by dividing by the (nonzero) leading coefficient of $p(x)$.

- Suppose that S is an integral extension of R and that I is an ideal in S. Then S/I is an integral ring extension of $R/(R\cap I)$ (reducing the monic polynomial over R satisfied by $s \in S$ modulo I gives a monic polynomial satisfied by $\overline{s} := s + I \in S/I$ over $R/(R \cap I)$).
- If R is a UFD, then R is integrally closed as follows: Suppose that a/b is an element in field of fractions of R (with $b \neq 0$ and a, b having no common factors), and satisfies

$$(a/b)^n + r_{n-1}(a/b)^{n-1} + \cdots + r_1(a/b) + r_0 = 0$$

with $r_0, r_1, \cdots, r_{n-1} \in R$. Then we have

$$a^n = b(-r_{n-1}a^{n-1} - \cdots - r_1 ab^{n-2} - r_0 b^{n-1})$$

which follows that any irreducible element dividing b divides a. Since a/b is in the lowest term, then b must be a unit, i.e.: $a/b \in R$.

- The polynomial ring $F[x, y]$ over the field F is integrally closed in its fraction field $F(x, y)$. The ideal $(x^2 - y^3)$ is prime and hence the quotient ring $R = F[x, y]/(x^2 - y^3) = F[\overline{x}, \overline{y}]$ is an ID. This domain is *not* integrally closed, however, since $\overline{x}/\overline{y}$ is an element of the fraction field of R that is integral over R ($(\overline{x}/\overline{y})^3 - \overline{x} = 0$), but is not an element of R. In particular, R is *not* a UFD by the above example.

Definition 2.6.5. Extension and Contraction of Ideals
Let $\varphi : R \to S$ be a homomorphism of commutative rings.

- If I is an ideal in R, then the *extension of I to S* is the ideal $\varphi(I)S$ of S generated by the image of I.
- If J is an ideal in S, then the *contraction in J of R* is the ideal $\varphi^{-1}(J)$.

Remark: In the special case where R is a subring of S and φ is the natural injection, the extension of $I \triangleleft R$ is the ideal IS in S and the contraction of $J \triangleleft S$ is the ideal $J \cap R$ of R. Furthermore, it is immediate from the def that

- $I \subseteq IS \cap R$, more generally, I is contained in the contraction of its extension to S, and
- $(J \cap R)S \subseteq J$, more generally, J contains the extension of its contraction in R.

Proposition 2.6.6. *Contraction-Prime Proposition*
Let $\varphi : R \to S$ be a homomorphism of commutative rings. If Q is a prime ideal in S, then its contraction is prime.

Proof. Since Q is proper, then $1_S \notin Q$ and hence $1_R \notin \varphi^{-1}(Q)$ which follows that $\varphi^{-1}(Q)$ is proper. $\forall a, b \in R$ with $ab \in \varphi^{-1}(Q)$, $\varphi(a)\varphi(b) = \varphi(ab) \in Q$. Since Q is prime, then WLOG, $\varphi(a) \in Q$ and hence $a \in \varphi^{-1}(Q)$. $\square$

Remark:

- The contraction of a maximal ideal need *not* be maximal. For instance, let $R = \mathbb{Z}, S = \mathbb{Q}$ and let $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Q}$ be the inclusion map. Since $S = \mathbb{Q}$ is a field, then $Q = (0) \subseteq \mathbb{Q}$ is the only maximal ideal in $\mathbb{Q}$. But $\varphi^{-1}((0)) = (0) \subseteq \mathbb{Z}$ is *not* maximal in $\mathbb{Z}$.
- The extension of a prime ideal P in R need *not* be prime (or even proper) in S. For instance,
  - Let $R = \mathbb{Z}, S = \mathbb{Q}$, let $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Q}$ be the inclusion map and let $P = (2) \subseteq \mathbb{Z}$ be a prime ideal in $R = \mathbb{Z}$. But $\varphi((2))\mathbb{Q} = \mathbb{Q}$ is *not* proper in $S = \mathbb{Q}$.
  - Let $R = \mathbb{Z}, S = \mathbb{Z}[i]$, let $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Z}[i]$ be the inclusion map and let $P = (5) \subseteq \mathbb{Z}$ be a prime ideal in $R = \mathbb{Z}$. But $\varphi((5))\mathbb{Z}[i] = 5\mathbb{Z}[i]$ is *not* prime in $\mathbb{Z}[i]$ since $2+i, 2-i \notin 5\mathbb{Z}[i]$ but $5 = (2+i)(2-i) \in 5\mathbb{Z}[i]$.
- It is *not* generally true that any prime ideal of R is the contraction of a prime ideal S. For instance, let $R = \mathbb{Z}, S = \mathbb{Q}$, let $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Q}$ be the inclusion map and let $P = (2) \subseteq \mathbb{Z}$ be a prime ideal in $R = \mathbb{Z}$. But the only prime ideal in $S = \mathbb{Q}$ is (0), so there's not any prime ideal Q of S, s.t.: $\varphi^{-1}(Q) = (2)$. For integral ring extensions, however, the situation is more controlled:

Theorem 2.6.7. *Integral Ring Extensions-Prime Ideals Theorem*
Let R be a subring of the ring S and suppose that S is integral over R.

1. *Assume that S is an ID. Then R is a field $\Leftrightarrow$ S is a field.*
2. *Let P be a prime ideal in R. Then there is a prime ideal Q in S with $P = Q \cap R$. Moreover, P is maximal $\Leftrightarrow$ Q is maximal.*
3. *(The Going-up Theorem) Let $P_1 \subseteq P_2 \subseteq \cdots \subseteq P_n$ be a chain of prime ideals in R and suppose that there are prime ideals $Q_1 \subseteq Q_2 \subseteq \cdots \subseteq Q_m$ of S with $P_i = Q_i \cap R, 1 \leq i \leq m$ and $m < n$. Then the ascending chain of ideals can be completed: there are prime ideals $Q_{m+1} \subseteq \cdots \subseteq Q_n$ in S such that $P_i = Q_i \cap R, \forall i$.*
4. *(The Going-down Theorem) Assume that S is an ID and that R is integrally closed in S. Let $P_1 \supseteq P_2 \supseteq \cdots \supseteq P_n$ be a chain of prime ideals in R and suppose that there are prime ideals $Q_1 \supseteq Q_2 \supseteq \cdots \supseteq Q_m$ of S with $P_i = Q_i \cap R, 1 \leq i \leq m$ and $m < n$. Then the descending chain of ideals can be completed: there are prime ideals $Q_{m+1} \supseteq \cdots \supseteq Q_n$ in S such that $P_i = Q_i \cap R, \forall i$.*

Proof. 1. • $(\Rightarrow)$: $\forall s \in S \setminus \{0\}$, since S is integral over R, then s is integral over R and hence

$$s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$$

for some $a_0, a_1, \cdots, a_{n-1} \in R$. Since S is an ID, then we may assume $a_0 \neq 0$ (otherwise we can apply cancellation law on s). Then

$$s(s^{n-1} + a_{n-1}s^{n-2} + \cdots + a_1) = -a_0 \overset{a_0 \in \text{ field } R}{\Longrightarrow} s \cdot (-1/a_0)(s^{n-1} + a_{n-1}s^{n-1} + \cdots + a_1) = 1$$

Hence, $s \in U(S)$. Therefore, S is a field.

- ($\Leftarrow$): $\forall r \in R \setminus \{0\}$, since $R \subseteq S$, then $r \in S$ and hence $r^{-1} \in S$. It suffices to show that $r^{-1} \in R$. Since r^{-1} is integral over R, then we have

$$r^{-m} + a_{m-1}r^{m-1} + \cdots + a_1 r + a_0 = 0$$

for some $a_0, a_1, \cdots, a_{m-1} \in R$. Then $r^{-1} = -(a_{m-1} + \cdots + a_1 r^{m-2} + a_0 r^{m-1}) \in R$.

2. The first statement will be prove later by the coro of Localization-on-Primary Decomposition Proposition. For the second statement, note that the ID S/Q is an integral extension of R/P. By 1, we're done.

3. It suffices by induction to show that: if $P_1 \subseteq P_2$ and Q_1 is a prime ideal of S with $Q_1 \cap R = P_1$, then there is a prime ideal Q_2 of S with $Q_1 \subseteq Q_2$ and $Q_2 \cap R = P_2$. Since $\overline{S} := S/Q_1$ is an integral extension of $\overline{R} := R/P_1$, then the first part of 2 shows that there exists a prime ideal $\overline{Q}_2$ of $\overline{S}$ with $\overline{Q}_2 \cap \overline{R} = P_2/P_1$. Then the preimage Q_2 of $\overline{Q}_2$ in S is the desired prime ideal of S.

4. The proof of the Going-down Theorem need the concept of *localization*, so we'll list the outline later after providing localization.

□

Corollary 2.6.8. *of Integral Ring Extensions-Prime Ideals Theorem*
Suppose that R is a subring of the ring S and that S is integral and finitely generated (as a ring) over R. If P is a maximal ideal in R, then there is a nonzero and finite number of maximal ideals Q of S with $Q \cap R = P$.

Proof. By the Integral Ring Extensions-Prime Ideals Thm.2, there exists at least one desired maximal ideal, so it suffices to prove the finiteness. Note that if Q is a maximal ideal of S with $Q \cap R = P$, then S/Q is a field containing the field R/P. Hence, it suffices to show that there are only finitely many homomorphisms from S to a field containing R/P that extend the homomorphism from R to R/P:
Let $S = R[s_1, \cdots, s_n]$, where s_i's are integral over R by assumption and let $p_i(x) \in R[x]$ be a monic polynomial satisfied by s_i. Note that $S/Q = (R/P)[\overline{s}_1, \cdots, \overline{s}_n]$ is the field extension of the field R/P with generators $\overline{s}_i$'s, where $\overline{s}_i$ is a root of the monic polynomial $\overline{p}_i(x) \in (R/P)[x]$ obtained by reducing the coefficients of of $p_i(x)$ mod P. Since there are only a finite number of possible roots of $\overline{p}_i(x)$ in a field algebraic closure of R/P, then only finitely many possible field extensions of the form $(R/P)[\overline{s}_1, \cdots, \overline{s}_n]$. □

2.7 Algebraic Integers

Intuition: We can use the concept of an integral ring extension to define the "integers" in extension fields of the rational numbers $\mathbb{Q}$.

Definition 2.7.1. Algebraic Integers
Let K be an extension field of $\mathbb{Q}$.

- An element $\alpha \in K$ is called an *algebraic integer* if α is integral over $\mathbb{Z}$, i.e.: if α is the root of some monic polynomial with integer coefficients.
- The integral closure of $\mathbb{Z}$ in K is called the *ring of integers of* K, denoted as $\mathcal{O}_K$.

Example: $\sqrt{2}, \sqrt{-1}, \sqrt[3]{5}$, etc. are certainly roots of monic polynomials with integer coefficients.
Remark:

- An algebraic integer is clearly algebraic over $\mathbb{Q}$ and hence the ring of all algebraic integers is the ring of integers in $K = \overline{\mathbb{Q}}$, an algebraic closure of $\mathbb{Q}$.
- The def of an algebraic integer α is that α be a root of some monic polynomial in $\mathbb{Z}[x]$, a condition which is difficult to check.

Proposition 2.7.2. *Algebraic Integers-Criterion Proposition*
An element α in some field extension K of $\mathbb{Q}$ is an algebraic integer $\Leftrightarrow$ α is algebraic over $\mathbb{Q}$ and its minimal polynomial $m_{\alpha,\mathbb{Q}}(x)$ has integer coefficients. In particular, the algebraic integers in $\mathbb{Q}$ are the integers $\mathbb{Z}$, i.e.: $\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$.

Proof.
- ($\Leftarrow$): holds immediately by def.
- ($\Rightarrow$): Let $f(x)$ be a monic polynomial in $\mathbb{Z}[x]$ of minimal degree having α as a root. If f is reducible in $\mathbb{Q}[x]$, then by the Gauss' Lemma, $f(x) = g(x)h(x)$ for some monic polynomials $g, h \in \mathbb{Z}[x]$ of degree smaller than the degree of f. But then α would be a root of either g or h, contradicting the minimality of degree of f. Hence, f is irreducible in $\mathbb{Q}[x]$, so $f(x) = m_{\alpha,\mathbb{Q}}(x)$ and thus the minimal polynomial for α has integer coefficients.

Finally, the minimal polynomial of $\alpha = a/b \in \mathbb{Q}$ where $a, b \in \mathbb{Z}$ with $\gcd(a, b) = 1$ is $bx - a$, which is monic $\Leftrightarrow$ $b = 1$. Therefore, $\alpha \in \mathbb{Q}$ is an algebraic integer $\Leftrightarrow$ $\alpha \in \mathbb{Z}$. $\square$

Definition 2.7.3. Number Fields
A field is called *number field* if it is an extension of finite degree over $\mathbb{Q}$.

Theorem 2.7.4. *Basic Structure-of-Rings of Integers-in Number Fields Theorem*
Let K be a number field of degree n over $\mathbb{Q}$.

1. *The ring $\mathcal{O}_K$ of integers in K is a Noetherian ring and is a free $\mathbb{Z}$-module of rank n.*
2. *For every $\beta \in K$, there is some nonzero $d \in \mathbb{Z} \setminus \{0\}$, s.t.: $d\beta$ is an algebraic integer. In particular, K is the field of fractions of $\mathcal{O}_K$.*
3. *If $\beta_1, \beta_2, \cdots, \beta_n$ is any $\mathbb{Q}$-basis of K, then there is an integer d such that $d\beta_1, d\beta_2, \cdots, d\beta_n$ is a basis for a free-$\mathbb{Z}$-submodule of $\mathcal{O}_K$ of rank n. Any basis of the $\mathbb{Z}$-module $\mathcal{O}_K$ is also a basis for K as a vector space over $\mathbb{Q}$.*

Definition 2.7.5. Integral Basis
An *integral basis* for the number field S is a basis of the ring of integers in S considered as a free $\mathbb{Z}$-module of rank $[K : \mathbb{Q}]$.

2.8 Noetherian's Normalization Lemma and Hilbert's Nullstellensatz

Definition 2.8.1. Algebraical Independence
Let F be a field. The elements $y_1, y_2, \cdots, y_q$ is some F-algebra are called algebraically independent over F if there is no non-zero polynomial p in q variables over F such that $p(y_1, y_2, \cdots, y_q) = 0$.

Remark:

- $y_1, y_2, \cdots, y_q$ are algebraically independent $\Leftrightarrow$ F-algebra homomorphism from the polynomial ring $F[x_1, \cdots, x_q]$ to $F[y_1, \cdots, y_q]$ given by $x_i \mapsto y_i$ is an isomorphism.
- Elements in a field extension of F are algebraically independent $\Leftrightarrow$ they are independent transcendentals over F.

Theorem 2.8.2. *Noetherian's Normalization Lemma*
Let F be a field and suppose that $A := F[r_1, \cdots, r_m]$ is a finitely generated F-algebra. Then for some q, with $0 \le q \le m$, there are algebraically independent elements $y_1, y_2, \cdots, y_q \in A$ such that A is integral over $F[y_1, \cdots, y_q]$.

Proof. If $r_1, \cdots, r_m$ are algebraically independent, then take $q = m, y_i = r_i$ and then we're done. Otherwise, $\exists 0 \neq f \in F[x_1, \cdots, x_m]$, s.t.: $f(r_1, \cdots, r_m) = 0$. Since $f \neq 0$, then WLOG, let x_m appear in the leading term, i.e.: we can consider f as:

$$f(x_1, \cdots, x_m) = c_t(x_1, \cdots, x_{m-1})x_m^t + c_{t-1}(x_1, \cdots, x_{m-1})x_m^{t-1} + \cdots + c_1(x_1, \cdots, x_{m-1})x_m + c_0(x_1, \cdots, x_{m-1})$$

where $c_i \in F[x_1, \cdots, x_{m-1}]$ with $c_t \neq 0, t \ge 1$. Then we have $c_t(r_1, \cdots, r_{m-1})r_m^t + \cdots + c_0(r_1, \cdots, r_{m-1}) = 0$. But c_t may not equal to 1, namely this polynomial over $F[r_1, \cdots, r_{m-1}]$ may not be monic, so we need to perform a change of variables to "normalize" f into a monic polynomial in x_m.
For $1 \le i \le m-1$, define $\alpha_i := (1+d)^i$, where d denotes the degree of f, and define $X_i := x_i - x_m^{\alpha_i}$.
Let $g \in F[X_1, \cdots, X_{m-1}, x_m]$ be defined by

$$g(X_1, X_2, \cdots, X_{m-1}, x_m) := f(X_1 + x_m^{\alpha_1}, X_2 + x_m^{\alpha_2}, \cdots, X_{m-1} + x_m^{\alpha_{m-1}}, x_m)$$

Now we can see that the highest power N of x_m is with a "constant" coefficient, namely, g can be written as:

$$g(X_1, X_2, \cdots, X_{m-1}, x_m) = cx_m^N + \sum_{i=0}^{N-1} h_i(X_1, \cdots, X_{m-1})x_m^i$$

where $0 \neq c \in F, h_i \in F[X_1, \cdots, X_{m-1}]$. If $s_i := r_i - r_m^{\alpha_i}$, then

$$\frac{1}{c}g(s_1, \cdots, s_{m-1}, r_m) = \frac{1}{c}f(r_1, \cdots, r_{m-1}, r_m) = 0$$

Hence, r_m is integral over $B = F[s_1, \cdots, s_{m-1}]$. Note that $\forall i \in \{1, \cdots, m-1\}$, r_i is a root of the monic polynomial $x - s_i - r_m^{\alpha_i}$. Hence, each r_i, for $1 \le i \le m-1$, is integral over $B[r_m]$, so

$A = F[r_1, \cdots, r_m]$ is integral over $B[r_m]$. Since $B[r_m]$ is integral over B, then by the transitivity of integrality (the coro.1 of the Integral Elements-Properties Prop.3), A i integral over B. So we now reduce the case from m into $m-1$. Repeat this procedure until we're done. $\square$

Theorem 2.8.3. *Hilbert's Nullstellensatz — Weak Form*
Let F be an algebraically closed field. Then M is a maximal ideal in the polynomial ring $F[x_1, \cdots, x_n]$ $\Leftrightarrow$ $M = (x_1 - a_1, \cdots, x_n - a_n)$ for some $a_1, \cdots, a_n \in F$. Equivalently, the maps $\mathcal{Z}$ and $\mathcal{I}$ give a bijective correspondence

$$\{\textit{points in } \mathbb{A}^n\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\textit{maximal ideals in } F[\mathbb{A}^n]\}$$

Moreover, if I is any proper ideal in $F[x_1, \cdots, x_n]$, then $\mathcal{Z}(I) \neq \emptyset$.

Proof.
- ($\Leftarrow$): Certainly $(x_1 - a_1, \cdots, x_n - a_n)$ is a maximal ideal in $F[x_1, \cdots, x_n]$.
- ($\Rightarrow$): Let $E := F[x_1, \cdots, x_n]/M$. Then E is a field containing F that is finitely generated over F (by $\overline{x}_1, \cdots, \overline{x}_n$). By the Noetherian's Normalization Lemma, E is integral over $F[y_1, \cdots, y_q]$. Since $F[y_1, \cdots, y_q]$ is an ID and $F[y_1, \cdots, y_q]$ is integral over F, then by the Integral Ring Extension-Prime Ideals Theorem.1, $F[y_1, \cdots, y_q]$ is a field. But since a polynomial ring in one or more variables is never a field, then $q = 0$. Therefore, E is integral over F and hence E is algebraic over F. Since F is algebraically closed, then $E = F$, i.e.: $\overline{x}_i \in F$, for $1 \leq i \leq n$. Hence, $\forall i = 1, \cdots, n, \exists a_i \in F$, s.t.: $x_i - a_i \in M$, which follows that $(x_1 - a_1, \cdots, x_n - a_n) \subseteq M$. Note that $(x_1 - a_1, \cdots, x_n - a_n)$ is maximal and M is proper. Therefore, $(x_1 - a_1, \cdots, x_n - a_n) = M$.

Finally, if $I = 0$, then $\mathcal{Z}(I) = \mathbb{A}^n \neq \emptyset$. Otherwise, $I \subseteq M$ for some maximal ideal $M = (x_1 - a_1, \cdots, x_n - a_n)$. Therefore, $(x_1 - a_1, \cdots, x_n - a_n) \in \mathcal{Z}(I) \neq \emptyset$. $\square$

Theorem 2.8.4. *Hilbert's Nullstellensatz*
Let F be an algebraically closed field. Then $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$ for every ideal I of $F[x_1, \cdots, x_n]$. Moreover, the maps $\mathcal{Z}$ and $\mathcal{I}$ define inverse bijections

$$\{\textit{affine algebraic sets}\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\textit{radical ideals}\}$$

Proof.
- $\forall f \in \text{rad } I, \exists m \in \mathbb{N}_+$, s.t.: $f^m \in I$. Hence, $(f(a))^m = f^m(a) = 0, \forall a \in \mathcal{Z}(I)$. Since $f(a) \in F$ and F has no zero-divisors, then $f(a) = 0, \forall a \in \mathcal{Z}(I)$ and hence $f \in \mathcal{I}(\mathcal{Z}(I))$.
- Since field$\Rightarrow$PID$\Rightarrow$Noetherian ring, then by the coro.1 of Hilbert's Basis Theorem, $F[x_1, \cdots, x_n]$ is Noetherian. Then by the Equivalent-Conditions-of-Noetherian Rings Prop.3, $I = (f_1, \cdots, f_m)$. Consider a new variable x_{n+1}. For any $g \in \mathcal{I}(\mathcal{Z}(I))$, consider the ideal I' generated by $f_1, \cdots, f_m, x_{n+1}g - 1$ in $F[x_1, \cdots, x_n, x_{n+1}]$. $\forall a = (a_1, \cdots, a_n, b) \in \mathbb{A}^{n+1}$ with $f_1(a) = \cdots = f_m(a) = 0$, since $g \in \mathcal{I}(\mathcal{Z}(I))$, then $g(a) = 0 \Rightarrow (x_{n+1}g - 1)(a) = 0 - 1 = -1 \neq 0$ and hence $\mathcal{Z}(I') = \emptyset$ in $\mathbb{A}^{n+1}$. By the Weak Form of the Hilbert's Nullstellensatz, I' cannot be a proper

ideal, i.e.: $1 \in I' = (f_1, \cdots, f_m, x_{n+1}g - 1)$ and hence we can write

$$1 = c_1 f_1 + \cdots + c_m f_m + c_{m+1}(x_{n+1}g - 1) \text{ for some } c_i \in F[x_1, \cdots, x_n, x_{n+1}]$$

Let $y = 1/x_{n+1}$ and let $N - 1$ be the highest power among $c_1, \cdots, c_m, c_{m+1}$. Multiplying by y^N in the above equation and then we have

$$y^N = d_1 f_1 + \cdots + d_m f_m + d_{m+1}(g - y) \text{ for some } d_i \in F[x_1, \cdots, x_n, y]$$

If $y = g$, then we have $g^N = d_1 f_1 + \cdots + d_m f_m \in I \Rightarrow g \in \text{rad } I$.

□

Corollary 2.8.5. *of Hilbert's Nullstellensatz*
If F is any field with algebraic closure $\overline{F}$ and I is an ideal in $F[x_1, \cdots, x_n]$, then $\mathcal{I}_F(\mathcal{Z}_{\overline{F}}(I)) = \text{rad } I$, where $\mathcal{Z}_{\overline{F}}(I) :=$ the zero set in $\overline{F}^n$ of the polynomials in I and $\mathcal{I}_F(\mathcal{Z}_{\overline{F}}(I))$ is the ideal of polynomials in $F[x_1, \cdots, x_n]$ vanishing at all the points in $\mathcal{Z}_{\overline{F}}(I)$. In particular, $I = (1) \Leftrightarrow$ there are no common zeros in $\overline{F}^n$ of the polynomials in I.

Remark:

- This (coro.) is the geometric interpretation of the following: By the Radicals-Prime Ideals Prop. and the Integral Ring Extension-Prime Ideals Thm.2, if S is an integral extension of R with $1 \in R$ and if I is an ideal of R, then

$$(\text{rad}_S\ IS) \cap R = \text{rad}_R\ I$$

- From Nullstellensatz, we now have a dictionary between geometry and ring-theoretic objects over the algebraically closed field F:

Geometry	**Algebra**
affine algebraic set V	coordinate ring $k[V]$
points of V	maximal ideals of $k[V]$
affine algebraic subsets in V	radical ideals of $k[V]$
subvarieties in V	prime ideals in $k[V]$
morphism $\varphi\colon V \to W$	k-algebra homomorphism $\widetilde{\varphi}\colon k[W] \to k[V]$

2.9 Localization

2.9.1 Localization of Rings

Intuition: Localization is an algebraic analogue of the familiar idea of localizing at a point when considering questions of, for example, the differentiability of a real-valued function $f(x) \in \mathcal{C}(\mathbb{R})$.

Definition 2.9.1. Multiplicative Closed Sets
A *multiplicatively closed set* $D \subseteq R$ is a subset, s.t.: $1 \in D$ and $a, b \in D \Rightarrow ab \in D$. Furthermore, let $U \subseteq R$ be any set. The *multiplicative closure* of U is the set

$$\{\Pi_{i=1}^n u_i \mid n \geq 0, u_i \in U\}$$

which is the smallest multiplicatively closed set containing U.

Example:

- Let R be an ID. Then $D = R \setminus \{0\}$ is multiplicatively closed.
- If P is a prime ideal in R, then $D = R \setminus P$ is multiplicatively closed.
- If $x \in R$, then the set $\{x^n \mid n \in \mathbb{N}_+\}$ is multiplicatively closed.

Method: We first consider a very general construction of "rings of fractions". Let D be a multiplicatively closed subset of R. Then we construct a new ring $D^{-1}R$ which is the "smallest" ring in which the elements of D become units.

Theorem 2.9.2. *Universal Property of Localization of Rings*
Let R be a ring and let D be a multiplicatively closed subset of R. Then there is a ring $D^{-1}R$ and a ring homomorphism $\pi : R \to D^{-1}R$ satisfying the following universal property:
For any unital homomorphism $\psi : R \to S$ of rings such that $\psi(d) \in U(S)$, $\forall f \in D \subseteq R$, there exists a unique homomorphism $\Psi : D^{-1}R \to S$, s.t.: $\Psi \circ \pi = \psi$, i.e.: the following diagram commutes:

$$\begin{tikzcd} R \arrow[r, "\pi"] \arrow[dr, "\psi"'] & D^{-1}R \arrow[d, dashed, "\exists!\Psi"] \\ & S \end{tikzcd}$$

Proof. Define a relation $\sim$ on $R \times D$ by

$$(r, d) \sim (s, e) \Leftrightarrow x(er - ds) = 0 \text{ for some } x \in D$$

It's clear that $\sim$ is reflexive and symmetric. If $(r, d) \sim (s, e)$ and $(s, e) \sim (t, f)$, then $x(er - ds) = 0$ and $y(fs - et) = 0$ for some $x, y \in D$. Multiplying the first equation by fy and the second equation by dx and adding them gives $exy(fr - dt) = 0$. Since D is multiplicatively closed, then $(r, d) \sim (t, f)$ and hence $\sim$ is transitive, so $\sim$ is an equivalence relation.

Let r/d denote the equivalence class of (r, d) under $\sim$ and let $D^{-1}R$ be the set of these equivalence classes. Define addition and multiplication in $D^{-1}R$ by

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd} \text{ and } \frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}$$

It's easy to check that these operations are well-defined and make $D^{-1}R$ into a commutative ring with $1_{D^{-1}R} = \frac{1}{1}$. For any $d \in D$, $\frac{d}{1} \in U(D^{-1}R)$ is a unit in $D^{-1}R$.
Define $\pi : R \to D^{-1}R$ by $\pi(r) := r/1$. It's easy to see that π is a ring homomorphism. Define $\Psi : D^{-1}R \to S$ by

$$\Psi(\frac{r}{d}) := \psi(r)\psi(d)^{-1}$$

which is well-defined since:

- by the def of ψ, $\psi(d)$ is a unit in S;
- if $r/d = s/e$, then $x(er - ds) = 0$ for some $x \in D$; then $\psi(x)(\psi(er) - \psi(ds)) = 0 \in S$ and hence $\psi(er) - \psi(ds) = 0$ since $\psi(x) \in U(S)$ and therefore, $\psi(r)\psi(d)^{-1} = \psi(s)\psi(e)^{-1}$.

Then it's immediately that Ψ is a ring homomorphism and $\Psi \circ \pi = \psi$.
As for the uniqueness of Ψ, note that $\forall$ element $r/d \in D^{-1}R$, it can be written as $(r/1)(d/1)^{-1}$ since

$$\frac{r}{d} = \frac{r}{1} \cdot \frac{1}{d} = (\frac{r}{1}) \cdot (\frac{d}{1})^{-1}$$

that the value of Ψ on each element of the form $x/1$ is uniquely determined by ψ, namely $\Psi(x/1) = \Psi(\pi(x)) = \psi(x)$ and that $\forall$ unit $u \in U(D^{-1}R)$, since Ψ is ring homomorphism, then its value on u^{-1} is uniquely determined by $\Psi(u)$. Therefore, Ψ is uniquely determined on every element of $D^{-1}R$. ☐

Corollary 2.9.3. *of Universal Property of Localization of Rings*
In the notation of the Universal Property of Localization of Rings,

1. $\ker \pi = \{r \in R \mid xr = 0$ *for some* $x \in D\}$. *In particular,* $\pi : R \to D^{-1}R$ *is injective* $\Leftrightarrow$ D *contains no zero-divisors of* R.
2. $D^{-1}R = 0 \Leftrightarrow 0 \in D \Leftrightarrow D$ *contains nilpotent elements.*

Proof.
1. $\pi(r) = 0 \Leftrightarrow r/1 = 0_{D^{-1}R} \Leftrightarrow (r, 1) \sim (0, 1) \Leftrightarrow xr = 0$ for some $x \in D$.
2. $D^{-1}R = 0 \Leftrightarrow (1, 1) \sim (0, 1) \Leftrightarrow x1 = 0$ for some $x \in D \Leftrightarrow 0 \in D$.

☐

Definition 2.9.4. Localization of Rings
The ring $D^{-1}R$ is called the *ring o fractions of* R *with respect to* D or the *localization of* R *at* D.

Example:

- Given any $f \in R$, let $D = \{f^n \mid n \in \mathbb{N}_+$. Define

$$R_f := D^{-1}R$$

Note that $R_f = 0 \Leftrightarrow f$ is nilpotent by the corollary of Universal Property of Localization of Rings. If f is *not* nilpotent, then $f = f/1$ becomes a unit in R_f. It's easy to see that

$$R_f \cong R[x]/(xf - 1)$$

Note that $R_f \cong R_{f^n}, \forall n \in \mathbb{N}_+$, since both f and f^n are units. If f is a zero-divisor, then $\pi : R \to R_f$ does *not* embed R into R_f. For instance, let $R = F[x,y]/(xy)$ and take $f = \overline{x} + (xy)$. Since $\overline{x}$ is *not* nilpotent in R, then $\overline{x} = \overline{x}/1$ is a unit in R_f. Since $\overline{x} \in D$ and $\overline{x} \cdot \overline{y} = 0_R$, then by the coro. of Universal Property of Localization of Rings.1, $\overline{y} \in \ker \pi$. Hence,

$$\pi(R) = F[x] \subseteq R_f = F[x, x^{-1}]$$

- *Localizing at a Prime:* Let P be a prime ideal in R and let $D = R \setminus P$. In this case, $D^{-1}R$ is called *localizing R at P*, which can be denoted as R_P. For instance, if $R = \mathbb{Z}$ and $P = (p)$ is a prime ideal, then
$$\mathbb{Z}_{(p)} = \{\frac{a}{b} \in \mathbb{Q} \mid p \not| b\} \subseteq \mathbb{Q}$$
and every integer not divisible by p is a unit.
- Let V be any non-empty set, let F be a field and let R be any ring of F-valued functions on V containing the constant functions (for instance, $\mathcal{C}([0,1])$). For any $a \in V$, let M_a be the ideal of functions in R that vanish at a. Then M_a is the kernel of the ring homomorphism from R to the field F given by evaluating each function in R at a. Since R contains the constant functions, then evaluation is surjective, so M_a is a maximal (hence also prime) ideal. The localization of R at M_a is
$$R_{M_a} = \{\frac{f}{g} \mid f, g \in R, g(a) \neq 0\}$$
Each function in R_{M_a} can then be evaluated at a by $(f/g)(a) = f(a)/g(a)$ and this value does *not* depend on the choice of representative for the class f/g and hence R_{M_a} becomes a ring of F-valued "rational functions" defined at a.

2.9.2 Extensions and Contractions of Ideals in Localization

Notation:

- If I is an ideal of R, then we denote by ${}^eI$ or by $D^{-1}I$ the extension of I to $D^{-1}R$ with respect to $\pi : R \to D^{-1}R$, namely,
$${}^eI := D^{-1}R\pi(I)$$
- If J is an ideal of $D^{-1}R$, then we denote by ${}^cJ$ the contraction of J to R with respect to $\pi : R \to D^{-1}R$, namely,
$${}^cJ := \pi^{-1}(J)$$

Proposition 2.9.5. *Extensions and Contractions-Localization-Properties-Proposition*

1. $\forall$ *ideal* J *of* $D^{-1}R$, *we have* $J =^e ({}^cJ)$. *In particular, every ideal of* $D^{-1}R$ *is the extension of some ideal of* R *and distinct ideals of* $D^{-1}R$ *have distinct contractions in* R.

2. $\forall$ *ideal* I *of* R, *we have*

$$^c({}^eI) = \{r \in R \mid dr \in I \text{ for some } d \in D\}$$

Furthermore, $^eI = D^{-1}R \Leftrightarrow I \cap D \neq \emptyset$.

3. *Extension and contraction give a bijective correspondence*

$$\left\{\begin{matrix} \text{prime ideals } P \text{ of } R \\ \text{with } P \cap D = \emptyset \end{matrix}\right\} \underset{c}{\overset{e}{\rightleftarrows}} \{\text{prime ideals of } D^{-1}R\}$$

4. *If* R *is Noetherian (or Artinian), then* $D^{-1}R$ *is Noetherian (respectively Artinian).*

Proof.

1. It's immediately by def that $^e({}^cJ) \subseteq J$. For the reverse inclusion, let $a/d \in J$. Then $a/1 = d(a/d) \in J$ and hence $a \in \pi^{-1}(J) =^c J$. Hence, $a/1 = \pi(a) =^e ({}^cJ)$, so we have

$$a/d = (a/1)(1/d) \in^e ({}^cJ)$$

Therefore, $^e({}^cJ) = J$.

2. Let $I' := \{r \in R \mid dr \in I \text{ for some } d \in D\}$.

   - $I' \subseteq^c ({}^eI)$: $\forall r \in I', \exists d \in D$, s.t.: $dr \in I$. Then $r/1 = (dr)/d \in^e I$ and hence $r \in^c ({}^eI)$.
   - $^c({}^eI) \subseteq I'$: $\forall r \in^c ({}^eI), \exists a \in I, d \in D$, s.t.: $r/1 = a/d \in^e I$. Then $x(dr - a) = 0$ for some $x \in D$ and hence $xdr = xa \in I$. Since $xd \in D$, then by the def of I', $r \in I'$.

   Therefore, $I' =^c ({}^eI)$ which proves the first statement of 2. For the further part,

$$^eI = D^{-1}R \Leftrightarrow 1/1 \in^e I \Leftrightarrow 1 \in^c ({}^eI) \overset{1}{=} I' \overset{\text{def of } I'}{\Longleftrightarrow} I \cap D \neq \emptyset.$$

3. Note that if Q is a prime ideal in $D^{-1}R$, then its preimage under any unital homomorphism is a prime ideal and hence c maps prime ideals of $D^{-1}R$ to prime ideals of R disjoint from D. In th reverse direction, let P be a prime ideal of R disjoint from D and let $Q :=^e P$. Suppose $(a/d_1)(b/d_2) \in Q$. Then

$$(ab)/(d_1d_2) \in Q \Rightarrow (ab)/(d_1d_2) = c/d \text{ for some } c \in P, d \in D.$$

   Hence, $x(dab - d_1d_2c) = 0$ for some $x \in D$. Since $c \in P$, then $xdab = xd_1d_2c \in P$. Since $P \cap D = \emptyset$, then $ab \in P \Rightarrow a \in P$ or $b \in P$ and hence $a/d_1 \in Q$ or $b/d_2 \in Q$.

4. By 1, every ascending (or descending) chain of distinct ideals in $D^{-1}R$ contrasts to an as-

cending (respectively descending) chain of distinct ideals in R.

□

Remark: By 2, we can easily see that $I \subseteq^c ({}^eI)$.

Definition 2.9.6. Saturation
Let R be a ring and D be a multiplicatively closed subset of R. The *saturation* of the ideal I in R with respect to D is the ideal ${}^c({}^eI)$ in R, where contraction and extension are computed with respect to $\pi : R \to D^{-1}R$. If $I =^c ({}^eI)$, then I is said to be *saturated* with respect to D.

2.9.3 Localization of Modules

Definition 2.9.7. Localization of Modules
Let M be an R-module and let D be a multiplicatively closed subset of R. Then we define an equivalence relation on $D \times M$ by

$$(d, m) \sim (e, n) \Leftrightarrow x(dn - em) = 0 \text{ for some } x \in D.$$

Let m/d denote the equivalence class of (d, m) and let $D^{-1}M$ denote the set of equivalence classes. Then it is easy to verify that the operations

$$\frac{m}{d} + \frac{n}{e} := \frac{em + dn}{de} \text{ and } (\frac{r}{d}) \cdot (\frac{m}{e}) := \frac{rm}{de}$$

are well-defined and give $D^{-1}M$ the structure of a $D^{-1}R$-module, called the *module of fractions of M with respect to D* or the *localization of M at D*.

Remark: Since each $r \in R$ can act by $r/1$ on $D^{-1}M$, then the localization $D^{-1}M$ is also an R-module. Hence, the map $\pi : M \to D^{-1}M$ given by $m \mapsto \frac{m}{1}$ is an R-module homomorphism. It follows immediately from the def of the equivalence relation on $D \times M$ that

$$\ker \pi = \{m \in M \mid dm = 0 \text{ for some } d \in D\}$$

similarly as the coro. of Universal Property of Localization of Rings.1. Also, similar as the Universal Property of Localization of Rings, we have a universal property of localization of modules:

Theorem 2.9.8. *Universal Property of Localization of Modules*
Let M be an R-module, let D be a multiplicatively closed subset of R and let $D^{-1}M$ be the localization of M at D. Suppose N is an R-module with the property that dN is a bijection of N for every $d \in D$. Consider the R-module homomorphism $\pi : M \to D^{-1}M$ with $m \mapsto \frac{m}{1}$. If $\psi : M \to N$ is any R-module homomorphism, then there is a unique R-module homomorphism $\Psi : D^{-1}M \to N$,

s.t.: $\Psi \circ \pi = \psi$, *i.e.: the following diagram commutes:*

$$\begin{tikzcd} M \arrow[r, "\pi"] \arrow[dr, "\psi"'] & D^{-1}M \arrow[d, dashed, "\exists!\Psi"] \\ & N \end{tikzcd}$$

Remark: If M and N are R-modules and $\varphi : M \to N$ is an R-module homomorphism, then for any multiplicatively closed subset D of R, it is easy to check that there is an induced $D^{-1}R$-module homomorphism from $D^{-1}M$ to $D^{-1}N$ defined by $m/d \mapsto \varphi(m)/d$.

Proposition 2.9.9. *Localization of Modules-Tensor Products Proposition*
Let D be a multiplicatively closed subset of R and let M be an R-module. Then

$$D^{-1}M \cong D^{-1}R \otimes_R M$$

as $D^{-1}R$-modules, i.e.: $D^{-1}M$ is the $D^{-1}R$-module obtained by extension of scalars from the R-module M.

Proof. The map from $D^{-1}R \times M$ to $D^{-1}M$ defined by $(r/d, m) \mapsto (rm)/d$ is clearly well-defined, which induces, by the def of tensor products, an R-module homomorphism from $D^{-1}R \otimes_R M$ to $D^{-1}M$ with $m/d \mapsto (1/d) \otimes m$, which is a well-defined inverse homomorphism (if $m/d = m'/d' \in D^{-1}M$, then $x(d'm - dm') = 0$ for some $x \in D$ and hence $(1/d) \otimes m = (1/(xd'd)) \otimes (xd'm) = (1/(xd'd)) \otimes (xdm') = (1/d') \otimes m'$). Hence, we're done. $\square$

Proposition 2.9.10. *Localization-Operations Proposition*
Let R be a ring and let $D^{-1}R$ be its localization with respect to the multiplicatively closed subset D of R.

1. *Localization commutes with finite sums and intersections of ideals: If I and J are ideals of R, then*

$$D^{-1}(I + J) = D^{-1}(I) + D^{-1}(J) \text{ and } D^{-1}(I \cap J) = D^{-1}(I) \cap D^{-1}(J)$$

   Localization commutes with quotients:

$$D^{-1}R/D^{-1}I \cong D^{-1}(R/I)$$

   where the localization on the right is with respect to the image of D in the quotient R/I.

2. *Localization commutes with taking ideals: If N is the nilradical of R, then $D^{-1}N$ is the nilradical of $D^{-1}R$. If I is an ideal in R, then rad $(D^{-1}I) = D^{-1}($rad $I)$.*

3. *Primary ideals correspond to primary ideals in the correspondence 3 of the Extensions and Contractions-Localization-Properties Prop. More precisely, suppose that Q is a P-primary*

ideal in R. If $D \cap P \neq \emptyset$, then $D^{-1}Q = D^{-1}R$. If $D \cap P = \emptyset$, then $D^{-1}P$ is a prime ideal, the extension $D^{-1}Q$ of Q is a $D^{-1}P$-primary ideal in $D^{-1}R$, and the contraction back to R of $D^{-1}Q$ is Q.

4. *Localization commutes with finite sums, intersections and quotients of modules: If L and N are submodules of an R-module M, then:*
   (a) $D^{-1}(L+N) = D^{-1}L + D^{-1}N$ *and* $D^{-1}(L \cap N) = D^{-1}L \cap D^{-1}N$.
   (b) $D^{-1}N$ *is a submodule of* $D^{-1}M$ *and* $D^{-1}M/D^{-1}N = D^{-1}(M/N)$.
5. *Localization commutes with finite direct sums of modules: If M and N are R-modules, then $D^{-1}(M \oplus N) \cong D^{-1}M \oplus D^{-1}N$.*
6. *Localization is exact (i.e.: $D^{-1}R$ is a flat R-module): If $0 \to L \xrightarrow{\psi} M \xrightarrow{\varphi} N \to 0$ is a short exact sequence of R-modules, then the induced sequence $0 \to D^{-1}L \xrightarrow{\psi'} D^{-1}M \xrightarrow{\varphi'} D^{-1}N \to 0$ of $D^{-1}R$-modules is also exact.*

Proof. We first prove 6:

- Every element of $D^{-1}N$ is of the form n/d for some $n \in N, d \in D$. Since φ is surjective, then $n = \varphi(m)$ for some $m \in M$, so $\varphi'(m/d) = \varphi(m)/d = n/d$ and hence φ' is surjective.
- If $m/d \in \ker \varphi'$, then $d_1\varphi(m) = 0$ for some $d_1 \in D$. Then $\varphi(d_1 m) = 0 \overset{\ker\varphi = \mathrm{im}\,\psi}{\Longrightarrow} d_1 m = \psi(l)$ for some $l \in L$ and hence $m/d = (d_1 m)/(d_1 d) = \psi(l)/(d_1 d) = \psi'(l/(d_1 d)) \Rightarrow \ker\varphi' \subseteq \mathrm{im}\,\psi'$. If $\psi(l)/d \in \mathrm{im}\,\psi'$, then $\varphi'(\psi(l)/d) = \varphi(\psi(l))/d \overset{\ker\varphi = \mathrm{im}\,\psi}{=} 0 \Rightarrow \mathrm{im}\,\psi' \subseteq \ker\varphi'$, so we have the exactness of the induced sequence at $D^{-1}M$.
- Suppose $\psi'(l/d) = 0$. Then $d_2\psi(l) = 0$ for some $d_2 \in D$, i.e.: $\psi(d_2 l) = 0 \Rightarrow d_2 l = 0$, by the injectivity of ψ. Hence, $l/d = (d_2 l)/(d_2 d) = 0$ and hence we're done for 6.

Next, we'll prove in order.

1. $(i+j)/d := i/d + j/d \in D^{-1}(I) + D^{-1}(J) \Rightarrow D^{-1}(I+J) \subseteq D^{-1}(I) + D^{-1}(J)$; $i/d_1 + j/d_2 := (d_2 i + d_1 j)/(d_1 d_2) \in D^{-1}(I+J) \Rightarrow D^{-1}(I) + D^{-1}(J) \subseteq D^{-1}(I+J)$.

   For the second statement, the inclusion $D^{-1}(I \cap J) \subseteq D^{-1}(I) \cap D^{-1}(J)$ is immediate. If $a/d \in D^{-1}(I) \cap D^{-1}(J)$, then $d_1 a \in I, d_2 a \in J$ for some $d_1, d_2 \in D \Rightarrow d_1 d_2 a \in I \cap J \Rightarrow a/d = (d_1 d_2 a)/(d_1 d_2 d) \in D^{-1}(I \cap J) \Rightarrow D^{-1}(I \cap J) \subseteq D^{-1}(I) \cap D^{-1}(J)$.

   The last statement in 1 follows by applying 6 to the exact sequence $0 \to I \xrightarrow{\psi} R \xrightarrow{\varphi} R/I \to 0$.
2. $a \in \mathrm{rad}\ I \Rightarrow a^n \in I$ for some $n \in \mathbb{N}_+ \Rightarrow (a/d)^n = a^n/d^n \in D^{-1}I \Rightarrow d_1 a^n \in I$ for some $d_1 \in D \Rightarrow (d_1 a)^n = d_1^n a^n = d_1^{n-1}(d_1 a^n) \in I \Rightarrow \mathrm{rad}\ (D^{-1}I) \subseteq D^{-1}(\mathrm{rad}\ I)$. This prove the second statement in 2, and then the first statement follows by applying 2 to $I = (0)$.
3. Note that $D \cap P = \emptyset \Leftrightarrow D \cap Q = \emptyset$ ("$\Rightarrow$" is obvious since $Q \subseteq \mathrm{rad}\ Q = P$; "$\Leftarrow$": if $d \in D \cap P$, then $d^n \in Q$ for some $n \in \mathbb{N}_+$, so by the multiplicative closed-ness of D, $d^n \in D$, contradiction). Hence, by the Extensions and Contractions-Localization-Properties Prop.2, we have: $D \cap P \neq \emptyset \Rightarrow D \cap Q \neq \emptyset \Leftrightarrow^e Q = D^{-1}Q = D^{-1}R$.

To see that $D^{-1}Q$ is a primary ideal in $D^{-1}R$, suppose that $(a/d_1)(b/d_2) \in D^{-1}Q$ and $a/d_1 \notin D^{-1}Q$. Then there is $d \in D$, s.t.: $dab \in Q$. Since $a \notin Q$ and Q is primary, then $(db)^n \in Q$ for some $n \in \mathbb{N}_+$. Then $(b/d_1)^n = (d^n b^n)/(d^n d_2^n) \in D^{-1}Q$ and hence we're done.

Finally, by the Extensions and Contractions-Localization-Properties Prop.2, the contraction of $D^{-1}Q$ is an ideal of R containing Q and consists of precisely of the elements $r \in R$ with $dr \in Q$ for some $d \in D$. Since Q is P-primary, then by assumption ($D \cap P = \emptyset \Rightarrow d \notin P$), $r \in Q$ and hence, ${}^c(D^{-1}Q) = Q$.

4. Similarly as 1.
5. Note that if the exact sequence $0 \to L \overset{\psi}{\to} M \overset{\varphi}{\to} N \to 0$ of R-modules splits, then the exact sequence $0 \to D^{-1}L \overset{\psi'}{\to} D^{-1}M \overset{\varphi'}{\to} D^{-1}N \to 0$ of $D^{-1}R$-modules also splits, which gives 5.

□

Proposition 2.9.11. *Localization-on-Primary Decomposition Proposition*
Let R be a Noetherian ring and let

$$I = Q_1 \cap \cdots \cap Q_m$$

be a minimal primary decomposition of the proper ideal I, where Q_i is a P_i-primary ideal. Suppose that D is a multiplicatively closed subset of R and the primary ideals $Q_1, \cdots, Q_m$ are numbered so that $D \cap P_i = \emptyset$ for $i \le i \le t$ and $D \cap P_i \neq \emptyset$ for $t+1 \le i \le m$. Then

$$D^{-1}I = D^{-1}Q_1 \cap \cdots \cap D^{-1}Q_t$$

is a minimal primary decomposition of $D^{-1}I$ in $D^{-1}R$ and $D^{-1}Q_i$ is a $D^{-1}P_i$-primary ideal. Furthermore, the contraction of $D^{-1}Q_i$ back to R is Q_i for $1 \le i \le t$ and

$${}^c(D^{-1}I) = Q_1 \cap \cdots \cap Q_t$$

is a minimal primary decomposition of the contraction of $D^{-1}I$ back to R.

Proof. By the Localization-Operations Prop.3, $D^{-1}Q_i = D^{-1}R$ for $t+1 \le i \le m$, and $D^{-1}Q_i$ is a $D^{-1}P_i$-primary ideal with pullback Q_i for $1 \le i \le t$. By the Localization-Operations Prop.1, we have

$$D^{-1}I = D^{-1}Q_1 \cap \cdots \cap D^{-1}Q_t$$

and then 3 shows that this is a primary decomposition. Contracting to R shows that ${}^c(D^{-1}I) = Q_1 \cap \cdots \cap Q_t$, which also implies that the decompositions are minimal. □

Corollary 2.9.12. *of Localization-on-Primary Decomposition Proposition*
The primary ideals belonging to the isolated primes in a minimal primary decomposition of I are uniquely defined by I.

Proof. Let P be a minimal element in $\{P_1, \cdots, P_m\}$ of primes belonging to I and take $D = R \setminus P$ in the Localization-on-Primary Decomposition Prop. Then $D \cap P_i = \emptyset$ only for $P = P_i$, so the contraction of the localization of I at D is precisely the primary ideal Q belonging to the minimal prime P. Since the prime ideals $\{P_1, \cdots, P_m\}$ of primes belonging to I are uniquely determined by I, which follows that the primary ideals Q belonging to the isolated primes of I are also uniquely determined by I. □

Remark: This coro. proves the Integral Ring Extensions-Prime Ideal Thm.2.

2.9.4 Local Rings and Localization at Prime Ideals

Definition 2.9.13. Local Rings
A ring is *local* if it has a unique maximal ideal.

Proposition 2.9.14. *Equivalent-Conditions-of-Local Rings Proposition*
Let R be a ring. The followings are equivalent:

1. *R is a local ring with unique maximal ideal M.*
2. *If M is the set of elements of R that are not units, then M is an ideal.*
3. *There is a maximal ideal M of R, s.t.: every element $1+m$ with $m \in M$ is a unit in R.*

Proof.

- $(1 \Leftrightarrow 2)$: Note that $\forall a \in R$, the ideal (a) is either R, in which case a is a unit, or is a proper ideal, in which case (a) is contained in a maximal ideal. Hence, we're immediately done.
- $(3 \Rightarrow 1)$: $\forall a \in R \setminus M$, since M is maximal, then $(a) + M = R$, so $ab + m = 1$ for some $b \in R, m \in M$. By 3, $ab = 1 - m \in U(R) \Rightarrow abc = 1$ for some $c \in R \Rightarrow a \in U(R)$ (with inverse $a^{-1} = bc$). Hence, M is the unique maximal ideal.
- $(1 \Rightarrow 3)$: $\forall m \in M$, since M is maximal, then $1 + m \notin M \Rightarrow 1 + m \in U(R)$.

□

Proposition 2.9.15. *Localization of Rings-at-Prime Ideals-Properties Proposition*
Let R_P be the localization of the ring R at the prime ideal P in R and let ${}^eP$ be the extension of P to R_P.

1. *R_P is a local ring with unique maximal ideal ${}^eP$. The contraction of ${}^eP$ to R is P, i.e.: ${}^c({}^eP) = P$, and the map from R to R_P induces an injection of the ID R/P into $R_P/{}^eP$. The quotient $R_P/{}^eP$ is a field and is isomorphic to the fraction of the ID R/P.*
2. *If R is an ID, then R_P is an ID. The ring R injects into the local ring R_P and, identifying R with its image in R_P, the unique maximal ideal of R_P is PR_P.*
3. *The prime ideals in R_P are in bijective correspondence with the prime ideals of R contained in P.*
4. *If P is a minimal nonzero prime ideal of R, then R_P has a unique nonzero prime ideal.*

5. *If $P = M$ is a maximal ideal in R and I is any M-primary ideal of R, then $R_M/{}^eI \cong R/I$. In particular,*

$$R_M/{}^eM \cong R/M \text{ and } ({}^eM)/({}^eM)^n \cong M/M^n \ \forall n \in \mathbb{N}_+.$$

Proof. If P' is a prime ideal in R, then $P' \cap (R \setminus P) = \emptyset \Leftrightarrow P' \subseteq P$, so by the Extensions and Contractions-Localization-Properties Prop.3, 3 holds, and 4 follows. Since ${}^eP \neq R_P$ by the Extensions and Contractions-Localization-Properties Prop.2, then by 3, R_P is a local ring with unique maximal ideal ${}^eP$, which proves the first statement in 1.

By the Extensions and Contractions-Localization-Properties Prop.2, the contraction ${}^c({}^eP)$ is the set $\{r \in R \mid dr \in P \text{ for some } d \in R \setminus P\}$ and since P is prime, then $dr \in P$ with $d \notin P$ implies $r \in P$, which follows that ${}^c({}^eP) = P$ and hence 1 follows.

For 2, note that the kernel of the map from R to $R_P/{}^eP$ is ${}^c({}^eP) = P$, so the induced map from R/P into $R_P/{}^eP$ is injective. The quotient $R_P/{}^eP$ is then a field by the first part of 1 and hence there is an induced homomorphism from the fraction field of the ID R/P into $R_P/{}^eP$. Then by the Universal Property of localization, there is an inverse homomorphism from $R_P/{}^eP$ to the fraction field of R/P (since every element of R not in P maps to a unit in R/P). It follows that $R_P/{}^eP \cong$ the fraction field of R/P. If R is an ID, then R/P has no zero-divisor, so by the coro of Universal Property of Localization of Rings, R injects into R_P; if R is identified with its image in R_P, then ${}^eP = PR_P$, and hence 2 follows.

To prove 5, by the Localization-Operations Prop.1, we may pass to the quotient R/I, so reduce to the case $I = 0$. In this case, the maximal ideal $P = M$ in R is the nilradical of R and hence is the unique maximal ideal of R. By the Equivalent-Conditions-of-Local Rings Prop.2, every element of R/M is a unit, so $R_P = R$ and then each of the statements in 5 follows immediately. □

Remark: The results of the Localization-at-Prime Ideals-Properties Prop.5 above *not* true in general if P is a prime ideal that is *not* maximal. For instance, $P = (0)$ in $R = \mathbb{Z}$ has $R/P = \mathbb{Z}$ and $R_P/PR_P = \mathbb{Q}$; in this case $(PR_P)/(PR_P)^n \cong P/P^n = 0, \forall n \in \mathbb{N}_+$.

Definition 2.9.16. Localization of Modules at Prime Ideals
Let M be an R-module, let P be a prime ideal in R and let $D = R \setminus P$. The R_P-module $D^{-1}M$ is called the *localization of M at P*, denoted as M_P.

Remark:

- By the Localization of Modules-Tensor Products Prop, M_P can be identified with the tensor product $R_P \otimes_R M$.
- If R is an ID and $P = (0)$, then $M_{(0)}$ is a module over the field of fractions F of R, i.e.: is a vector space over F.

- $m/1 = 0_{M_P} \Leftrightarrow rm = 0$ for some $r \in R \setminus P$, so localizing at P annihilates the P'-torsion elements of M for prime P' not contained in P. In particular, *localizing at* (0) *over an ID annihilates the torsion subgroup of* M.

Definition 2.9.17. Rank of Modules
If R is an ID, then the *rank* of an R-module M is the dimension of the localization $M_{(0)}$ as a vector spave over the field of fractions of R.

Remark: It's easy to see that this def of rank agrees with the *free module of rank over* R given in the example of def of submodules.

2.9.5 Local-Global Principle

Proposition 2.9.18. *Local-Global Principle for Zero Modules*
Let M be an R-modules. The followings are equivalent:

1. $M = 0$.
2. $M_P = 0$ *for all prime ideals* P *of* R.
3. $M_N = 0$ *for all maximal ideals* N *of* R.

Proof. $(1 \Rightarrow 2)$ and $(2 \Rightarrow 3)$ are obvious, so it suffices to prove $(3 \Rightarrow 1)$: Suppose on the contrary that there is a non-zero element m in M. Consider the annihilator I of m in R, i.e.: $I := \{r \in R \mid rm = 0\}$. Since $m \neq 0$, then I is proper. Let N be a maximal ideal of R containing I and then consider $m/1$ in the corresponding localization M_N of M. If $m/1 \neq 0$, then $rm = 0$ for some $r \in R \setminus N \Leftrightarrow I \cap (R \setminus N) \neq \emptyset$, contradicting $I \subseteq N$. Hence, $m/1 \neq 0 \Rightarrow M_N \neq 0$, contradicting 3. □

Remark: It's *not* generally true that a property shared by all of the localizations of a module M is also shared by M. For instance, all of the localizations of a ring R can be IDs, without R itself being an ID. But this technique — localization — is still useful: If R is an ID, for example, then each of the localizations R_P can be considered as a subring of the fraction field F of R that contains R. The next proposition shows that the elements of the ID R are the only elements of F contained in every localization.

Proposition 2.9.19. *Local-Global Principle for Integral Domain*
Let R be an ID. Then R is the intersection of the localizations of R:

$$R = \bigcap_P R_P$$

In fact, $R = \bigcap_M R_M$ *is the intersection of the localizations of* R *at the maximal ideals* M *of* R.

Proof. It's clear that $R \in \bigcap_{M \text{ maximal}} R_M$. For the reverse inclusion, we consider an element a of the fraction field F of R that is contained in R_M for any maximal ideal M and consider

$$I_a := \{r \in R \mid ra \in R\}$$

which is easy to check that I_a is an ideal of R and that $a \in R \Leftrightarrow 1 \in I_a \Leftrightarrow I_a = R$. Suppose that $I_a \neq R$. Then there is a maximal ideal M containing I_a. Since $a \in R_M$, then $a = r/d$ for some $r \in R, d \in R \setminus M$. But then $d \in I_a$ and $d \notin M$, contradiction. Hence, $a \in R$, so we have $\bigcap_{M \text{ maximal}} R_M \subseteq R$. Therefore, the second part holds and the first then follows. □

Proposition 2.9.20. *Local-Global Principle for Normality Propersition*
Let R be an ID. The followings are equivalent:

1. *R is normal, i.e.: R is integrally closed (in its field of fractions).*
2. *R_P is normal for all prime ideals P of R.*
3. *R_M is normal for all maximal ideals M of R.*

Proof. Let F be the field of fractions of R, so all of the localizations of R may be considered as subrings of F.

- $(1 \Rightarrow 2)$: $\forall y \in F$ integral over R_P, let $f(x) = x^n + (a_{n-1}/d_{n-1})x^{n-1} + \cdots + (a_1/d_1)x + a_0/d_0$ be the monic polynomial with a root y, where $a_i \in R, d_i \in R \setminus P$. The element $y' := y(d_0 d_1 \cdots d_{n-1})^n$ is then a root of a monic polynomial of degree n with coefficients from R, so y' is integral over R and hence by 1, $y' \in R$. Therefore, $y = y'/(d_0 d_1 \cdots d_{n-1})^n \in R_P$.
- $(2 \Rightarrow 3)$: Obvious.
- $(3 \Rightarrow 1)$: $\forall y \in F$ integral over R, since $R \subseteq R_M$ for every maximal ideal M of R, then y is integral over R_M and hence by 3, $y \in R_M$ for every maximal ideal M of R. Therefore, by the Local-Global Principle for ID, we have

$$y \in \bigcap_{M \text{ maximal}} R_M = R$$

□

Corollary 2.9.21. *of Local-Global Principle for Normality*
Let R be a subring of a ring S and assume that S is integral over R. If P is a prime ideal in R, then there is a prime ideal Q of S with $P = Q \cap R$.

Proof. Let $D = R \setminus P$, which is a multiplicatively closed subset of both R and S. Then the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\pi} & D^{-1}R = R_P \\ \big\downarrow & & \big\downarrow \\ S & \xrightarrow{\pi} & D^{-1}S \end{array}$$

It's easy to see that $D^{-1}S$ is integral over R_P. Let M be any maximal ideal of $D^{-1}S$. Then by the second statement of the Integral Ring Extensions-Prime Ideals Thm.2, $M \cap R_P$ is a maximal ideal in R_P. By the Extensions and Contractions-localization-Properties Prop.1, $M \cap R_P$ is the extension of P to the local ring R_P and the contraction of this ideal to R is just P. Namely, the preimage of M by the maps along $R \xrightarrow{\pi} R_P \hookrightarrow D^{-1}S$ is P. If $Q \subseteq S$ denotes the preimage of M by the map along $R \hookrightarrow S \xrightarrow{\pi} D^{-1}S$, then Q is a prime ideal by the Extensions and Contractions-Localization-Properties Prop.3. Since $Q \cap R$ is the pullback of Q by the map along $R \hookrightarrow S$, then the commutativity of the diagram shows that $Q \cap R = P$. □

Remark: This coro. gives a more easy proof of the first part of the Going-up Thm — the Integral Ring Extensions-Prime Ideals Thm.

2.9.6 Local Rings of Affine Algebraic Varieties

Convention: For the remainder of this section, let F be an algebraically closed field and let V be an affine variety over F with coordinate ring $F[V]$. Then by the coro. of Affine Algebraic Sets-Decomposition Prop, $F[V]$ is an ID, so we may form its field of fractions:

$$F(V) := \{f/g \mid f, g \in F[V], g \neq 0\}$$

Definition 2.9.22. Rational Functions
The elements of $F(V)$ are called *rational functions* on V and $F(V)$ is called the *field of rational functions* on V.

Remark:

- Since $F[V]$ is an ID, then $f_1/g_1 = f_2/g_2 \Leftrightarrow f_1g_2 = f_2g_1$.
- If $F[V]$ is a UFD, then there is an essentially unique representative for f/g that is in "lowest terms", but in general, each fraction $f/g \in F(V)$ has many representatives as a ratio of two elements of $F[V]$.

Definition 2.9.23. Regularity
We say that f/g is *regular at* v or *defined at the point* $v \in V$ if there is some $f_1, g_1 \in F[V]$ with

$$f/g = f_1/g_1 \text{ and } g_1(v) \neq 0.$$

Example: The variety $V = \mathcal{Z}(xz - yw)$ in $\mathbb{A}^4$ has coordinate ring $F[V] = F[x, y, z, w]/(xz - yw)$. Consider $f = \overline{x}/\overline{y} \in F(V)$. Since $\overline{x}\,\overline{z} = \overline{y}\,\overline{w}$, then f can also be written as $\overline{w}/\overline{z}$. From the first expression for $f = \overline{x}/\overline{y}$, f is regular ar all points of V where $\overline{y} \neq 0$, and from the second expression for $f = \overline{w}/\overline{z}$, f is regular at all points of V where $\overline{z} \neq 0$. It's easy to show that these are all the points of V where f is regular. Furthermore, there is no single expression $f = a/b$ for f with $a, b \in F[V]$ such that $b(v) \neq 0, \forall v$ where f is regular.
Remark:

- If $f/g \in F(V)$ is regular at the point v, say $f/g = f_1/g_1$ with $g_1(v) = 0$, then f/g is also regular at all the points v in the Zariski open neighborhood V_{g_1} of v where $0 \notin g_1(V_{g_1})$. Note that any non-empty open set of an affine algebraic variety is Zariski dense. Hence, every rational function on V is defined at a dense set of points in V. Furthermore, each pair f_1/g_1

and f_2/g_2 representing f/g agree as functions on the open neighborhood $V_{g_1} \cap V_{g_2}$ of v, but the "size" of this neighborhood depends on g_1 and g_2. In fact, there is generally *not* a common open neighborhood of v where all representatives of f/g with nonzero denominator at v are defined.

- Fix $v \in V$. A rational function f/g is regular at $v \Leftrightarrow f/g = f_1/g_1$ for some $f_1, g_1 \in F[V]$ with $g_1 \notin \mathcal{I}(v)$. Hence, the set of rational functions that are defined at v is the same as the localization of $F[V]$ at the maximal ideal $\mathcal{I}(v)$:

Definition 2.9.24. Local Rings of Affine Algebraic Sets
For each point $v \in V$, the collection of rational functions on V that are defined at v,

$$\mathcal{O}_{v,V} := \{f/g \in F(V) \mid f/g \text{ is regular at } v\}$$

is called the *local ring of V at v*. Equivalently, the local ring of V at v is th localization of $F[V]$ at the maximal ideal $\mathcal{I}(v)$.

Remark:

- In particular, $\mathcal{O}_{v,V}$ is a local ring with unique maximal ideal $M_{v,V}$ where

$$M_{v,V} := \{f/g \in \mathcal{O}_{v,V} \mid f/g = f_1/g_1 \text{ with } f_1(v) = 0, g_1(v) = 0\}$$

is the set of rational functions on V that are defined and equal to 0 at v.

- Since $\mathcal{O}_{v,V}$ is a localization of the Noetherian ID $F[V]$ at a prime ideal, then $\mathcal{O}_{v,V}$ is also a Noetherian ID. Then by the Localization of Rings-at-Prime Ideals-Properties Prop.5,

$$\mathcal{O}_{v,V}/M_{v,V} \cong F[V]/\mathcal{I}(v) \cong F$$

Proposition 2.9.25. *Local-Global Principle-for-Regularity Proposition*
If V is an affine variety over an algebraically closed field F, then the rational functions on V that are regular at all points of V are precisely the polynomial functions $F[V]$.

Proof. By the Local-Global Principle for ID Prop, the intersection (in $F(V)$) of all of the localizations of $F[V]$ at the maximal ideals of $F[V]$ is precisely $F[V]$. Then we're done. □

Remark: Now we can use the language of localization to build geometric intuition for algebraic geometry:

- *The intrinsic nature of the local ring $\mathcal{O}_{v,V}$:* Since by the weak form Hilbert's Nullstellensatz, we have a bijective correspondence between

$$\{\text{maximal ideals of } F[V]\} \leftrightarrow \{\text{points in } V\}$$

then the fact, that the local ring $\mathcal{O}_{v,V}$ is the same as the localization of $F[V]$ at the maximal ideal corresponding to v, shows that $\mathcal{O}_{v,V}$ depends intrinsically on the ring $F[V]$ and is independent of the embedding of V in a particular affine algebraic space.

- *A morphism of affine varieties induces a local homomorphism of local rings:* Suppose $\varphi : V \to W$ is a morphism of affine varieties with associated F-algebra homomorphism $\widetilde{\varphi} : F[W] \to F[V]$. If $\varphi(v) = w$ with $v \in V, w \in W$, then it is straightforward to show that $\widetilde{\varphi}$ induces a homomorphism (also denoted as $\widetilde{\varphi}$) between the corresponding local rings:

$$\widetilde{\varphi} : \mathcal{O}_{w,W} \to \mathcal{O}_{v,V} \text{ with } \widetilde{\varphi}(h/k) = \widetilde{\varphi}(h)/\widetilde{\varphi}(k)$$

  and that under this homomorphism, $\widetilde{\varphi}^{-1}(M_{v,V}) = M_{w,W}$ (a homomorphism of local rings having this property is called a *local homomorphism*). Note that $\widetilde{\varphi}$ does *not* in general extend to a field homomorphism from all of $F(W)$ into $F(V)$, since elements of $F[W]$ lying in the kernel of $\widetilde{\varphi}$ do not map to invertible elements in $F(V)$. It's also easy to check that if $\psi \circ \varphi$ is a composition of morphisms, then on the local rings $\widetilde{\psi \circ \varphi} = \widetilde{\varphi} \circ \widetilde{\psi}$.

- *Define geometric smoothness and tangent spaces via local rings:* Suppose first that $V = \mathcal{Z}(f)$ is the hypersurface variety in $\mathbb{A}^n$ defined by the zeros of an irreducible polynomial f in $F[x_1, \cdots, x_n]$. $\forall v = (v_1, \cdots, v_n) \in V$, let $D_v(f)(x_1, \cdots, x_n)$ be the linear polynomial:

$$D_v(f)(x_1, \cdots, x_n) := \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(v) x_i$$

  where the partial derivative of f with respect to x_i is given by the usual formal rule for the derivative of a polynomial x_i (with all other variables considered constant). The polynomial $D_v(f)(x_1 - v_1, \cdots, x_n - v_n)$ is the first order Taylor polynomial of the function f at v, so gives the approximation to $f(x_1, \cdots, x_n) \in F[x_1, \cdots, x_n]$ at v. Hence, if $\mathbf{T}$ is the linear variety $\mathcal{Z}(D_v(f)(x_1, \cdots, x_n))$ consisting of those points where $D_v(f)$ is zero, then the translate $v + \mathbf{T}$ is "tangent" to the hypersurface $\mathcal{Z}(f)$ at v.

Definition 2.9.26. Tangent Spaces to Affine Varieties / Zariski Tangent Spaces
The *tangent space to* V *at* v is defined by the linear variety

$$\mathbb{T}_{v,V} := \mathcal{Z}(\{D_v(f)(x_1, \cdots, x_n) \mid f \in \mathcal{I}(V)\})$$

Remark:

- The formal partial derivatives are F-linear and obey the usual product rule for derivatives, so the tangent space may be computed from the generators for $\mathcal{I}(V)$:

$$\mathcal{I}(V) = (f_1, f_2, \cdots, f_m) \Rightarrow \mathbb{T}_{v,V} = \bigcap_{i=1}^{m} \mathcal{Z}(D_v(f_i))$$

- Since $\mathbb{T}_{v,V}$ is an intersection of vector spaces, then $\mathbb{T}_{v,V}$ is a vector subspace of F^n.

- This def of the tangent space $\mathbb{T}_{v,V}$, while making apparent the connection with tangents to the variety V, seems to depend on the embedding of V in $\mathbb{A}^n$:

Proposition 2.9.27. *Independence of Embedding for Tangent Spaces Proposition*
Let V be an affine variety over the algebraically closed field F and let v be a point on V with local ring $\mathcal{O}_{v,V}$ and corresponding maximal ideal $M_{v,V}$. Then there is a F-vector space isomorphism

$$(\mathbb{T}_{v,V})^* \cong M_{v,V}/M_{v,V}^2$$

where $(\mathbb{T}_{v,V})^*$ denotes the vector space dual of the tangent space $\mathbb{T}_{v,V}$ to V at v.

Proof. Denote by $(F^n)^*$ the n-dimensional vector space dual to F^n. Since each $D_v(f)$ is a linear function, then D_v is a linear transformation from $F[x_1, \cdots, x_n]$ to $(F^n)^*$.
Let M_v be the maximal ideal in $F[x_1, \cdots, x_n]$ generated by the set $\{x_1 - v_1, \cdots, x_n - v_n\}$. The image $M_v/\mathcal{I}(V)$ of M_v in $F[V]$ is the ideal $\mathcal{I}(v)$ of functions on V that vanishes at v and

$$\mathcal{I}(v) = M_v^2 + \mathcal{I}(V)$$

Then $\mathcal{O}_{v,V}$ is the localization of $F[V]$ at $\mathcal{I}(v)$ and identifying $\mathcal{I}(v)$ with its image in $\mathcal{O}_{v,V}$, we have $M_{v,V} = \mathcal{I}(v)\mathcal{O}_{v,V}$ by the Localization of Rings-at-Prime Ideals-Properties Prop.2. By the def of D_v, we have $D_v(x_i - v_i) = x_i$. Since these linear functions form a basis of $(F^n)^*$, then D_v maps M_v surjective onto $(F^n)^*$. The kernel of D_v consists of the elements of $F[x_1, \cdots, x_n]$ whose Taylor expansion at v starts in degree at least 2 and these are the elements in M_v^2. Hence, by the first isomorphism theorem, D_v defines an isomorphism between

$$M_v/M_v^2 \cong (F^n)^*$$

Since tangent space $\mathbb{T}_{v,V}$ is a vector space over F^n, then every linear function on F^n restricts to a linear function on $\mathbb{T}_{v,V}$. Composing D_v with this restriction map, say res, gives a linear transformation

$$D^v := M_v \overset{D_v}{\to} (F^n)^* \overset{\text{res}}{\to} (\mathbb{T}_{v,V})^*$$

which is surjective since so are D_v and res. Since $\mathcal{I}(v)^2 = M_v^2 + \mathcal{I}(V)$, we have

$$\mathcal{I}(v)/\mathcal{I}(v)^2 \cong M_v/(M_v^2 + \mathcal{I}(V))$$

which follows, by the Localization of Rings-at-Prime Ideals-Properties Prop.5, that $M_{v,V}/M_{v,V}^2 \cong \mathcal{I}(v)/\mathcal{I}(v)^2$. Therefore, it suffices to show that $\ker D^v = M_v^2 + \mathcal{I}(V)$, since then

$$M_{v,V}/M_{v,V}^2 \cong M_v/(M_v^2 + \mathcal{I}(V)) = M_v/\ker D^v \cong (\mathbb{T}_{v,V})^*$$

$f \in \ker D^v \Leftrightarrow D_v(f) = 0 \in \mathbb{T}_{v,V} \Leftrightarrow$ the linear term of the Taylor polynomial of expanded about v lies in $\mathcal{I}(\mathbb{T}_{v,V})$. Since the linear terms of the functions in $\mathcal{I}(V)$ generate the ideal $\mathcal{I}(\mathbb{T}_{v,V})$, then

$f \in \ker D^v \Leftrightarrow f - g$ has zero linear term for some $g \in \mathcal{I}(V) \Leftrightarrow f \in \mathcal{I}(V) + M_v^2$. □

Definition 2.9.28. Singularity

- We say V is *nonsingular* at the point $v \in V$ (or v is a *nonsingular point* of V) if $\dim_F \mathbb{T}_{v,V} = \dim V :=$ the transcendence degree of $F(V)$ over $F =$ the greatest number of {algebraic independent functions}, or equivalently, by the Independence of Embedding for Tangent Spaces Proposition, $\dim_F(M_{v,V}/M_{v,V}^2) = \dim V$. Otherwise, the point v is called a *singular point*.
- V is *nonsingular* or *smooth* if it is nonsingular at every point.

2.10 Prime Spectra of Rings

2.10.1 Prime Spectra of Rings

Intuition: Geometrically, if there is a morphism $\varphi : V \to W$ between two geometric spaces, then algebraically it induces a reversed ring homomorphism $\widetilde{\varphi} : F[W] \to F[V]$. But when we try to use the ring homomorphism to pullback a maximal ideal: if M is a maximal ideal of R, then its preimage $\widetilde{\varphi}^{-1}(M)$ is *not* necessarily a maximal ideal in general. This means that: if the points in the space are only defined using maximal ideals, then the ring homomorphism can *not* even induce a continuous map between the spaces, which severs the correspondence between geometry and algebra.
Redemption: Prime ideals step up! Although the pullback of a maximal ideal is *not* necessarily a maximal ideal, however:
the preimage of a prime ideal under a ring homomorphism can *not* is always a prime ideal.

Definition 2.10.1. Prime Spectra of Rings
Let R be a ring. The *spectrum* or *prime spectrum* of R, denoted as Spec R is the set of all prime ideals of R. Furthermore, the set of all maximal ideals of R, denoted as mSpec R, is called the *maximal spectrum* of R.

Example:

- If R is a field, then Spec $R =$ mSpec $R = \{(0)\}$.
- $R = \mathbb{Z}$. Spec $\mathbb{Z}$ consists of the prime ideal (0) and the prime ideals (p) where $p > 0$ is a prime; mSpec $R =$ (Spec $R) \setminus \{(0)\}$.
- $R = \mathbb{Z}[x]$. The elements of Spec $\mathbb{Z}[x]$ are the followings:
  1. (0).
  2. (p) where p is a prime in $\mathbb{Z}$.
  3. (f) where $f \neq 1$ is a polynomial of content 1, i.e.: the g.c.d. of its coefficients is 1, that is irreducible in $\mathbb{Q}[x]$.
  4. (p, g) where p is a prime in $\mathbb{Z}$ and g is a monic polynomial that is irreducible $\mod p$.

The elements of mSpec $\mathbb{Z}[x]$ are the primes in 4 above.

Definition 2.10.2. Values of Ring Elements
If $f \in R$, then the *value of f at the point $P \in$ Spec R* is the element $f(P) := f + P \in R/P$.

Remark:

- The values of f at different points P in general lie in different IDs (P is prime in $R \Leftrightarrow R/P$ is an ID).
- In general, $f \in R$ is *not* uniquely determined by its values, while f is determined up to an element in the nilradical of R:
  two elements $f, g \in R$ have the same values at all elements P in Spec $R \Leftrightarrow f+P = g+P, \forall P \in$ Spec $R \Leftrightarrow f - g \in R, \forall P \in$ Spec $R \Leftrightarrow f - g \in \bigcap_{P \in \text{Spec } R} P = \text{Nil } R$, where the last equal sign is by the Radicals-Prime Ideals Prop.
  In particular, an element in an affine F-algebra is uniquely determined by its values.

Definition 2.10.3. Maps $\mathcal{Z}$ and $\mathcal{I}$ between Rings and Spectra

- For any subset A of R, define the collection of prime ideals containing A by
$$\mathcal{Z}(A) := \{P \in \text{Spec } R \mid A \subseteq P\} \subseteq \text{Spec } R$$
- For any subset Y of Spec R, define the intersection of the prime ideals in Y by
$$\mathcal{I}(Y) := \bigcap_{P \in Y} P$$

Remark:

- It is immediate that $\mathcal{Z}(A) = \mathcal{Z}(I)$, where $I = (A)$ denotes the ideal generated by A.
- $P \in \mathcal{Z}(I) \Leftrightarrow I \subseteq P \Leftrightarrow f \in P, \forall f \in I \Leftrightarrow f(P) := f + P = 0_{R/P}, \forall f \in I$.
  Hence, $\mathcal{Z}(I)$ consists of the points in Spec R at which all the elements (which could be viewed as functions on Spec R) in I have the value 0.

Proposition 2.10.4. *$\mathcal{Z}$ and $\mathcal{I}$-Spectrum-Properties Proposition*
Let R be a ring. The maps $\mathcal{Z}$ and $\mathcal{I}$ between R and Spec R satisy the followings:

1. *For any ideals I of R, $\mathcal{Z}(I) = \mathcal{Z}(\text{rad } I) = \mathcal{Z}(\mathcal{I}(I))$ and $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$.*
2. *For any ideals I, J of R, $\mathcal{Z}(I \cap J) = \mathcal{Z}(IJ) = \mathcal{Z}(I) \cup \mathcal{Z}(J)$.*
3. *If $\{I_j\}$ is an arbitrary collection of ideals of R, then $\mathcal{Z}(\bigcup_j I_j) = \bigcap_j \mathcal{Z}(I_j)$.*

Proof. 1. If $I \subseteq P \in$ Spec R, then it's clearly that rad $I \subseteq P$, which proves that $\mathcal{Z}(I) = \mathcal{Z}(\text{rad } I)$. By the Radicals-Prime Ideals Prop, rad $I = \bigcap_{I \subseteq P \in \text{Spec } R} P$. Hence, by the def of $\mathcal{I}$
$$\mathcal{I}(\mathcal{Z}(I)) = \bigcap_{P \in \mathcal{Z}(I)} P = \bigcap_{I \subseteq P \in \text{Spec } R} P = \text{rad } I$$

2. It's immediate that $\mathcal{Z}(I \cap J) = \mathcal{Z}(I) \cup \mathcal{Z}(J)$. Suppose that $P \in \text{Spec } R$ contains IJ. $I \nsubseteq P$, then $i \notin P$ for some $i \in I$. Since $iJ \subseteq P$ and P is prime, then $J \subseteq P$.
3. Omitted.

$\square$

Remark:

- The $\mathcal{Z}$ and $\mathcal{I}$-Spectrum-Properties Prop.1 shows that every set $\mathcal{Z}(I)$ in Spec R occurs for some radical ideal and since $\mathcal{I}(\mathcal{Z}(I)) = \text{rad } I$, then this radical ideal is unique.
- The $\mathcal{Z}$ and $\mathcal{I}$-Spectrum-Properties Prop.2.3 shows that the collection

$$\{\mathcal{Z}(I) \mid I \lhd R\}$$

satisfies the axioms for the closed sets of a topology on Spec R.

Definition 2.10.5. Zariski Topology on Spectrum
The topology on Spec R defined by the closed sets $\mathcal{Z}(I)$ for the ideals I of R is called the *Zariski topology on Spec R*.

Remark: By def, the closure in the Zariski topology on Spec R of the singleton set $\{P\}$ in Spec R consists of all the prime ideals of R that contain P. In particular, a point P in Spec R is closed in Zariski topology on Spec $R \Leftrightarrow$ the prime ideal P of R is *not* contained in any other prime ideal of $P \Leftrightarrow P$ is a maximal ideal of R. (So tthe Zariski topology on Spec R is *not* generally Hausdorff.) These points are given a name:

Definition 2.10.6. Closed Points in Spectrum
The maximal ideals of R are call the *closed points in Spec R*.

Remark: the points in Spec R that are closed in the Zariski topology on Spec R are precisely the points in mSpec R.

Proposition 2.10.7. *$\mathcal{Z}$ and $\mathcal{I}$-Spectrum-Bijection Proposition*
The maps $\mathcal{Z}$ and $\mathcal{I}$ define inverse bijections

$$\{\textit{Zariski closed subsets of Spec } R\} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} \{\textit{radical ideals of } R\}$$

Under this correspondence, the closed points in Spec R correspond to the maximal ideals in R, and the irreducible subsets of Spec R correspond to the prime ideals in R.

Proof. The bijective correspondence is clear and the "closed points" part is proved by the above remark. As for the final statement, recall that a closed subset of a topological space is irreducible if it is not the union of two proper closed subsets, or equivalently, if every nonempty open set is dense. Arguments similar to those used to prove the Affine Algebraic Sets-Decomposition Prop. show that the closed subset $Y = \mathcal{Z}(I)$ in Spec R is irreducible $\Leftrightarrow \mathcal{I}(Y) = \text{rad } I$ is prime. $\square$

Example:

- If $X = \text{Spec } \mathbb{Z}$, then X is irreducible and the nonzero primes give closed points in X. The point (0) is not a closed point, in fact the closure of (0) is all of X, i.e.: (0) is dense in Spec $\mathbb{Z}$. For this reason, the element (0) is called a *generic point* in Spec $\mathbb{Z}$.
  Since every ideal of $\mathbb{Z}$ is principal, then the Zariski closed sets in Spec $\mathbb{Z}$ are $\emptyset$, Spec $\mathbb{Z}$ and any finite set of nonzero prime ideals in $\mathbb{Z}$.
- Suppose $X = \text{Spec } \mathbb{Z}[x]$. For each integer prime p, the Zariski closure of the element $(p) \in X$ consists if the maximal ideal (p, g) of type 4. Likewise for each $\mathbb{Q}$-irreducible polynomial f of type 3, the Zariski closure of the element (f) is the collection of prime ideals of type 4 where g is some divisor of f in $\mathbb{Z}/p\mathbb{Z}[x]$.

Proposition 2.10.8. *Ring Homomorphism-Induces-Continuous Maps Proposition*
Every homomorphism $\varphi : R \to S$ induces a map $\varphi^* :$ Spec $S \to$ Spec R that is continuous with respect to the Zariski topologies on Spec R and Spec S.

Proof. Note that if P is a prime ideal in S, then $\varphi^{-1}(P)$ is a prime ideal in R, which gives the well-defined-ness of φ^*, i.e.: $\varphi^*(P) := \varphi^{-1}(P)$. If $\mathcal{Z}(I) \subseteq \text{Spec } R$ is a Zariski closed subset of Spec R, then $(\varphi)^{-1}(\mathcal{Z}(I)) = \mathcal{Z}(\varphi(I)S)$, which is a Zariski closed subset of Spec S. Then we're done. □

2.10.2 Principal Open Sets

Intuition: The space Spec R together with its Zariski topology gives a geometric generalization for arbitrary rings of points in a variety V. We now consider the question of generalizing the ring of rational functions on V.
Recall that: each element $\alpha \in F(V)$ can in general be written as a/f where $a, f \in F[V]$, in many different ways; $\forall v \in V$, this representative, a/f, defines a local ring $\mathcal{O}_{v,V}$, open in V. So $\forall 0 \neq f \in F[V]$, we can also define an open subset of V by

$$V_f := \{w \in V \mid f(w) \neq 0\}$$

If we denote by U the set of all points which α is regular, then $\{V_f \mid f \in F[V] \setminus \{0\}\}$ is an open cover of U.

Definition 2.10.9. Principal Open Sets in Spectrum
For any $f \in R$,let X_f denote the collection of prime ideals in $X = \text{Spec } R$ that do not contain f. Equivalently, X_f is the set of points in Spec R at which the value of $f \in R$ is nonzero, i.e.:

$$X_f := \{P \in \text{Spec } R \mid f(P) \neq 0\}$$

The set X_f is called a *principal* (or *basic*) *open set* in $X = \text{Spec } R$.

Remark: It's easy to see that $X_f = \text{Spec } R \setminus \mathbb{Z}(f)$ and hence X_f is indeed an open set in Spec R as the name implies.

Proposition 2.10.10. *Principal Open Sets-Properties Proposition*
Let $f \in R$ and let X_f be the corresponding principal open set in $X = Spec\ R$.

1. *$X_f = X \Leftrightarrow f \in U(R)$ is a unit in R; $X_f = \emptyset \Leftrightarrow f$ is nilpotent.*
2. *$X_f \cap X_g = X_{fg}$.*
3. *$X_f \subseteq X_{g_1} \cup \cdots \cup X_{g_n} \Leftrightarrow f \in rad\ (g_1, \cdots, g_n)$. In particular, $X_f = X_g \Leftrightarrow rad\ (f) = rad\ (g)$.*
4. *the principal open sets form a basis for the Zariski topology on Spec R, i.e.: every Zariski open set in X is the union of some collections principal open sets X_f.*
5. *The natural map from R to R_f induces a homeomorphism from Spec R_f to X_f, where R_f is the localization of R at f.*
6. *The spectrum of any ring is quasicompact (i.e.: every open cover has a finite subcover). In particular, X_f is quasicompact.*
7. *If $\varphi : R \to S$ is any homomorphism, then under the induced map $\varphi^* : Y = Spec\ S \to X = Spec\ R$, the full preimage of the principal open set X_f in X is the principal open set $Y_{\varphi(f)}$ in Y.*

Proof. 1. For the first statement,

- $(\Rightarrow)$: Suppose on the contrary that $f \notin U(R)$, so the ideal (f) generated by f is proper in R. Consider $S := \{1, f, f^2, \cdots\}$. By the def of principal open sets, $f \neq 0$ and hence $0 \notin S$. then consider the set

$$\mathcal{S} := \{I \triangleleft R \mid I \cap S = \emptyset\}$$

Since $0 \notin S$, then $(0) \in \mathcal{S}$, $\mathcal{S} \neq \emptyset$ and hence we can apply Zorn's Lemma to get a maximal element P in $\mathcal{S}$.
Claim: P is a prime ideal in R.

Proof. of claim: Suppose on the contrary that there exist $a, b \in R$, s.t.: $ab \in P$ but $a \notin P, b \notin P$. Since $a \notin P$, then $P \subsetneq P + (a)$. By the maximality of P in $\mathcal{S}$, $P + (a) \notin \mathcal{S}$, i.e.: $P + (a) \cap S = \emptyset \Leftrightarrow \exists n \in \mathbb{N}_+$, s.t.: $f^n \in P + (a)$. Similarly, $\exists m \in \mathbb{N}_+$, s.t.: $f^m \in P + (b)$. Hence,

$$f^{m+n} \in (P + (a))(P + (b)) \subseteq P + (ab) \overset{ab \in P}{=} P$$

contradicting $P \in \mathcal{S}$. □

By the def of $\mathcal{S}$, since $f \in S$, then $f \notin P$, contradicting our assumption.

- ($\Leftarrow$): $f \in U(R) \Rightarrow fg = 1$ for some $g \in R$. By the def of prime ideals, $fg = 1 \notin P, \forall P \in$ Spec R, which follows by the def of ideals that $f \notin P, \forall P \in$ Spec $R \Leftrightarrow X_f = X$

For the second statement,

$$X_f = \emptyset \Leftrightarrow f \in P, \forall P \in \text{Spec } R \Leftrightarrow f \in \bigcap_{P \in \text{Spec } R} P = \text{Nil } R.$$

2.

$$X_f \cap X_g := \{P \in \text{Spec } R \mid f(P) \neq 0, g(P) \neq 0\} = \{P \in \text{Spec } R \mid fg(P) \neq 0\} = X_{fg}$$

3. Note that $X_{g_1} \cup \cdots \cup X_{g_n}$ consists of the primes P not containing at least one of $g_1, \cdots, g_n$. Hence, $X_{g_1} \cup \cdots \cup X_{g_n} = (\text{Spec } R) \backslash \mathcal{Z}(g_1, \cdots, g_n)$. If $(g_1, \cdots, g_n) = X$, then $X_{g_1} \cup \cdots \cup X_{g_n} = X$ and there's nothing to prove. Otherwise, $X_f \subseteq X_{g_1} \cup \cdots \cup X_{g_n} \Leftrightarrow$ every prime P with $f \notin P$ also satisfies $P \notin \mathcal{Z}(g_1, \cdots, g_n) \Leftrightarrow$ every prime P containing $(g_1, \cdots, g_n)$ also contains $f \Leftrightarrow f \in \bigcap_{(g_1, \cdots, g_n) \subseteq P \subseteq \text{Spec } R} P = \text{rad } (g_1, \cdots, g_n)$ where the last equal sign is by the Radicals-Prime Ideals Prop.
4. Note that every Zariski open subset of X can be written as

$$U = X \setminus \mathcal{Z}(I) := \{P \in \text{Spec } R \mid I \not\subseteq P$$

for some ideal I of R. Hence, $U = \bigcup_{f \in I} X_f$, which proves 4.

5. By the Extensions and Contractions-Localization-Properties Prop.3, the natural ring homomorphism from R to R_f establishes a bijection between the prime ideals in R_f and the prime ideals in R not containing (f). Hence, the corresponding map from Spec R_f to Spec R is continuous and bijective. By the Extensions and Contractions-Localization-Properties Prop.1, every ideal of R_f is the extension of some ideal of R. Hence, inverse map is also continuous.
6. By 4, every open set in X is the union of principal open sets, so it suffices to show that: if X is considered by principal open sets $\{X_{g_i}\}_{i \in \mathcal{J}}$, then X is a finite union of some of the X_{g_i}. Consider an ideal I generated by $\{g_i \mid i \in \mathcal{J}\}$. If I is proper in R, then $I \subseteq M$ for some maximal ideal M of R, which then is prime, but in this case, $M \in X = \text{Spec } R$ could *not* be contained by any of X_{g_i}, contradicting the assumption. Hence, $I = R$, so $1 \in I \Leftrightarrow 1 = a_1 g_{i_1} + \cdots + a_n g_{i_n}$ for some $a_j \in R, i_j \in \mathcal{J}$. If there is a point $P \in X \setminus (X_{g_1} \cup \cdots \cup X_{g_n})$, which proves the first part of 6 and the second part then by 5 follows.
7.

$$P \in (\varphi^*)^{-1}(X_f) \Leftrightarrow \varphi^{-1}(P) = \varphi^*(P) \in X_f \Leftrightarrow f \notin \varphi^{-1}(P) \Leftrightarrow \varphi(f) \notin P \Leftrightarrow P \in Y_{\varphi(f)}.$$

$\square$

2.10.3 Introduction to Sheaf Theory

Structure Sheaves

Intuition: We now define an analogue for $X = \text{Spec } R$ of the rational functions on a variety V: As we observed in our previous intuition for principal open sets, let U be some open subset of V on which $\alpha \in F(V)$ is a regular function. $\forall v \in U, \exists$ a representative a/f for α, s.t.: $f(v) \neq 0$. This representative is an element in the localization $\mathcal{O}_{v,V} = F[V]_{\mathcal{I}(v)}$. In this way, the regular function α on U can be considered as a function from U to the disjoint union of these localizations, namely, $\alpha : v \in U \mapsto a/f \in F[V]_{\mathcal{I}(v)}$. Furthermore, the same representative can be used not only at v but on the Zariski neighborhood of v, so, "locally near v", α is given by a single quotient of elements from $F[V]$. Note that a/f is an element in the localization $F[V]_f$, which is contained in each of the localizations $F[V]_{\mathcal{I}(w)}$ for $w \in V_f$.

Motivation: Now we generalize this to Spec R by considering the collection of functions s from the Zariski open subset U of Spec R to the disjoint union of the localizations R_P for $P \in U$, s.t.: $s(P) \in R_P$ and that s is given locally by quotients of elements of R:

Definition 2.10.11. Let U be a Zariski open subset of Spec R. If $U = \emptyset$, then define $\mathcal{O}(U) = 0$. Otherwise define $\mathcal{O}(U)$ by the set of functions $s : U \to \bigsqcup_{Q \in U} R_Q$, where R_Q denotes the localization for Q, with the following two properties:

1. $s(Q) \in R_Q, \forall Q \in U$, and
2. $\forall P \in U, \exists$ am open neighborhood $X_f \in U$ of P in U and an element a/f^n in the localization R_f defining s on X_f, i.e.: $s(Q) = a/f^n \in R_Q, \forall Q \in X_f$.

Remark:

- $\forall s, t \in \mathcal{O}(U)$, s.t.: $s + t$ and st are elements in $\mathcal{O}(U)$, so $\mathcal{O}(U)$ is a ring.
- $a \in R$ gives an element in $\mathcal{O}(U)$ defined by $s(Q) = a \in R_Q$, and in particular, $1 \in R$ gives an identity element for thhe ring $\mathcal{O}(U)$. If $U' \subseteq U$ is an open subset, then there is a natural restriction map from $\mathcal{O}(U)$ to $\mathcal{O}(U')$ which is easy to check to be a homomorphism.

Definition 2.10.12. Structure Sheaves
Let $X = \text{Spec } R$.

- The collection of rings $\mathcal{O}(U)$ for the Zariski open subset U of X together with the restriction maps $\mathcal{O}(U) \to \mathcal{O}(U')$ for Zariski open $U' \subseteq U$ is called the *structure sheaf* on X, denoted as $\mathcal{O}_X$ (or simply $\mathcal{O}$).
- The elements s of $\mathcal{O}(U)$ are called the *sections of $\mathcal{O}$ over U* and the elements of $\mathcal{O}(X)$ are called the *global sections of $\mathcal{O}$*.

Proposition 2.10.13. *Generalization-of-the Local-Global Principle-for-Resulting Proposition*
Let $X = \text{Spec } R$ and let $\mathcal{O} = \mathcal{O}_X$ be its structure sheaf. The global sections of $\mathcal{O}$ are the elements

of R, i.e.: $\mathcal{O}(X) \cong R$. More generally, if X_f is a principal open set in X for some $f \in R$, then $\mathcal{O}(X_f)$ is isomorphic to the localization R_f.

Proof. It suffices to prove the general part, namely, $R_f \cong \mathcal{O}(X_f)$ and the first statement is the particular case when $f = 1$.

- **Step 1** *Constructing map:* $\forall a/f^n \in R_f$, define a function $s : X_f \to \bigsqcup_{Q \in X_f} R_Q$ by $Q \mapsto a/f^n \in R_Q, \forall Q \in X_f$, which is an element in $\mathcal{O}(X_f)$ and is well-defined since: if if $Q \in X_f$, then $f \notin Q$, so f^n is invertible. Then we construct a map $\psi : R_f \to \mathcal{O}(X_f)$ given by $a/f^n \mapsto s$.
- **Step 2** *Injectivity of ψ:* Suppose that $a/f^n = b/f^m \in R_Q$ for every $Q \in X_f$, i.e.: $g(af^m - bf^n) = 0 \in R$ for some $g \notin Q$. Consider an ideal I in R consisting of all the elements $r \in R$ with $r(af^m - bf^n) = 0$. Then g is immediately an element of I. Hence, $I \nsubseteq Q, \forall Q \in X_f \Leftrightarrow$ if P is a prime ideal containing I, then $P \notin X_f \Leftrightarrow$ if P is a prime ideal containing I, then $f \notin P$. Hence, by the Radicals-Prime Ideals-Prop, $f \in \text{rad } I$, i.e.: $f^N \in I$ for some $N \in \mathbb{N}_+$, so $f^N(af^m - bf^n) = 0 \in R$. Therefore, $a/f6n = b/f^m$ in R_f, which gives the injectivity of ψ.
- **Step 3** *Surjectivity of ψ:* $\forall s \in \mathcal{O}(X_f)$, by the def, X_f can be covered by principal open subsets X_{g_1} on which $s(Q) = a_i/g_i^{n_i} \in R_Q$ for every $Q \in X_{g_i}$. By the Principal Open Sets-Properties Prop.6, we may take a finite number of g_i and then by taking different a_i, we may assume all the n_i are equal, say n ($a_i/g_i^{n_i} = (a_i g_i^{n-n_i})/g^n$ if n is the maximum of n_i). Since $s(Q) = a_i/g_i^n = a_j/g_j^n$ in $R_Q, \forall Q \in X_{g_1} \cap X_{g_j} = X_{g_i g_j}$, then the injectivity of ψ gives that $a_i/g_i^n = a_j/g_j^n \in R_{g_i g_j}$. Hence, $g_i^N g_j^N (a_i g_j^n - a_j g_i^n) = 0$, i.e.:

  $$a_i g_i^N g_j^{n+N} = a_j g_i^{n+N} g_j^N \tag{*}$$

  for some $N \in \mathbb{N}_+$. By the Principal Open Sets-Properties Prop.2, we have $X_{g_i} = X_{g_i^k}$ for any $k \in \mathbb{N}_+$, so

  $$X = \bigcup_i X_{g_i} = \bigcup_i X_{g_i^{n+N}}$$

  Hence, by the Principal Open Sets-Properties Prop.3, $f \in \text{rad } (\{g_i^{n+N}\})$, say $f^M = \sum_i b_i g_i^{n+N}$ for some $M \in \mathbb{N}_+, b_i \in R$. Define $a := \sum_i b_i a_i g_i^N$. Then for any fixed index j,

  $$g_j^N a_j f^M = g_j^N a_j \sum_i b_i g_i^{n+N} = \sum_i b_i a_j g_i^{n+N} g_j^N \overset{(*)}{=} \sum_i b_i a_i g_i^N g_j^{n+N} = g_j^{n+N} a$$

  Therefore, $a/f^M = a_j/g_j^n \in R_{g_j}$, so the element in $\mathcal{O}(X_f)$ defined by a/f^M in R_f agrees with s on every X_{g_j} and hence on all of X_f since X_{g_j} cover X_f.

□

Stalk

Intuition: In the case of affine varieties V, the local ring $\mathcal{O}_{v,V}$ at the point $v \in V$ is the collection of all the rational functions in $F(V)$ that are defined at v. Put another way, $\mathcal{O}_{v,V}$ = the union of

the rings of regular functions on $U \ni v$.
However, in the more general case of $X = \text{Spec } R$, for the different open sets U_1 and U_2, the rings $\mathcal{O}(U_1)$ and $\mathcal{O}(U_2)$ are *not* generally belonging to a common same "big ring". Hence, we can no longer directly cram all these rings into one function field to find the union as we did in the case of affine varieties. So we need to introduce an algebraic tool to forcibly determine: When are two functions defined on different open sets considered the same function at point P?

Motivation: Consider the collection of pairs (s, U) with U an open set of X containing P and $s \in \mathcal{O}(U)$. We identify two pairs (s, U) and (s', U') if there is an open set $U'' \subseteq U \cap U'$ containing P on which s and s' restrict to the same element of $\mathcal{O}(U'')$, i.e.: we define an equivalence relation by

$$(s, U) \sim (s', U') \Leftrightarrow \exists \text{ open } U'' \subseteq U \cap U', \text{ s.t.: } s|_{U''} = s'|_{U''} \in \mathcal{O}(U'').$$

(In the situation of affine varieties, this says that two functions defined in Zariski neighborhood of the point v define the same regular function at v if they agree in some common neighborhood of v.)

Definition 2.10.14. Stalk
Let $P \in X = \text{Spec } R$. The collection of equivalence classes of pairs (s, U) defines the *direct limit* of the rings $\mathcal{O}(U)$, denoted as $\varinjlim_{U \ni P} \mathcal{O}(U)$, i.e.:

$$\varinjlim_{U \ni P} \mathcal{O}(U) := \{[(c, U)] \mid U \text{ is an open subset of } X \text{ containing } P, s \in \mathcal{O}(U)\}$$

And the direct limit is called the *stalk* of the structure sheaf at P, denoted as $\mathcal{O}_P$, i.e.: $\mathcal{O}_P := \varinjlim_{U \ni P} \mathcal{O}(U)$.

Proposition 2.10.15. *Stalk-Localization-Isomorphism Proposition*
Let $X = \text{Spec } R$ and let $\mathcal{O} = \mathcal{O}_X$ be its structure sheaf. The stalk of $\mathcal{O}$ at the point $P \in X$ is isomorphic to the localization R_P of R at P:

$$\mathcal{O}_P \cong R_P$$

In particular, the stalk $\mathcal{O}_P$ is a local ring.

Proof.

- **Step 1** *Constructing map:* $\forall [(x, U)] \in \mathcal{O}_P, s(P) \in R_P$, so we can define a map $\varphi : \mathcal{O}_P \to R_P$ by $[(s, U)] \mapsto s(P)$, where the well-defined-ness of φ and φ a ring homomorphism are given by the def of equivalence relation.
- **Step 2** *Surjectivity of φ:* If $a, f \in R$ with $f \notin P$, then the map $s(Q) = a/f \in R_Q$ defines an element s in $\mathcal{O}(X_f)$. Then $[(s, X_f)] \in \mathcal{O}_P$ is mapped to $a/f \in R_P$ by φ.
- **Step 3** *Injectivity of φ:* Suppose that $(s, U) \sim (s', U')$ satisfy $s(P) = s'(P) \in R_P$. By the def of $\mathcal{O}(U)$, $s = a/g^n$ on X_g for some $g \notin P$ and $s' = b/(g')^m$ on $X_{g'}$ for some $g' \notin P$. Since $a/g^n = b/(g')^m \in R_P$, then there is $h \notin P$ with $h(a(g')^m - bg^n) = 0 \in R$. If $Q \in X_{gg'h} =$

$X_g \cap X_{g'} \cap X_h$, then $a/g^n = b/(g')^m$ in R_Q, so that s and s' agree when restricted to $X_{gg'h}$. By the def of direct limit, (s, U) and (s', U') define the same element in $\mathcal{O}_P$.

□

Remark: This prop. gives us two geometric "scenes":

- *Algebraic Tangent Space and Smoothness.*
  The algebraically defined localization R_P for $P \in \text{Spec } R$ plays the role of the local ring $\mathcal{O}_{v,V}$ of regular functions at v for the affine variety V. If M_P denotes the maximal ideal PR_P in R_P and $F(P) := R_P/M_P$ denotes the corresponding quotient field (which, by the Localization of Rings-at-Prime Ideals-Properties Prop.1, is also the fraction field of R/P), then the *tangent space* at P is defined as the $F(P)$-vector space dual of M_P/M_P^2. This is an algebraic def that generalizes the def of the tangent space $\mathbb{T}_{v,V}$ to a variety V at a point v (by the Independence of Embedding for Tangent Spaces Prop.). This can now be used to define that it means for a point in Spec R to be nonsingular: the point $P \in \text{Spec } R$ is *nonsingular* or *smooth* if the local ring R_P is what is called a "regular local ring".

- *Visualizing a Sheaf: "A Sheaf of Wheat".*
  If we view each point $P \in \text{Spec } R$ as having the local ring R_P above it, then above the open set U in $X = \text{Spec } R$ is a "sheaf" (in the sense of "bundle") of these "stalks" (in the sense of a "stalk of wheat"), which helps explain some of the terminology. A section s in the structure sheaf $\mathcal{O}(U)$ is a map from U to this bundle of stalks. The image of U under such a section s is indicated by the shade region in the following figure.

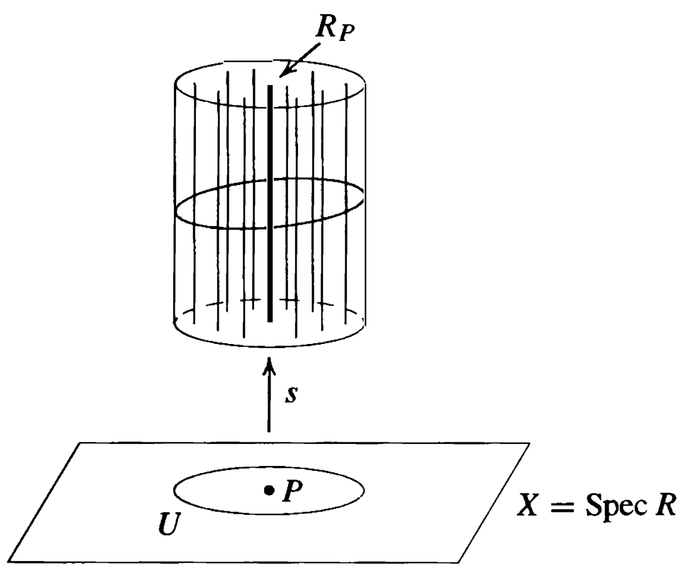

Figure 2.1: Geometric intuition of stalks and sections

Affine Scheme

Definition 2.10.16. Affine Scheme
Let R be a ring. The pair (Spec $R, \mathcal{O}_{\text{Spec } R}$), consisting of the space Spec R with the Zariski topology together with the structure sheaf $\mathcal{O}_{\text{Spec } R}$, is called an *affine schemes*.

Remark: The notion of affine schemes gives a completely algebraic generalization of the geometry of affine algebraic sets valid for arbitrary rings.
Example: Now we give some examples of the two notions of structure sheaves and of stalks and hence of the concept of affine scheme:

- If F is a field, then $X = \text{Spec } F = \{(0)\}$. In this case, there are only two open sets X and $\emptyset$, both of which are principal open sets: $X = X_1$ and $\emptyset = X_0$. The global sections are $\mathcal{O}(X) = F$. There is only stalk: $\mathcal{O}_{(0)} = F_0 = F$.
- If $R = \mathbb{Z}$, then R is a PID, so every open set in $X = \text{Spec } \mathbb{Z}$ is principal open:

$$X_n = \{(p) \mid p \nmid n\} \quad \text{and}$$
$$\mathcal{O}(X_n) = \mathbb{Z}_n = \mathbb{Z}[1/n] = \{a/b \in \mathbb{Q} \mid \text{if the prime } p \mid b \text{ then } p \mid n\}.$$

  For nonzero p, the stalk at (p) is the local ring $\mathbb{Z}_{(p)}$ and the stalk at (0) is $\mathbb{Q}$. All the restriction maps as well as the maps from sections to stalks are the natural inclusions.
- For a general ID R with quotient field F, the stalks and sections are

$$\mathcal{O}(U) = \{a/b \in F \mid b \notin P, \forall P \in U\}$$
$$\mathcal{O}_P = R_P = \{a/b \in F \mid b \notin P\}$$

  where the stalk at (0) is F, i.e.: $\mathcal{O}_{(0)} = F$. Again, the restriction maps and the maps to the stalks are all inclusions.
- For the local ring $R = \mathbb{Z}_{(2)} = \{a/b \in \mathbb{Q} \mid b \text{ odd}\}$, we have Spec $R = \{(0), (2)\}$ with (2) the only closed point and $\{(0)\} = X_2$ a principal open set. The sections $\mathcal{O}(\{0\})$ are $R_2 = \mathbb{Q}$ and the stalks are $\mathcal{O}_{(0)} = R_{(0)} = \mathbb{Q}$ and $\mathcal{O}_{(2)} = R_{(2)} = R$.

Intuition: Now we consider the relationship of the affine schemes corresponding to rings R and S with respect to a ring homomorphism from R to S:
Suppose that $\varphi : R \to S$ is a ring homomorphism. Then by the Principal Open Sets-Properties Prop.7, there is an induced continuous map $\varphi^* : Y = \text{Spec } S \to X = \text{Spec } R$ and under this map the full preimage of the principal open set X_g for $g \in R$ is the principal open set $Y_{\varphi(g)}$.
Also, φ induces a map on corresponding sections, as follows: Let $Q' \in Y$ be any element in Spec S and let $Q = \varphi^*(Q') = \varphi^{-1}(Q') \in X$ be the corresponding element in Spec R. If U is a Zariski open set in Y containing Q'. Note that φ induces a natural ring homomorphism, say φ_Q, from the

localization R_Q to the localization $S_{Q'}$ defined by

$$\varphi_Q(a/f) := \varphi(a)/\varphi(f) \in S_{Q'}$$

for $f \notin Q$. Let $s \in \mathcal{O}_X(U)$ be a section of the structure sheaf of X given locally in the neighborhood X_g of $P \in X$ by a/g^n. It is easy to check that the composite

$$s' : U' \xrightarrow{\varphi^*} U \xrightarrow{s} \coprod_{Q \in U} R_Q \xrightarrow{\varphi} \coprod_{Q' \in U'} S_{Q'}$$

defines a map given locally in the neighborhood $Y_{\varphi(g)}$ by the element $\varphi(g)/\varphi(g)^n$, so that $s' \in \mathcal{O}_Y(U')$ is a section of the structure sheaf of Y. It is then straightforward to check that the resulting map $\varphi^{\#} : \mathcal{O}_{X,P} \to \mathcal{O}_{Y,P'}$ for any point $P' \in \text{Spec } S$ and corresponding point $P = \varphi^*(P') \in \text{Spec } R$. Under the isomorphism in the Stalk-Localization-Isomorphism Prop, the homomorphism $\varphi^{\#} : R_P \cong \mathcal{O}_{X,P} \to S_{P'} \cong \mathcal{O}_{Y,P'}$ is just the natural ring homomorphism φ_P on the localizations induced by the homomorphism φ. In particular, the inverse image under $\varphi^{\#}$ of the maximal ideal in the local ring $\mathcal{O}_{Y,P'}$ is the maximal ideal in the local ring $\mathcal{O}_{X,P}$.

Definition 2.10.17. Morphisms of Affine Schemes
Let $(\text{Spec } R, \mathcal{O}_{\text{Spec } R})$ and $(\text{Spec } S, \mathcal{O}_{\text{Spec } S})$ be two affine schemes. A *morphism of affine schemes* from $(\text{Spec } R, \mathbb{O}_{\text{Spec } R})$ to $(\text{Spec } S, \mathcal{O}_{\text{Spec } S})$ is a pair $(\varphi^*, \varphi^{\#})$ satisfying the following three conditions:

1. $\varphi^* : \text{Spec } S \to \text{Spec } R$ is Zariski continuous.
2. There are ring homomorphisms $\varphi^{\#} : \mathcal{O}(U) \to \mathcal{O}((\varphi^*)^{-1}(U))$ for every Zariski open subset U in Spec R that commute with the restriction maps.
3. If $P' \in \text{Spec } S$ with corresponding point $P = \varphi^{\#}(P') \in \text{Spec } R$, then under the induced homomorphism on stalks $\varphi^{\#} : \mathcal{O}_{\text{Spec } R,P} \to \mathcal{O}_{\text{Spec } S,P'}$, the preimage of the maximal ideal of $\mathcal{O}_{\text{Spec } S,P'}$ is the maximal ideal of $\mathcal{O}_{\text{Spec } R,P}$.

Remark: A homomorphism $\psi : A \to B$ between the local rings with the property that the preimage of the maximal ideal of B is the maximal ideal of A is called a *local homomorphism* of local rings. The third condition in the definition is then the statement that the induced homomorphism on stalks is required to be a local homomorphism.

Theorem 2.10.18. *Equivalence-of-Categories-between-Affine Schemes-and0Rings Theorem*
Every ring homomorphism $\varphi : R \to S$ induces a morphism

$$(\varphi^*, \varphi^{\#}) : (Spec\ S, \mathcal{O}_{Spec\ S}) \to (Spec\ R, \mathcal{O}_{Spec\ R})$$

of affine schemes. Conversely, every morphism of affine schemes arises from such a ring homomorphism φ.

Theorem 2.10.19 (Equivalence of Rings and Affine Schemes, Theorem 59)**.** *Every ring homomorphism $\varphi : R \to S$ induces a morphism*

$$(\varphi^*, \varphi^\#) : (Spec\ S, \mathcal{O}_{Spec\ S}) \to (Spec\ R, \mathcal{O}_{Spec\ R})$$

of affine schemes. Conversely, every morphism of affine schemes arises from such a ring homomorphism φ.

Proof. The proof consists of two directions. The fact that a ring homomorphism induces a morphism of affine schemes was already shown in the preceding discussion. We focus on thoroughly proving the converse: extracting a ring homomorphism from a morphism of affine schemes and showing they are naturally equivalent.

Step 1: Extracting the global ring homomorphism $\varphi : R \to S$.
Suppose we are given a morphism of affine schemes, denoted by $(f, f^\#) : (\text{Spec } S, \mathcal{O}_{\text{Spec } S}) \to (\text{Spec } R, \mathcal{O}_{\text{Spec } R})$. (Note: we use f for the continuous map to avoid confusion with the φ we are trying to construct).
By the definition of a morphism of ringed spaces, $f^\#$ is a morphism of sheaves from $\mathcal{O}_{\text{Spec } R}$ to the pushforward sheaf $f_*\mathcal{O}_{\text{Spec } S}$. Evaluating these sheaves on the global topological space $U = \text{Spec } R$, we get a ring homomorphism on the global sections:

$$f^\#(\text{Spec } R) : \mathcal{O}_{\text{Spec } R}(\text{Spec } R) \to (f_*\mathcal{O}_{\text{Spec } S})(\text{Spec } R).$$

By the definition of the pushforward sheaf, $(f_*\mathcal{O}_{\text{Spec } S})(\text{Spec } R) = \mathcal{O}_{\text{Spec } S}(f^{-1}(\text{Spec } R)) = \mathcal{O}_{\text{Spec } S}(\text{Spec } S)$.
By Proposition 57 (the structure sheaf evaluated on the whole spectrum recovers the original ring), we have natural ring isomorphisms:

$$\mathcal{O}_{\text{Spec } R}(\text{Spec } R) \cong R \quad \text{and} \quad \mathcal{O}_{\text{Spec } S}(\text{Spec } S) \cong S.$$

Composing $f^\#(\text{Spec } R)$ with these isomorphisms gives us a well-defined ring homomorphism between the underlying rings:

$$\varphi : R \to S.$$

Step 2: Checking the local behavior on stalks (The Commutative Diagram).
Pick any prime ideal $P' \in \text{Spec } S$. Let its image under the continuous map be $P = f(P') \in \text{Spec } R$. Because $(f, f^\#)$ is a morphism of *locally ringed spaces* (which affine schemes are), $f^\#$ induces a **local homomorphism** on the stalks at P'. By Proposition 58, the stalk of $\mathcal{O}_{\text{Spec } R}$ at P is the localization R_P, and the stalk of $\mathcal{O}_{\text{Spec } S}$ at P' is $S_{P'}$. Thus, we have a local ring homomorphism:

$$\varphi^\#_{P'} : R_P \to S_{P'}.$$

Because the sheaf morphism $f^\#$ is compatible with the restriction maps (going from global sections to local stalks), the following diagram must commute:

$$\begin{tikzcd} R \arrow[r, "\varphi"] \arrow[d, "\text{nat}"'] & S \arrow[d, "\text{nat}"] \\ R_P \arrow[r, "\varphi^{\#}_{P'}"'] & S_{P'} \end{tikzcd}$$

where the two vertical maps are the natural localization homomorphisms (e.g., $r \mapsto r/1$).

Step 3: Proving the topological map f is exactly φ^*.
We need to show that the given continuous map $f(P')$ is exactly the pullback of prime ideals under our constructed φ, i.e., $f(P') = \varphi^{-1}(P')$.
Recall that a local ring homomorphism (like $\varphi^{\#}_{P'}$) must map the unique maximal ideal of the domain into the unique maximal ideal of the codomain.

- The unique maximal ideal of the local ring R_P is PR_P.
- The unique maximal ideal of the local ring $S_{P'}$ is $P'S_{P'}$.

Therefore, taking the pre-image under $\varphi^{\#}_{P'}$, we have:

$$(\varphi^{\#}_{P'})^{-1}(P'S_{P'}) = PR_P.$$

Now, let's trace this equality back up through the commutative diagram to the global rings R and S. If we pull back $P'S_{P'}$ along the natural map $S \to S_{P'}$, we get the original prime ideal P' in S. Pulling this further back to R via φ gives $\varphi^{-1}(P')$. On the other hand, going the other way around the diagram, pulling back PR_P along the natural map $R \to R_P$ recovers P (since P is prime). Because the diagram commutes, the results of pulling back through both paths must be identical. Hence:

$$\varphi^{-1}(P') = P.$$

Since we initially defined $P = f(P')$, this proves $f(P') = \varphi^{-1}(P')$. By definition, $\varphi^{-1}(P')$ is exactly $\varphi^*(P')$. So the continuous map f is identical to φ^*.

Step 4: Conclusion.
We have shown that $f = \varphi^*$. Because the structural sheaf morphism $f^{\#}$ on affine schemes is entirely determined by its action on the global sections (which we defined as φ) and its compatibility with localizations, it follows that $f^{\#}$ is exactly the induced sheaf morphism $\varphi^{\#}$.
Consequently, the arbitrary morphism $(f, f^{\#})$ we started with is precisely $(\varphi^*, \varphi^{\#})$, meaning every morphism of affine schemes is uniquely induced by a ring homomorphism. $\square$

Chapter 3

Homological Algebra

3.0 Introduction and Motivation

Intuition: As we saw in the Module Theory chapter, short exact sequences are fundamental tools for understanding the structure of modules. However, applying functors (like the Hom functor or the tensor product) to a short exact sequence often "breaks" the exactness at one end. Homological algebra provides the machinery to "repair" this loss of exactness by extending the sequence infinitely.

Let R be a ring with 1. In previous chapters, we saw that a short exact sequence of R-modules

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0 \tag{3.1}$$

gives rise to an exact sequence of abelian groups when applying the contravariant functor $\mathrm{Hom}_R(-, D)$ for any R-module D:

$$0 \longrightarrow \mathrm{Hom}_R(N, D) \xrightarrow{\varphi'} \mathrm{Hom}_R(M, D) \xrightarrow{\psi'} \mathrm{Hom}_R(L, D) \tag{3.2}$$

Remark *(The Extension Problem):* The crucial observation in sequence (3.2) is that the homomorphism ψ' is *in general not surjective*. This means we cannot simply append "$\longrightarrow 0$" to the right side of the sequence to make it a short exact sequence.

Equivalently, what does it mean for ψ' to not be surjective? It means that a homomorphism from L to D cannot always be "lifted" or "extended" to a homomorphism from M into D. We can visualize this geometric obstruction using a commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & L \arrow[r, "\psi"] \arrow[d, "f"'] & M \arrow[r, "\varphi"] \arrow[dl, dashed, "F"] & N \arrow[r] & 0 \\
 & D & & &
\end{tikzcd}$$

Given a homomorphism $f \in \mathrm{Hom}_R(L, D)$, does there always exist an extension $F \in \mathrm{Hom}_R(M, D)$ such that $f = F \circ \psi$? The lack of surjectivity of ψ' tells us the answer is *no* in general.

Solution provided by Homological Algebra: In this chapter, we introduce techniques of "homological algebra" which provide a natural method for extending these broken exact sequences. For the situation above, one obtains an infinite long exact sequence involving the *cohomology groups* $\mathrm{Ext}^n_R(-, D)$:

$$0 \to \mathrm{Hom}_R(N, D) \to \mathrm{Hom}_R(M, D) \to \mathrm{Hom}_R(L, D) \xrightarrow{\delta} \mathrm{Ext}^1_R(N, D) \to \dots \tag{3.3}$$

These Ext groups provide a precise mathematical measure of the set of homomorphisms from L into D that *cannot* be extended to M.

We will also consider analogous questions for the other two functors discussed previously:

- Taking homomorphisms *from* D into the terms of sequence (3.1) (covariant Hom functor).
- Tensoring the sequence (3.1) with D (which will introduce the Tor functors).

Roadmap for this Chapter: In the subsequent sections, we will concentrate on an important special case of this general homological construction: the *cohomology of finite groups.* We will make explicit computations in this case and indicate some applications to establish new results in group theory, serving as an explicit "example" illustrating the general abstract theory.

3.1 Basic Notions

Definition 3.1.1. Cohomology Modules
Let $\mathcal{C}$ be a sequence of module homomorphisms:

$$0 \to C^0 \xrightarrow{d_1} C^1 \xrightarrow{d_2} \cdots \xrightarrow{d_{n-1}} C^{n-1} \xrightarrow{d_n} C^n \xrightarrow{d_{n+1}} \cdots$$

- The sequence $\mathcal{C}$ is called a *cochain complex* if the composition of any two successive maps is zero: $d_{n+1} \circ d_n = 0, \forall n \in \mathbb{N} \cup \{0\}$, i.e.: im $d_n \subseteq$ ker $d_{n+1}, \forall n \in \mathbb{N} \cup \{0\}$.
- If $\mathcal{C}$ is a cochain complex, then its *n-th cohomology module*, denoted as $H^n(\mathcal{C})$, is the quotient module
$$H^n(\mathcal{C}) := \ker d_{n+1}/\mathrm{im}\ d_n$$

Remark: There is a completely analogous "dual" version in which the homomorphisms are between groups in decreasing order:

Definition 3.1.2. Homology Modules
Let $\mathcal{C}$ be a sequence of module homomorphisms:

$$\cdots \xrightarrow{d_{n+1}} C_n \xrightarrow{d_n} \cdots \xrightarrow{d_1} C_0 \to 0$$

- The sequence $\mathcal{C}$ is called a *chain complex* if the composition of any two successive homomorphisms is zero, i.e.: $d_n \circ d_{n+1} = 0, \forall n \in \mathbb{N} \cup \{0\}$.
- If $\mathcal{C}$ is a chain complex, then its *n-th homology module*, denoted as $H_n(\mathcal{C})$, is the quotient

module

$$\ker d_n / \mathrm{im}\ d_{n+1}$$

Remark:

- *Convention:* For chain complexes, the notation is often chosen so that the indices appear as subscripts and are decreasing, while for cochain complexes, the indices are superscripts and increasing.
- d_n, ker d_n and im d_n are often called *differentials*, *n-cycles* and *n-boundaries* of $\mathcal{C}$: Consider

  $$\cdots P(\Delta^{n+1}) \xrightarrow{\partial_{n+1}} P(\Delta_n) \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_1} P(\Delta^0) \to 0$$

  Then ∂_n is the behavior of choosing $(n-1)$-skeleton. As in differential form, we have $\partial^2 = 0$. In particular, $\mathbb{S}^1 \in \ker \partial_1$, which is why $\ker d_n$ are called n-cycles, and $\{\{1\},\{2\},\{3\},\{1,2\},\{1,3\},\{2,3\}\} = \partial_2\Delta^2 \in \mathrm{im}\ \partial_2$, which is that why im d_n are call n-boundaries.

Proposition 3.1.3. *Complex-Exact Proposition*
Let $\mathcal{C}$ be a cochain complex. Then $\mathcal{C}$ is an exact sequence iff all its cohomology modules are zero. Same for chain complex.

Remark:

- This prop. gives a perspective that the n-th cohomology (or homology) module measures the failure of the exactness of a cochain (respectively chain) complex at the n-th stage.
- From now on, we shall mainly discuss on the situation regarding cochain complex.

Definition 3.1.4. Homomorphism of (Co)Chain Complexes
Let $\mathcal{A} = \{A^n\}$ and $\mathcal{B} = \{B^n\}$ be two cochain complexes. A *homomorphism of complexes* $\alpha : \mathcal{A} \to \mathcal{B}$ is a set of homomorphisms $\alpha_n : A^n \to B^n$ such that for every n, the following diagram commutes:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & A^n & \xrightarrow{d_{\mathcal{A}}^{n+1}} & A^{n+1} & \longrightarrow & \cdots \\
 & & \downarrow{\scriptstyle \alpha_n} & & \downarrow{\scriptstyle \alpha_{n+1}} & & \\
\cdots & \longrightarrow & B^n & \xrightarrow{d_{\mathcal{B}}^{n+1}} & B^{n+1} & \longrightarrow & \cdots
\end{array}$$

Remark: There is a category $\mathcal{C}h_R$ of chain complexes of R-modules with $\mathrm{Ob}(\mathcal{C}h_R)$ consisting of all chain complexes of R-modules and $\mathrm{Mor}_{\mathcal{C}h_R}(\mathcal{A}, \mathcal{B})$ consisting of all homomorphism of chain complexes $\alpha : \mathcal{A} \to \mathcal{B}$.

Proposition 3.1.5. *Homomorphism of Complexes-Inducing-Homomorphisms of Cohomology Modules Proposition*
A homomorphism $\alpha : \mathcal{A} \to \mathcal{B}$ of cochain complexes induces module homomorphisms $H^n(\alpha) : H^n(\mathcal{A}) \to H^n(\mathcal{B})$ for $n \in \mathbb{N} \cup \{0\}$ on their respective cohomology modules.

Proof. We only provide the explicit construction and the remaining checks will be omitted:

$$H^n(\alpha)(\overline{z}) := \overline{\alpha^n(z)}, \forall z \in A^n \text{ with } d^n_{\mathcal{A}}(z) = 0.$$

□

Remark: Note that $0 = f : (\cdots \to \mathbb{Z} \xrightarrow{\mathrm{id}_{\mathbb{Z}}} \mathbb{Z} \to 0 \to \cdots) \to 0$ is not an isomorphism but it induces an isomorphism $H_n(f) : H_n(B) \to H_n(C)$ between zeros, so we need a concept called quasi-isomorphism:

Definition 3.1.6. Quasi-Isomorphism of Chain Complexes
A homomorphism $\alpha : \mathcal{A} \to \mathcal{B}$ of chain complexes is called a *quasi-isomorphism* of *homologism* if every $H^n(\alpha) : H^n(\mathcal{A}) \to H^n(\mathcal{B})$ is an isomorphism.

Definition 3.1.7. Short Exact Sequence of Complexes
Let $\mathcal{A} = \{A^n\}, \mathcal{B} = \{B^n\}$ and $\mathcal{C} = \{C^n\}$ be cochain complexes. A *short exact sequence of complexes* $0 \to \mathcal{A} \overset{\alpha}{\to} \mathcal{B} \overset{\beta}{\to} \mathcal{C} \to 0$ is a sequence of homomorphisms of complexes such that for every n, the sequence of cohomology modules $0 \to A^n \xrightarrow{\alpha_n} B^n \xrightarrow{\beta_n} C^n \to 0$ is short exact.

Theorem 3.1.8. *Long Exact Sequence-in-Cohomology Theorem*
Let $0 \to \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \to 0$ be a short exact sequence of cochain complexes. Then there is a long exact sequence of cohomology modules:

$$0 \to H^0(\mathcal{A}) \xrightarrow{H^0(\alpha)} H^0(\mathcal{B}) \xrightarrow{H^0(\beta)} H^0(\mathcal{C}) \xrightarrow{\delta_0} H^1(\mathcal{A}) \xrightarrow{H^1(\alpha)} H^1(\mathcal{B}) \xrightarrow{H^1(\beta)} H^1(\mathcal{C}) \xrightarrow{\delta_1} \cdots$$

Proof. For each n, the sequence $H^n(\mathcal{A}) \xrightarrow{H^n(\alpha)} H^n(\mathcal{B}) \xrightarrow{H^n(\beta)} H^n(\mathcal{C})$ is clearly exact. Now we define a module homomorphism $\delta_n : H^n(\mathcal{C}) \to H^{n+1}(\mathcal{A})$:

- **Claim 1:** If $c \in C^n$ represents the class $x \in H^n(\mathcal{C})$, then there is some $b \in B^n$ with $\beta_n(b) = c$.
- *Proof. of claim 1:* Since the short sequence $0 \to \mathcal{A} \overset{\alpha}{\to} \mathcal{B} \overset{\beta}{\to} \mathcal{C} \to 0$ is exact, then by the Complex-Exact Prop, β is surjective. □
- **Claim 2:** $d^{n+1}_{\mathcal{B}}(b) \in \ker \beta_{n+1}$ where this b is the b mentioned in claim 1.
- *Proof. of claim 2:* By the commutativity of the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r] & B^n \arrow[r, "d^{n+1}_{\mathcal{B}}"] \arrow[d, "\beta_n"] & B^{n+1} \arrow[r] \arrow[d, "\beta_{n+1}"] & \cdots \\
\cdots \arrow[r] & C^n \arrow[r, "d^{n+1}_{\mathcal{C}}"] & C^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

  we have $\beta_{n+1}(d^{n+1}_{\mathcal{B}}(b)) = d^{n+1}_{\mathcal{C}}(\beta_n(b)) \overset{\text{claim 1}}{=} d^{n+1}_{\mathcal{C}}(c)$. Since c represents the class $x \in H^n(\mathcal{C})$, then $c \in \ker d^{n+1}_{\mathcal{C}}$, so $\beta_{n+1}(d^{n+1}_{\mathcal{B}}(b)) = 0$. □

By the exactness of the sequence $0 \to A^{n+1} \xrightarrow{\alpha_{n+1}} B^{n+1} \xrightarrow{\beta_{n+1}} C^{n+1} \to 0$, we have $\ker \beta_{n+1} =$

im α_{n+1} and α_{n+1} is injective, so there exists a unique $a \in\in A^{n+1}$, s.t.:

$$\alpha_{n+1}(a) = d_{\mathcal{B}}^{n+1}(b) \tag{*}$$

- **Claim 3:** $d_{\mathcal{A}}^{n+2}(a) = 0$, i.e.: $a \in \ker d_{\mathcal{A}}^{n+1}$.
- *Proof. of claim 3:* By the injectivity of α_{n+2}, it suffices to show that $\alpha_{n+2}(d_{\mathcal{A}}^{n+2}(a)) = 0$. Then by the commutativity of the following diagram:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & A^{n+1} & \xrightarrow{d_{\mathcal{A}}^{n+2}} & A^{n+2} & \longrightarrow & \cdots \\ & & \downarrow{\scriptstyle \alpha_{n+1}} & & \downarrow{\scriptstyle \alpha_{n+2}} & & \\ \cdots & \longrightarrow & B^{n+1} & \xrightarrow{d_{\mathcal{B}}^{n+2}} & B^{n+2} & \longrightarrow & \cdots \end{array}$$

we have $\alpha_{n+2}(d_{\mathcal{A}}^{n+2}(a)) = d^{n+2}(\alpha_{n+1}(a)) \overset{(*)}{=} d_{\mathcal{B}}^{n+2}(d_{\mathcal{B}}^{n+1}(b)) = 0$, where the last equation is because $\mathcal{B}$ is a cochain complex. □

Hence, a can define a class $\overline{a} \in H^{n+1}(\mathcal{A})$.

- **Claim 4:** $\overline{a}$ is independent of the choice of b, i.e.: if $b' \neq b$ with $\beta_n(b') = c$ and if a' is the unique preimage in A^{n+1} of b', then $\overline{a} = \overline{a'}$.

  Proof. of claim 4: Since $\beta_n(b-b') = c-c = 0$, then $b-b' \in \ker \beta_n = \text{im } \alpha_n$, so $\alpha_n(a_0) = b-b'$ for some $a_0 \in A^n$. Then we have

  $$\alpha_{n+1}(a-a') \overset{(*)}{=} d_{\mathcal{B}}^{n+1}(b) - d_{\mathcal{B}}^{n+1}(b') = d_{\mathcal{B}}^{n+1}(b-b') = d_{\mathcal{B}}^{n+1}(\alpha_n(a_0)) \overset{\text{commutativity}}{=} \alpha_{n+1}(d_{\mathcal{B}}^{n+1}(a_0))$$

  By the injectivity of α_{n+1}, $a - a' = d_{\mathcal{B}}^{n+1}(a_0) \in \text{im } d_{\mathcal{B}}^{n+1}$ and hence by the def of cohomology modules $\overline{a} = \overline{a'} \in H^{n+1}(\mathcal{A})$. □

- **Claim 5:** $\overline{a}$ is independent of the choice of c representing the class x.
- *Proof. of claim 5:* Suppose that $\overline{c'} = \overline{c} = x$, which follows that $c - c' \in \text{im } d_{\mathcal{C}}^n$, i.e.: $e \in C^{n-1}$, s.t.: $d_{\mathcal{C}}^n(e) = c - c'$. By the surjectivity of β_{n-1}, there is $f \in B^{n-1}$, s.t.: $\beta_{n-1}(f) = e$. Then by commutativity,

  $$c - c' = d_{\mathcal{C}}^n(\beta_{n-1}(f)) = \beta_n(d_{\mathcal{B}}^n(f))$$

  Consider $b' := b - d_{\mathcal{B}}^n(f)$. Then

  $$\beta_n(b') = \beta_n(b) - \beta_n(d_{\mathcal{B}}^n(f)) = c - \beta_n(d_{\mathcal{B}}^n(f)) = c'$$

  so b' is a preimage of c'. By the def of cohomology modules, we then have

  $$d_{\mathcal{B}}^{n+1}(b') := d_{\mathcal{B}}^{n+1}(b) - d_{\mathcal{B}}^{n+1}(d_{\mathcal{B}}^n(f)) = d_{\mathcal{B}}^{n+1}(b) \tag{**}$$

Hence, by claim 2 and $(*)$, there exists a unique $a' \in A^{n+1}$, s.t.:

$$\alpha_{n+1}(a) \overset{(*)}{=} d_{\mathcal{B}}^{n+1}(b) \overset{(**)}{=} d_{\mathcal{B}}^{n+1}(b') = \alpha_{n+1}(a')$$

so by the injectivity of α_{n+1}, $a = a'$. Therefore, $\overline{a} = \overline{a'}$. $\square$

Now we define $\delta_n(x) := \overline{a}$. To prove that δ_n is a module homomorphism, we need to show that for any $x_1, x_2 \in H^n(\mathcal{C})$, $\delta_n(x_1 + x_n) = \delta_n(x_1) + \delta_n(x_2)$:

Let $c_1, c_2 \in C^n$ represent x_1 respectively x_2. Then $c_1 + c_2$ represents $x_1 + x_2$. By claim 1, there exist $b_1, b_2 \in B^n$, s.t.: $\beta_n(b_1) = c_1, \beta_n(b_2) = c_2$. Then $\beta_n(b_1 + b_2) = \beta_n(b_1) + \beta_n(b_2) = c_1 + c_2$. By claim 2, there exist unique $a_1, a_2 \in A^n$, s.t.:

$$\alpha_{n+1}(a_1) = d_{\mathcal{B}}^{n+1}(b_1), \alpha_{n+1}(a_2) = d_{\mathcal{B}}^{n+1}(b_2)$$

Then $\alpha_{n+1}(a_1 + a_2) = \alpha_{n+1}(a_1) + \alpha_{n+1}(a_2) = d_{\mathcal{B}}^{n+1}(b_1) + d_{\mathcal{B}}^{n+1}(b_2) = d_{\mathcal{B}}^{n+1}(b_1 + b_2)$. Since by claim 4,5, a_1, a_2 are independent of the choice of b_1, c_1 respectively b_2, c_2, then by the def of δ_n, we're done. $\square$

Remark:

- We call δ_n here the *connected homomorphisms.*
- The Long Exact Sequence-in-Cohomology Theorem gives a consequence that: if any two of the cochain complexes $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are exact, then so is the third.

Definition 3.1.9. Subcomplex
A (co)chain complex $\mathcal{D}$ is called a *subcomplex* of $\mathcal{C}$ if B_n are submodules of C_n and $d_n^{\mathcal{B}} = d_n^{\mathcal{C}}|_{B_n}$.

Remark: In this case, we have a cochain complex (similar for chain complex)

$$0 \to C^0/B^0 \xrightarrow{d_1} \cdots \xrightarrow{d_{n-1}} C^{n-1}/B^{n-1} \xrightarrow{d_n} C^n/B^n \to \cdots$$

denoted as C/B and called the *quotient complex.*

3.2 Homomorphisms and Ext

Intuition: To apply the Long Exact Sequence-in-Cohomology Thm to analyze the sequence (3.2), we try to produce a cochain complex whose first few cohomology groups in the long exact sequence in the Long Exact Sequence-in-Cohomology Thm agree with the terms in (3.2). To do this, we introduce the notion of a "resolution" of an R-module:

Definition 3.2.1. Projective Resolution
Let A be an R-module. A *projective resolution* of A is an exact sequence

$$\cdots \xrightarrow{d_{n+1}} P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} A \to 0$$

such that each P_i is a projective R-module.

Proposition 3.2.2. *Existence-of-Projective Resolution Proposition*
Every R-module has a projective resolution.

Proof. Let A be any R-module and then let P_0 be any free module (hence projective) R-module on a set of generators of A. Define an R-module homomorphism ϵ from P_0 onto A by the Universal Property of Free Modules. This begins the resolution $\epsilon : P_0 \to A \to 0$. The surjectivity of ϵ ensures that this sequence is exact. Next, let P_1 be any free module mapping onto the submodule $\ker \epsilon$ of P_0. This gives the second stage $P_1 \to P_0 \to A$ which, by construction, is exact. Continue this way, taking at the n-th stage a free R-mofule P_{n+1} that maps surjectively onto the submodule $\ker d_n$ of P_n, obtaining a free resolution of A. □

Remark:

- In general, a projective resolution is infinite in length, but if A is itself projective, then it has a very simple projective resolution of finite length, namely $0 \to A \xrightarrow{\mathrm{id}_A} A \to 0$ given by the identity map on A.
- Given a projective resolution on R-module $A \cdots \xrightarrow{d_{n+1}} P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} A \to 0$, we may form a related sequence by taking homomorphisms of each of the terms into D. This yields the sequence

$$\begin{aligned} 0 \to \mathrm{Hom}_R(A, D) \xrightarrow{\epsilon} \mathrm{Hom}_R(P_0, D) \xrightarrow{d_1} \mathrm{Hom}_R(P_1, D) \xrightarrow{d_2} \cdots \\ \xrightarrow{d_{n-1}} \mathrm{Hom}_R(P_{n-1}, D) \xrightarrow{d_n} \mathrm{Hom}_R(P_n, D) \xrightarrow{d_{n+1}} \cdots \end{aligned}$$

  where to simplify notation, we've denoted again by d_n the induced maps from $\mathrm{Hom}_R(P_{n-1}, D)$ to $\mathrm{Hom}_R(P_n, D)$ for $n \in \mathbb{N}_+$ and similarly for the map induced by ϵ. By the Associated $\mathrm{Hom}(\cdot, D)$ Sequence Thm, this sequence is not necessarily exact, but it is a cochain complex.

Definition 3.2.3. Ext
Let A and D be R-modules. For any projective resolution of A

$$\cdots \to P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} A \to 0$$

let $d_n : \mathrm{Hom}_R(P_{n-1}, D) \to \mathrm{Hom}_R(P_n, D)$ denote the induced map for $n \in \mathbb{N}_+$. Define

$$\mathrm{Ext}^n_R(A, D) := \ker\, d_{n+1}/\mathrm{im}\; d_n$$

where in particular $\mathrm{Ext}^0_R(A, D) = \ker\, d_1$. The module $\mathrm{Ext}^n_R(A, D)$ is called the *n-th cohomology module derived from the functor $Hom_R(\cdot, D)$*. In particular, when $R = \mathbb{Z}$, the module $\mathrm{Ext}^n_{\mathbb{Z}}(A, D)$ is also denoted by simply $\mathrm{Ext}^n(A, D)$.

Remark: The groups $\mathrm{Ext}^n_R(A, D)$ are also the cohomology modules of the cochain complex obtained from the projective resolution of A by replacing the term $\mathrm{Hom}_R(A, D)$ with zero (which does not effect the cochain property), i.e.: they are the cohomology modules of the cochain complex $0 \to \mathrm{Hom}_R(P_0, D) \to \cdots$.

Proposition 3.2.4. *Identification-of-Ext^0_R Proposition*
For any R-module A, we have

$$Ext^0_R(A, D) \cong Hom_R(A, D)$$

Proof. By the exactness of the sequence $P_1 \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} A \to 0$, the corresponding sequence $0 \to \mathrm{Hom}_R(A, D) \xrightarrow{\epsilon} \mathrm{Hom}_R(P_0, D) \xrightarrow{d_1} \mathrm{Hom}_R(P_1, D)$ is also exact by the Associated $\mathrm{Hom}_R(\cdot, D)$ Sequence Thm. Hence,

$$\mathrm{Ext}^0_R(A, D) := \ker\, d_1 = \mathrm{im}\ \epsilon \cong \mathrm{Hom}_R(A, D)/\ker \epsilon \cong \mathrm{Hom}_R(A, D)$$

□

Example:

- Let $R = \mathbb{Z}$ and let $A = \mathbb{Z}/n\mathbb{Z}$ with $m \geq 2$. By the Identification-of-Ext^0_R Prop, we have

  $$\mathrm{Ext}^0(\mathbb{Z}/m\mathbb{Z}, D) \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, D) \cong D_m$$

  where $D_m := \{d \in D \mid md = 0 \in D\}$ are the set of elements of D that have order dividing m. For the higher cohomology modules, we use the simple projective resolution

  $$0 \to \mathbb{Z} \xrightarrow{z \mapsto mz} \mathbb{Z} \xrightarrow{z \mapsto \overline{z}} \mathbb{Z}/m\mathbb{Z} \to 0$$

  for A. Taking homomorphisms into a field $\mathbb{Z}$-module D gives the cochain complex

  $$0 \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, D) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, D) \cong D$$

  Under the isomorphism $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, D) \cong D$, we have:

  $$\mathrm{Ext}^1(\mathbb{Z}/m\mathbb{Z}, D) \cong D/mD$$

  for any abelian group D. Then by def and by the cochain complex above, for any $n \geq 2$, any abelian group D, we have

  $$\mathrm{Ext}^n(\mathbb{Z}/m\mathbb{Z}, D) = 0$$

  We summarize as:

  $$\begin{aligned}
  &\mathrm{Ext}^0(\mathbb{Z}/m\mathbb{Z}, D) \cong D_m \\
  &\mathrm{Ext}^1(\mathbb{Z}/m\mathbb{Z}, D) \cong D/mD \\
  &\mathrm{Ext}^n(\mathbb{Z}/m\mathbb{Z}, D) \cong= 0, \forall n \geq 2
  \end{aligned}$$

- The same abelian groups may be modules over several different rings R and the Ext cohomology modules depend on R. For example, let $R = \mathbb{Z}/m\mathbb{Z}$ for some $m \in \mathbb{N}_+$. An R-module D is the same as an abelian group D with exponent dividing m, i.e.: $D_m = 0$. In particular, for any divisor $d \in \mathbb{Z}$ of m, the group $\mathbb{Z}/d\mathbb{Z}$ is an R-module and

$$\cdots \xrightarrow{z\mapsto(m/d)z} \mathbb{Z}/m\mathbb{Z} \xrightarrow{z\mapsto dz} \mathbb{Z}/m\mathbb{Z} \xrightarrow{z\mapsto(m/d)z} \mathbb{Z}/m\mathbb{Z} \xrightarrow{z\mapsto dz} \mathbb{Z}/m\mathbb{Z} \xrightarrow{[x]_m\mapsto[x]_d} \mathbb{Z}/d\mathbb{Z} \to 0$$

is a projective resolution of $\mathbb{Z}/d\mathbb{Z}$ as a $\mathbb{Z}$-module, which in fact is a free sequence. Taking homomorphisms into the $\mathbb{Z}/m\mathbb{Z}$-module D, using the isomorphism $\mathrm{Hom}_{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, D) \cong D$ and removing the first term gives the cochain complex

$$0 \to D \xrightarrow{z\mapsto dz} D \xrightarrow{z\mapsto(m/d)z} D \xrightarrow{z\mapsto dz} D \xrightarrow{z\mapsto(m/d)z} D \to \cdots$$

Hence, we can summarize as:

$$\begin{aligned}
&\mathrm{Ext}^0_{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/d\mathbb{Z}, D) \cong D_d\\
&\mathrm{Ext}^n_{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/d\mathbb{Z}, D) \cong (D/dD)_{(m/d)} \text{ for odd } n \geq 1\\
&\mathrm{Ext}^n_{\mathbb{Z}/m\mathbb{Z}}(\mathbb{Z}/d\mathbb{Z}, D) \cong (D_d/(m/d)D) \text{ for even } n \geq 2
\end{aligned}$$

where $M_k := \{r \in M \mid kr = 0 \in M\}$ denotes the set of elements of module M killed by integer k. In particular,

$$\mathrm{Ext}^n_{\mathbb{Z}/p^2\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$$

for all $n \geq 0$.

Lemma 3.2.5. *Lifting Lemma for Projective Resolutions*
Let $f : A \to A'$ be an R-module homomorphism. Taking projective resolutions of A respectively A'. Then for each $n \geq 0$, there is a lift f_n of f such that the following diagram commutes:

$$\begin{tikzcd}
\cdots \arrow[r, "d_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "f_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "f_0"] & A \arrow[r] \arrow[d, "f"] & 0\\
\cdots \arrow[r, "d_2'"] & P_1' \arrow[r, "d_1'"] & P_0 \arrow[r, "\epsilon'"] & A' \arrow[r] & 0
\end{tikzcd}$$

where the rows are the projective resolutions of A respectively A'.

Proof. Given the two rows and the map f with the following commutative diagram

$$\begin{tikzcd}
\cdots \arrow[r, "d_2"] & P_1 \arrow[r, "d_1"] & P_0 \arrow[r, "\epsilon"] & A \arrow[r] \arrow[d, "f"] & 0\\
\cdots \arrow[r, "d_2'"] & P_1' \arrow[r, "d_1'"] & P_0 \arrow[r, "\epsilon'"] & A' \arrow[r] & 0
\end{tikzcd}$$

since P_0 is projective, then we can lift the map $f \circ \epsilon : P_0 \to A'$ to a map $f_0 : P_0 \to P_0'$ in such a way that $\epsilon' \circ f_0 = f \circ \epsilon$, by the second def of projective modules, which gives the first lift of f. Proceeding inductively in this way, assume f_n has been defined to make the diagram commutative to that point. Hence, im $(f_n \circ d_{n+1}) \subseteq$ ker d'_n. The projectivity of P_{n+1} implies that we can lift the map $f_n \circ d_{n+1} : P_{n+1} \to P'_n$ to a map $f_{n+1} : P_{n+1} \to P'_{n+1}$ to make the diagram commute at the next stage. □

Remark: The commutative diagram in the Lifting Lemma for Projective Resolutions implies the induced diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Hom}_R(A, D) \arrow[r] & \mathrm{Hom}_R(P_0, D) \arrow[r] & \mathrm{Hom}_R(P_1, D) \arrow[r] & \cdots \\
0 \arrow[r] & \mathrm{Hom}_R(A', D) \arrow[r] \arrow[u, "f"] & \mathrm{Hom}_R(P'_0, D) \arrow[r] \arrow[u, "f_0"] & \mathrm{Hom}_R(P_1, D) \arrow[r] \arrow[u, "f_1"] & \cdots
\end{tikzcd}$$

is also commutative. The two rows of these cochain complexes and this commutative diagram depicts a homomorphism of Cohomology Modules Prop, we have an induced map on their cohomology modules:

Lemma 3.2.6. *Induced Maps on Ext-and-Homotopy Independence Lemma*
Let $f : A \to A'$ be an R-module homomorphism. Take projective resolution of A respectively A' as in the Lifting Lemma for Projective Resolutions. Then for every n, there is an induced group homomorphism $\varphi_n : Ext^n_R(A', D) \to Ext^n_R(A, D)$ on the cohomology modules obtained via these resolutions and the maps φ_n depend only on f, not on the choice of lifts f_n in the Lifting Lemma for Projective Resolutions.

Proof. The existence of the map on the cohomology modules Ext follows from the Homomorphisms of Cohomology Modules Prop applied to the homomorphism of cochain complexes in the above remark.

Then it remains to show the independence of lifts f_n. It remains to show the independence of lifts f_n. It suffices to show that for two different lift f^1_n, f^2_n inducing the same φ_n, their difference $f^1_n - f^2_n$ induces zero map. Hence, it suffices to show that: if f is the zero map, then the induced maps on cohomology modules are all zeros, so assume then that $f = 0$.

Since $\epsilon' \circ f_0 = f \circ \epsilon = 0$, then im $f_0 \subseteq$ ker ϵ'. By the exactness of the lower row, we have ker $\epsilon' =$ im d'_1. Then by the projectivity of P_0, there is a lift $s_0 : P_0 \to P'_1$, s.t.: the following diagram commutes:

$$\begin{tikzcd}
 & P_0 \arrow[dl, dashed, "s_0"'] \arrow[d, "f_0"] & \\
P'_1 \arrow[r, "d'_1"] & \mathrm{im}\ d'_1 \arrow[r] & 0
\end{tikzcd}$$

i.e.: $f_0 = d'_1 \circ s_0$. Inductively, suppose that we've constructed $s_0, s_1, \cdots, s_{n-1}$ with $f_k = d'_{k+1} \circ s_k +$

$s_{k-1} \circ d_k$. Note that

$$
\begin{aligned}
d'_n \circ (f_n - s_{n-1} \circ d_n) &= d'_n \circ f_n - d'_n \circ s_{n-1} \circ d_n \\
&\overset{\text{commutative diagram of cochain complexes}}{=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=} f_{n-1} \circ d_n - d'_n \circ s_{n-1} \circ d_n \\
&= (f_{n-1} - d'_n \circ s_{n-1}) \circ d_n \overset{\text{inductive hypothesis}}{=\!=\!=\!=\!=\!=} s_{n-2} \circ d_{n-1} \circ d_n = 0
\end{aligned}
$$

Hence, im $(f_n - s_{n-1} \circ d_n) \subseteq \ker d'_n = \operatorname{im} d'_{n+1}$. Again, by the projectivity of P_n, we can get a lift $s_n : P_n \to P'_{n+1}$, s.t.: $f_n = d'_{n+1} \circ s_n + s_{n-1} \circ d_n$. So now we construct R-module homomorphisms $s_n : P_n \to P_{n+1}$ with the property that for all n,

$$f_n = d'_{n+1} \circ s_n + s_{n-1} \circ d_n \tag{*}$$

Then for any $\alpha \in \operatorname{Hom}_R(P'_n, D)$, we can get:

$$\alpha \circ f_n \overset{(*)}{=} \alpha \circ (d'_{n+1} \circ s_n + s_{n-1} \circ d_n) = (\alpha \circ d'_{n+1}) \circ s_n + (\alpha \circ s_{n-1}) \circ d_n \tag{**}$$

Now if $\alpha \in \operatorname{Hom}_R(P'_n, D)$ represents the class $\overline{\alpha} \in \operatorname{Ext}^n_R(A, D)$, then $\alpha \circ d'_{n+1} = 0$, so

$$\alpha \circ f_n \overset{(**)}{=} 0 \circ s_n + (\alpha \circ s_{n-1}) \circ d_n = (\alpha \circ s_{n-1}) \circ d_n$$

Since $\alpha \circ s_{n-1}$ is a map from P_{n-1} to D, then $(\alpha \circ s_{n-1}) \circ d_n \in \operatorname{im} d_n$ and hence $\overline{\alpha \circ f_n} = 0 \in \operatorname{Ext}^n_R(A, D)$. Therefore, $\varphi_n : \operatorname{Ext}^n_R(A', D) \to \operatorname{Ext}^n_R(A, D)$ is zero map. $\square$

Remark: The name of this lemma and the intuition for s_n in the proof comes from the notion of *chain homotopy* between chain homomorphism given by f_n and $f = 0$, namely, the collection of maps $\{s_n\}$. Furthermore, f is chain homotopic to g, denoted as $f \simeq g$, if $f - g = ds + sd$ for some chain homotopy s, understanding from the following commutative diagram:

$$
\begin{tikzcd}
\cdots \arrow[r] & P_n \arrow[r, "d_n^P"] \arrow[d, shift right, "g_n"'] \arrow[d, shift left, "f_n"] & P_{n-1} \arrow[r] \arrow[d, shift right, "g_{n-1}"'] \arrow[d, shift left, "f_{n-1}"] & \cdots \\
\cdots \arrow[r] & Q_n \arrow[r, "d_n^Q"'] \arrow[ur, "h_{n-1}"] & Q_{n-1} \arrow[r] & \cdots
\end{tikzcd}
$$

Theorem 3.2.7. *Ext-Independence-of-Projective Resolutions Theorem*
The groups $Ext^n_R(A, D)$ only depend on A and D, i.e.: they are independent of the choice of projective resolution of A.

Proof. In the notation of the Lifting Lemma for Projective Resolutions, let $A' = A$, let $f = \mathrm{id}_A : A \to A'$ be the identity map on A and let the zero rows be two projective resolutions of A. Then by the Lifting Lemma for Projective Resolutions, we can get a lift $f_n : P_n \to P'_n$ of $f = \mathrm{id}_A$ and by the Induced Maps on Ext-and-Homotopy Independence Lemma, we can also get an induced map

$\varphi_n : \mathrm{Ext}_R^n(A', D) \to \mathrm{Ext}_R^n(A, D)$.

Now add a third row to the diagram, in the Lifting Lemma for Projective Resolutions, by copying the projective resolution in the top row below the second row. Then let $g = \mathrm{id}_A : A' \to A$ be the identity map on A, lift g to maps $g_n : P'_n \to P_n$ and let $\psi_n : \mathrm{Ext}_R^n(A, D) \to \mathrm{Ext}_R^n(A', D)$ be the induced map corresponding to g_n.

Note that the maps $g_n \circ f_n : P_n \to P_n$ are now a lift of the identity map $g \circ f$ and that they induce the homomorphisms $\varphi_n \circ \psi_n$ on the cohomology modules.

Since the first and third rows are identical, then the maps id_{P_n} form a obvious lift of $g \circ f$ and this choice clearly induces the identity homomorphisms $\mathrm{id}_{\mathrm{Ext}_R^n(A,D)}$ on cohomology modules.

Now, for $g \circ f = \mathrm{id}_A$, we have two lifts $g_n \circ f_n$ and id_{P_n}. By the Induced Maps on Ext-and-Homotopy Independence Lemma, the two lifts must induce the same homomorphisms of the cohomology modules, i.e.:

$$\varphi_n \circ \psi_n = \mathrm{id}_{\mathrm{Ext}_R^n(A,D)}$$

Similarly, we can also get $\psi_n \circ \varphi_n = \mathrm{id}_{\mathrm{Ext}_R^n(A',D)}$. Therefore, φ_n and ψ_n are isomorphisms between

$$\mathrm{Ext}_R^n(A, D) \cong \mathrm{Ext}_R^n(A', D)$$

□

Remark:

- The Ext-Independence-of-Projective Resolutions Thm intimates the well-defined-ness of Ext.
- For a fixed R-module D and fixed $n \in \mathbb{N} \cup \{0\}$, the Induced Maps on Ext-and-Homotopy Independence Lemma and the Ext-Independence-of-Projective Resolutions Thm show that $\mathrm{Ext}_R^n(\cdot, D)$ defines a (contravariant) functor from the category of R-modules to the category of abelian groups.

Lemma 3.2.8. *Horseshow Lemma*
Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules, let $L = A$ have a projective resolution denoted by P_n and let N have a similar projective resolution where the projective modules are denoted by $\overline{P}_n$. Then there is a resolution of M by the projective modules $P_n \oplus \overline{P}_n$ such that

the following diagram commutes:

$$\begin{array}{ccccccccc}
 & & \vdots & & \vdots & & \vdots & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & P_1 & \longrightarrow & P_1 \oplus \overline{P}_1 & \longrightarrow & \overline{P}_1 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & P_0 & \longrightarrow & P_0 \oplus \overline{P}_0 & \longrightarrow & \overline{P}_0 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & L & \longrightarrow & M & \longrightarrow & N & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

Moreover, the rows and columns of this diagram are exact and rows are split.

Proof. Recall the examples of projective modules that $P_n \oplus \overline{P}_n$ are projective module. We define the horizontal maps by:

$$0 \to P_n \xrightarrow{x \mapsto (x,0)} P_n \oplus \overline{P}_n \xrightarrow{(x,y) \mapsto y} \overline{P}_n \to 0$$

for all rows exact except for the bottom row, which can be verified that these sequences are split exact. Then it remains to define the vertical maps in the middle column to make the diagram commute:

Let $\mu : \overline{P}_0 \to M$ be a lift of th map $\overline{P}_0 \to N$ (which exists since $\overline{P}_0$ is projective), and let λ be the composition $P_0 \to L \to M$ in the diagram. Define $\pi : P_0 \oplus \overline{P}_0 \to M$ by

$$\pi(x,y) := \lambda(x) + \mu(y)$$

Now we verify the commutativity of the two diagrams:

$$\begin{array}{ccc}
P_0 & \xrightarrow{x \mapsto (x,0)} & P_0 \oplus \overline{P}_0 \\
\downarrow & & \downarrow{\scriptstyle \pi} \\
L & \longrightarrow & M
\end{array}
\quad \text{and} \quad
\begin{array}{ccc}
P_0 \oplus \overline{P}_0 & \xrightarrow{(x,y) \mapsto y} & \overline{P}_0 \\
\downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \overline{\epsilon}} \\
M & \longrightarrow & N
\end{array}$$

- For the left diagram above, it suffices to show that $\lambda = \pi \circ (x \mapsto (x,0))$ since the commutativity of downleft part of the left diagram above after adding the diagonal $P_0 \xrightarrow{\lambda} M$, which is clearly directly by the def of μ.
- Denote the unnamed map in the right diagram above by $p : M \to N$. For any $(x,y) \in P_0 \oplus \overline{P}_0$, note that $p(\lambda(x)) = 0$ by the exactness of the sequence $0 \to L \to M \xrightarrow{p} N \to 0$. Hence,

$$(p \circ \pi)(x,y) = p(\lambda(x)) + p(\mu(y)) = 0 + \overline{\epsilon}(y) = (\overline{\epsilon} \circ ((x,y) \mapsto y))(x,y)$$

Now we've constructed a vertical map $d_0 = \pi : P_0 \oplus \overline{P}_0 \to M$. Then inductively, we can similarly construct d_n by replacing the bottom row $0 \to L \to M \to N \to 0$ with $0 \to P_{n-1} \to P_{n-1} \oplus \overline{P}_{n-1} \to \overline{P}_{n-1} \to 0$ when constructing d_n and then applying the same argument. $\square$

Remark: Now by the Long Exact Sequence in Cohomology Thm and the Horseshoe Lemma, the exact sequence (3.2) can be extended to the long exact sequence for Ext:

Theorem 3.2.9. *Long Exact Sequence-for-Ext Theorem*
Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules. Then there is a long exact sequence of abelian groups

$$0 \to Hom_R(N, D) \to Hom_R(M,D) \to Hom_R(L, D) \xrightarrow{\delta_0}$$
$$Ext^1_R(N, D) \to Ext^1_R(M, D) \to Ext^1_R(L, D) \xrightarrow{\delta_1} Ext^2_R(N, D) \to \cdots$$

where the maps between groups at the same level n are as in the Induced Maps on Ext-and-Homotopy Independence Lemma and the connecting homomotphism δ_n are given by the Long Exact Sequence in Cohomology Thm.

Proof. By the Horseshoe Lemma, we have a huge two-dimensional diagram with projective resolution in the columns and with split short exact sequence in the row as in the Horseshoe Lemma. We act Hom on this diagram and then we get a new diagram:

$$\begin{tikzcd}
 & \vdots & \vdots & \vdots & \\
0 \arrow[r] & \mathrm{Hom}_R(\overline{P}_1, D) \arrow[r] \arrow[u] & \mathrm{Hom}_R(P_1 \oplus \overline{P}_1, D) \arrow[r] \arrow[u] & \mathrm{Hom}_R(P_1, D) \arrow[r] \arrow[u] & 0 \\
0 \arrow[r] & \mathrm{Hom}_R(\overline{P}_0, D) \arrow[r] \arrow[u] & \mathrm{Hom}_R(P_0 \oplus \overline{P}_0, D) \arrow[r] \arrow[u] & \mathrm{Hom}_R(P_0, D) \arrow[r] \arrow[u] & 0 \\
 & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] &
\end{tikzcd}$$

where the right "$\to 0$" of every row is guarenteed by the stability of split short exact sequence. Hence, we get a short exact sequence of cochain complexes with $\mathcal{A} = \{\mathrm{Hom}_R(\overline{P}_n, D)\}, \mathcal{B} = \{\mathrm{Hom}_R(P_n \oplus \overline{P}_n, D)\}$ and $\mathcal{C} = \{\mathrm{Hom}_R(P_n, D)\}$. Apply then the Long Exact Sequence in Cohomology Thm and we're done. $\square$

Remark:

- The Long Exact Sequence-for-Ext Thm show that module $\mathrm{Ext}^1_R(N, D)$ is the first measure of the failure of (3.2) to be exact on the right. In fact, (3.2) can be extended to be a short exact sequence on the right $\Leftrightarrow$ the connecting homomorphism δ_0 in the Long Exact Sequence-for-Ext Thm is the zero homomorphism. In particular, if $\mathrm{Ext}^1_R(N, D) = 0, \forall R$-module N, then (3.2) will be exact on the right for *every* exact sequence (3.1), which implies that the R-module D is injective. Part of the next prop shows that the converse is also true.

- For a fixed R-module D, the result in the Long Exact Sequence-for-Ext Thm can be viewed as explaining what happens to the short exact sequence $0 \to L \to M \to N \to 0$ on the right after applying the left exact functor $\mathrm{Hom}_R(\cdot, D)$. This is why the (contravariant) functors $\mathrm{Ext}^n_R(\cdot, D)$ are called the *right derived functors* for the functor $\mathrm{Hom}_R(\cdot, D)$. In the contrast, one can also consider the effect of applying the left exact functor $\mathrm{Hom}_R(D, \cdot)$, i.e.: by taking homomorphisms from D rather than into D. We'll look into this later in the next-next theorem which shows that in fact the same Ext_R groups define the (covariant) right derived functors for $\mathrm{Hom}_R(D, \cdot)$ as well and we now first take a look at the previous remark:

Proposition 3.2.10. *Ext-Criterion-for-Injective Modules Proposition*
Let Q be an R-module. The following are equivalent:

1. *Q is injective.*
2. *$Ext^1_R(A, Q) = 0$ for all R-modules A.*
3. *$Ext^n_R(A, Q) = 0$ for all R-modules A, and all $n \in \mathbb{N}_+$.*

Proof. We've proved $(2 \Rightarrow 1)$ in the above remark and $(3 \Rightarrow 2)$ is direct, so it remains to prove $(1 \Rightarrow 3)$. Take a projective resolution

$$\cdots \to P_n \to P_{n-1} \to \cdots \to P_0 \to A \to 0$$

for A. By the injectivity of Q, the sequence

$$0 \to \mathrm{Hom}_R(A, Q) \to \mathrm{Hom}_R(P_0, Q) \to \cdots \to \mathrm{Hom}_R(P_{n-1}, Q) \to \mathrm{Hom}_R(P_n, Q) \to \cdots$$

is still exact, so all of the cohomology modules for this cochain complex are 0. $\square$

Remark: We've chosen to define the cohomology group $\mathrm{Ext}^n_R(A, B)$ using a projective resolution of A. There is a parallel development using an *injective resolution* of B:

$$0 \to B \to Q_0 \to Q_1 \to \cdots$$

where each Q_i is injective. In this situation, one defines $\mathrm{Ext}^n_R(A, B)$ as the n-th cohomology module of the cochain sequence obtained by applying $\mathrm{Hom}_R(A, \cdot)$ to the resolution for B.
Example:

- $R = \mathbb{Z}$. By the Injectivity PID-Modules Prop.1, we know that $\mathbb{Z}$-module is injective $\Leftrightarrow$ it is divisible. by the coro of Injective PID-Modules Prop, $\mathbb{Z}$-module B can be embedded in an injective $\mathbb{Z}$-module Q_0 and the quotient, Q_1, of Q_0 by the image of B is again injective.

Hence, we have an injective resolution

$$0 \to B \to Q_0 \to Q_1 \to 0$$

of B. Applying $\mathrm{Hom}_{\mathbb{Z}}(A, \cdot)$ to this sequence gives the cochain complex

$$0 \to \mathrm{Hom}_{\mathbb{Z}}(A, B) \to \mathrm{Hom}_{\mathbb{Z}}(A, Q_0) \to \mathrm{Hom}_{\mathbb{Z}}(A, Q_1) \to 0 \to \cdots$$

which follows that $\mathrm{Ext}^n(A, B) = 0$ for all abelian groups A, B and all $n \geq 2$.

- Suppose A is a torsion abelian group, i.e.: a group whose elements a with $ka = 0$ for some $k \in \mathbb{Z}$. Then we have $\mathrm{Ext}^0(A, \mathbb{Z}) \cong \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Z}) = 0$, since $\mathbb{Z}$ is torsion-free. The sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ gives an injective resolution of $\mathbb{Z}$. Applying $\mathrm{Hom}_{\mathbb{Z}}(A, \cdot)$ gives the cochain complex

$$0 \to \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Z}) \to \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}) \to \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z}) \to 0 \to \cdots$$

Since $\mathbb{Q}$ is also torsion-free, thn $\mathrm{Ext}^1(A, \mathbb{Z}) \cong \mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$. This group, $\mathrm{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z})$ is called the *Portriagin dual group* to A.

Theorem 3.2.11. *Exact-Left Exact Function-Long Exact Sequence Theorem*
Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules. Then there is a long exact sequence of abelian groups

$$\begin{aligned} 0 \to \mathit{Hom}_R(D, L) \to \mathit{Hom}_R(D, M) \to \mathit{Hom}_R(D, N) &\xrightarrow{\gamma_0} \\ \mathit{Ext}^1_R(D, L) \to \mathit{Ext}^1_R(D, M) \to \mathit{Ext}^1_R(D, N) &\xrightarrow{\gamma_1} \mathit{Ext}^2_R(D, L) \to \cdots \end{aligned}$$

Proof. Take a projective resolution of D and then apply $\mathrm{Hom}_R(\cdot, L), \mathrm{Hom}_R(\cdot, M)$ and $\mathrm{Hom}_R(\cdot, N)$ to this resolution. Then we can obtains the columns in a commutative diagram similar to the diagram in the proof of Long Exact Sequence-for-Ext Thm, but with L, M, N in the second positions rather than the first. Appying the Long Exact Sequence in Cohomology Thm to this array gives the desired sequence. □

Remark: The Exact-Left Exact Functor-Long Exact Sequence Thm shows that module $\mathrm{Ext}^1_R(D, L)$ measures whether the exact sequence

$$0 \to \mathrm{Hom}_R(D, L) \to \mathrm{Hom}_R(D, M) \to \mathrm{Hom}_R(D, N)$$

can be extended to a short exact sequence. In fact, it can be extended iff γ_0 is the zero homomorphism. In particular, this will always be the case if the module D has the property that $\mathrm{Ext}^1_R(D, B) = 0$ for all modules B, say D is a projective R-module.

Proposition 3.2.12. *Ext-Criterion-for-Projective Modules Proposition*
Let P be an R-module. The followings are equivalent:

1. *P is projective.*
2. *$Ext^1_R(P,B)=0$ for all R-modules B.*
3. *$Ext^n_R(P,B)=0$ for all R-modules B, and all $n\geq 1$.*

Proof. We've proved $(2\Rightarrow 1)$ in the above remark and $(3\Rightarrow 2)$ is trivial. For $(1\Rightarrow 3)$, consider the simple exact sequence

$$0\to P\xrightarrow{\mathrm{id}_P} P\to 0$$

which is clearly a projective resolution of P with respect to $\epsilon=\mathrm{id}_P$. Then taking homomorphisms into B gives the simple cochain complex

$$0\to \mathrm{Hom}_R(P,B)\xrightarrow{\mathrm{id}} \mathrm{Hom}_R(P,B)\to 0\to\cdots$$

which gives 3. □

Example:

- Since $\mathbb{Z}^m$ is a free, hence projective, $\mathbb{Z}$-module, then by the Ext-Criterion-for-Projective Modules Prop, $\mathrm{Ext}^n_{\mathbb{Z}}(\mathbb{Z}^m,B)=0$ for all abelian groups B, all $m\geq 1$ and all $n\geq 1$.
- It's easy to show that $\mathrm{Ext}^1_R(A_1\oplus A_2,B)\cong \mathrm{Ext}^n_R(A_1,B)\oplus\mathrm{Ext}^n_R(A_2,B)$ for all $n\geq 1$, (in fact, the general case
  $$\mathrm{Ext}^n_R(\bigoplus_{i\in I}A_i,B)\cong \Pi_{i\in I}\mathrm{Ext}^n_R(A_i,B)$$
  also holds,) so the previous example together with the example following the Identification-of-Ext^0_R Prop determines $\mathrm{Ext}^n(A,B)$ for all finitely generated abelian groups A. In particular, $\mathrm{Ext}^n(A,B)=0$ for all finitely generated groups A, all abelian groups B and all $n\geq 2$.

Theorem 3.2.13. *Classification Theorem for Extensions via Ext^1_R*
For any R-modules N,L, there is a bijection between $Ext^1_R(N,L)$ and the set of equivalence classes of extensions of N by L.

3.3 Tensor Products and Tor

Intuition: Ext determines what happens to short exact sequences on the right after applying the left exact functors $\mathrm{Hom}_R(D,\cdot)$ and $\mathrm{Hom}_R(\cdot,D)$. Similarly, we ask for the behavior of short exact sequences on the left after applying the right exact functor $D\otimes_R\cdot$ or $\cdot\otimes_R D$.
Motivation: We concentrate on the situation for $D\otimes_R\cdot$: For the sequence (3.2), applying $D\otimes_R\cdot$ gives an exact sequence of abelian groups

$$D\otimes_R L\to D\otimes_R M\to D\otimes_R N\to 0$$

which may be extended at the left end to a long exact sequence as follows.

Let $\cdots \to P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} B \to 0$ be a projective resolution of B and take tensor products with D to obtain

$$\cdots \to D \otimes_R P_n \xrightarrow{\mathrm{id}_D \otimes d_n} D \otimes_R P_{n-1} \to \cdots \xrightarrow{\mathrm{id}_D \otimes d_1} D \otimes_R P_0 \xrightarrow{\mathrm{id}_D \otimes \epsilon} D \otimes_R B \to 0$$

It follows from the Tensor-Exact Sequences Thm that the above sequence is a chain complex, so we may form its homology modules:

Definition 3.3.1. Tor
Let B, D be R-modules. For any projective resolution of B, let $\mathrm{id}_D \otimes d_n : D \otimes_R P_n \to D \otimes_R P_{n-1}$ for $n \in \mathbb{N}_+$. Then we define

$$\mathrm{Tor}_n^R(D, B) := \ker\,(\mathrm{id}_D \otimes d_n)/\mathrm{im}\,(\mathrm{id}_D \otimes d_{n+1})$$

where $\mathrm{Tor}_0^R(D, B) := (D \otimes_R P_0)/\mathrm{im}\,(\mathrm{id}_D \otimes d_1)$. The module $\mathrm{Tor}_n^R(D, B)$ is called the *n-th homology module desired from the functor $D \otimes_R \cdot$*. In particular, when $R = \mathbb{Z}$, the module $\mathrm{Tor}_n^R(D, B)$ is also clearly the n-th homology module obtained from the chain complex in motivation above by removing the term $D \otimes_R B$.

Proposition 3.3.2. *Identification-of-Tor_0^R Proposition*
For any R-module B, we have

$$Tor_0^R(D, B) \cong D \otimes_R B$$

Proof. Similar as the Identification-of-Ext_0^R Prop. □

Example:

- Let $R = \mathbb{Z}$ and let $B = \mathbb{Z}/m\mathbb{Z}$ for some $m \geq 2$. By the Identification-of-Tor_0^R Prop, we have

  $$\mathrm{Tor}_0(D, \mathbb{Z}/m\mathbb{Z}) \cong D \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z} \cong D/mD$$

  For the higher groups, we apply $D \otimes_{\mathbb{Z}} \cdot$ to the projective resolution $0 \to \mathbb{Z} \xrightarrow{z \mapsto mz} \mathbb{Z} \twoheadrightarrow \mathbb{Z}/m\mathbb{Z} \to 0$ of B to get the chain complex

  $$\cdots \to 0 \to D \otimes_{\mathbb{Z}} \mathbb{Z}(\cong D) \to D \otimes_{\mathbb{Z}} \mathbb{Z}(\cong D) \to D \otimes_{\mathbb{Z}} \mathbb{Z}/m\mathbb{Z}(\cong D/mD) \to 0$$

  which follows that $\mathrm{Tor}_1(D, \mathbb{Z}/m\mathbb{Z}) \cong D_m$ and that $\mathrm{Tor}_n(D, \mathbb{Z}/m\mathbb{Z}) = 0$ for all $n \geq 2$. In summary,

  $$\begin{aligned} &\mathrm{Tor}_0(D, \mathbb{Z}/m\mathbb{Z}) \cong D/mD \\ &\mathrm{Tor}_1(D, \mathbb{Z}/m\mathbb{Z}) \cong D_m \\ &\mathrm{Tor}_n(D, \mathbb{Z}/m\mathbb{Z}) = 0 \text{ for all } n \geq 2 \end{aligned}$$

- As for Ext, the Tor groups also depend on the ring R.

Theorem 3.3.3. *Tor-Independence-of-Projective Resolutions Theorem*
The homology modules $Tor_n^R(D,B)$ are independent of the choice projective resolution of B.

Proposition 3.3.4. *Induced Maps on Tor-and-Independence Proposition*
For every R-module homomorphism $f : B \to B'$, there are induced maps $\psi_n : Tor_n^R(D,B) \to Tor_n^R(D,B')$ on homology modules (depending only on f).

Remark: The Tor-Independence-of-Projective Resolutions Thm and the Induced Maps on Tor-and-Independence Prop give the well-defined-ness of Tor and the following Thm:

Theorem 3.3.5. *Long Exact Sequence-for-Tor Theorem*
Let $0 \to L \to M \to N \to 0$ be a short exact sequence of R-modules. Then there is a long exact sequence of abelian groups

$$\cdots \to Tor_2^R(D,N) \xrightarrow{\delta_1} Tor_1^R(D,L) \to Tor_1^R(D,M) \to Tor_1^R(D,N)$$
$$\xrightarrow{\delta_0} D \otimes_R L \to D \otimes_R M \to D \otimes_R N \to 0$$

where the maps between groups at the same level n are as in the Induced Maps on Tor-and-Independence Prop and the maps δ_n are called connecting homomorphisms.

Proposition 3.3.6. *Tor-Criterion-for-Flat Modules Prop*
Let D be an R-module. The followings are equivalent:

1. *D is a flat R-module.*
2. *$Tor_1^R(D,B) = 0$ for all R-modules B.*
3. *$Tor_n^R(D,B) = 0$ for all R-modules B and all $n \geq 1$.*

Example:

- If $R = \mathbb{Z}$, then $\mathrm{Tor}_n(A, \mathbb{Z}^m) = 0$ for all $n \geq 1$ and all abelian groups A since $\mathbb{Z}^m$ is free hence flat.
- It's easy to show that $\mathrm{Tor}_n^R(A, B_1 \oplus B_2) \cong \mathrm{Tor}_n^R(A, B_1) \oplus \mathrm{Tor}_n^R(A, B_2)$ for all $n \geq 1$ (in fact, the general case

  $$\mathrm{Tor}_n^R(A, \bigoplus_{j \in J} B_j) \cong \bigoplus_{j \in J} \mathrm{Tor}_n^R(A, B_j)$$

  also holds). Then the previous two examples determine $\mathrm{Tor}_n^R(A,B)$ for all abelian groups A and all finitely generated abelian groups B.

  In particular, $\mathrm{Tor}_1(A,B)$ is a torsion group and $\mathrm{Tor}_n(A,B) = 0$ for every abelian group A, every finitely generated abelian group B and all $n \geq 2$. In fact, these results hold without the condition that B is finitely generated.
- The exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$ gives the long exact sequence

$$\cdots \to \mathrm{Tor}_1(D, \mathbb{Q}) \to \mathrm{Tor}_1(D, \mathbb{Q}/\mathbb{Z}) \to D \otimes \mathbb{Z} \to D \otimes \mathbb{Q} \to D \otimes \mathbb{Q}/\mathbb{Z} \to 0$$

Since $\mathbb{Q}$ is a flat $\mathbb{Z}$-module, then by the Tor-Criterion-for-Flat Modules Prop, we have an exact sequence

$$0 \to \mathrm{Tor}_1(D, \mathbb{Q}/\mathbb{Z}) \to D \to D \otimes \mathbb{Q}$$

so $\mathrm{Tor}_1(D, \mathbb{Q}/\mathbb{Z}) \cong \ker(D \to D \otimes \mathbb{Q})$, which is the torsion subgroup of D.

Proposition 3.3.7. *$Tor_1^{\mathbb{Z}}$-Torsion Proposition*
Let A, B be $\mathbb{Z}$-modules and let $t(A), t(B)$ denote their respective torsion submodules. Then

$$Tor_1(A, B) \cong Tor_1(t(A), t(B))$$

Proof.

- **Claim:** $\mathrm{Tor}_1(A, t(B)) \cong \mathrm{Tor}_1(t(A), t(B))$.
- *Proof. of claim:* Consider a short exact sequence $0 \to t(B) \to B \to B/t(B) \to 0$. Apply the left functor $A \otimes_{\mathbb{Z}} \cdot$ to get a long exact sequence

$$\cdots \to \mathrm{Tor}_2(A, B/t(B)) \to \mathrm{Tor}_1(A, t(B)) \to \mathrm{Tor}_1(A, B) \to \mathrm{Tor}_1(A, B/t(B)) \to \cdots$$

Since $\mathbb{Z}$ is a PID, then $B/t(B)$ is torsion-free hence flat. Then by the Tor-Criterion-for-Flat Modules Prop, we have $\mathrm{Tor}_n(A, B/t(B)) = 0$ for all $n \geq 1$. Hence, we can get a long exact

$$0 \to \mathrm{Tor}_1(A, t(B)) \to \mathrm{Tor}_1(A, B) \to 0$$

Therefore, we're done. □

Similarly, by the symmetry of Tor, we can prove that $\mathrm{Tor}_1(t(A), t(B)) \cong \mathrm{Tor}_1(A, t(B))$. □

Corollary 3.3.8. *of $Tor_1^{\mathbb{Z}}$-Torsion Proposition*
If A is an abelian group, then A is torsion-free iff $Tor_1(A, B) = 0$ for every abelian group B (in which case A is flat as $\mathbb{Z}$-module).

Proof.

- ($\Rightarrow$): A is torsion-free $\Rightarrow \mathrm{Tor}_1(A, B) \cong \mathrm{Tor}_1(t(A), B) = 0$.
- ($\Leftarrow$): Let $B = \mathbb{Q}/\mathbb{Z}$. Then by assumption, $\mathrm{Tor}_1(A, \mathbb{Q}/\mathbb{Z}) = 0$. By the previous example, we have $t(A) \cong \mathrm{Tor}_1(A, \mathbb{Q}/\mathbb{Z})$ and then we're done.

□

Remark: We can see in proof of the $\mathrm{Tor}_1^{\mathbb{Z}}$-Torsion Prop and its coro that they hold for any PID R in place of $\mathbb{Z}$.

3.4 Abelian Category

3.4.1 Definitions

Intuition: For (co)chain complexes, we need to do some operations on them, so we introduce the notion of abelian category to strip away the specific forms of elements while retaining all the "good" properties of the module category $\mathcal{M}od_R$.

Motivation: We construct the abelian category through the following steps:

$$\begin{array}{c} \text{Category} \\ \Big\downarrow{\scriptstyle \text{Hom}(\cdot,\cdot) \text{ with abelian group structure}} \\ \text{Preadditive Category/Ab-Category} \\ \Big\downarrow{\scriptstyle \text{Zero Objects \& Product}} \\ \text{Additive Category} \\ \Big\downarrow{\scriptstyle \text{kernel \& cokernel and monic \& epi}} \\ \text{Abelian Category} \end{array}$$

Definition 3.4.1. Preadditive Categories/Ab-Categories
A category $\mathcal{A}$ is called a *preadditive category* or an *Ab-category* if the followings are satisfied:

1. $\forall A, B \in \text{Ob}(\mathcal{A})$, $\text{Hom}_{\mathcal{A}}(A, B)$ is an abelian group.
2. The composition of morphisms are additively bilinear, i.e.: for any $f, f' : A \to B$, any $g, g' : B \to C$ where $A, B, C \in \text{Ob}(\mathcal{A})$, we have

$$(g + g') \circ (f + f') = g \circ f + g \circ f' + g' \circ f + g' \circ f'$$

Example: $\mathcal{AbGroup}, \mathcal{M}od_R, {}_R\mathcal{M}od, \mathcal{C}h_R, \mathcal{S}h(X)$.

Definition 3.4.2. Additive Functors
An additive functor $F : \mathcal{B} \to \mathcal{A}$ of Ab-categories is a functor such that each $\text{Hom}_{\mathcal{B}}(B,' B) \to \text{Hom}_{\mathcal{A}}(F(B), F(B'))$ is a group homomorphism.

Definition 3.4.3. Additive Categories
An *additive category* is an Ab-category $\mathcal{A}$ with

- an object in $\mathcal{A}$ is both initial and terminal, called a *zero object* of $\mathcal{A}$;
- a product $A \times B \in \text{Ob}(\mathcal{A})$ for every $A, B \in \text{Ob}(\mathcal{A})$.

Example: In $\mathcal{C}h_R$, we define the zero object and product as follows:

- $0_{\mathcal{C}h_R} := (\cdots \to 0 \xrightarrow{0} 0 \xrightarrow{0} 0 \to \cdots)$
- Given a family $\{A_\alpha\}$ of complexes of R-modules, we define the product $\Pi_\alpha A_\alpha$ and coproduct (direct sum) $\bigoplus_\alpha A_\alpha$ degreewise:
  - $(\Pi_\alpha A_\alpha)_n := \Pi_\alpha A_{\alpha,n}$, $(\Pi_\alpha d_\alpha)_n := \Pi_\alpha d_{\alpha,n} : (\Pi_\alpha A_\alpha)_n \to (\Pi_\alpha A_\alpha)_{n-1}$
  - $(\bigoplus_\alpha A_\alpha)_n := \bigoplus_\alpha A_{\alpha,n}$, $(\bigoplus_\alpha d_\alpha)_n := \bigoplus_\alpha d_{\alpha,n} : (\bigoplus_\alpha A_\alpha)_n \to (\bigoplus_\alpha A_\alpha)_{n-1}$

Then we make $\mathcal{C}h_R$ into an additive category $(\mathcal{C}h_R, 0_{\mathcal{C}h_R}, \Pi)$.

Definition 3.4.4. Abelian Categories
Before defining abelian category, we first define some prerequisite concepts: In an additive category $\mathcal{A}$,

- A *kernel* of a morphism $f : B \to C$ is defined to be a map $i : A \to B$ such that $f \circ i = 0$ and that is universal with respect to this property, i.e.: if there is any morphism $i' : A' \to B$ with $f \circ i' = 0$, then $\exists! g : A' \to A$, s.t.: $i' = i \circ g$.
- A *cokernel* of a morphism $f : B \to C$ is defined to be a map $e : C \to D$ which is universal with respect to having the property $e \circ f = 0$.
- A map $i : A \to B$ is *monic* if $i \circ g = 0$ implies $g = 0$ for every $g : A' \to A$.
- A map $e : C \to D$ is an *epi* if $h \circ e = 0$ implies $h = 0$ for every map $h : D \to D'$.

Now we can define abelian category: An *abelian category* is an additive category $\mathcal{A}$ such that the following three conditions are satisfied:

1. Every maps in $\mathcal{A}$ has a kernel and a cokernel.
2. Every monic map in $\mathcal{A}$ is the kernel of its cokernel.
3. Every epi in $\mathcal{A}$ is the cokernel of its kernel.

Definition 3.4.5. Abelian Subcategories
A *subcategory* $\mathcal{B}$ of $\mathcal{A}$ is called an *abelian subcategory* if it is abelian and any exact sequence in $\mathcal{B}$ is also exact in $\mathcal{A}$.

3.4.2 Chain Complexes in Abelian Categories

Motivation: If $\mathcal{A}$ is any abelian category, then we can define chain complexes in $\mathcal{A}$ by repeating the discussion of the def of chain complexes of modules, just replacing $\mathcal{M}od_R$ by $\mathcal{A}$.

Theorem 3.4.6. *Abelianness of Chain Complex Categories Theorem*
the category $\mathcal{C}h_{\mathcal{A}}$ is an abelian category.

Proof.

- Given any chain map(, i.e.: homomorphism of chain complexes) $f : B \to C$ in $\mathcal{C}h_{\mathcal{A}}$, since $\mathcal{A}$ is an abelian category, then every $f_n : B_n \to C_n$ in $\mathcal{A}$ has a kernel $i_n : \ker f_n \to B_n$. Note that the following diagram commutes:

$$\begin{tikzcd} B_n \arrow[r, "d_n^B"] \arrow[d, "f_n"] & B_{n-1} \arrow[d, "f_{n-1}"] \\ C_n \arrow[r, "d_n^C"] & C_{n-1} \end{tikzcd}$$

Hence, $f_{n-1} \circ d_n^B \circ i_n = d_n^C \circ f_n \circ i_n = d_n^C \circ 0 = 0$. By the universal property of i_{n-1}, $\exists! d_n^K : \ker f_n \to \ker f_{n-1}$, s.t.: $i_{n-1} \circ d_n^K = d_n^B \circ i_n$. Since by the universal property of i_{n-2}, we immediately have i_{n-2} is monic, then

$$i_{n-2} \circ d_{n-1}^K \circ d_n^K = d_{n-1}^B \circ i_{n-1} \circ d_n^K = d_{n-1}^B \circ d_n^B \circ i_n = 0$$

can imply that $d_{n-1}^K \circ d_n^K = 0$, so $K := \{\ker f_n, d_n^K\}$ is a chain complex and hence $i := \{i_n :$

$\ker f_n \to B_n\}$ is a chain map, which is a kernel of f. Dually, we can similarly get a cokernel e of f.

- **Lemma 3.4.7.** *$f : B \to C$ in $\mathcal{C}h_{\mathcal{A}}$ is monic $\Leftrightarrow$ $f_n : B_n \to C_n$ in $\mathcal{A}$ is monic.*

  Proof.
  - ($\Leftarrow$) is obvious.
  - ($\Rightarrow$): Note that f is monic $\Leftrightarrow$ $i =$ (the kernel of f) $= 0$. By the degreewise construction, $i = 0$ implies that $i_n = 0 \Leftrightarrow f_n$ are monic.

  ☐

  If f is monic, then by lemma, f_n are monic. Since $\mathcal{A}$ is an abelian category, then in $\mathcal{A}$ $f_n =$ the cokernel of i_n. Then degreewisely in $\mathcal{C}h_{\mathcal{A}}$, $f =$ the cokernel of i.

- Dually to the second step.

☐

3.4.3 Chain Complexes in the Category of Chain Complexes

Motivation (*Def of double complexes*): Consider a lattice

$$\begin{tikzcd}
 & \vdots \arrow[d] & \vdots \arrow[d] & \vdots \arrow[d] & \\
\cdots & C_{m-1,n+1} \arrow[l] \arrow[d, "d^v_{m-1,n+1}"'] & C_{m,n+1} \arrow[l, "d^h_{m,n+1}"'] \arrow[d, "d_{d^v_{m,n+1}}"'] & C_{m+1,n+1} \arrow[l, "d^h_{m+1,n+1}"'] \arrow[d, "d^v_{m+1,n+1}"'] & \cdots \arrow[l] \\
\cdots & C_{m-1,n} \arrow[l] \arrow[d, "d^v_{m-1,n}"'] & C_{m,n} \arrow[l, "d^h_{m,n}"'] \arrow[d, "d_{d^v_{m,n}}"'] & C_{m+1,n} \arrow[l, "d^h_{m+1,n}"'] \arrow[d, "d^v_{m+1,n}"'] & \cdots \arrow[l] \\
\cdots & C_{m-1,n-1} \arrow[l] \arrow[d] & C_{m,n+1} \arrow[l, "d^h_{m,n-1}"'] \arrow[d] & C_{m+1,n-1} \arrow[l, "d^h_{m+1,n-1}"'] \arrow[d] & \cdots \arrow[l] \\
 & \vdots & \vdots & \vdots &
\end{tikzcd}$$

where $C_{p,q} \in \mathrm{Ob}(\mathcal{A})$ and the maps

$$d^h_{p,q} : C_{p,q} \to C_{m-1,n} \text{ and } d^v_{p,q} : C_{p,q} \to C_{p,q-1}$$

are such that $d^h \circ d^h = 0, d^v \circ d^v = 0$ and $d^v \circ d^h + d^h \circ d^v = 0$, i.e.: each square anticommutes. (This lattice $\{C_{p,q}, d^h, d^v\}$ is called a *double complex* or *bicomplex* in $\mathcal{A}$.) But by the anticommutativity, d^v are not maps in $\mathcal{C}h_{\mathcal{A}}$:

Definition 3.4.8. Sign-Adjusted Vertical Differentials in Double Complexes
Let $\mathcal{A}$ be any abelian category and let $\{C_{p,q}, d^h, d^v\}$ be a double complex in $\mathcal{A}$. Define

$$f_{p,q} := (-1)^p d^v_{p,q} : C_{p,q} \to C_{p,q-1}$$

Remark:

- Now we get a chain map $\{f_{p,q}\}_q : C_{p,q} \to C_{p,q-1}$ for any fixed p.
- By this trick, we get an equivalence of two categories:

$$\mathcal{D}ouble\mathcal{C}h_{\mathcal{A}} =_{\mathrm{i}} \mathcal{C}h_{\mathcal{C}h_{\mathcal{A}}}$$

Definition 3.4.9. Total Complexes
Given a double complex $C = \{C_{p,q}, d^h, d^v\}$ in some abelian category $\mathcal{A}$, define the *total complexes* (two types $\mathrm{Tot}^{\Pi}(C)$ and $\mathrm{Tot}^{\oplus}(C)$) by

$$\mathrm{Tot}^{\Pi}(C)_n := \prod_{p+q=n} C_{p,q} \text{ and } \mathrm{Tot}^{\oplus}(C)_n := \bigoplus_{p+q=n} C_{p,q}$$

together with the *total differential* $d := d^h + d^v$.

Remark:

- *Check:* Total differential is a differential, i.e.: $d \circ d = 0$:

$$d \circ d := (d^h + d^v) \circ (d^h + d^v) \xlongequal{\text{additive category}} d^h \circ d^h + d^h \circ d^v + d^v \circ d^h + d^v \circ d^h \xlongequal{\text{def of double cplx}} 0$$

- When C is bounded, i.e.: C has only finitely many nonzero terms along each diagonal line $p + q = n$, we have:

$$\mathrm{Tot}^{\Pi}(C)_n \cong \mathrm{Tot}^{\oplus}(C)_n$$

- $\mathrm{Tot}^{\Pi}(C)$ and $\mathrm{Tot}^{\oplus}(C)$ do *not* exist in all abelian categories, for example, they don't exist when $\mathcal{A} = \mathcal{A}b\mathcal{G}roup_{\mathrm{fin}}$ is the category of all finite abelian groups since $\bigoplus_{i=1}^{\infty} \mathbb{Z}/2\mathbb{Z} \notin \mathcal{A} = \mathcal{A}b\mathcal{G}roup_{\mathrm{fin}}$.
  To ensure the well-defined-ness of total complex, we hence need to impose higher completeness requirements on the abelian category $\mathcal{A}$:
  - $\mathcal{A}$ is *complete* if all infinite direct products exist.
  - $\mathcal{A}$ is *cocomplete* if all infinite direct sums exist.

3.4.4 Operations on Chain Complexes: Trunctions and Translations

Truncations

Motivation: When studying a complicated chain complex, we sometimes are only interested in the information that is above or below a certain degree n, so we need a way to "sut off" the chain complex while trying not to destroy its homology groups.

Definition 3.4.10. Good Truncations / Canonical Truncations of Chain Complexes
Let C be a chain complex and let n be an integer. Then we define

- the *good truncation of* C *below* n by the subcomplex of C

$$(\tau_{\geq n}C)_i := \begin{cases} 0 & \text{, if } i < n \\ \ker(d_n : C_n \to C_{n-1}) & \text{, if } i = n \\ C_i & \text{, if } i > n \end{cases}$$

  i.e.: $\tau_{\geq n}C := (\cdots \to C_{n+2} \xrightarrow{d_{n+2}} C_{n+1} \xrightarrow{d_{n+1}} \ker d_n \xrightarrow{d_n} 0 \to \cdots);$
- the *good truncation of* C *above* n by the quotient complex $\tau_{<n}C := C/(\tau_{\geq n}C)$.

Remark: This truncation is "good" since it perfectly retains all the homology groups on one side of the "incision", clears all the homology on the other side of the "incision" and precisely retains the homology at the "incision", while we have an extremely brutal and straightforward approach: simply cut the "incision" and one side of it into zero:

Definition 3.4.11. Brutal Truncations of Chain Complexes

$$(\sigma_{<n}C)_i := \begin{cases} C_i & \text{, if } i < n \\ 0 & \text{, if } i \geq n \end{cases} \text{ and } \sigma_{\geq n}C := C/(\sigma_{<n}C)$$

Remark: This truncation "destroys" the homology at the "incision" but it's form is simple.

Translation

Definition 3.4.12. Translation of Chain Complexes
Let C be a chain complex in an abelian category $\mathcal{A}$ and let $p \in \mathbb{Z}$. The *p-th translate* (or *shift*) of C, denoted as $C[p]$, is defined by the chain complex:

$$(C[p])_n := C_{n+p}, \forall n \in \mathbb{Z} \text{ together with } d_n^{C[p]} := (-1)^p d_{n+p}^C : (C[p])_n \to (C[p])_{n-1}$$

Similarly for cochain complex.

Remark: Why is the sign factor $(-1)^p$? Similarly as the first rank of the definition of total complexes.

Definition 3.4.13. Translation of Chain Maps
Let $f : B \to C$ be a homomorphism of chain complexes and let $p \in \mathbb{Z}$. The *p-th translate of* f, denoted as $f[p] : B[p] \to C[p]$, is defined degreewise by:

$$(f[p])_n := f_{n+p} : B_{n+p} \to C_{n+p}$$

Proposition 3.4.14. *Translation-Properties Proposition*
Let $\mathcal{C}h_{\mathcal{A}}$ be the category of chain complexes and let $p \in \mathbb{Z}$.

1. *Functoriality: We have an endofunctor $[p] : \mathcal{C}h_{\mathcal{A}} \to \mathcal{C}h_{\mathcal{A}}$ defined by $C \mapsto C[p]$ and $f \mapsto f[p]$.*
2. *Category Automorphism: The functor $[p]$ is an isomorphism of categories, with inverse $[-p]$:*

$$[p] \circ [-p] = id_{\mathcal{C}h_{\mathcal{A}}} = [-p] \circ [p]$$

3. *Homology Shift: $H_n(C[p]) \cong H_{n+p}(C), \forall n \in \mathbb{Z}$.*

3.5 Derived Functors

3.5.1 δ-Functors

Definition 3.5.1. δ-Functors
A (cocariant) *homological* or *cohomological δ-functor* between $\mathcal{A}$ and $\mathcal{B}$ is a collection of additive functors $T_n : \mathrm{Ob}(\mathcal{A}) \to \mathrm{Ob}(\mathcal{B})$ respectively $T^n : \mathrm{Ob}(\mathcal{A}) \to \mathrm{Ob}(\mathcal{B})$ for $n \in \mathbb{N} \cup \{0\}$, together with morphisms

$$\delta_n : T_n(C) \to T_{n-1}(A) \text{ respectively } \delta^n : T^n(C) \to T^{n+1}(A)$$

defined for each short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$, where we make the convention that $T_n = 0, T^n = 0$ for $n < 0$, satisfies the following conditions:

- For any short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$, there is a long exact sequence

  $$\cdots \to T_{n+1}(C) \xrightarrow{\delta_{n+1}} T_n(A) \to T_n(B) \to T_n(C) \xrightarrow{\delta_n} T_{n-1}(A) \to \cdots$$

  respectively

  $$\cdots \to T^{n-1}(C) \xrightarrow{\delta^{n-1}} T^n(A) \to T^n(B) \to T^n(C) \xrightarrow{\delta^n} T^{n+1}(A) \to \cdots$$

  In particular, T_0 is right exact and T^0 is left exact.
- For each morphism of short exact sequences from $0 \to A' \to B' \to C' \to 0$ to $0 \to A \to B \to$

$C \to 0$, the following diagram commutes:

$$\begin{array}{ccc} T_n(C') & \xrightarrow{\delta_n} & T_{n-1}(A') \\ \downarrow & & \downarrow \\ T_n(C) & \xrightarrow{\delta_n} & T_{n-1}(A) \end{array} \quad \text{respectively} \quad \begin{array}{ccc} T^n(C') & \xrightarrow{\delta^n} & T^{n+1}(A') \\ \downarrow & & \downarrow \\ T^n(C) & \xrightarrow{\delta^n} & T^{n+1}(A) \end{array}$$

Definition 3.5.2. Morphisms of δ-Functors
A *morphism* $f : S \to T$ of δ-functors is a system of natural transformations $f_n : S_n \to T_n$ (respectively $f^n : S^n \to T^n$) that commute with δ, i.e.: for any short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$, the following diagram commutes:

$$\begin{array}{ccc} S_n(C) & \xrightarrow{\delta_n^S} & S_{n-1}(A) \\ \downarrow{\scriptstyle f_n(C)} & & \downarrow{\scriptstyle f_{n-1}(A)} \\ T_n(C) & \xrightarrow{\delta_n^T} & T_{n-1}(A) \end{array} \quad \text{respectively} \quad \begin{array}{ccc} S^n(C) & \xrightarrow{\delta_S^n} & S^{n+1}(A) \\ \downarrow{\scriptstyle f^n(C)} & & \downarrow{\scriptstyle f^{n+1}(A)} \\ T^n(C) & \xrightarrow{\delta_T^n} & T^{n+1}(A) \end{array}$$

Definition 3.5.3. Universal δ-Functors

- A homological δ-functor T is *universal* if given any δ-functor S and a natural transformation $f_0 : S_0 \to T_0$, there is a unique morphism $\{f_n : S_n \to T_n\}$ of δ-functors that extend f_0.
- A cohomological δ-functor T is *universal* if given any S and $f^0 : T^0 \to S^0$, there is a unique morphism $\{f^n : T^n \to S^n\}$ of δ-functors that extend f^0.

3.5.2 Left Derived Functors

Intuition: Let $F : \mathcal{A} \to \mathcal{B}$ be a *right exact functor* between two abelian categories, i.e.: F is an additive functor between $\mathcal{A}$ and $\mathcal{B}$ satisfying that for any short exact sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ in $\mathcal{A}$, applying F on it we can get an exact sequence

$$F(A) \xrightarrow{F(f)} F(B) \xrightarrow{F(g)} F(C) \to 0$$

For instance, $\cdot \otimes_R D : \mathcal{M}od_R \to \mathcal{M}od_R$ is a right exact functor in category of modules. But note that applying F may lose the information on the left, namely $\ker f$.

Construction and Well-defined-ness

Definition 3.5.4. Left Derived Functors
Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor between two abelian categories, let $A \in \mathrm{Ob}(\mathcal{A})$ be an object of $\mathcal{A}$ and take a projective resolution

$$\cdots \to P_n \xrightarrow{d_n} P_{n-1} \to \cdots \to P_1 \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} A \to 0$$

of A. Consider a chain complex $P := (\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \to 0)$. Then we define the *left derived functors* L_iF $(i \in \mathbb{N} \cup \{0\})$ of F by the homology groups

$$L_iF(A) := H_i(F(P)) = \ker F(d_i)/\text{im } F(d_{i+1})$$

Proposition 3.5.5. 0-*th Left Derived Functor Proposition*
Given any $F : \mathcal{A} \to \mathcal{B}$ right exact functor between two abelian categories and any $A \in Ob(\mathcal{A})$, we always have

$$L_0F(A) \cong F(A)$$

Proof. By the def of right exact functors, we have an exact sequence

$$\cdots \to F(P_1) \xrightarrow{F(d_1)} F(P_0) \xrightarrow{F(\epsilon)} F(A) \to 0$$

Hence, we get: $(*) : \ker F(\epsilon) = \text{im } F(d_1)$ and $(**) : F(\epsilon)$ is surjective. Therefore,

$$L_0F(A) := F(P_0)/\text{im } F(d_1) \overset{(*)}{=\!=} F(P_0)/\ker F(\epsilon) \overset{\text{1st isom thm}}{\cong} \text{im } F(\epsilon) \overset{(**)}{\cong} F(A)$$

$\square$

Lemma 3.5.6. *Comparison Theorem*
Let M, N be objects of an abelian category $\mathcal{A}$, take P to be a projective resolution of M and let $f' \in Hom_{\mathcal{A}}(M, N)$. Then for every projective resolution Q of N, there is a chain map $f : P \to Q$ lifting f' in the sense that $\epsilon_Q \circ f_0 = f' \circ \epsilon_P$. The chain map f is unique up to chain homotopy.

$$\begin{tikzcd}
\cdots \arrow[r] & P_n \arrow[r] \arrow[d, "f_n"] & P_{n-1} \arrow[r] \arrow[d, "f_{n-1}"] & \cdots \arrow[r] & P_1 \arrow[r] \arrow[d, "f_1"] & P_0 \arrow[r, "\epsilon_P"] \arrow[d, "f_0"] & M \arrow[r] \arrow[d, "f'"] & 0 \\
\cdots \arrow[r] & Q_n \arrow[r] & Q_{n-1} \arrow[r] & \cdots \arrow[r] & Q_1 \arrow[r] & Q_0 \arrow[r, "\epsilon_Q"] & N \arrow[r] & 0
\end{tikzcd}$$

Remark: The Comparison Theorem is the general version of the Induced Maps on Ext-and-Homotopy Independence Lemma.

Proposition 3.5.7. *Well-defined-ness of Left Derived Functors Proposition*
The objects $L_iF(A)$ of $\mathcal{B}$ are well-defined up to natural isomorphism, namely, if we take another projective resolution

$$\cdots \to Q_n \xrightarrow{d_n^Q} Q_{n-1} \to \cdots \to Q_1 \xrightarrow{d_1^Q} Q_0 \xrightarrow{\epsilon^Q} A \to 0$$

of A with $Q = (\cdots \to Q_2 \xrightarrow{d_2^Q} Q_1 \xrightarrow{d_1^Q} Q_0 \to 0)$, then there is a canonical isomorphism

$$L_iF(A) = H_i(F(P)) \xrightarrow{\cong} H_i(F(Q))$$

where "canonical" means that uniquely determined and independent of choice of the projective resolutions. In particular, a different choice of would yield new functors $\hat{L}_iF$, which are naturally isomorphic to L_iF.

Proof. By the Comparison Theorem, there is a chain map $f : P \to Q$ lifting the identity map $f' = \mathrm{id}_A$, yielding a map $f_i : H_iF(P) \to H_iF(Q)$, and any other such chain map $\hat{f} : P \to Q$ is a chain homotopic to f, so $f_i = \hat{f}_i$. Hence, the map f_i is canonical. Similarly, there is a chain map $g : Q \to P$ lifting id_A and a map g_i. Since $g \circ f$ and id_P are both chain map $P \to P$ lifting id_A, then we have

$$g_i \circ f_i = (g \circ f)_i = \mathrm{id}_{H_iF(P)}$$

and similarly $f_i \circ g_i = \mathrm{id}_{H_iF(Q)}$, which shows that f_i and g_i are isomorphisms. □

Remark: This prop shows that no matter which resolution we choose, the resulted homology groups are the same in the sense of isomorphism, so we have the well-defined-ness of our construction of left derived functors $L_iH(A)$.

Corollary 3.5.8. *of Well-defined-ness of Left Derived Functors Proposition*
If A is projective, thn $L_iF(A) = 0$ for $i \neq 0$.

Proof. By the Well-defined-ness of Left Derived Functors Prop, we can focus on the trivial projective resolution $0 \to A \xrightarrow{\mathrm{id}_A} A \to 0$ of A, so for $i \neq 0$, we have

$$L_iF(A) := H_i(F(P)) = \ker\ (0 \to 0)/\mathrm{im}\ (0 \to 0) = 0$$

□

Functoriality

Lemma 3.5.9. *Induction-of-Morphisms-on-Left Derived Functors Lemma*
For any map $f : A' \to A$ in any abelian category $\mathcal{A}$, there is induced map $L_iF(f) : L_iF(A') \to L_iF(A)$ for each i.

Proof. Let P, P' be projective resolutions of A respectively A'. By thw Comparison Theorem, there is a lift f of a chain map $\tilde{f} : P' \to P$, hence a map $\tilde{f}_i : H_i(F(P')) \to H_i(F(P))$, and the map $\tilde{f}_i$ is independent of the choice of f. Then $L_iF(f) := \tilde{f}_i$. □

Proposition 3.5.10. *Additivity-of-Left Derived Functors Proposition*
Each L_iF is an additive functor from $\mathcal{A}$ to $\mathcal{B}$.

Proof.
- To prove the functoriality of L_iF, by the Induction-of-Morphisms-on-Left Derived Functors Lemma, it suffices to show that L_iF preserves identity morphism and composition:
  - $L_iF(\mathrm{id}_A) := (\mathrm{id}_P)_i = \mathrm{id}_{L_iF(A)}$

- Given $A' \xrightarrow{f} A \xrightarrow{g} A''$ with $A, A', A'' \in \mathrm{Ob}(\mathcal{A})$, f, g morphisms, take their projective resolutions P, P', P'' and chain maps $\tilde{f} : P' \to P', \tilde{g} : P \to P''$ lifting f, g. Then $\tilde{g} \circ \tilde{f} : P' \to P''$ lifts $g \circ f$, so

$$L_iF(g \circ f) = (\tilde{g} \circ \tilde{f})_i = \tilde{g}_i \circ \tilde{f}_i = L_iF(g) \circ L_iF(f)$$

- $\forall f_1, f_2 \in \mathrm{Hom}_{\mathcal{A}}(A, A')$, where $A, A' \in \mathrm{Ob}(\mathcal{A})$ are with projective resolutions P respectively P', by the Comparison Theorem, there are chain maps $\tilde{f}_1, \tilde{f}_2 : P' \to P$ lifting f_1 respectively f_2. Then
$$(\tilde{f}_1 + \tilde{f}_2)_i \overset{\text{def}}{=\!=} (\tilde{f}_1)_i + (\tilde{f}_2)_i : P'_i \to P_i$$
is a well-defined chain map where $\tilde{f}_1 + \tilde{f}_2$ lifts $f_1 + f_2$. Applying the additive functor F, we can get $F(\tilde{f}_1 + \tilde{f}_2) = F(\tilde{f}_1) + F(\tilde{f}_2)$. Hence,

$$(F(\tilde{f}_1) + F(\tilde{f}_2))_i = F(\tilde{f}_1)_i + F(\tilde{f}_2)_i$$

□

Theorem 3.5.11. *Long Exact Sequence-in-Left Derived Functors Theorem*
The left derived functors L_iF form a homological δ-fuctor.

Proof. Given a short sequence $0 \to A' \to A \to A'' \to 0$ in $\mathcal{A}$ with projective resolutions P' of A' and P'' of A''. By the general version of Horseshoe Lemma, there is a projective resolution P of A with a short exact sequence $0 \to P' \to P' \to P'' \to 0$ and each short exact sequence $0 \to P'_n \to P_n \to P''_n \to 0$ is split. Since F is an additive functor, then applying F on a split exact sequence still can retain the splitness and exactness, i.e.:each sequence $0 \to F(P'_n) \to F(P_n) \to F(P''_n) \to 0$ is split exact in $\mathcal{B}$. Therefore, $0 \to F(P') \to F(P) \to F(P'') \to 0$ is a short exact sequence of chain complexes. Then by the dual version for homology of the Long Exact Sequence-in-Cohomology Theorem, there is a long exact homology sequence

$$\cdots \xrightarrow{\partial_{i+1}} L_iF(A') \to L_iF(A) \to L_iF(A'') \xrightarrow{\partial_i} L_{i-1}F(A') \to L_{i-1}F(A) \to L_{i-1}F(A'') \xrightarrow{\partial_{i-1}} \cdots$$

with corresponding connecting homomorphisms.

Not yet, we still need to prove that the connecting homomorphisms ∂_n are natural in A, that is, if we have two short exact sequences and a morphism between them:

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "i_A"] \arrow[d, "f'"] & A \arrow[r, "\pi_A"] \arrow[d, "f"] & A'' \arrow[r] \arrow[d, "f''"] & 0 \\
0 \arrow[r] & B' \arrow[r, "i_B"] & B \arrow[r, "\pi_B"] & B'' \arrow[r] & 0
\end{tikzcd}$$

then the blocks corresponding to the long exact sequence they induce must be commutative:

$$\begin{array}{ccc} L_iF(A'') & \xrightarrow{\partial_i^A} & L_{i-1}F(A') \\ \downarrow{\scriptstyle L_iF(f'')} & & \downarrow{\scriptstyle L_{i-1}F(f')} \\ L_iF(B'') & \xrightarrow{\partial_i^B} & L_{i-1}F(B') \end{array}$$

Consider the projective resolutions P' of A', P'' of A'', Q' of B' and Q'' of B'' with $\epsilon' : P_0' \to A', \epsilon'' : P_0'' \to A'', \eta' : Q_0' \to B'$ and $\eta'' : Q_0'' \to B''$ and use the general case of the Horseshoe Lemma to get the projective resolutions P of A and Q of B with $\epsilon : P_0 \to A$ and $\eta : Q_0 \to B$. Then by the Comparison Theorem, we have chain map $g' : P' \to Q'$ and $g'' : P'' \to Q''$ lifting the maps f' respectively f''.

Now we need to show that there is a chain map $g : P \to Q$ lifting f and giving a commutative diagram of chain complexes with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & P' & \longrightarrow & P & \longrightarrow & P'' & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle g''} & & \\ 0 & \longrightarrow & Q' & \longrightarrow & Q & \longrightarrow & Q'' & \longrightarrow & 0 \end{array}$$

which shall give the naturality of ∂_n and finish the proof:

In order to produce g, we will construct maps (*not* chain maps) $\gamma_n : P_n'' \to Q_n'$ such that g_n is

$$g_n = \begin{pmatrix} g_n' & \gamma_n \\ 0 & g_n'' \end{pmatrix} : P_n = P_n' \oplus P_n'' \to Q_n = Q_n' \oplus Q_n'' \text{ given by } g_n(p', p'') = (g_n'(p') + \gamma_n(p''), g''(p''))$$

We first construct γ_0: Note that for our desired form, $g_0|_{P_0'} = g_0'$ since γ_0 is on P_0'', so we only focus on $g_0|_{P_0''}$. Let $\lambda_P'' : P_0'' \to A$ and $\lambda_Q : Q_0'' \to B$ be the restrictions of ϵ respectively η, i.e.: $\lambda_P := \epsilon|_{P_0''}$ and $\lambda_Q := \eta|_{Q_0''}$. Note that

$$\begin{aligned} \pi_B \circ (f \circ \lambda_P - \lambda_Q \circ g_0'') &= \pi_B \circ f \circ \lambda_P - \pi_B \circ \lambda_Q \circ g_0'' \\ &= f'' \circ \pi_A \circ \lambda_P - \eta'' \circ g_0'' \\ &= f'' \circ \epsilon'' - \eta'' \circ g_0'' = 0 \end{aligned}$$

Since also $\pi_B \circ i_B = 0$, then by the universal property of kernel, $\exists! \beta_0 : P_0'' \to B'$, s.t.: $i_B \circ \beta_0 = f \circ \lambda_P - \lambda_Q \circ g_0''$. Therefore, by the projectivity of P_0'', we can define γ_0 to be a lift of β_0 to Q_0':

$$\begin{tikzcd} & P_0'' \arrow[dl, dashed, "\gamma_0"'] \arrow[d, "\beta_0"] & \\ Q_0' \arrow[r, "\eta'"] & B' \arrow[r] & 0 \end{tikzcd}$$

In order for g to be a chain map, we must have the following commutative diagram

$$\begin{tikzcd} P_n \arrow[r, "d_n^P"] \arrow[d, "g_n"] & P_{n-1} \arrow[d, "g_{n-1}"] \\ Q_n \arrow[r, "d_n^Q"] & Q_{n-1} \end{tikzcd}$$

Similarly for defining λ_P and λ_Q, we define $\lambda_n^P : P_n'' \to P_{n-1}'$ and $\lambda_n^Q : Q_n'' \to Q_{n-1}'$ by

$$d_n^P = \begin{pmatrix} d_n^{P'} & \lambda_n^P \\ 0 & d_n^{P''} \end{pmatrix} : P_n = P_n' \oplus P_n'' \to P_{n-1} = P_{n-1}' \oplus P_{n-1}''$$
$$d_n^Q = \begin{pmatrix} d_n^{Q'} & \lambda_n^Q \\ 0 & d_n^{Q''} \end{pmatrix} : Q_n = Q_n' \oplus Q_n'' \to Q_{n-1} = Q_{n-1}' \oplus Q_{n-1}''$$

of which easily to prove the existence. Then the commutative diagram above can be "translated" as

$$\begin{aligned} 0 = d_n^Q \circ g_n - g_{n-1} \circ d_n^P &= \begin{pmatrix} d_n^{Q'} & \lambda_n^Q \\ 0 & d_n^{Q''} \end{pmatrix} \circ \begin{pmatrix} g_n' & \gamma_n \\ 0 & g_n'' \end{pmatrix} - \begin{pmatrix} g_{n-1}' & \gamma_{n-1} \\ 0 & g_{n-1}'' \end{pmatrix} \circ \begin{pmatrix} d_n^{P'} & \lambda_n^P \\ 0 & d_n^{P''} \end{pmatrix} \\ &= \begin{pmatrix} d_n^{Q'} \circ g_n' - g_{n-1}' \circ d_n^{P'} & d_n^{Q'} \circ \gamma_n - \gamma_{n-1} \circ d_n^{P''} + \lambda_n^Q \circ g_n'' - g_{n-1}' \circ \lambda_n^P \\ 0 & d_n^{Q''} \circ g_n'' - g_{n-1}'' \circ d_n^{P''} \end{pmatrix} \\ &= \begin{pmatrix} 0 & d_n^{Q'} \circ \gamma_n - \gamma_{n-1} \circ d_n^{P''} + \lambda_n^Q \circ g_n'' - g_{n-1}' \circ \lambda_n^P \\ 0 & 0 \end{pmatrix} \end{aligned}$$

To guarentee this, we need: $d_n^{Q'} \circ \gamma_n = \gamma_{n-1} \circ d_n^{P''} - \lambda_n^Q \circ g_n'' + g_{n-1}' \circ \lambda_n^P$. Since we've already obtained γ_0, then we can construct γ_n, inductively: Suppose γ_i defined for $i < m$, and denote $h_n := \gamma_{n-1} \circ d_n^{P''} - \lambda_n^Q \circ g_n'' + g_{n-1}' \circ \lambda_n^P$. Note that by the def of λ_k^P, λ_k^Q, we have

$$\begin{aligned} 0 = d_{n-1}^P \circ d_n^P &= \begin{pmatrix} d_{n-1}^{P'} & \lambda_{n-1}^P \\ 0 & d_{n-1}^{P''} \end{pmatrix} \circ \begin{pmatrix} d_n^{P'} & \lambda_n^P \\ 0 & d_n^{P''} \end{pmatrix} = \begin{pmatrix} d_{n-1}^{P'} \circ d_n^{P'} & d_{n-1}^{P'} \circ \lambda_n^P + \lambda_{n-1}^P \circ d_n^{P''} \\ 0 & d_{n-1}^{P''} \circ d_n^{P''} \end{pmatrix} \\ &= \begin{pmatrix} 0 & d_{n-1}^{P'} \circ \lambda_n^P + \lambda_{n-1}^P \circ d_n^{P''} \\ 0 & 0 \end{pmatrix} \end{aligned}$$

Hence, we have

$$d_{n-1}^{P'} \circ \lambda_n^P = -\lambda_{n-1}^P \circ d_n^{P''} \text{ and similarly } d_{n-1}^{Q'} \circ \lambda_n^Q = -\lambda_{n-1}^Q \circ d_n^{Q''} \tag{*}$$

Therefore, we see that:

$$
\begin{aligned}
d_{n-1}^{Q'} \circ h_n &:= d_{n-1}^{Q'} \circ \gamma_{n-1} \circ d_n^{P''} - d_{n-1}^{Q'} \circ \lambda_n^Q \circ g_n'' + d_{n-1}^{Q'} \circ g_{n-1}' \circ \lambda_n^P \\
&\overset{1}{=} h_{n-1} \circ d_n^{P''} + \lambda_{n-1}^Q \circ d_n^{Q''} \circ g_n'' + g_{n-2}' \circ d_{n-1}^{P'} \circ \lambda_n^P \\
&\overset{2}{=} \gamma_{n-2} \circ d_{n-1}^{P''} \circ d_n^{P''} - \lambda_{n-1}^Q \circ g_{n-1}'' \circ d_n^{P''} + g_{n-2}' \circ \lambda_n^P \circ d_n^{P''} \\
&\qquad\qquad\qquad\qquad + \lambda_{n-1}^Q \circ d_n^{Q''} \circ g_n'' - g_{n-2}' \circ \lambda_n^P \circ d_n^{P''} \\
&\overset{3}{=} 0
\end{aligned}
$$

where the explanation for each equation is:

1. By the inductive hypothesis, we have: $d_{n-1}^{Q'} \circ \gamma_{n-1} = g_{n-1}$; by $(*)$, we have $d_{n-1}^{Q'} \circ \lambda_n^Q = -\lambda_{n-1}^Q \circ d_n^{Q''}$; since g' is a chain map, then we have $d_{n-1}^{Q'} \circ g_{n-1}' = g_{n-2}' \circ d_{n-1}^{P'}$.
2. By $(*)$, we have $d_{n-1}^{P'} \circ \lambda_n^P = -\lambda_n^P \circ d_n^{P''}$; by the def of h_{n-1}, $h_{n-1} = \gamma_{n-2} \circ d_{n-1}^{P''} - \lambda_{n-1}^Q \circ g_{n-1}'' + g_{n-2}' \circ \lambda_n^P$.
3. $d_{n-1}^{P''} \circ d_n^{P''} = 0$; since g'' is a chain map, then we have $g_{n-1}'' \circ d_n^{P''} = d_n^{Q''} \circ g_n''$.

Therefore, $\text{im } g_n \subseteq \ker d_{n-1}^{Q'} = \text{im } d_n^{Q'}$, so $\exists! \beta_n : P_n'' \to \text{im } d_n^{Q'}$, s.t.: the following diagram commutes:

$$
\begin{tikzcd}
 & \text{im } d_n^{Q'} \arrow[d, hook] \\
P_n'' \arrow[ur, "\exists! \beta_n"] \arrow[r, "h_n"] & Q_{n-1}'
\end{tikzcd}
$$

and we then define γ_n to be any lift of β_n to Q_n'. □

Dimension Shifting / Syzygy Theory

Intuition: The calculation for $L_iF(A)$ is complicated when taking projective resolutions, so we introduce a way to calculating $L_iF(A)$.

Definition 3.5.12. Acyclic Objects
Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor between two abelian categories. An object $Q \in \text{Ob}(\mathcal{A})$ is called *F-acyclic* if $L_iF(Q) = 0$ for all $i \neq 0$.

Example:

- By the coro of the Well-defined-ness of Left Derived Functors Prop, projectives are F-acyclic for every right exact functor F.
- Flat modules are acyclic for tensor products.

Definition 3.5.13. Acyclic Resolutions
An *F-acyclic resolution* of A is a left resolution $\cdots \to Q_2 \to Q_1 \to Q_0 \to A \to 0$ for each Q_i is F-acyclic.

Proposition 3.5.14. *One-Step Dimension Shifting Proposition*
Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor between abelian categories. If $0 \to M \to P \to A \to 0$ is an exact sequence in $\mathcal{A}$ with P F-acyclic (e.g. projective), then

1. *$L_iF(A) \cong L_{i-1}F(M)$, for all $i \geq 2$, and*
2. *$L_1F(A) \cong \ker(F(M) \to F(P))$.*

Proof. By the Long Exact Sequence-in-Left Derived Functors Theorem, apply L_kF to the short exact sequence $0 \to M \to P \to A \to 0$ to get a long exact sequence

$$\cdots \to L_iF(P) \to L_iF(A) \xrightarrow{\partial} L_{i-1}F(M) \to L_{i-1}F(P) \to \cdots$$

Since P is F-acyclic, then $L_kF(P) = 0$ for all $k \geq 1$.

- For $i \geq 2$, the exactness of sequence $0 \to L_iF(A) \xrightarrow{\partial} L_{i-1}F(M) \to 0$ gives

$$L_iF(A) \cong L_{i-1}F(M)$$

- For $i = 1$, the sequence $0 \to L_1F(A) \to L_0F(M) \to L_0F(P)$ is exact. By the 0-th Left Derived Functor Prop, we have an exact sequence $0 \to L_1F(A) \to F(M) \to F(P)$ and hence we're done.

□

Theorem 3.5.15. *General Dimension Shifting*
Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact functor between categories. If $0 \to M_m \to P_m \to P_{m-1} \to \cdots \to P_0 \to A$ is exact with P_j F-acyclic, then

1. *$L_iF(A) \cong L_{i-m-1}F(M_m)$ for all $i \geq m+2$, and*
2. *$L_{m+1}F(A) \cong \ker(F(M_m) \to F(P_m))$*

Proof. Consider $K_0 := \ker(P_0 \to A)$ and $K_j := \ker(P_j \to P_{j-1})$ for $1 \leq j \leq m-1$. Then we have the short exact sequences

$$\begin{aligned} &0 \to K_0 \to P_0 \to A \to 0 \\ &0 \to K_j \to P_j \to K_{j-1} \to 0 \text{ for } 1 \leq j \leq m-1 \\ &0 \to M_m \to P_m \to K_{m-1} \to 0 \end{aligned}$$

Apply the One-Step Dimension Shifting Prop.1 to these short exact sequences and then we get

$$\begin{aligned} &L_iF(A) \cong L_{i-1}F(K_0) \text{ for } i \geq 2 \\ &L_iF(K_0) \cong L_{i-2}F(K_1) \text{ for } i-1 \geq 2 \\ &\quad\vdots \\ &L_{i-j}F(K_{j-1}) \cong L_{i-j-1}F(K_j) \text{ for } i-j \geq 2 \end{aligned}$$

Hence, iteratively, $L_iF(A) \cong L_{i-1}F(K_0) \cong \cdots \cong L_{i-m}F(K_{m-1}) \cong L_{i-(m+1)}F(M_m)$ for $i - m \geq 2 \Leftrightarrow i \geq m+2$. For $i = m+1$, we have $L_{m+1}F(A) \cong L_1F(K_{m-1})$. On the last step, apply the One-Step Dimension Shifting Prop.2 to the short exact sequence $0 \to M_m \to P_m \to K_{m-1} \to 0$ and then we get

$$L_1F(K_{m-1}) \cong \ker(F(M_m) \to F(P_m))$$

□

Definition 3.5.16. Syzygy
The object M_m is called the *m-th syzygy* of A.

Theorem 3.5.17. *Acyclic Resolution Theorem*
Let $F : \mathcal{A} \to \mathcal{B}$ be a right exact sequence between abelian categories. If $\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} A \to 0$ is a F-acyclic resolution of A, then for any $i \geq 0$,

$$L_iF(A) \cong H_i(F(P))$$

where $P = (\cdots \to P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \to 0)$.

Proof.

- For $i = 0$, since F is a right exact functor, then $F(P_1) \xrightarrow{F(d_1)} F(P_0) \xrightarrow{F(\epsilon)} F(A) \to 0$ is exact. Hence,

$$H_0(F(P)) = \ker\ F(\epsilon)/\text{im}\ F(d_1) = F(P_0)/\ker\ F(\epsilon) \cong \text{im}\ F(\epsilon) = F(A) \cong L_0F(A)$$

- $i \geq 1$, consider $M_m := \ker d_m = \text{im}\ d_{m+1}$ for $m \geq 0$ and we have a long exact sequence $0 \to M_{i-1} \hookrightarrow P_{i-1} \to P_{i-2} \to \cdots \to P_0 \to A \to 0$. By the General Dimension Shifting Theorem, we can get

$$L_iF(A) \cong \ker\ (F(M_{i-1}) \xrightarrow{\iota := F(M_{i-1} \hookrightarrow P_{i-1})} F(P_{i-1}))$$

  Note that $0 \to M_i \to P_i \to M_{i-1} \to 0$ is exact (since $M_i = \text{im}\ d_{i+1}$). Since F is a right exact functor, then $F(M_i) \to F(P_i) \xrightarrow{\pi} F(M_{i-1}) \to 0$ is exact, which gives π is surjective and that $\ker\ \pi = \text{im}\ (F(M_i) \to F(P_i))$. Note that $F(d_i) = \iota \circ \pi$. By the First Isomorphism Theorem, $\ker \iota = \ker(\iota \circ \pi)/\ker \pi = \ker F(d_i)/\ker \pi$.
  Consider the short exact sequence $0 \to M_{i+1} \to P_{i+1} \to M_i \to 0$ and then similarly we have an exact sequence $F(P_{i+1}) \to F(M_i) \to 0$, which gives $F(P_{i+1}) \twoheadrightarrow F(M_i)$ is surjective. Therefore,

$$\text{im}\ F(d_{i+1}) = \text{im}\ (F(P_{i+1}) \twoheadrightarrow F(M_i) \to F(P_i)) = \text{im}\ (F(M_i) \to F(P_i))$$

  By the exactness of $F(M_i) \to F(P_i) \xrightarrow{\pi} F(M_{i-1})$, we then have $\text{im}\ F(d_{i+1}) = \ker\ \pi$, so we're done:

$$L_iF(A) \cong \ker \iota \cong \ker F(d_i)/\text{im}\, F(d_{i+1})$$

□

Universality

Theorem 3.5.18. *Universal Property of Left Derived Functors*
Assume that $\mathcal{A}$ has enough projectives. Then for any righta exact functor $F : \mathcal{A} \to \mathcal{B}$, the left derived functors L_iF form a universal δ-functor.

Proof. Suppose that T is a homological δ-functor and that $\varphi_0 : T_0 \to L_0F \cong F$ is a given natural transformation, i.e.: $\forall A \in \mathrm{Ob}(\mathcal{A})$, φ_0 induces a morphism $\varphi_{0,A} \in \mathrm{Hom}_{\mathcal{B}}(T_0(A), F(A))$ in $\mathcal{B}$, this transformation must be commutative for all morphisms in category, namely, $\forall f \in \mathrm{Hom}_{\mathcal{A}}(A, B)$, the following diagram commutes:

$$\begin{tikzcd} T_0(A) \arrow[r, "\varphi_{0,A}"] \arrow[d, "T_0(f)"] & F(A) \arrow[d, "F(f)"] \\ T_0(B) \arrow[r, "\varphi_{0,B}"] & F(B) \end{tikzcd}$$

We need to show that φ_0 admits a unique extension to a morphism $\varphi : T \to \{L_iF\}$ of δ-functors. Do induction on n: Suppose that $\forall 0 \leq k < n$, we can uniquely construct a natural transformation $\varphi_k : T_k \to L_kF$ and that they commute with all δ_i's. Given $A \in \mathrm{Ob}(\mathcal{A})$, consider an exact sequence $0 \to K := \ker \pi \to P \xrightarrow{\pi} A \to 0$ with $P \in \mathrm{Ob}(\mathcal{A})$ projective, and hence $L_nF(P) = 0$. Then we have a commutative diagram with exact rows

$$\begin{tikzcd} & T_n(A) \arrow[r, "\delta_n^T"] & T_{n-1}(K) \arrow[r] \arrow[d, "\varphi_{n-1}(K)"] & T_{n-1}(P) \arrow[d, "\varphi_{n-1}(P)"] \\ 0 = L_nF(P) \arrow[r] & L_nF(A) \arrow[r, "\delta_n^L"] & L_{n-1}F(K) \arrow[r] & L_{n-1}F(P) \end{tikzcd}$$

Hence, $\delta_n^L : L_nF(A) \to L_{n-1}F(K)$ is injective. Consider $h := \varphi_{n-1}(K) \circ \delta_n^T : T_n(A) \to L_{n-1}F(K)$. By the commutativity,

$$(L_{n-1}F(K) \to L_{n-1}F(P)) \circ h = (T_n(A) \xrightarrow{\delta_n^T} T_{n-1}(K) \to T_{n-1}(P) \xrightarrow{\varphi_{n-1}(P)} L_{n-1}F(P)) \overset{\text{exactness}}{=\!=\!=} 0$$

Therefore, im $h \subseteq$ ker $(L_{n-1}F(K) \to L_{n-1}F(P)) =$ im δ_n^L. By the projectivity of P, δ_n^L is injective and then by the universal property of kernel, there is a unique $\varphi_n(A) : T_n(A) \to L_nF(A)$, s.t.:

$$\delta_n^L \circ \varphi_n(A) = \varphi_{n-1}(K) \circ \delta_n^T$$

Now for every $A \in \mathrm{Ob}(\mathcal{A})$, we've constructed a unique $\varphi_n(A)$, so we need to show, for the well-defined-ness of $\varphi_n(A)$, that $\varphi_n(A)$ is independent of the choice of $P \xrightarrow{\pi} A \to 0$. Before this, we shall first prove the naturality if $\varphi_n(A)$, i.e.: $\forall f \in \mathrm{Hom}_{\mathcal{A}}(A', A)$, φ_n commutes with f. Take $0 \to K' := \ker \pi' \to P' \xrightarrow{\pi'} A' \to 0$ and $0 \to K \to P \xrightarrow{\pi} A \to 0$ with P, P' projective. Then by the projectivity of P and P', we can lift f:

$$\begin{tikzcd}
0 \arrow[r] & K' \arrow[r] \arrow[d, "h"] & P' \arrow[r] \arrow[d, "g"] & A' \arrow[r] \arrow[d, "f"] & 0 \\
0 \arrow[r] & K \arrow[r] & P \arrow[r] & A \arrow[r] & 0
\end{tikzcd}$$

This induces the commutativity of the innermost square of the following diagram:

$$\begin{tikzcd}
T_n(A') \arrow[rrr, "T_n(f)"] \arrow[dr, "\delta'^T_n"] \arrow[ddd, "\varphi_n(A')"'] & & & T_n(A) \arrow[dl, "\delta^T_n"'] \arrow[ddd, "\varphi_n(A)"] \\
& T_{n-1}(K') \arrow[r, "T_{n-1}(h)"] \arrow[d, "\varphi_{n-1}(K')"] & T_{n-1}(K) \arrow[d, "\varphi_{n-1}(K)"] & \\
& L_{n-1}F(K') \arrow[r, "L_{n-1}F(h)"] & L_{n-1}F(K) & \\
L_nF(A') \arrow[rrr, "L_nF(f)"] \arrow[ur, "\delta'^L_n"] & & & L_nF(A) \arrow[ul, "\delta^L_n"']
\end{tikzcd}$$

We now need to show that outmost square commutes, i.e.: $\varphi_n(A) \circ T_n(f) = L_nF(f) \circ \varphi_n(A')$. Note that

$$\begin{aligned}
\delta^L_n \circ (\varphi_n(A) \circ T_n(f)) = (\delta^L_n \circ \varphi_n(A)) \circ T_n(f) &\overset{\text{def of } \varphi_n(A)}{=\!=\!=} \varphi_{n-1}(K) \circ \delta^T_n \circ T_n(f) \\
&\overset{T \text{ homological } \delta\text{-functor}}{=\!=\!=\!=\!=} \varphi_{n-1}(K) \circ T_{n-1}(h) \circ \delta'^T_n \\
&\overset{\text{innermost square commutes}}{=\!=\!=\!=\!=} L_{n-1}F(h) \circ \varphi_{n-1}(K') \circ \delta'^T_n \\
&\overset{\text{def of } \varphi_n(A')}{=\!=\!=} L_{n-1}F(h) \circ \delta'^L_n \circ \varphi_n(A') \\
&\overset{T \text{ homological } \delta\text{-functor}}{=\!=\!=\!=\!=} \delta^L_n \circ L_nF(f) \circ \varphi_n(A') = \delta^L_n \circ (L_nF(f) \circ \varphi_n(A'))
\end{aligned}$$

By the injectivity of δ^L_n, we have $\varphi_n(A) \circ T_n(f) = L_nF(f) \circ \varphi_n(A')$, which gives the naturality of $\varphi_n(A)$. Then for the well-defined-ness of $\varphi_n(A)$, we take $A' = A$ and $f = \mathrm{id}_A$ and then we're done.

Next, we need to verify that φ_n commutes with δ_n: Given a short exact sequence $0 \to A' \xrightarrow{\alpha} A \xrightarrow{\beta} A'' \to 0$, take an exact sequence $0 \to K'' := \ker \pi'' \to P'' \xrightarrow{\pi''} A'' \to 0$ with P'' projective. By the projectivity of P'', we can lift $\mathrm{id}_{A''}$:

$$\begin{tikzcd}
0 \arrow[r] & K'' \arrow[r] \arrow[d, "g"] & P'' \arrow[r] \arrow[d, "f"] & A'' \arrow[r] \arrow[d, "\mathrm{id}_{A''}"] & 0 \\
0 \arrow[r] & A' \arrow[r] & A \arrow[r] & A'' \arrow[r] & 0
\end{tikzcd}$$

This yields a commutative diagram:

$$\begin{array}{ccccc}
T_n(A'') & \xrightarrow{\delta_n{}''^T} & T_{n-1}(K) & \xrightarrow{T_{n-1}(g)} & T_{n-1}(A') \\
\Big\downarrow{\scriptstyle \varphi_n(A'')} & & \Big\downarrow{\scriptstyle \varphi_{n-1}(K'')} & & \Big\downarrow{\scriptstyle \varphi_{n-1}(A')} \\
L_nF(A'') & \xrightarrow{\delta_n{}''^L} & L_{n-1}F(K'') & \xrightarrow{L_{n-1}F(g)} & L_{n-1}F(A')
\end{array}$$

Then we have:

$$\begin{aligned}
\varphi_{n-1}(A') \circ \delta_n^T &\overset{T \text{ homological } \delta\text{-functor}}{=\!=\!=\!=\!=} \varphi_{n-1}(A') \circ T_{n-1}(g) \circ \delta_n''^T \\
&\overset{\text{inductive hypothesis for naturality of } \varphi_{n-1}}{=\!=\!=\!=\!=} L_{n-1}F(g) \circ \varphi_{n-1}(K'') \circ \delta_n''^T \\
&\overset{\text{def of } \varphi_n(A'')}{=\!=\!=} L_{n-1}F(g) \circ \delta_n''^L \circ \varphi(A'') \overset{L_{n-1}F \text{ homological } \delta\text{-functor}}{=\!=\!=\!=\!=} \delta_n''^L \circ \varphi_n(A'')
\end{aligned}$$

Finally, for the uniqueness of φ, suppose that there is another δ-functor $\psi = (\psi_n : T_n \to L_nF)_{n\geq 0}$ with $\psi_0 = \varphi_0$. Inductively suppose $\psi_{n-1} = \varphi_{n-1}$. Then for any $A \in \mathrm{Ob}(\mathcal{A})$, take $0 \to \ker \pi \to P \xrightarrow{\pi} A \to 0$ with P projective. Since ψ is a δ-functor, then

$$\delta_n^L \circ \psi_n(A) = \psi_{n-1}(K) \circ \delta_n^T = \varphi_{n-1}(K) \circ \delta_n^T = \delta_n^T \circ \varphi_n(A) \overset{\delta_n^L \text{ injective}}{\Rightarrow} \psi_n(A) = \varphi$$

□

3.5.3 Right Derived Functors

Method: Dually to the left derived functors, we can construct the right derived functor by using left exact functor and injective resolution, namely, if $F : \mathcal{A} \to \mathcal{B}$ is a left exact functor between abelian categories, if $\mathcal{A}$ has enough injectives, and if A is an object of $\mathcal{A}$ with an injective resolution $0 \to A \xrightarrow{\epsilon} I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} \cdots$ where $I := (0 \to I^0 \xrightarrow{d^0} I^1 \to \cdots)$, then we can define the right derived functor by

$$R^iF(A) := H^i(F(I))$$

Proposition 3.5.19. *Relations-between-Left and Right Derived Functors Proposition*
Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor between abelian categories and let $\mathcal{A}$ have enough injectives. Then $\forall A \in Ob(\mathcal{A})$, we have

$$R^iF(A) = (L_iF^{op})^{op}(A)$$

Proof. Note that $F^{op} : \mathcal{A}^{op} \to \mathcal{B}^{op}$ is a right exact functor and that $\mathcal{A}^{op}$ has enough projectives. □

Remark: Hence, all the results about right exact functors can apply to left derived functors: The right derived functors are well-defined, satisfy $R^0F(A) \cong F(A)$, form a universal cohomological δ-functor and $R^iF(I) = 0$ for all $i \neq 0$ whenever I is injective. Also, in Syzygy Theory, if $R^iF(Q) = 0$ $(i \neq 0)$, then we call the object Q F-acyclic and we can compute the right derived functors of F form F-acyclic resolutions.

Example: Ext functors are right derived functors of $\mathrm{Hom}_R(A, \cdot)$, say

$$\mathrm{Ext}^i_R(A, B) = R^i\mathrm{Hom}_R(A, \cdot)(B)$$

3.6 Homological Dimensions

3.6.1 Global Dimension and Tor-Dimension

Definition 3.6.1. Projective, Injective and Flat Dimensions
Let A be a right R-module.

- The *projective dimension* $pd(A)$ is minimum integer n (if it exists) such that there is a resolution of A by projective modules

$$0 \to P_n \to \cdots \to P_1 \to P_0 \to A \to 0$$

- The *injective dimension* $id(A)$ is the minimum integer n (if it exists) such that there is a resolution of A by injective modules

$$0 \to A \to E^0 \to E^1 \to \cdots \to E^n \to 0$$

- The *flat dimension* $fd(A)$ is the minimum integer n (if it exists) such that there is a resolution of A by flat modules

$$0 \to F_n \to \cdots \to F_1 \to F_0 \to A \to 0$$

If no finite resolution exists, then we set $pd(A), id(A)$, or $fd(A)$ equal to ∞.

Lemma 3.6.2. *pd Lemma*
Let A be a right R-module. The followings are equivalent:

1. $pd(A) \leq d$.
2. $Ext^n_R(A, B) = 0$ *for all* $n > d$ *and all* R*-modules* B.
3. $Ext^{d+1}_R(A, B) = 0$ *for all* R*-modules* B.
4. *If* $0 \to M_d \to P_{d-1} \to P_{d-2} \to \cdots \to P_1 \to P_0 \to A \to 0$ *is any resolution with* P_i *projective, then the syzygy* M_d *is also projective.*

Proof. It's clear that $(4 \Rightarrow 1 \Rightarrow 2 \Rightarrow 3)$, so it suffices to prove $(3 \Rightarrow 4)$. By the General Dimension Shifting Theorem, $\mathrm{Ext}^{d+1}_R(A, B) \cong \mathrm{Ext}^1_R(M_d, B)$. Hence, by 3, $\mathrm{Ext}^1_R(M_d, B) = 0$ for all R-modules B, which, by the Ext-Criterion-for-Projective Modules Prop, is equivalent to that M_d is projective. $\square$

Remark: Similarly, we have the following two lemmas, where these three lemmas are key tools for characterizing dimensions.

Lemma 3.6.3. *id Lemma*
Let B be a right R-module B. The followings are equivalent:

1. *$id(B) \leq d$.*
2. *$Ext_R^n(A, B) = 0$ for all $n > d$ and all R-modules A.*
3. *$Ext_R^{d+1}(A, B) = 0$ for all R-modules B.*
4. *If $0 \to B \to E^0 \to \cdots \to E^{d-1} \to M^d \to 0$ is a resolution with the E^j injective, then M^d is also injective.*

Lemma 3.6.4. *fd Lemma*
Let A be a right R-module. The followings are equivalent:

1. *$fd(A) \leq d$.*
2. *$Tor_n^R(A, B) = 0$ for all $n > d$ and all left R-module B.*
3. *$Tor_{d+1}^R(A, B) = 0$ for all left R-module B.*
4. *If $0 \to M_d \to F_{d-1} \to F_{d-2} \to \cdots \to F_0 \to A \to 0$ is a resolution with F_i flat, then M_d is also a flat R-module.*

Lemma 3.6.5. *Ext Form of Baer's Criterion*
A left R-module B is injective iff $Ext_R^1(R/I, B) = 0$ for all left ideals I.

Proof. Applying $\mathrm{Hom}_R(\cdot, B)$ to $0 \to I \hookrightarrow R \twoheadrightarrow R/I \to 0$, we have a long exact sequence

$$0 \to \mathrm{Hom}_R(R/I, B) \to \mathrm{Hom}_R(R, B) \xrightarrow{\tau} \mathrm{Hom}_R(I, B) \xrightarrow{\delta_0} \mathrm{Ext}_R^1(R/I, B) \to \mathrm{Ext}_R^1(R, B) \to \cdots$$

is exact. By the *Baer's Criterion*, B is injective $\Leftrightarrow$ τ is surjective $\Leftrightarrow$ $\mathrm{Ext}_R^1(R/I, B) = 0$. $\square$

Theorem 3.6.6. *Global Dimension Theorem*
Let R be any ring. The following numbers are the same:

1. *sup $\{id(B) \mid B \in Ob(\mathcal{M}od_R)\}$*
2. *sup $\{pd(A) \mid A \in Ob(\mathcal{M}od_R)\}$*
3. *sup $\{pd(R/I) \mid I$ right ideal of $R\}$*
4. *sup $\{d \mid Ext_R^d(A, B) \neq 0$ for some $A, B \in (\mathcal{M}od_R)\}$*

Proof. By the *pd* Lemma and the *id* Lemma, we can immediately get sup(1) = sup(2) = sup(4). Note that sup(2) $\geq$ sup(3). Hence, we may assume that $d := \sup(3) < \infty$ and that $id(B) > d$ for some module B. Then take an resolution

$$0 \to B \to E^0 \to E^1 \to \cdots \to E^{d-1} \to M \to 0$$

with E^i injective, where M denotes the cokernel of $E^{d-2} \to E^{d-1}$. Then by the General Dimension Shifting Theorem, we have $\mathrm{Ext}_R^{d+1}(R/I, B) \cong \mathrm{Ext}_R^1(R/I, M)$. By the *id* Lemma, we have $0 =$

$\text{Ext}_R^{d+1}(R/I, B)$. Hence,

$$\text{Ext}_R^1(R/I, M) = 0$$

Therefore, by the Ext Form of Baer's Criterion, M is injective, contradicting the assumption that $id(B) > d$. □

Definition 3.6.7. Global Dimension

The common number (possible ∞) in the Global Dimension Theorem is called the *right global* (or *homological*) *dimension* of R, denoted as $r.gl.\text{dim}(R)$.

Remark: Similarly, one may define the left global dimension $l.gl.\text{dim}(R)$. If R is commutative, we clearly have $l.gl.\text{dim}(R) = r.gl.\text{dim}(R)$.

Theorem 3.6.8. *Tor-Dimension Theorem*

Let R any ring. The following numbers are the same:

1. *sup* $\{fd(A) \mid A \in Ob(\mathcal{M}od_R)\}$
2. *sup* $\{fd(R/J) \mid J \text{ right ideal of } R\}$
3. *sup* $\{fd(B) \mid B \in Ob({}_R\mathcal{M}od)\}$
4. *sup* $\{fd(R/I) \mid I \text{ left ideal of } R\}$
5. *sup* $\{d \mid Tor_d^R(A, B) \neq 0 \text{ for some } R\text{-modules } A, B\}$

Proof. The fd Lemma characterizing $fd(A)$ over R shows that $\sup(5) = \sup(1) \geq \sup(2)$, and the fd Lemma characterizing $fd(A)$ over R^{op} shows that $\sup(5) = \sup(3) \geq \sup(4)$. Suppose that $d := \sup(2) \leq \sup(4)$. If $d = \infty$, then we're done. Otherwise, if $d < \infty$, then we have $d = \sup(2) \leq \sup(4) \leq \sup(3)$, so we can suppose on the contrary that $fd(B) > d$ for some left R-module B. Then take a resolution

$$0 \to M \to F_{d-1} \to \cdots \to F_0 \to B \to 0$$

with F_i flat, where M denots the kernel of $F_{d-1} \to F_{d-2}$. Then by the General Dimension Shifting Theorem, we have $\text{Tor}_{d+1}^R(R/J, B) \cong \text{Tor}_1^R(R/J, M)$. By the Tor-Criterion-for-Flat Modules Prop, $\text{Tor}_{d+1}^R(R/J, B) = 0$. Hence, $\text{Tor}_1^R(R/J, M) = 0 \Leftrightarrow M$ is flat, contradicting $fd(B) > d$. □

Definition 3.6.9. Tor-Dimension

The common number (possible ∞) in the Tor-Dimension Theorem is called the *Tor-dimension* of R (or the *weak dimension* of R).

Example:

- For every field F, since any module V over F is a vector space, which must be free with its basis, then any resolution of V stops in step 0, so F has both global dimension and Tor-dimension zero.

- $R = \mathbb{Z}$. Since $\mathbb{Z}$ is a PID having the property that every submodule of free module over a PID is again free. For any $\mathbb{Z}$-module (abelian group) A, take a resolution $0 \to K := \ker \pi \to F \xrightarrow{\pi} A \to 0$ of A with F free and $K \subseteq F$ submodule of F. Hence, K is free, which shows that $\mathbb{Z}$ has global dimension 1 and Tor-dimension 1.
- $R = \mathbb{Z}/m\mathbb{Z}$ with $p^2 \mid m$ where p is a prime integer. Consider $A = \mathbb{Z}/p\mathbb{Z}$ which is a $\mathbb{Z}/m\mathbb{Z}$-module. Let $m = p^2 k$ where $k \in \mathbb{Z}_+$.
  - Consider a map $d_1 : R \to R$ given by $x \mapsto px$. Then
$$\begin{aligned}\ker\, d_1 &= \{x \in R \mid px = 0\} = (pk)\\ \text{im}\, d_1 &= (p)\end{aligned}$$
  - Consider a map $d_2 : R \to R$ given by $x \mapsto pkx$. Then
$$\begin{aligned}\ker\, d_1 &= \{x \in R \mid pkx = 0\} = (p)\\ \text{im}\, d_1 &= (pk)\end{aligned}$$

  Note that $\ker\, d_1 = \text{im}\, d_2$ and $\text{im}\, d_1 = \ker\, d_2$. Hence, we can construct an infinitely long and periodic *free* resolution of A over R
$$\cdots \xrightarrow{d_2} R \xrightarrow{d_1} R \xrightarrow{d_2} R \xrightarrow{d_1} R \to A \to 0$$
  Applying $\text{Hom}_R(\cdot, A)$ to $\cdots \xrightarrow{d_2} R \xrightarrow{d_1} R \xrightarrow{d_2} R \xrightarrow{d_1} R \to 0$, we get, by $\text{Hom}_R(R, A) \cong A$, a cochain complex
$$0 \to A \xrightarrow{d_1^*} A \xrightarrow{d_2^*} A \xrightarrow{d_1^*} \cdots$$
  Note that $d_1^* : a \mapsto pa = 0 \in A$ and $d_2^* : a \mapsto kpa = 0 \in A$. Hence, all the differentials in this cochain complex are zero maps. Therefore, $\forall n \geq 0$, $\ker\, d_{n+1}^* = A, \text{im}\, d_n^* = 0$, so
$$\text{Ext}_R^n(A, A) := \ker\, d_{n+1}^*/\text{im}\, d_n^* = A/0 \cong A = \mathbb{Z}/p\mathbb{Z} \neq 0$$
  Similarly, applying $\cdot \otimes_R A$ to $\cdots \xrightarrow{d_2} R \xrightarrow{d_1} R \xrightarrow{d_2} R \xrightarrow{d_1} R \to 0$, we can get $\text{Tor}_n^R(A, A) \cong A \neq 0, \forall n \geq 0$. Thus, $R = \mathbb{Z}/m\mathbb{Z}$ with $p^2 \mid m$, has global dimension ∞ and Tor-dimension ∞.
  But in contrast, if m has no square-divisors (e.g.: $m = 6 = 2 \times 3$), then by the Chinese Remainder Theorem, $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ is a direct product of fields, having global dimension 0 and Tor-dimension 0.

Proposition 3.6.10. *Equality-of-Dimension-over-Noetherian Rings Proposition*
Let R be a right Noetherian, i.e.: R satisfies ACCP on right ideals, or equivalently, every right ideal in R is finitely generated.

1. *$fd(A) = pd(A)$ for every finitely generated R-module A.*

2. *Tor-dim*$(R) = r.gl.dim(R)$.

Proof.

- **Definition 3.6.11.** Finitely Presented Modules
  An R-module M is finitely presented if there is an exact sequence $R^m \to R^n \to M \to 0$.

- **Lemma 3.6.12.** *Govorov-Lacard Theorem*
  Every finitely presented flat R-module M is projective.

1. Since projective modules are flat, then $fd(A) \leq pd(A)$, so it suffices to suppose that $fd(A) = n < \infty$ and prove that $pd(A) \leq n$. Since R is Noetherian, then there is a resolution

$$0 \to M \to P_{n-1} \to P_{n-2} \to \cdots \to P_1 \to P_0 \to A \to 0$$

where P_i are finitely generated free modules and M denotes the kernel of $P_{n-1} \to P_{n-2}$, which is finitely generated. By the fd Lemma, M is a flat R-module. Hence, by the Govorov-Lacard Theorem, M is projective, so $pd(A) \leq n$.

2. Since by the Global Dimension Theorem and the Tor-Dimension Theorem, we can compute $r.gl.\text{dim}(R)$ and Tor-dim(R) using the modules R/I, then 1 holding can immediately prove 2 holding.

□

3.7 Group Cohomology

3.7.1 Basics

Definition 3.7.1. Modules over Groups
Let G be a group. An abelian group A on which G acts (on the left) as automorphisms is called a G-module. Precisely, the action on A of G satisfies:

- $(gh) \cdot a = g \cdot (h \cdot a)$ and $e \cdot a = a$
- $g \cdot (a + b) = g \cdot a + g \cdot b$

where $g, h, e \in G$ with e identity of G and $a, b \in A$.

Remark: A G-module is the same as an abelian group A together with a homomorphism $\varphi : G \to \text{Aut}(A)$. Since an abelian group is the same as a $\mathbb{Z}$-module, then it's also easy to see that a G-module A is the same as $\mathbb{Z}G$-module where

$$\mathbb{Z}G := \{a_1 g_1 + \cdots + a_n g_n \mid a_i \in \mathbb{Z}, g_i \in G, n \in \mathbb{N}_+\}$$

is the *integral group ring of G*.

Definition 3.7.2. Fixed Points Subgroup
If A is a G-module, then let $A^G := \{a \in A \mid ga = a, \forall g \in G\}$ be the set of elements of A fixed by all the elements of G.

Example:

- If $ga = a$ for all $g \in G, a \in A$, i.e.: $A^G = A$, then G is said to act *trivially* on A. The abelian group $A = \mathbb{Z}$ will be assumed to have trivial G-action for any group G unless otherwise stated.
- For any G-module A, the fixed points A^G of A under the action of G is clearly a $\mathbb{Z}G$-submodule of A on which G acts trivially.
- If V is an n-dim'l vector space over the field F and $G = \mathrm{GL}_n(F)$, then V is a G-module. Since any nonzero element in V can be taken to any other nonzero element in V by some linear transformation, then $V^G = \{0\}$.

Remark: It's easy to see that a short exact sequence $0 \to A \to B \to C \to 0$ of G-modules can induce an exact sequence

$$0 \to A^G \to B^G \to C^G \to 0 \tag{3.4}$$

that in general can *not* be extended to a short exact sequence. One way to see that (3.4) is exact is to observe that A^G can be related to a Hom group:

Proposition 3.7.3. *Identification-of-Fixed Points Proposition*
Let A be a G-module and let $Hom_{\mathbb{Z}G}(\mathbb{Z}, A)$ be the group of all $\mathbb{Z}G$-module homomorphisms from $\mathbb{Z}$ (with trivial G-action) to A. Then

$$A^G \cong Hom_{\mathbb{Z}G}(\mathbb{Z}, A)$$

Proof. Note that any $\mathbb{Z}G$-module homomorphism $\alpha : \mathbb{Z} \to A$ is uniquely determined by evaluation on 1. Let α_a denote the $\mathbb{Z}G$-module homomorphism with $\alpha(1) = a$. Since α_a is a $\mathbb{Z}G$-module homomorphism, then we have

$$a = \alpha_a(1) = \alpha_a(g \cdot 1) = g \cdot \alpha_a(1) = g \cdot a, \forall g \in G$$

and hence $a \in A^G$. Then $\alpha_a \mapsto a$ gives a G-module isomorphism between $\mathrm{Hom}_{\mathbb{Z}G}(\mathbb{Z}, A) \cong A$. $\square$

Definition 3.7.4. Standard Resolution
The *standard resolution* or *bar resolution* of $\mathbb{Z}$ is a projective resolution of $\mathbb{Z}$:

$$\cdots \to F_n \xrightarrow{d_n} F_{n-1} \to \cdots \xrightarrow{d_1} F_0 \xrightarrow{\text{aug}} \mathbb{Z} \to 0$$

where:

- $F_n := \overbrace{\mathbb{Z}G \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} \mathbb{Z}G}^{(n+1)\text{-types}} = \mathbb{Z}G^{\otimes(n+1)}$ for $n \in \mathbb{N} \cup \{0\}$, which is a G-module under the action defined on simple tensor by

$$g \cdot (g_0 \otimes g_1 \otimes \cdots \otimes g_n) := (gg_0) \otimes g_1 \otimes \cdots \otimes g_n$$

- aug : $F_0 \to \mathbb{Z}$ is the *augmentation map* defined by $\mathrm{aug}(\sum \alpha_g g) := \sum \alpha_g$.
- We define d_n as follows:
  - **Claim:** F_n is a free $\mathbb{Z}G$-module of rank $|G|^n$ with $\mathbb{Z}G$-basis $1 \otimes g_1 \otimes \cdots \otimes g_n$ with $g_i \in G$.
  - *Proof. of claim:* By the G-action on F_n, every pure tensor $g_0 \otimes g_1 \otimes \cdots \otimes g_n \in F_n$ can be written as $g_0 \cdot (1 \otimes g_1 \otimes \cdots \otimes g_n)$. Since $g_0 \in G$ generates the free $\mathbb{Z}$-module $\mathbb{Z}G$, then every element in F_n can be uniquely expressed as $\mathbb{Z}G$-linear combination of the set $\mathcal{B} = \{1 \otimes g_1 \otimes \cdots \otimes g_n \mid g_i \in G\}$ with $|\mathcal{B}| = |G|^n$. □

  Now we define the G-module homomorphisms d_n for $n \geq 1$ on the basis $\{1 \otimes g_1 \otimes \cdots \otimes g_n \mid g_i \in G\}$ by $d_1(1 \otimes g_1) := g_1 - 1$ and for $n \geq 2$,

$$\begin{aligned} d_n(1 \otimes g_1 \otimes \cdots \otimes g_n) :=& g_1 \cdot (g_2 \otimes \cdots \otimes g_n) \\ &+ \sum_{i=1}^{n-1} (-1)^i (g_1 \otimes \cdots \otimes g_{i-1} \otimes (g_i g_{i+1}) \otimes g_{i+2} \otimes \cdots \otimes g_n) \\ &+ (-1)^n (g_1 \otimes \cdots \otimes g_{n-1}) \end{aligned}$$

Remark:

- The def of d_n is quite strange, so we here give the intuition for the construction of d_n: *Homogeneous Resolution.*

  Let P_n denote the free $\mathbb{Z}$-module with basis $\{(g_0, g_1, \cdots, g_n) \mid g_i \in G\}$ and define an action of G on P_n by $g \cdot (g_0, g_1, \cdots, g_n) := (gg_0, gg_1, \cdots, gg_n)$. For $n \geq 0$, we define $d_n^P : P_n \to P_{n-1}$ by

$$d_n^P(g_0, g_1, \cdots, g_n) := \sum_{i=0}^{n} (-1)^i (g_0, \cdots, \hat{g}_i, \cdots, g_n)$$

  Consider a $\mathbb{Z}G$-module homomorphism $\psi_n : P_n \to F_n$ given by $\psi_n(g_0, g_1, \cdots, g_n) := g_0 \otimes (g_0^{-1} g_1) \otimes (g_1^{-1} g_2) \otimes \cdots \otimes (g_{n-1}^{-1} g_n)$, which is easy to check that ψ_n is an isomorphism with inverse $g_0 \otimes g_1 \otimes \cdots \otimes g_n \mapsto (g_0, g_0 g_1, \cdots, g_0 g_1 \cdots g_n)$, so we get $P_n \cong F_n$. Hence, d_n is induced by d_n^P.

- Applying $\mathrm{Hom}_{\mathbb{Z}G}(\cdot, A)$ to the standard resolution we get a cochain complex

$$0 \to \mathrm{Hom}_{\mathbb{Z}G}(F_0, A) \xrightarrow{d_1} \mathrm{Hom}_{\mathbb{Z}G}(F_1, A) \xrightarrow{d_2} \mathrm{Hom}_{\mathbb{Z}G}(F_2, A) \xrightarrow{d_3} \cdots$$

  the cohomology groups of which are $\mathrm{Ext}^n_{\mathbb{Z}G}(\mathbb{Z}, A)$. Then, as in the Long Exact Sequence-in-Cohomology Thm, the short exact sequence $0 \to A \to B \to C \to 0$ of G-modules can induce a long exact sequence whose first terms are $0 \to A^G \to B^G \to C^G$, of fixed points groups, and whose higher terms are the cohomology groups $\mathrm{Ext}^n_{\mathbb{Z}G}(\mathbb{Z}, A)$.

Definition 3.7.5. Cochains of Groups

Let G be a finite group and let A be a G-module. We define $C^0(G, A) := A$ and for $n \geq 1$, define

$C^n(G, A) :=$ the collection of all maps from $G^n = \underbrace{G \times \cdots \times G}_{n\text{-copies}}$ to A. The elements of $C^n(G, A)$ are called *n-cochains* (*of G with values in A*).

Remark: Each $C^n(G, A)$ is an abelian group.

Definition 3.7.6. Coboundary Homomorphisms
For $n \geq 0$, we define the *n-th coboundary homomorphism* from $C^n(G, A)$ to $C^{n+1}(G, A)$ by

$$\begin{aligned} d_n(f)(g_1, \cdots, g_{n+1}) :=& g_1 \cdot f(g_2, \cdots, g_{n+1}) \\ &+ \sum_{i=1}^{n} (-1)^i f(g_1, \cdots, g_{i-1}, g_i g_{i+1}, g_{i+2}, \cdots, g_{n+1}) \\ &+ (-1)^{n+1} f(g_1, \cdots, g_n) \end{aligned}$$

Remark: It's clear that d_n are group homomorphisms with property that $d_n \circ d_{n-1} = 0$ for $n \geq 1$.

Definition 3.7.7. Cohomology Groups

- Let $Z^n(G, A) := \ker d_n$ for $n \geq 0$. The elements of $Z^n(G, A)$ are called *n-cocycles*.
- Let $B^n(G, A) := \operatorname{im} d_{n-1}$ for $n \geq 1$ where $B^0(G, A) = 1$. The elements of $B^n(G, A)$ are called *n-boundaries*.
- Since $d_n \circ d_{n-1} = 0$ for $n \geq 1$, then $B^n(G, A) \subseteq Z^n(G, A)$, so we define the *$n$-th cohomology group of G with coefficients in A*, denoted as $H^n(G, A)$ for $n \geq 0$, by

$$H^n(G, A) := Z^n(G, A)/B^n(G, A)$$

Example:

- For $f = a \in C^0(G, A)$, we have $d_0(f)(g) = g \cdot a - a$, so

$$\ker d_0 = \{a \in A \mid g \cdot a = a, \forall g \in G\}$$

  i.e.: $Z^0(G, A) = A^G$ and hence $H^0(G, A) = A^G, \forall$ group G and G-module A.
- Suppose $G = 1$ is the trivial group. Then $G^n = \{(1, 1, \cdots, 1)\}$ is also trivial and hence $f \in C^n(G, A)$ is completely determined by $f(1, \cdots, 1) = a \in A$. Identifying $f = a$, we obtain $C^n(G, A) = A$ for all $n \geq 0$, and

$$d_n(f)(1, \cdots, 1) = a + \sum_{i=1}^{n} (-1)^i a + (-1)^{n+1} a = \begin{cases} 0 & \text{, if } n \text{ is even} \\ 1 & \text{, if } n \text{ is odd} \end{cases}$$

so $d_n = 0$ if n is even and $d_n = \mathrm{id}$ if n is odd. Therefore,

$$H^0(1, A) = A^G = A$$
$$H^n(1, A) = 0 \text{ for all } n \geq 1$$

- (*Cohomology of a Finite Cyclic Group*) Suppose G is cyclic of order m with generator σ. Let $N = 1 + \sigma + \sigma^2 + \cdots + \sigma^{m-1} \in \mathbb{Z}G$. Then $N(\sigma - 1) = (\sigma - 1)N = \sigma^m - 1 = 0 \in \mathbb{Z}G$, so we have a particularly simple free resolution

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{\sigma-1} \cdots \xrightarrow{N} \mathbb{Z}G \xrightarrow{\sigma-1} \mathbb{Z}G \xrightarrow{\text{aug}} \mathbb{Z} \to 0$$

where aug denotes the augmentation map given by $\text{aug}(\sum \alpha_g g) = \sum \alpha_g$. Applying $\mathrm{Hom}_{\mathbb{Z}G}(\cdot, A)$ to this resolution ($\cdots \xrightarrow{N} \mathbb{Z}G \xrightarrow{\sigma-1} \mathbb{Z}G \to 0$) and using the identification $\mathrm{Hom}_{\mathbb{Z}G}(\mathbb{Z}G, A) = A$, we can get a chain complex

$$0 \to A \xrightarrow{\sigma-1} A \xrightarrow{N} A \xrightarrow{\sigma-1} A \xrightarrow{N} \cdots$$

whose cohomology computes the groups $H^n(G, A)$:

$$H^0(G, A) = A^G \text{ and } H^n(G, A) = \begin{cases} A^G/NA & \text{, if } n \text{ is even, } n \geq 2 \\ A_N/(\sigma - 1)A & \text{, if } n \text{ is odd, } n \geq 1 \end{cases}$$

where $A_N := \{a \in A \mid Na = 0\}$ is the subgroup of A annihilated by N, since the kernel of multiplication by $\sigma - 1$ is A^G.

In particular, if $G = \langle \sigma \rangle$ acts trivially on A, then $Na = ma$, so, in this case, $H^0(G, A) = A$ and $H^n(G, A) = \begin{cases} A/mA & \text{, if } n \text{ is even, } n \geq 2 \\ A_m & \text{, if } n \text{ is odd, } n \geq 1 \end{cases}$

Proposition 3.7.8. *Annihilator Inheritance Property Proposition*
Suppose $mA = 0$ for some integer $m \geq 1$, i.e.: the G-module A has exponent dividing m as an abelian group. Then for all $n \geq 0$

$$mZ^n(G, A) = mB^n(G, A) = mH^n(G, A) = 0$$

In particular, if A has exponent p for some prime p, then the abelian groups $Z^n(G, A), B^n(G, A)$ and $H^n(G, A)$ have exponent dividing p, so these groups are all vector spaces over the finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

Proof. If $f \in C^n(G, A)$ is an n-cochain, then $f \in A$ (if $n = 0$), in which case $mf = 0$, or f is a function from G^n to A (if $n \geq 1$), in which case mf is a function from G^n to $mA = 0$, so again

$mf = 0$. Hence, we immediately have

$$mZ^n(G, A) = mB^n(G, A) = 0$$

so by def, $mH^n(G, A) = 0$. ☐

Theorem 3.7.9. *Long Exact Sequence-in-Group Cohomology Theorem*
Suppose $0 \to A \to B \to C \to 0$ is a short exact sequence of G-modules. Then there is a long exact sequence of abelian groups:

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta_0} H^1(G, A) \to H^1(G, B) \to H^1(G, C) \xrightarrow{\delta_1} H^2(G, A) \to \cdots$$

Remark: This theorem gives a technique called *dimension shifting* for cohomology groups, which is one of the particular versions of One-Step Dimension Shifting Prop.

Definition 3.7.10. Cohomologically Trivial Modules over Groups
A G-module M is called *cohomologically trivial for G* if $H^n(G, M) = 0, \forall n \geq 1$.

Corollary 3.7.11. *of Long Exact Seq.-in-Group Cohomology Thm: Dimension Shfting for Cohomology Groups*
Suppose $0 \to A \to M \to C \to 0$ is a short exact sequence of G-modules and that M is cohomologically trivial for G. Then there is an exact sequence

$$0 \to A^G \to B^G \to C^G \to H^1(G, A) \to 0$$

and

$$H^{n+1}(G, A) \cong H^n(G, C), \forall n \geq 1$$

Proof. The first statement is clear since $H^1(G, M) = 0$. For the second statement, since M is cohomologically trivial for G, then we have the portion of the long exact sequence in the Long Exact Sequence-in-Group Cohomology Thm

$$0 = H^n(G, M) \to H^n(G, C) \to H^{n+1}(G, A) \to H^{n+1}(G, M) = 0$$

for $n \geq 1$. ☐

3.7.2 Induced Modules

Intuition: For using the above tool, dimension shfting, of computing cohomology groups, we need to ensure that we can find the cohomologically trivial group M.

Definition 3.7.12. Induced Modules
Let G be a group, let H be a subgroup of G and let A be an H-module. Define the *induced G-module*, $M_H^G(A)$, by

$$M_H^G(A) := \mathrm{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, A)$$

In other words, $M_H^G(A)$ is the set of maps f from G to A satisfying $f(hx) = hf(x)$ for every $x \in G$ and $h \in H$.

Remark:

- The action of an element $g \in G$ on $f \in M_H^G(A)$ is given by $(g \cdot f)(x) = f(xg)$ for $x \in G$.
- Note that the module $\mathbb{Z}G \otimes_{\mathbb{Z}H} A$, obtained by extension of scalars from $\mathbb{Z}H$ to $\mathbb{Z}G$, is a G-module. For a finite group G, or more generally if H has finite index in G, we have

  $$M_H^G(A) \cong \mathbb{Z}G \otimes_{\mathbb{Z}H} A$$

  where the module $\mathbb{Z}G \otimes_{\mathbb{Z}H} A$ is sometimes called the *induced G-module* and the module $M_H^G(A)$ is sometimes referred to as the *coinduced G-module*. Then the associativity of the tensor product shows that for subgroups $K \leq H \leq G$

  $$M_H^G(M_K^H(A)) = M_K^G(A)$$

  In fact, the above equation holds in general.

Example:

- If H is a subgroup of G and $0 \to A \to B \to C \to 0$ is a short exact sequence of H-modules, then $0 \to M_H^G(A) \to M_H^G(B) \to M_H^G(C) \to 0$ is a short exact sequence of G-modules.
- For the trivial group $H = 1$, any abelian group A is an H-module. The induced G-module $M_1^G(A)$ is just the collection of all maps f from G into A.
- Suppose that A is a G-module. Then there is a natural map $\varphi : A \to M_1^G(A)$ given by $a \mapsto f_a$ with $f_a(x) := xa, \forall x \in G$. It's clear that φ is a group homomorphism. Furthermore, φ is a G-module homomorphism as well since

  $$f_{ga}(x) := x(ga) = (xg)a := f_a(xg) = (g \cdot f_a)(x)$$

  Since $f_a(1) = a$, then $f_a = 0 \Leftrightarrow a = 0$, so φ is injective. Hence, we can identify A as a G-submodule of the induced G-module $M_1^G(A)$.
  More generally, if A is a G-module and H is any subgroup of G, then $\varphi(a) := f_a \in M_H^G(A)$ since

  $$f_a(hx) := (hx)a = h(xa) = hf_a(x) \ \forall x \in G, h \in H$$

  say, φ can be regarded as an injective G-module homomorphism from A to $M_H^G(A)$.
- The fixed points are

  $$\begin{aligned} (M_H^G(A))^G &= \{f : G \to A \mid gf = f, \forall g \in G\} \\ &= \{f : G \to A \mid (gf)(x) = f(x), \forall g, x \in G\} \\ &= \{f : G \to A \mid f(xg) = f(x), \forall g, x \in G\} \end{aligned}$$

Taking $x = 1, f(g) = f(1), \forall g \in G$, so f is constant on all of G. Note that the constant function $f = a$ is an element of $M_H^G(A)$ iff $\forall h \in H$,

$$a = f(hx) = hf(x) = ha$$

Hence, $(M_H^G(A))^G \cong A^H$. Furthermore, an element $f_a(x)$ in the previous example is in $(M_H^G(A))^G$ iff xa is constant $\forall x \in G$ iff $a \in A^G$.

Theorem 3.7.13. *Shapiro's Lemma*
For any subgroup H of G, any H-module A and any $n \geq 0$, we have

$$H^n(G, M_H^G(A)) \cong H^n(H, A)$$

Proof. Let $\cdots \to P_1 \to P_0 \to \mathbb{Z} \to 0$ be any resolution of $\mathbb{Z}$ bt projective G-modules. The cohomology groups on the LHS $C_{LHS}^n := H^n(G, M_H^G(A))$ are computed by taking homomorphisms from this resolution into $M_H^G(A) := \mathrm{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, A)$. On the other hand, since $\mathbb{Z}G$ is a free $\mathbb{Z}H$-module by Lagrange's, then this G-module resolution of $\mathbb{Z}$ is also a resolution of $\mathbb{Z}$ by projective H-modules, so by taking homomorphisms into A, the same resolution may be used to compute the cohomology groups on the RHS $C_{RHS}^n := H^n(H, A)$. To see that these two collections of cohomology groups, C_{LHS}^n and C_{RHS}^n, are isomorphic, we use the natural isomorphism of abelian groups

$$\Phi_n : \mathrm{Hom}_{\mathbb{Z}G}(P_n, \mathrm{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, A)) \xrightarrow{\cong} \mathrm{Hom}_{\mathbb{Z}H}(P_n, A)$$

given by $\Phi_n(f)(p) := f(p)(1)$, for all $f \in \mathrm{Hom}_{\mathbb{Z}G}(P_n, \mathrm{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, A))$ and $p \in P_n$. The inverse isomorphism Ψ_n is defined by $\Psi_n(f')(p)(g) := f'(gp) \in A$ for all $f' \in \mathrm{Hom}_{\mathbb{Z}H}(P_n, A), p \in P_n$ and $g \in G$. For the well-defined-ness of Φ_n and Ψ_n, $\forall h \in H$,

$$\Phi_n(f)(hp) := f(hp)(1) = (h \cdot f(p))(1) = f(p)(h) = f(p)(h \cdot 1) = h \cdot f(p)(1) = h \cdot \Phi_n(f)(p)$$
$$\Psi_n(f')(p)(hg) := f'(hgp) = h \cdot f'(gp) = h \cdot \Psi_n(f')(p)(g)$$

Note that the following diagram commutes:

$$\begin{array}{ccc} C_{LHS}^n & \xrightarrow{d_n^L} & C_{LHS}^{n+1} \\ \Big\downarrow{\scriptstyle \Phi_n} & & \\ C_{RHS}^n & \xrightarrow{d_n^R} & C_{RHS}^{n+1} \end{array}$$

since $\forall f \in C_{LHS}^n, p \in P_n$, we have

$$\Phi_{n+1}(d_n^L(f))(p) = d_n^L(f)(p)(1) = f(d_{n+1}^R(p))(1) = \Phi_n(f)(d_{n+1}^R(p)) = (d_{n+1}^R \circ \Phi_n)(f)(p)$$

Then we're done. $\square$

Corollary 3.7.14. *1 of Shapiro's Lemma*
For any G-module A, the module $M_1^G(A)$ is cohomologically trivial for G, i.e.: $H^n(G, M_1^G(A)) = 0$ for all $n \geq 1$.

Proof. This follows immediately from the Shapiro's Lemma by applying $H = 1$ and from the secong example of cohomology groups. □

Remark: This coro shows that the third example of induced modules gives us a short exact sequence of G-modules

$$0 \to A \xrightarrow{\varphi} M \to C \to 0$$

where $M = M_1^G(A)$ is cohomologically trivial for G and where $C = M/\text{im } \varphi$. Then the Dimension Shifting for Cohomology Groups then becomes:

Corollary 3.7.15. *2 of Shapiro's Lemma*
For any G-module A and anny $n \geq 1$, we have

$$H^{n+1}(G, A) \cong H^n(G, M_1^G(A)/A)$$

3.7.3 Important Maps between Cohomology Groups

Intuition: In general, suppose we have two groups G and G' and that A is a G-module and A' is a G'-module. If $\varphi : G' \to G$ is a group homomorphism, then A becomes a G'-module by defining $g' \cdot a := \varphi(g')a$ for $g' \in G'$ and $a \in A$. If now $\psi : A \to A'$ is a homomorphism of abelian groups, then we consider whether ψ is a G'-module homomorphism:

Definition 3.7.16. Compatible Group Homomorphisms
Let A be a G-module and let A' be a G'-module. Th group homomorphisms $\varphi : G' \to G$ and $\psi : A \to A'$ are said to be *compatible* if ψ is a G'-module homomorphism when A is made into a G'-module by means of φ, i.e.: if $\forall g' \in G', a \in A$,

$$\psi(g' \cdot a) = \psi(\varphi(g')a) = g'\psi(a)$$

Remark: The group homomorphism $\varphi : G' \to G$ induces a group homomorphism $\varphi^n : (G')^n \to G^n$ and hence a group homomorphism from $C^n(G, A)$ to $C^n(G', A)$ given by $f \mapsto f \circ \varphi^n$. Similarly, the group homomorphism $\psi : A \to A'$ induces a group homomorphism from $C^n(G', A)$ to $C^n(G', A')$ given by $f \mapsto \psi \circ f$. Therefore, taking the two induced group homomorphisms together, we obtain a final induced homomorphism

$$\lambda_n : {}_{f \mapsto \psi \circ f \circ \varphi^n}^{C^n(G,A) \to C^n(G',A')}$$

If in addition φ and ψ are *compatible* homomorphisms, then it's easy to check that the induced maps λ_n commute with the coboundary homomorphisms:

$$\lambda_{n+1} \circ d_n = d'_n \circ \lambda_n \quad \forall n \geq 0$$

which follows that λ_n maps cocycles to cocycles and coboundaries to coboundaries, and hence induces a group homomorphism on cohomology groups

$$\lambda_n : H^n(G, A) \to H^n(G', A')$$

for $n \geq 0$.

Example:

- Suppose $G = G'$ and $\varphi = \mathrm{id}_G$ is the identity map on G. Then the group homomorphism $\psi : A \to A'$ is immediately compatible with φ. Hence, any G-module homomorphism from A to A' induces a group homomorphism

  $$H^n(G, A) \to H^n(G, A') \text{ for } n \geq 0$$

  In particular, if $0 \to A \to B \to C \to 0$ is a short sequence of G-modules, then we obtain induced homomorphisms from $H^n(G, A)$ to $H^n(G, B)$ and from $H^n(G, B)$ to $H^n(G, C)$, for $n \geq 0$.

- (*Restriction Homomorphism*) If A is a G-module, then A is also an H-module for any subgroup H of G. Consider the inclusion map $\varphi : H \hookrightarrow G$ and the identity map $\psi = \mathrm{id}_A : A \to A$ on A. Then it's clear that φ and ψ are compatible homomorphisms. The corresponding induced group homomorphism on cohomology is called the *restriction homomorphism*:

  $$\mathrm{Res} : H^n(G, A) \to H^n(H, A), n \geq 0$$

- (*Inflation Homomorphism*) Suppose H is a normal subgroup of G and A is a G-module. Note that the elements A^H of A that are fixed by H are naturally a (G/H)-module under the action given by

  $$(gH) \cdot a = g \cdot a, \forall g \in G, a \in A$$

  Hence, consider the projection map $\varphi : G \to G/H$ and this inclusion map $\psi : A^H \hookrightarrow A$, which are compatible homomorphisms. Then the corresponding group homomorphism on cohomology groups is called the *inflation homomorphism*:

  $$\mathrm{Inf} : H^n(G/H, A^H) \to H^n(G, A), n \geq 0$$

- (*Corestriction Homomorphism*) Suppose that H is a subgroup of G of index $m < \infty$ and that A is a G-module. Let $g_1, \cdots, g_m$ be representatives for the left cosets of H in G, i.e.: $G = g_1 H \cup \cdots \cup g_m H$. Define a map

  $$\psi : \substack{M_H^G(A) \to A \\ f \mapsto \sum_{i=1}^m g_i \cdot f(g_i^{-1})}$$

  Since $(g_i h) f((g_i h)^{-1}) = (g_i h) f(h^{-1} g_i^{-1}) = (g_i h h^{-1}) f(g_i^{-1}) = g_i f(g_i^{-1})$, then the map ψ is independent of the choice of coset representatives. It's easy to see that ψ is a G-module homomorphism and that it is surjective, so we obtain a group homomorphism $\widetilde{\psi}_n : H^n(G, M_H^G(A)) \to H^n(G, A), \forall n \geq 0$. Since A is also an H-module, then by the Shapiro's Lemma, we have an isomorphism $\Psi_n : H^n(H, A) \xrightarrow{\cong} H^n(G, M_H^G(A))$. Then the composition of these two homomorphisms is called the *corestriction homomorphism*:

$$\mathrm{Cor} := \widetilde{\psi}_n \circ \Psi_n : H^n(H, A) \to H^n(G, A)$$

More explicitly, we can write

$$\mathrm{Cor}(f)(g) = \sum_{i=1}^{m} g_i \cdot \Psi(f)(p)(g_i^{-1})$$

where Ψ is the isomorphism Ψ in the proof of Shapiro's Lemma and hence

$$\mathrm{Cor}(f)(p) = \sum_{i=1}^{m} g_i f(g_i^{-1} p)$$

for $p \in P_n$.

Proposition 3.7.17. *Fundamental Relation-between-Res-and-Cor Proposition*
Suppose H is a subgroup G of index m. Then $Cor \circ Res = m$, i.e.: if c is a cohomology class in $H^n(G, A)$ for some G-module A, then

$$Cor(Res(c)) = mc \in H^n(G, A), \forall n \geq 0$$

Proof. This follows from the explicit formula for corestriction, as follows. If $f \in \mathrm{Hom}_{\mathbb{Z}H}(P_n, A)$ is an element of $\mathrm{Hom}_{\mathbb{Z}G}(P_n, A)$, i.e.: if f is also a G-module homomorphism, then $g_i f(g_i^{-1} p) = g_i g_i^{-1} f(p) = f(p)$, for $1 \leq i \leq m$, where g_i are the representatives for the left cosets of H in G. Hence,

$$\begin{array}{ccc} \mathrm{Hom}_{\mathbb{Z}G}(P_n, A) \xrightarrow{\mathrm{Res}} & \mathrm{Hom}_{\mathbb{Z}H}(P_n, A) & \xrightarrow{\mathrm{Cor}} \mathrm{Hom}_{\mathbb{Z}G}(P_n, A) \\ & f \mapsto f \mapsto mf & \end{array}$$

$\square$

Corollary 3.7.18. *1 of Fundamental Relation-between-Res-and-Cor Proposition*
Suppose the finite group G has order m. Then $mH^n(G, A) = 0$ for all $n \geq 1$ and all G-modules A.

Proof. Let $H = 1$ be the trivial subgroup of G. Then $[G : H] = m$, so by the Fundamental Relation-between-Res-and-Cor Prop, $\forall c \in H^n(G, A)$, we have

$$mc = \mathrm{Cor}(\mathrm{Res}(c))$$

Since $\forall n \geq 1, \mathrm{Res}(c) \in H^n(H, A) = H^n(1, A) = 0$, then $\mathrm{Res}(c) = 0$. Hence, $mc = 0$. $\square$

Corollary 3.7.19. *2 of Fundamental Relation-between-Res-and-Cor Proposition*
If G is a finite group, then $H^n(G, A)$ is a torsion abelian group for all $n \geq 1$ and all G-modules A.

Corollary 3.7.20. *3 of Fundamental Relation-between-Res-and-Cor Proposition*
Suppose G is a finite group whose order is relatively prime to the exponent of the G-module A. Then $H^n(G, A) = 0$ for all $n \geq 1$. In particular, if A is a finite abelian group with $(|G|, |A|) = 1$, then $H^n(G, A) = 0$ for all $n \geq 1$.

Proof. By the coro.1 of Fundamental Relation-between-Res-and-Cor Prop, $H^n(G, A)$ is annihilated by $|G|$ and by the Annihilator Inheritance Property Prop, $H^n(G, A)$ is annihilated by the exponent of A. $\square$